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Diabetes, hypertension, body mass index, smoking and COVID-19-related mortality: A systematic review and meta-analysis of observational studies

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-052777
Article Type:	Original research
Date Submitted by the Author:	26-Apr-2021
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Keywords:	COVID-19, Hypertension < CARDIOLOGY, General diabetes < DIABETES & ENDOCRINOLOGY, EPIDEMIOLOGY

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Diabetes, hypertension, body mass index, smoking and COVID-19-related mortality: A systematic review and meta-analysis of observational studies

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Word: Abstract: 300; Full text: 3999

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Abstract

Objectives: We conducted a systematic literature review and meta-analysis of observational studies to investigate the association between diabetes, hypertension, body mass index (BMI) or smoking with the risk of death in patients with COVID-19 and to estimate the proportion of deaths attributable to these conditions.

Methods: Relevant observational studies were identified by searches in the PubMed and Embase databases through November 14, 2020. Random-effects models were used to estimate summary relative risks (SRRs) and 95% confidence intervals (CIs). Certainty of evidence was assessed using the Cochrane methods and the GRADE framework.

Results: A total of 186 studies representing 210,447 deaths among 1,304,587 patients with COVID-19 were included in this analysis. The SRR for death in COVID-19 patients was 1.54 (95% CI=1.44-1.64, I^2=92%, n=145, low certainty) for diabetes and 1.42 (95% CI=1.30-1.54, I^2=90%, n=127, low certainty) for hypertension compared to patients without each of these comorbidities. Regarding obesity, the SSR was 1.45 (95% CI=1.31-1.61, I^2=91%, n=54, high certainty) for patients with BMI ≥30kg/m^2 compared to those with BMI<30kg/m^2 and 1.12 (95%CI=1.07-1.17, I^2=68%, n=25) per 5 kg/m^2 increase in BMI. There was evidence of a J-shaped non-linear dose-response relationship between BMI and mortality from COVID-19, with the nadir of the curve at a BMI of around 22-24, and a 1.5-2 fold increase in COVID-19 mortality with extreme obesity (BMI of 40-50). The SRR was 1.28 (95% CI=1.29-1.50, I^2=74.0, n=28, low certainty) for ever, 1.29 (95% CI=1.03-1.62, I^2=84%) for current and 1.26 (95% CI=1.11-1.42, I^2=84%) for former smokers compared to never smokers. The proportion of deaths attributable to diabetes, hypertension, obesity, and smoking was 8%, 7%, 11%, and 2%, respectively.

Conclusion: Our findings suggest that diabetes, hypertension, obesity and smoking are major contributors to COVID-19 mortality accounting for nearly 30% of COVID-19 deaths.

Systematic review registration This study is registered with PROSPERO, CRD42020218115.

Keywords: diabetes; hypertension; obesity; smoking; COVID-19; mortality; meta-analysis

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Strengths and limitations of this study

- Our meta-analysis is, to the best of our knowledge, the first to estimate proportion of COVID-19 death attributable to diabetes, hypertension, obesity and smoking
- We did a systematic review and meta-analysis of 186 observational studies involving 210,447 deaths among 1,304,587 patients with COVID-19. We assessed certainty of evidence using the Cochrane methods and the GRADE framework.
- However, we did not investigate the association between presence of two or more coexisting comorbidities and risk of death in patients with COVID-19.

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Introduction

Coronavirus disease 2019 (COVID-19) is a viral infectious disease caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), which was first reported in Wuhan City, China, in December 2019[1]. SARS-CoV-2 has since spread to all countries worldwide and COVID-19 has been declared a pandemic by the World Health Organization (WHO)[2]. As of 24 April 2021, over 144,3 million cases and 3.1 million deaths have been reported globally since the start of the pandemic[3]. Age is the main risk factor for poor outcome in people with COVID-19 infection[4,5], as it is correlated with more comorbidities. About 70 to 87% of COVID-19 deaths are among people aged 70 years or older[6,7]. Patients with comorbidities, including diabetes, cardiovascular disease, respiratory disease, chronic kidney disease and others chronic diseases are at increased risk of developing severe or critical COVID-19[8,9], which may partly explain a greater mortality in hospital[10–14]. Studies suggest that about 20 to 51% of patients hospitalized with COVID-19 have at least one comorbidity[10,15].

Previous meta-analyses reported a higher mortality rate from COVID-19 in patients with comorbidities[16–22]. Ssentongo et al, based on 25 studies published from December 2019 to July 9th, 2020, suggested that diabetes and hypertension were respectively associated with a 1.48-fold and 1.82-fold greater risk of COVID-19 death compared to those without these comorbidities[16]. Based on studies published during the same period, Luo et al reported similar results[17]. Du et al found that patients with obesity had a 2.68-fold risk of dying from COVID-19 compared to non-obese patients[18]. Most of the published meta-analyses did not investigate the shape of the dose-response relationship between Body-Mass Index (BMI) and risk of death in order to clarify whether the association is dose-dependent or if there are threshold effects[20,23–25]. In addition, evidence suggests that smoking may increase risk of severe disease and death from COVID-19[19]. However, it is not clear whether such increase in COVID-19 mortality is different in current and past smokers since previous meta-analyses have not performed separate analyses[19,26,27].

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However, since the publication of these meta-analyses, several observational studies have been published on diabetes, hypertension, obesity or smoking and risk of death in patients with COVID-19[4,28–122] and the strength of the associations differed greatly between studies. Moreover, the proportion of deaths attributable to diabetes, hypertension, obesity or smoking habits has not been estimated. This last aspect may help adapting public health measures and vaccination strategies to populations at risk of severe COVID-19.

Given the rapidly increasing death from COVID-19 globally, and since diabetes, hypertension, obesity and tobacco smoking represent the most important public health problems worldwide, which contributed to higher risk of death globally; we thus conducted a systematic review and meta-analysis of published observational studies to investigate the association between diabetes, hypertension, smoking and obesity and risk of death in patient with COVID-19 and to clarify the strength of these associations. We further estimated the proportion of deaths attributable to these conditions.

Materials and methods

Search strategy and selection criteria

The meta-analysis was reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement[123] and Meta-analysis of Observational Studies in Epidemiology (MOOSE)[124]. This study was registered and accepted in the International Prospective Register of PROSPERO (registration number CRD42020218115) in October 2020. PubMed (MEDLINE) and Embase databases were searched to identify relevant articles published in English from December 2019 to 14th November 2020. The search terms that we used are provided in the **supplementary text**. In addition, we searched the reference lists of the relevant publications, reviews and meta-analyses to identify additional potentially relevant studies. We only included observational studies (cohort studies and cross-sectional) that reported relative risk estimates (such as hazard ratios (HR), risk ratios (RR) or odds ratios (OR)) with the 95% confidence intervals (CIs) with or without adjustment for potential confounders. The

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search was independently screened by two researchers (Y.MS and T.F) and discrepancies were resolved by discussion with a third researcher (M.ER).

Data collection

From each included publication, we extracted results and study characteristics which included first author's last name, publication year, country where the research was conducted in, study design, study description or name, study period, sample size with number of deaths, exposure, categories, risk estimate and 95% CIs, and adjustment factors. Data was extracted by Y.MS and extractions were checked for accuracy by T.F. Discrepancies were resolved through discussion with a third researcher (M.ER).

Quality assessment and risk of bias

The quality of individual studies was assessed independently by two researchers (Y.MS and T.F) using the Cochrane risk of bias tool ROBINS-I, which grades studies on a scale from critical risk of bias to low risk of bias considering bias due to confounding, selection of study participants, exposure measurement, misclassification of exposure during follow-up, missing data, measurement of outcomes, and bias due to selection of reported results[125]. Following the assessment of risk of bias, the body of evidence for each comorbidity and risk of death was rated independently using the GRADE (Grading of Recommendations, Assessment, Development and Evaluations) approach[126]. Discrepancies were resolved through discussion with a third researcher (D.A).

Data analysis

We used random effects models that consider both within study and between-study variation to calculate summary RRs (SRR) (95% CIs) of COVID-19 mortality for patients with diabetes compared to those without diabetes, for patients with hypertension versus those without hypertension, for obese versus non-obese and for current, former and ever smoker compared to never smokers. The natural logarithm of the RRs was weighted using random effects weights[127]. Statistical heterogeneity between studies was assessed by the Cochrane Q test and the I^2 statistic[128]. We calculated the absolute Risk difference (RD) from the

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baseline risk of mortality (BR) from Docherty et al[129], large cohort and Relative Risk (RR) using the formula RD=BR×(RR−1)[130].

We further performed a dose-response analysis for the associations between BMI and COVID-19 mortality using the method described by Greenland and Longnecker to compute the linear trend from the natural logs of the RRs and CIs across categories of BMI[131]. We calculated SRRs and 95% CIs for a 5-unit increment in BMI using random effects models. This method required mean or median of BMI, RRs and 95% CIs for at least three categories. The mean or median BMI level per category was used if provided in the publication, and if not, the midpoint of the upper and lower boundaries was estimated as a range in each category. When the highest and lowest categories were open-ended, we used the width of the adjacent interval to estimate the upper and lower boundaries for the category. For studies that reported results separately for young and adults, for current and former smoker, but not overall, we pooled the results using a fixed-effects model as reported by the Hamling procedure to obtain an overall estimate to be used in the meta-analysis[132].

To explore the potential nonlinear dose-response relation between BMI and mortality among patients with COVID-19, we used restricted cubic splines with 3 knots at 10th, 50th, and 90th percentiles of the distribution[133,134]. Subgroup and meta-regression analyses were conducted to investigate potential sources of heterogeneity. Small-study effects, such as publication bias, were visually assessed by examining funnel plots for asymmetry, and with Egger's test[135], and the results were considered to indicate potential small-study bias when p-values were<0.10. We conducted sensitivity analyses excluding one study at a time to clarify whether the results were driven by one large study or a study with an extreme result.

We finally calculated the population attributable fraction (PAF) of mortality among patients with COVID-19 due to diabetes, hypertension, obesity and smoking, worldwide using the following formula[136]:

$$PAF = \frac{\mathrm{p} \times (RR - 1)}{[\mathrm{p} \times (RR - 1)] + 1}$$

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Where RR was the relative risk, p was the prevalence of the exposure in patient with COVID-19. The prevalence of diabetes (11.5%), hypertension (22.9 %), obesity (29%) and smoking (9%) were obtained from previous meta-analyses[137–139].

Patient and public involvement

Patients or the public were not involved in any aspect of the study design, conduct or in the development of the research question or outcome measures.

Results

A total of 3,896 records were identified in MEDLINE and in EMBASE (**Figure 1)**. We excluded 3,118 publications after reading title and abstract. Among 778 full-text articles retrieved, 430 were excluded as not meeting the inclusion criteria, leaving a total of 348 publications. Of these, 162 articles were not eligible because they lacked sufficient data[140,141], reported no risk estimate[140,141], were duplicates, because they had identical populations[142–152] or were retracted[153]. Finally, a total of 186 observational studies were included in this meta-analysis. Of the included studies, 58 were from Europe, 58 from North America, 60 from Asia, 6 from South America and 4 from Africa (**Table S1)**. From the 186 publications assessed using the ROBINS-I tool, 92 were evaluated as being at low risk of bias, 49 at moderate risk of bias, 21 at serious risk of bias and 24 at critical risk of bias because of insufficient adjustment of relevant confounders (**Table S2**).

Diabetes and mortality in patient with COVID-19

A total of 145 studies[4,28–119,121,122,129,154–202] were included in the analysis of the association between diabetes and mortality, including a total of 198,491 deaths among 1,165,897 patients with COVID-19. The SRR for diabetes patients compared to those without diabetes was 1.54 (95% CI=1.44-1.64) and there was high heterogeneity among studies (I^2=92%). We found that diabetes increased the absolute risk of death by 14% (**Table 1**). The funnel plot did not provide evidence of publication bias (**Figure S1**) and we found no evidence of small study effects (Egger's test, P-value=0.54).

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The positive association persisted across all subgroups analyses stratified by study design, number of patients, geographic location, and adjustment for some confounding factors (**Table 2**). However, meta-regression analyses suggested some indication of heterogeneity between studies that adjusted for multiple factors such as age, sex and other comorbidities versus studies without such adjustment or adjusted for age only ($P_{heterogeneity}$=0.003), with a stronger association for the studies with no adjustment factors or those adjusted for age only. The positive association was also stronger among studies with critical or serious risk of bias compared to those with moderate or low risk of bias, with heterogeneity detected across studies ($P_{heterogeneity=}$0.001).

The influence analysis showed no substantial influence of any of the included studies on the global estimate of diabetes and mortality. The SRR ranged from 1.51 (95% CI=1.42-1.61) when excluding the Barbu et al. study[35] to 1.55 (95% CI=1.45-1.64) when excluding the Meng et al. study[114] (**Figure S2)**.

Hypertension and mortality in patient with COVID-19

We included 127 observational studies[4,29,30,32,33,35–39,42–45,47–50,52–56,58,59,61–63,65,66,69,71–76,78,81–84,86,88–90,92–95,97,103,105,105,106,106,108,110–112,114–118,120–122,154,155,157,157,157–160,162–168,170,174–181,183–189,194–196,196–201,203–209,209–222] in the analysis of the association between hypertension and mortality, including a total of 113,243 deaths among 934,958 patients with COVID-19. The SRR for hypertension patients versus those without hypertension was 1.42 (95% CI=1.30-1.54) with high evidence of heterogeneity (I^2=90%) (**Table 1**). The absolute risk of death for COVID-19 patients with hypertension compared to without hypertension was increased by 11%. Small study effects, such as publication bias, were not indicated with the funnel plot (**Figure S3**) or Egger's test (P-value=0.26).

Here again, the positive association persisted in most subgroup analyses (**Table 2**). Our meta-regression analysis showed that the study design and the geographic location did not significantly influence the magnitude of the overall association. However, heterogeneity between subgroup analyses was observed in analyses stratified by risk of bias with higher association for studies with critical compared to those with low, moderate or serious risk of bias ($P_{heterogeneity}$=<0.0001) and by adjustment for

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confounding factors with higher association for studies without adjustment compared to those that adjusted for age only or for multiple factors ($P_{heterogeneity}$=<0.0001).

When excluding the most influential studies, we found no substantial influence of any of the included studies (**Figure S4**).

BMI and mortality in patient with COVID-19

Fifty-four studies [29,32,38,44,45,48,49,52,61,62,64,65,67,69,73,74,79,84–86,88–90,93,99,103,107,118,120–122,129,155,165,166,179,182,183,185,193,197,198,203,208,223–231] investigated BMI and mortality risk including 145,605 deaths among 858,374 patients with COVID-19 were identified. The analysis yielded a SRR of 1.45 (95% CI=1.31-1.61) for obese (BMI≥30kg/m^2) versus non-obese (BMI<30kg/m^2) patients, with high heterogeneity detected between studies (I^2=91%) (**Figure 2**) (**Table 1**). We found that obesity increased the absolute risk of death by 12%. There was no statistically significant evidence of publication bias (P-value Egger's test=0.92) (**Figure S5**).The meta-regression analysis showed that study characteristics did not significantly alter the overall estimate (**Table S3**). When excluding the most influential studies one by one, the global estimate did not substantially change (**Figure S6**).

Twenty-five studies (32,072 deaths among 95,852 patients with COVID-19) [38,56,61,74,86,89,91–93,97,108,111,112,162,163,176,178,224,225,232–238] were included in the dose-response meta-analysis of BMI and mortality risk. The summary RR for a 5 kg/m^2 increment in BMI was 1.12 (95% CI=1.07-1.17, I^2=68%) (**Figure 3**) and no statistically significant evidence of publication bias (P-value=0.11) or by inspection of the funnel plot was observed (**Figure S7**). However, evidence of a J-shaped nonlinear relation between BMI and mortality risk was observed ($P_{nonlinearity}$=<0.0001), suggesting a flat dose–response curve at a BMI around 22-24 kg/m^2 with a slight increase in risk of death below that range and a 1.5-2 fold increase in risk of death with a BMI of 40-50 vs. 22-24 kg/m^2 (**Figure 4**).

In sensitivity analyses excluding one study at a time from the analysis, the summary for a 5 kg/m^2 increment in BMI ranged from 1.11 (95% CI=1.06-1.15) when excluding the Czernichow et al. Study[108] to 1.13 (95% CI=1.08–1.19) when excluding the Ferrando-Vivas et al. study[233] (**Figure S8**).

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Smoking and mortality in patient with COVID-19

Twenty-eight studies[38,58,61,63,66,67,76,93,97,103,108,110,111,118,121,129,162,165–167,176,178,184,205,235,239–241] were included in the analysis of ever smoking versus never smoking and mortality with a total of 11,333 deaths among 47,096 COVID-19 patients. The SRR for hospital death in patient with COVID-19 was 1.28 (95% CI=1.17-1.40, I^2=74%) for ever smokers versus never smokers (**Figure 5**). The absolute risk of death for smoking was increased by 7% (**Table 1**). There was no evidence of publication or small study bias (P-value=0.91) (**Figure S9**). In sensitivity analyses, the results persisted when excluding one study at a time (**Figure S10**).

For current smoking versus never smoking, nineteen studies [4,58,66,76,97,103,110,111,117,118,121,129,162,163,166,178,204,239,242], which included 9,845 deaths among 33 147 patients with COVID-19, were identified. The SRR of current smoking was 1.29 (95%CI=1.03-1.62, I^2=84%) (**Figure 6**) and no statistically significant evidence of publication bias was observed (P-value=0.86) (**Figure S11**).

A total of 14 studies[58,66,76,97,103,110,111,118,121,129,162,166,178,239] were included in the analysis of former smoking versus never smoking and mortality risk, including 8,121 deaths among 25,340 patients with COVID-19. The SRR was 1.25 (95% CI=1.11-1.42) with moderate to high heterogeneity (I^2=75%) (**Figure 7**). There was no evidence of publication bias with Egger's test (P-value=0.70). In sensitivity analyses excluding one study at a time from the analyses of current and former smoking, the results were not materially altered (**Figures S12-13**). Globally, results did not change in nearly all subgroup analyses (**Table S4**).

Finally, a total of 15 studies[29,44,52,58,58,107,175,187,193,210,221,243–246] did not provide a definition of the smoking variable (>76,400 deaths, 682,310 patients) and the SRR was 1.31 (95% CI=1.07-1.62, I^2=88%) (**Figure S14**) which seemed to be similar to current and ever smoking.

Population attributable fraction of deaths and assessment of certainty of the body of evidence

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The estimated PAF was 10.6 %, 11.8 %, 11.5 % and 2.5 % for diabetes, hypertension, obesity and ever smoking, respectively when considering all studies included in this meta-analysis (**Table 1)**. Based on studies that adjusted for multiple risk factors, attributable death was 8% for diabetes, 7% for hypertension, 11% for obesity and 2% for ever smoking.

Regarding the body of evidence, certainty of evidence was rated was "low" for diabetes, hypertension and smoking and "high" for obesity (**Table S5**).

Discussion

To our knowledge, this study is one of the largest meta-analyses of association between diabetes, hypertension, obesity, smoking and COVID-19 mortality. We found that patients with diabetes had a 54% higher risk of death from COVID-19 compared to patients without diabetes; those with hypertension had a 42% increase in the relative risk of death from COVID-19 compared to patients without hypertension and those with obesity have a 45% greater in the relative risk of death compared to non-obese patients. In addition, we found that ever, current and former smoking was associated with 28%, 29% and 25% increases in the relative risk of death in COVID-19 patients. Our linear dose-response meta-analysis suggested that each 5 kg/km^2 increment in BMI was associated with a 12% greater risk of death. However, evidence of nonlinearity was observed in the analysis of BMI and risk of death, with a J-shaped dose-response relation with flattening of the dose–response curve between 22 and 24 of BMI level and a slight increase below that range and a 1.5-2 fold increase in risk of death with a BMI of 40-50. While there was no publication bias, study heterogeneity was high for all exposure and this persisted in most of the subgroup analyses. However, the heterogeneity appeared to be driven to a larger extent by differences in the strength of the associations, than differences in the direction of the effect, as the vast majority of studies reported significant or non-significant positive associations between these exposures and increased mortality, and relatively few studies reported risk estimates in the direction of an inverse association.

Since the first reports of COVID-19, several studies have shown that COVID-19 patients with comorbidities have a higher risk of death[4,247–249]. However, these studies have differed greatly in term of

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sample size, and results are conflicting and heterogeneous. Previous meta-analyses have shown that patients with diabetes, hypertension and obesity had an increased risk of mortality[16,18,21,249–251]. Unfortunately, these previous meta-analyses were limited by the lack of subgroup analyses, which is crucial to evaluate heterogeneity and no previous meta-analysis has estimated the number of deaths attributed to these comorbidities or conditions.

This meta-analysis summarizes the results of 186 observational studies published up to November 2020, including 210,447 deaths among 1,304,587 million patients with COVID-19. Our findings are similar to results from previously published systematic reviews, suggesting a higher mortality rate of COVID-19 in patients with cardiovascular or chronic condition[16,21,22,249]. We found a 1.54-fold greater mortality from COVID-19 among patient with diabetes compared with those without (n=145 studies), which is similar to those yielded in previous meta-analyses[16] whereas, the estimate magnitude is weaker than those in Almeida-Pititto et al., 2020[21], Kumar et al., 2020[250], Shang et al.[252] and Luo et al., 2020[17]. In our subgroup analyses, we found that the positive association was stronger in studies without any adjustment or adjusted for age only compared to studies that adjusted for multiple factors. Thus, the higher magnitude observed in the previous meta-analyses may be in part due to the important number of studies without adjustment for confounding factors such as age and comorbidities, which are mostly with critical or serious risk of bias. In addition, we estimated that 8% of deaths by COVID-19 were attributed to diabetes; this aspect has never been investigated previously. The absolute risk of death associated with diabetes was increased by 14%. The mechanism underlying the increased mortality from COVID-19 in patients with diabetes may be explained by chronic inflammatory conditions. COVID-19 patients with diabetes have a significantly higher inflammatory markers such as C-reactive protein, interleukin-6 (IL-6) compared to patients without diabetes[253]. Inflammatory markers such as IL-6, CRP, IL-10, LDH and TNF-α, which are indicative of different aspects of COVID-19 severity, requirements of intensive care support including dialysis and ventilation, are associated with higher risk of death[254,255].

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Regarding hypertension, our findings yielded a 1.42-fold higher risk of death from COVID-19 in patient with hypertension (n=127 studies), which also is slightly weaker than the results from previous meta-analyses[16,22,251]. Nearly 7% of death in patients with COVID-19 could be attributed to hypertension and we found that hypertension increased the absolute risk of death by 11%. Previous clinical studies showed that hypertension is a major risk factor for worse outcome in patients infected with SARS and Middle East respiratory syndrome[256,257]. Although, the exact mechanism by which hypertension increase mortality rate remains unclear, chronic inflammation may play an active role in increasing risk of death. Studies suggested that cytokine storm is associated with COVID-19 severity[10].

Du et al suggested that patients with obesity had a 2.68-fold risk for COVID-19 mortality compared with non-obese patients (n=7 studies)[18]. In dose-response analysis, they showed that for each 1 kg/m^2 increase in BMI, the risk of death increased by 6%. Partly consistent with this study, we found that a 5 kg/m^2 increase in BMI level was associated with a 13% increase in the risk of death in patient with COVID-19 (n=25 studies). In addition, we found evidence of a nonlinear J-shaped association between BMI and mortality from COVID-19, with a flattening of the dose–response curve for BMI values between 22 and 24 kg/m^2 and with a slightly higher risk below that range and a moderate to strong increase in mortality with severe obesity (BMI 40-50). Obesity is associated with a low-grade systemic inflammation, which plays a major role in the pathogenesis of respiratory conditions. Patients with COVID-19 and with preexisting obesity may have an overactivated inflammation response, which may induce excessive inflammatory response[258].

In addition, our finding showed that ever, current and former smoking was associated with 28%, 29% and 26% increases in the relative risk of death in patients with COVID-19 compared to never smokers. Tobacco smoking is known to alter the function of the immune system; therefore smokers are more likely to get a severe infection of COVID-19 due to their poor mucociliary clearance, which could lead to the release of pro-inflammatory markers and oxidative stress and thereby contribute to higher mortality rates. The risk of death in former smokers was higher than in never smokers and only slightly

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lower than risk in current smoker. However, more studies are required to clarify the impact of longer durations of smoking cessation in former smokers on risk of death in COVID-19 patients.

This present meta-analysis of observational studies on diabetes, hypertension, obesity and smoking and risk of death in patients with COVID-19 has several strengths, including the large sample size and number of COVID-19 deaths, the detailed subgroup and sensitivity analyses, as well as the linear and nonlinear dose-response analyses, which clarified the strength and shape of the dose-response relationship. Original aspects of our study included the estimation of death attributable to these conditions. This is, to our knowledge, the first meta-analysis that perform a separate analysis of ever, current and former smoking versus never smoking in relation to COVID-19 mortality. As any previous published meta-analyses, the current analysis has some limitations that should be considered in the interpretation of our findings. First, we did not investigate the association between presence of two or more coexisting comorbidities and risk of death in patients with COVID-19. Finally, subgroup analyses stratified by clinical or lifestyle factors such as medications, diabetes type and duration, adherence to specific diet, or physical activity were not possible because of the lack of such data from the studies included.

Despite these limitations, our findings may have important public health implications in the context of increasing numbers of severe COVID-19 cases, overburdened hospitals and leading to higher hospital death due to COVID-19 and suggest that people with cardiovascular risk factors, in particular those with diabetes, hypertension and obesity, should be considered as a high priority to get vaccinated. In addition, since smoking is a risk factor for several chronic diseases, including cancer and cardiovascular disease, our finding lend support to the importance of smoking prevention and smoking cessation and support policies and public health efforts to reduce the prevalence smoking in the general population.

In conclusion, our finding suggests that presence of diabetes, hypertension, obesity, and smoking in patients with COVID-19 is associated with a 1.54-fold, 1.42 fold, 1.45 fold and 1.28 fold greater risk of mortality, respectively. We have found that the proportion of death attributable to diabetes, hypertension, obesity, and smoking was 8%, 7%, 11%, and 2% respectively. These findings support that people with

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diabetes, hypertension, obesity should be prioritized for vaccination in order to limit the higher death rates in hospital. Public policies should promote a healthier lifestyle including healthier diets and regular physical activity to reduce patient risk factors and comorbidities.

Author Contributions: Y.MS conceived and designed the research. Y.MS performed statistical analysis and wrote the first draft of the paper. YMS, TF and MR performed the literature search and literature screening. Y.MS and TF assessed the risk of bias of the studies and assessed the certainty of evidence of the associations. MM assisted with the statistical analysis. All of the authors contributed to data interpretation and revision of the manuscript for intellectual content. Y.MS takes responsibility for the integrity of the data and the accuracy of the data analysis.

Conflict of Interest Disclosures: We declare no competing interests

Funding: There was no funding source for this study

Data sharing statement: All data relevant to the study are included in the article or uploaded as supplementary information.

Provenance and peer review: Not commissioned; externally peer reviewed.

Patient consent for publication: Not required

Ethics approval statement: We adhered to the Preferred Reporting Items for Systemic Reviews and Meta-Analyses (PRISMA) guidelines. Ethical committee approval was not required for this study.

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Table 1: Summary relative risk, risk difference and fraction of deaths attributable to diabetes, hypertension, obesity and smoking in patients with COVID-19

	Number of studies	RR (95% CI)	I^2 (%)	Risk difference (95% CI) (%)	PAF (95% CI) (all included studies) (%)	PAF (95% CI) (for fully adjusted studies) (%)	Number of patients	Number of deaths
Prognostic factors								
Diabetes	145	1.54 (1.44-1.64)	92	+14 (+11 to +17)	10.6 (8.8 to 12.4)	8.1 (6.2 to 9.7)	1,165,897	198,491
Hypertension	127	1.42 (1.30-1.54)	90	+11 (+8 to +14)	11.8 (8.7 to 14.7)	6.9 (4.0 to 9.6)	934,958	113,243
Obesity	54	1.45 (1.31-1.61)	91	+12 (+8 to +16)	11.5 (8.2 to 15.0)	11.1 (7.1 to 15.0)	858,374	145,605
Smoking	28	1.28 (1.17-1.40)	74	+7 (+4 to +10)	2.45 (1.5 to 3.5)	2.0 (0.8 to 3.3)	47,096	11,333

Footnote: RR summary relative risk; CI confidence interval, PAF population attributable fraction of mortality

I^2 (%) is a measure of the proportion of the heterogeneity attributed to between study variation rather than due to chance. I^2-valuesof 25%, 50% and 75% indicates low, moderate and high between study heterogeneity, respectively.

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Table 2. Subgroup analyses of association between diabetes and hypertension and mortality risk in patients with COVID-19

	Diabetes					Hypertension				
	n	RR (95% CI)	I^2 (%)	P_{within}[a]	$P_{between}$[b]	n	RR (95% CI)	I^2 (%)	P_{within}[a]	$P_{between}$[b]
All studies	145	1.54 (1.44-1.64)	91.7	<0.0001		127	1.42 (1.30-1.54)	90.1	<0.0001	
Study design										
Retrospective	129	1.54 (1.44-1.64)	91.7	<0.0001		112	1.42 (1.30-1.56)	90.9	<0.0001	
Prospective	11	1.28 (1.12-1.48)	70.0	<0.0001		9	1.21 (0.98-1.50)	73.4	<0.0001	
Cross-sectional	5	2.47 (1.56-3.94)	81.4	<0.0001	0.05	6	1.70 (1.18-2.44)	61.8	0.02	0.50
Geographical location										
North America	45	1.33 (1.19-1.48)	90.3	<0.0001		40	1.30 (1.12-1.50)	93.4	<0.0001	
Europe	46	1.53 (1.36-1.72)	93.8	<0.0001		43	1.36 (1.20-1.54)	82.8	<0.0001	
Asia	43	1.94 (1.65-2.29)	75.1	<0.0001		38	1.62 (1.28-2.04)	86.1	<0.0001	
South America	6	1.54 (1.27-1.87)	93.7	<0.0001		2	2.03 (0.99-4.18)	69.1	0.072	
Africa	5	1.56 (0.92-2.62)	87.3	<0.0001	0.01	4	1.08 (0.86-1.35)	14.9	0.318	0.33
Number of patients										
<1000	79	1.73 (1.52-1.97)	74.8	<0.0001		78	1.56 (1.36-1.79)	74.8	<0.0001	
≥1000	66	1.43 (1.32-1.55)	95.4	<0.0001	0.001	49	1.28 (1.14-1.44)	94.8	<0.0001	<0.0001
Risk of bias										
Low	72	1.46 (1.34-1.59)	92.4	<0.0001		63	1.17 (1.08-1.29)	80.2	<0.0001	
Moderate	37	1.28 (1.19-1.39)	64.9	<0.0001		34	1.34 (1.13-1.59)	88.9	<0.0001	
Serious	17	1.90 (1.44-2.51)	86.4	<0.0001		15	1.83 (1.42-2.37)	84.4	<0.0001	
Critical	19	2.11 (1.61-2.77)	93.4	<0.0001	0.001	15	2.64 (1.61-4.34)	96.3	<0.0001	<0.0001
Adjustment for confounders										
No	21	2.09 (1.62-2.69)	89.4	<0.0001		17	2.54 (1.62-3.99)	95.9	<0.0001	
Age only	18	1.90 (1.56-2.31)	85.4	<0.0001		14	1.78 (1.36-2.32)	85.1	<0.0001	
Multiple	106	1.40 (1.30-1.49)	87.5	<0.0001	0.003	96	1.23 (1.13-1.33)	85.4	<0.0001	<0.0001

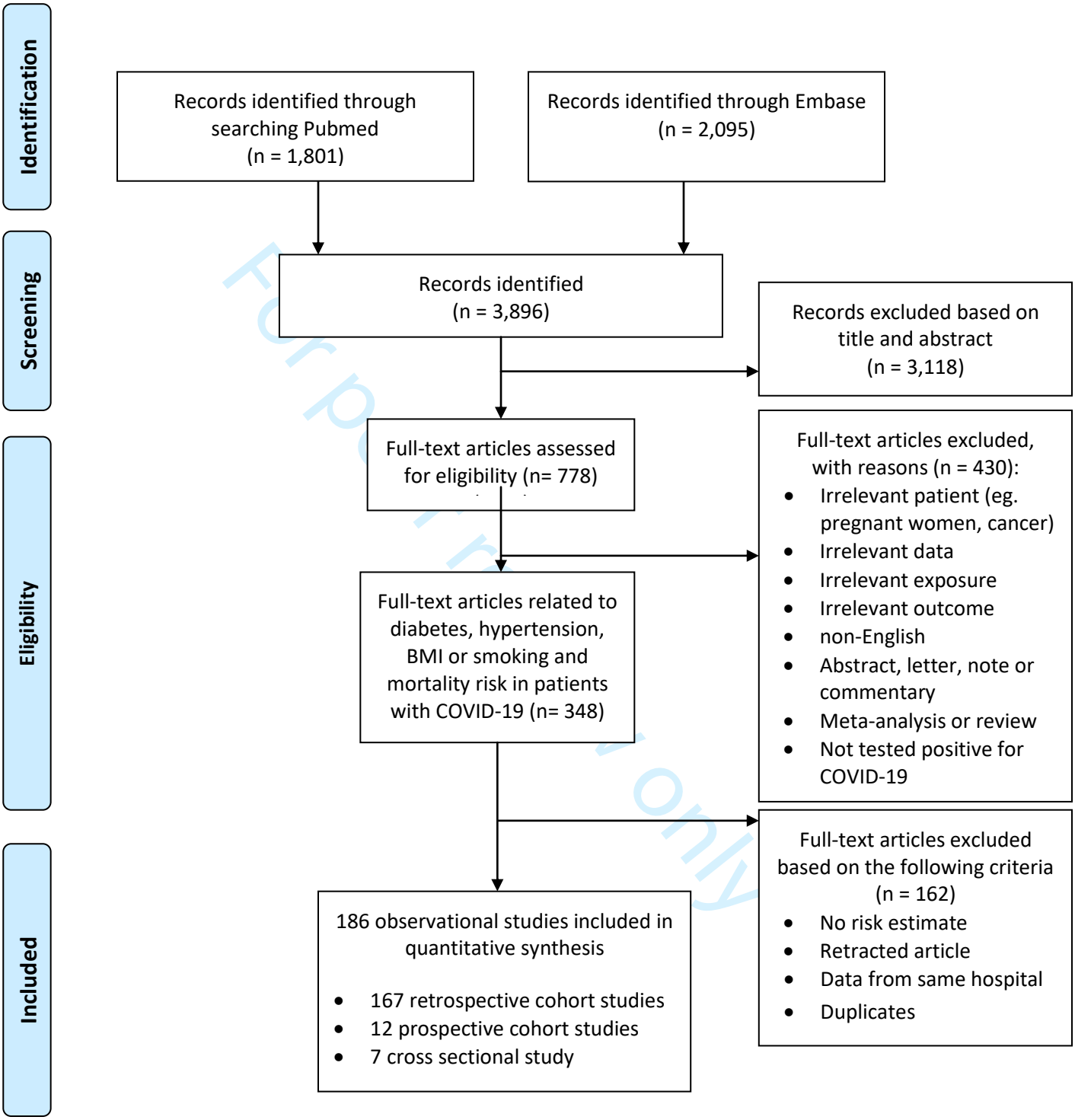
Footnote: RR summary relative risk; CI confidence interval

I^2 (%) is a measure of the proportion of the heterogeneity attributed to between study variation rather than due to chance. I^2-valuesof 25%, 50% and 75% indicates low, moderate and high between study heterogeneity, respectively.

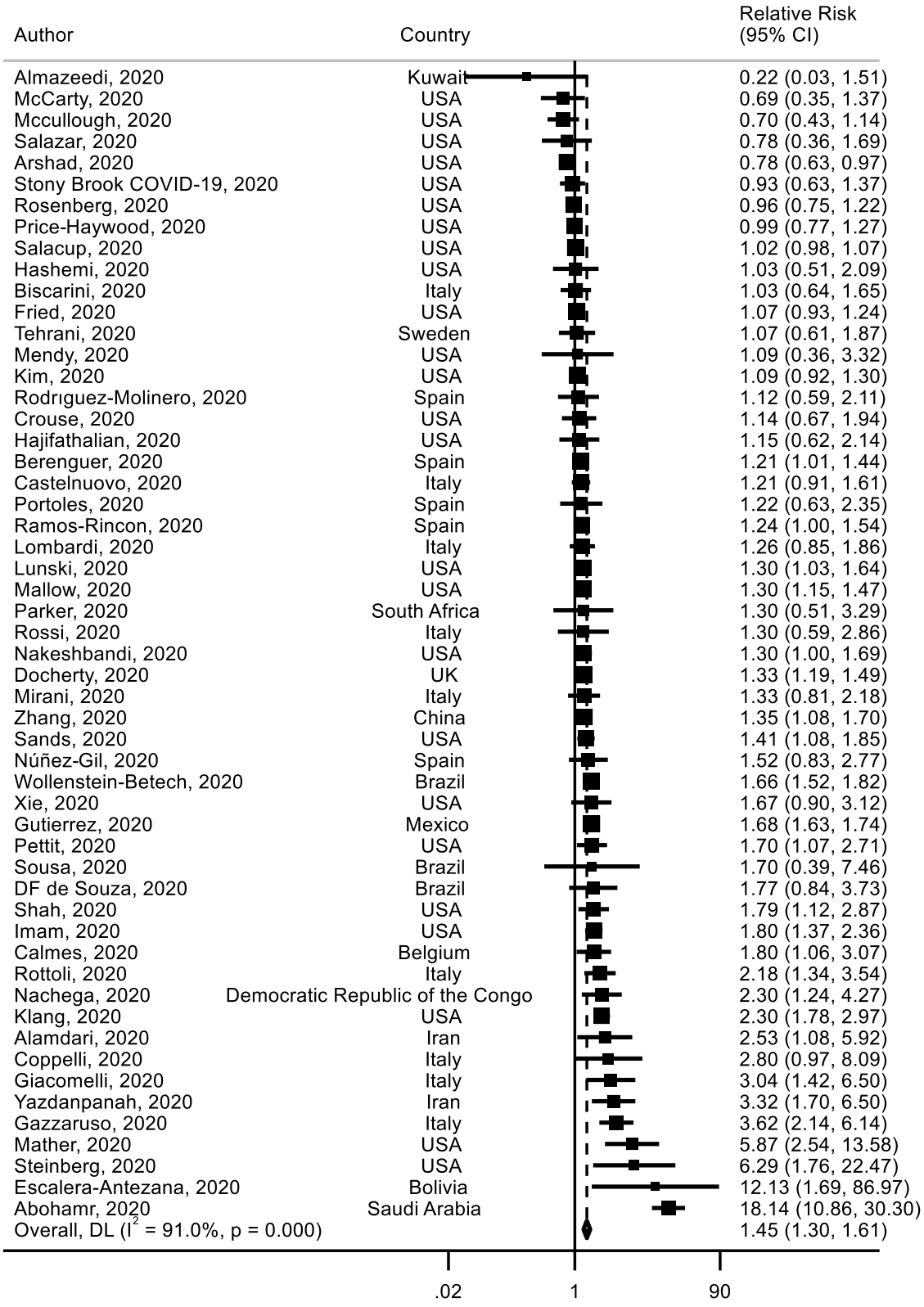
[a]P-value for heterogeneity among the studies within each cancer type

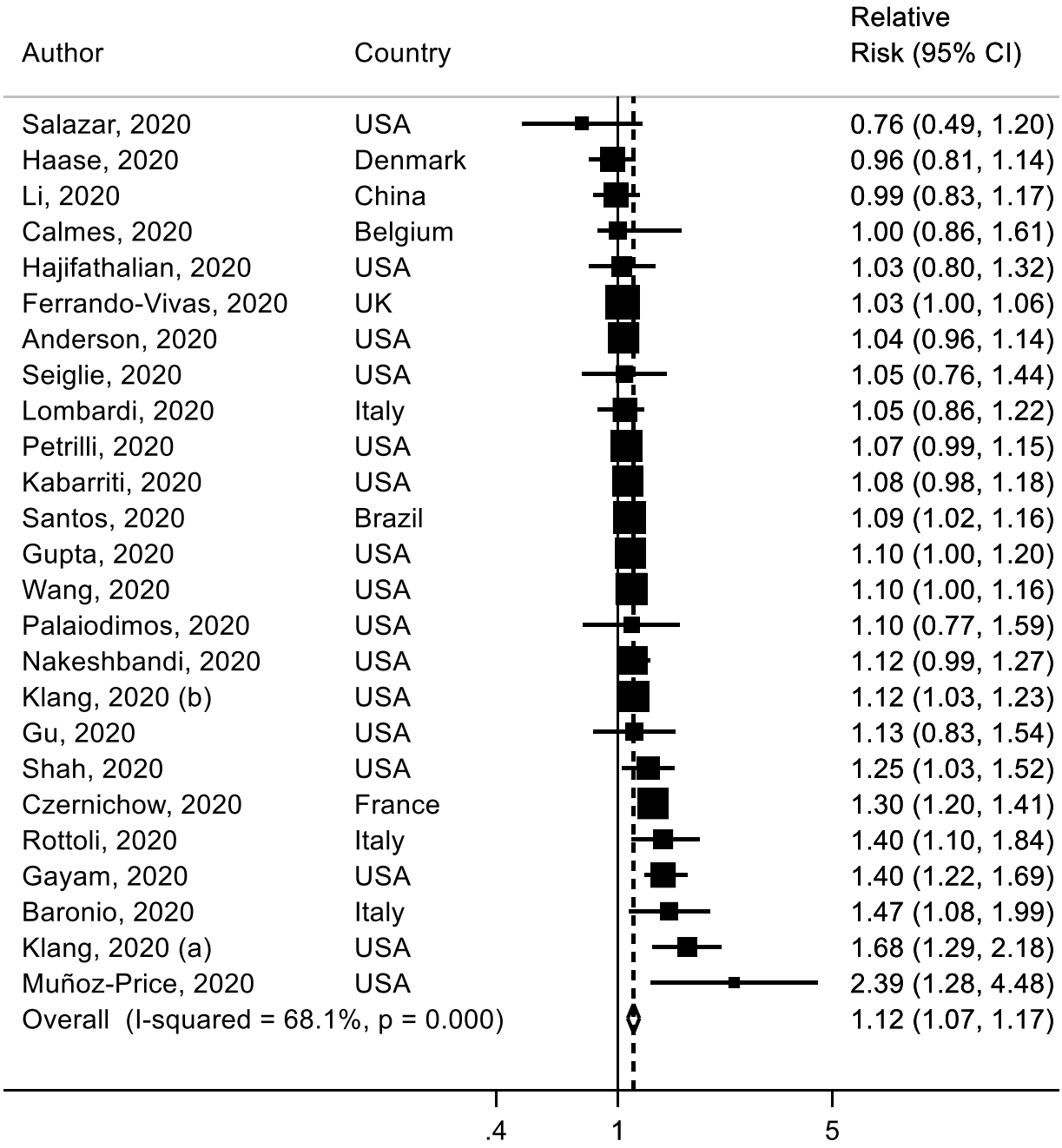
[b]P-value for between subgroup or category heterogeneity generated from meta-regression analysis

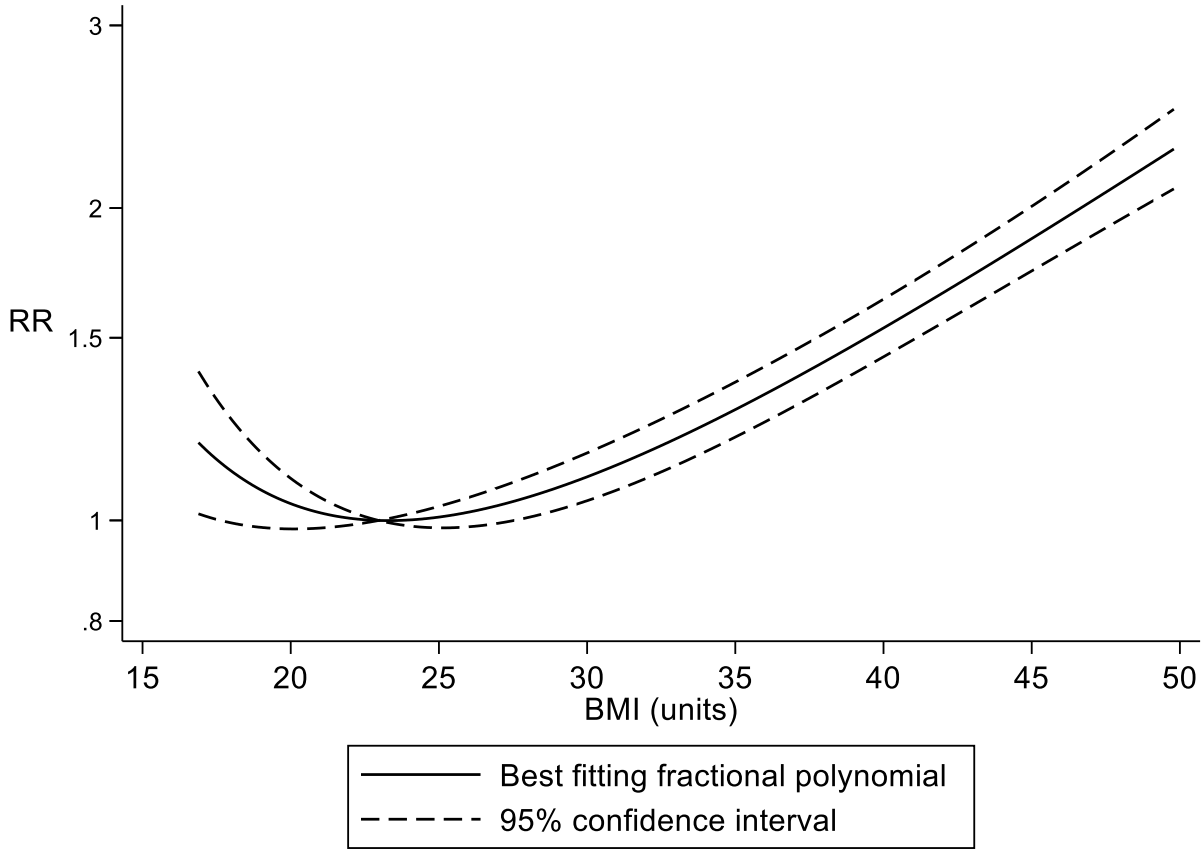
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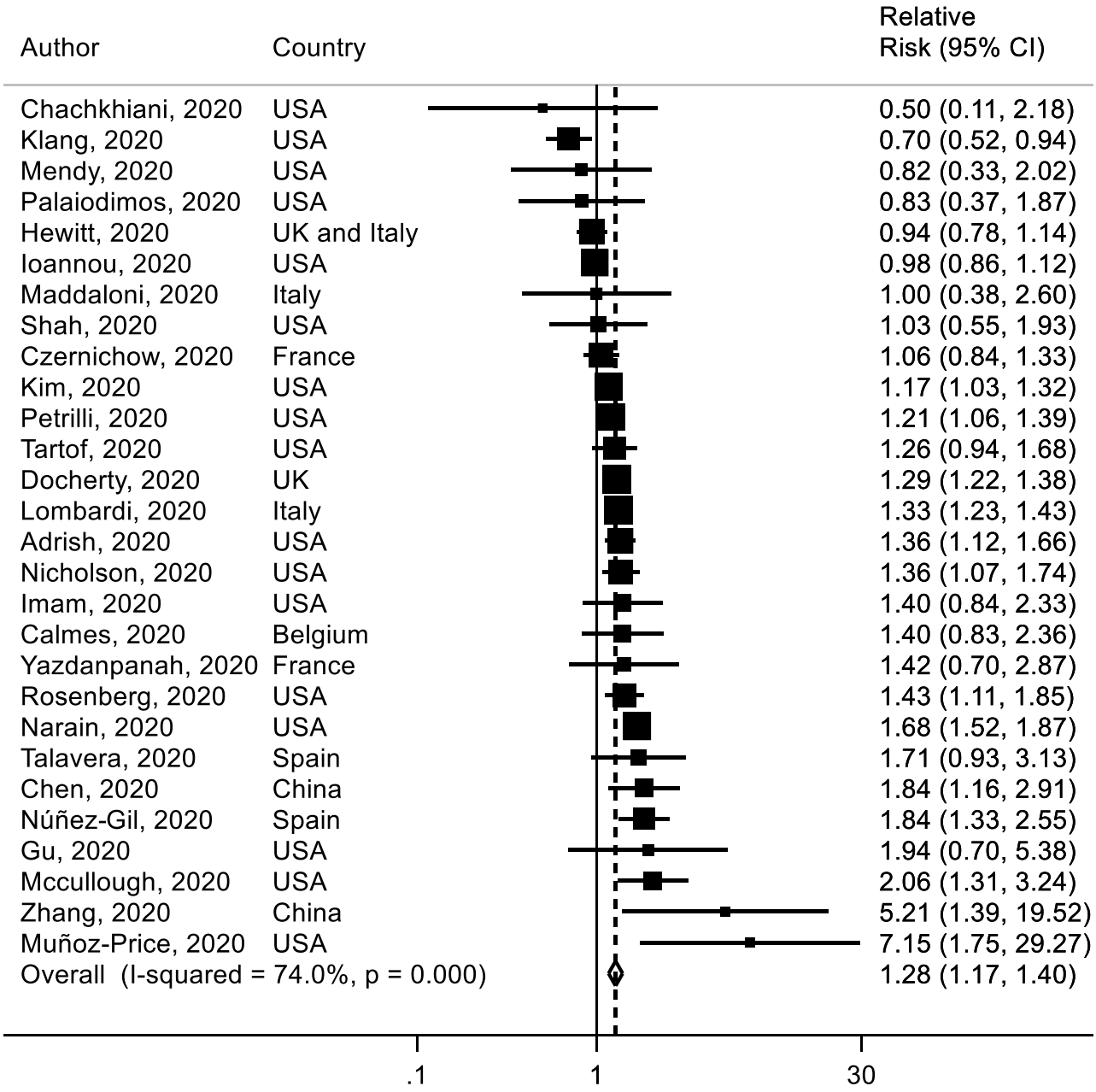
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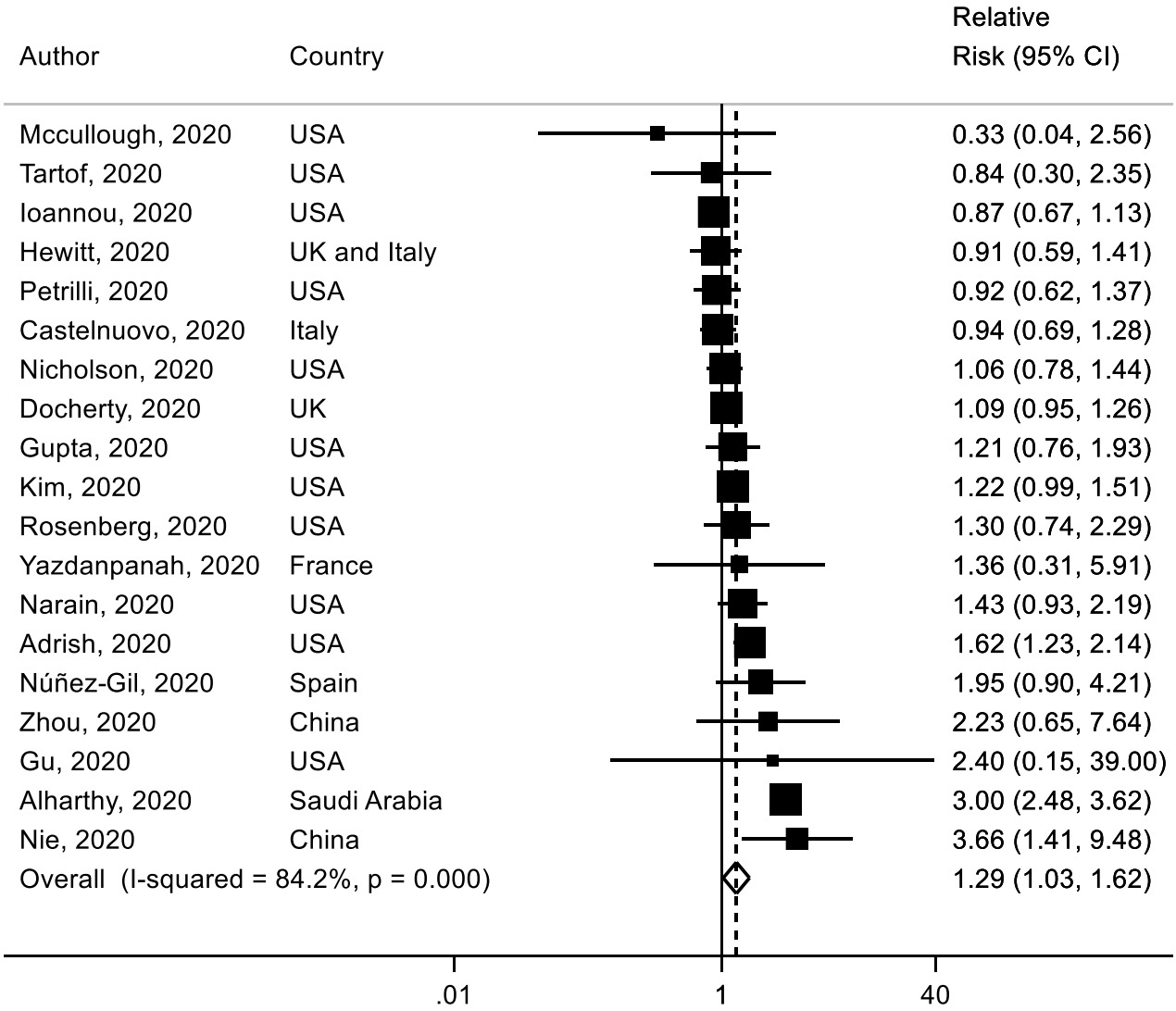

Author
Country
Relative Risk (95% CI)
Almazeedi, 2020 Kuwait 0.22 (0.03, 1.51)
McCarty, 2020 USA 0.69 (0.35, 1.37)
Mccullough, 2020 USA 0.70 (0.43, 1.14)
Salazar, 2020 USA 0.78 (0.36, 1.69)
Arshad, 2020 USA 0.78 (0.63, 0.97)
Stony Brook COVID-19, 2020 USA 0.93 (0.63, 1.37)
Rosenberg, 2020 USA 0.96 (0.75, 1.22)
Price-Haywood, 2020 USA 0.99 (0.77, 1.27)
Salacup, 2020 USA 1.02 (0.98, 1.07)
Hashemi, 2020 USA 1.03 (0.51, 2.09)
Biscarini, 2020 Italy 1.03 (0.64, 1.65)
Fried, 2020 USA 1.07 (0.93, 1.24)
Tehrani, 2020 Sweden 1.07 (0.61, 1.87)
Mendy, 2020 USA 1.09 (0.36, 3.32)
Kim, 2020 USA 1.09 (0.92, 1.30)
Rodrıguez-Molinero, 2020 Spain 1.12 (0.59, 2.11)
Crouse, 2020 USA 1.14 (0.67, 1.94)
Hajifathalian, 2020 USA 1.15 (0.62, 2.14)
Berenguer, 2020 Spain 1.21 (1.01, 1.44)
Castelnuovo, 2020 Italy 1.21 (0.91, 1.61)
Portoles, 2020 Spain 1.22 (0.63, 2.35)
Ramos-Rincon, 2020 Spain 1.24 (1.00, 1.54)
Lombardi, 2020 Italy 1.26 (0.85, 1.86)
Lunski, 2020 USA 1.30 (1.03, 1.64)
Mallow, 2020 USA 1.30 (1.15, 1.47)
Parker, 2020 South Africa 1.30 (0.51, 3.29)
Rossi, 2020 Italy 1.30 (0.59, 2.86)
Nakeshbandi, 2020 USA 1.30 (1.00, 1.69)
Docherty, 2020 UK 1.33 (1.19, 1.49)
Mirani, 2020 Italy 1.33 (0.81, 2.18)
Zhang, 2020 China 1.35 (1.08, 1.70)
Sands, 2020 USA 1.41 (1.08, 1.85)
Núñez-Gil, 2020 Spain 1.52 (0.83, 2.77)
Wollenstein-Betech, 2020 Brazil 1.66 (1.52, 1.82)
Xie, 2020 USA 1.67 (0.90, 3.12)
Gutierrez, 2020 Mexico 1.68 (1.63, 1.74)
Pettit, 2020 USA 1.70 (1.07, 2.71)
Sousa, 2020 Brazil 1.70 (0.39, 7.46)
DF de Souza, 2020 Brazil 1.77 (0.84, 3.73)
Shah, 2020 USA 1.79 (1.12, 2.87)
Imam, 2020 USA 1.80 (1.37, 2.36)
Calmes, 2020 Belgium 1.80 (1.06, 3.07)
Rottoli, 2020 Italy 2.18 (1.34, 3.54)
Nachega, 2020 Democratic Republic of the Congo 2.30 (1.24, 4.27)
Klang, 2020 USA 2.30 (1.78, 2.97)
Alamdari, 2020 Iran 2.53 (1.08, 5.92)
Coppelli, 2020 Italy 2.80 (0.97, 8.09)
Giacomelli, 2020 Italy 3.04 (1.42, 6.50)
Yazdanpanah, 2020 Iran 3.32 (1.70, 6.50)
Gazzaruso, 2020 Italy 3.62 (2.14, 6.14)
Mather, 2020 USA 5.87 (2.54, 13.58)
Steinberg, 2020 USA 6.29 (1.76, 22.47)
Escalera-Antezana, 2020 Bolivia 12.13 (1.69, 86.97)
Abohamr, 2020 Saudi Arabia 18.14 (10.86, 30.30)
Overall, DL (I^2 = 91.0%, p = 0.000) 1.45 (1.30, 1.61)
.02
1
90

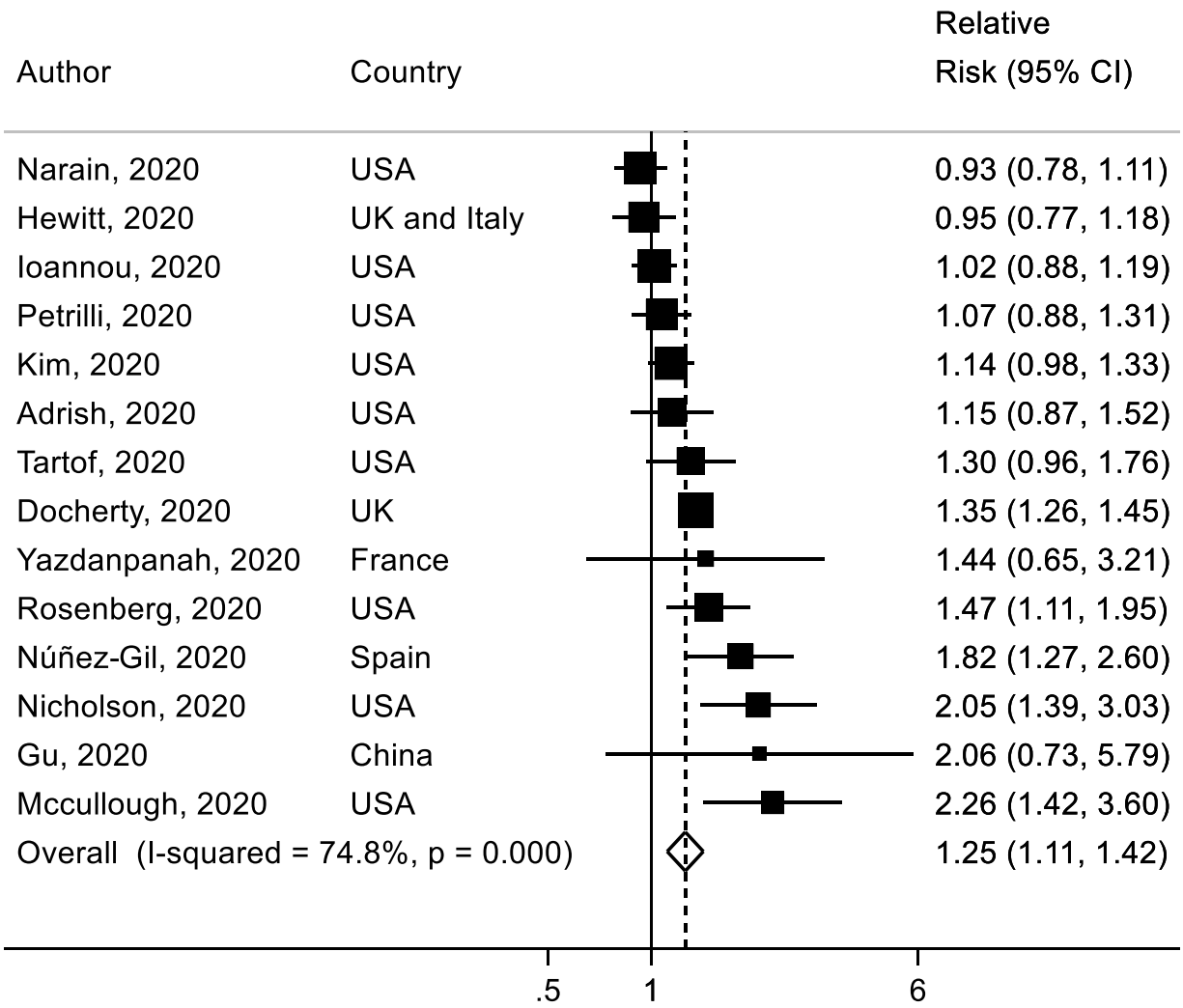

Author
Country
Relative Risk (95% CI)
Salazar, 2020 USA 0.76 (0.49, 1.20)
Haase, 2020 Denmark 0.96 (0.81, 1.14)
Li, 2020 China 0.99 (0.83, 1.17)
Calmes, 2020 Belgium 1.00 (0.86, 1.61)
Hajifathalian, 2020 USA 1.03 (0.80, 1.32)
Ferrando-Vivas, 2020 UK 1.03 (1.00, 1.06)
Anderson, 2020 USA 1.04 (0.96, 1.14)
Seiglie, 2020 USA 1.05 (0.76, 1.44)
Lombardi, 2020 Italy 1.05 (0.86, 1.22)
Petrilli, 2020 USA 1.07 (0.99, 1.15)
Kabarriti, 2020 USA 1.08 (0.98, 1.18)
Santos, 2020 Brazil 1.09 (1.02, 1.16)
Gupta, 2020 USA 1.10 (1.00, 1.20)
Wang, 2020 USA 1.10 (1.00, 1.16)
Palaiodimos, 2020 USA 1.10 (0.77, 1.59)
Nakeshbandi, 2020 USA 1.12 (0.99, 1.27)
Klang, 2020 (b) USA 1.12 (1.03, 1.23)
Gu, 2020 USA 1.13 (0.83, 1.54)
Shah, 2020 USA 1.25 (1.03, 1.52)
Czernichow, 2020 France 1.30 (1.20, 1.41)
Rottoli, 2020 Italy 1.40 (1.10, 1.84)
Gayam, 2020 USA 1.40 (1.22, 1.69)
Baronio, 2020 Italy 1.47 (1.08, 1.99)
Klang, 2020 (a) USA 1.68 (1.29, 2.18)
Muñoz-Price, 2020 USA 2.39 (1.28, 4.48)
Overall (I-squared = 68.1%, p = 0.000) 1.12 (1.07, 1.17)
.4
1
5


RR
3
2
1.5
1
.8
15
20
25
30
35
40
45
50
BMI (units)
Best fitting fractional polynomial
95% confidence interval

only


Author
Country
Relative Risk (95% CI)
Chachkhiani, 2020 USA 0.50 (0.11, 2.18)
Klang, 2020 USA 0.70 (0.52, 0.94)
Mendy, 2020 USA 0.82 (0.33, 2.02)
Palaiodimos, 2020 USA 0.83 (0.37, 1.87)
Hewitt, 2020 UK and Italy 0.94 (0.78, 1.14)
Ioannou, 2020 USA 0.98 (0.86, 1.12)
Maddaloni, 2020 Italy 1.00 (0.38, 2.60)
Shah, 2020 USA 1.03 (0.55, 1.93)
Czernichow, 2020 France 1.06 (0.84, 1.33)
Kim, 2020 USA 1.17 (1.03, 1.32)
Petrilli, 2020 USA 1.21 (1.06, 1.39)
Tartof, 2020 USA 1.26 (0.94, 1.68)
Docherty, 2020 UK 1.29 (1.22, 1.38)
Lombardi, 2020 Italy 1.33 (1.23, 1.43)
Adrish, 2020 USA 1.36 (1.12, 1.66)
Nicholson, 2020 USA 1.36 (1.07, 1.74)
Imam, 2020 USA 1.40 (0.84, 2.33)
Calmes, 2020 Belgium 1.40 (0.83, 2.36)
Yazdanpanah, 2020 France 1.42 (0.70, 2.87)
Rosenberg, 2020 USA 1.43 (1.11, 1.85)
Narain, 2020 USA 1.68 (1.52, 1.87)
Talavera, 2020 Spain 1.71 (0.93, 3.13)
Chen, 2020 China 1.84 (1.16, 2.91)
Núñez-Gil, 2020 Spain 1.84 (1.33, 2.55)
Gu, 2020 USA 1.94 (0.70, 5.38)
Mccullough, 2020 USA 2.06 (1.31, 3.24)
Zhang, 2020 China 5.21 (1.39, 19.52)
Muñoz-Price, 2020 USA 7.15 (1.75, 29.27)
Overall (I-squared = 74.0%, p = 0.000) 1.28 (1.17, 1.40)
.1
1
30


Author
Country
Relative Risk (95% CI)
Mccullough, 2020 USA 0.33 (0.04, 2.56)
Tartof, 2020 USA 0.84 (0.30, 2.35)
Ioannou, 2020 USA 0.87 (0.67, 1.13)
Hewitt, 2020 UK and Italy 0.91 (0.59, 1.41)
Petrilli, 2020 USA 0.92 (0.62, 1.37)
Castelnuovo, 2020 Italy 0.94 (0.69, 1.28)
Nicholson, 2020 USA 1.06 (0.78, 1.44)
Docherty, 2020 UK 1.09 (0.95, 1.26)
Gupta, 2020 USA 1.21 (0.76, 1.93)
Kim, 2020 USA 1.22 (0.99, 1.51)
Rosenberg, 2020 USA 1.30 (0.74, 2.29)
Yazdanpanah, 2020 France 1.36 (0.31, 5.91)
Narain, 2020 USA 1.43 (0.93, 2.19)
Adrish, 2020 USA 1.62 (1.23, 2.14)
Núñez-Gil, 2020 Spain 1.95 (0.90, 4.21)
Zhou, 2020 China 2.23 (0.65, 7.64)
Gu, 2020 USA 2.40 (0.15, 39.00)
Alharthy, 2020 Saudi Arabia 3.00 (2.48, 3.62)
Nie, 2020 China 3.66 (1.41, 9.48)
Overall (I-squared = 84.2%, p = 0.000) 1.29 (1.03, 1.62)
.01
1
40



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Supplementary text

PubMed search terms:

1) “Body mass index” OR body mass index [MeSH] OR BMI OR overweight OR overweight [MeSH] OR obesity OR obesity[MeSH] OR anthropometry OR anthropometry[MeSH] OR fatness OR “body fatness” OR “abdominal fatness” OR “abdominal obesity” OR abdominal obesity[MeSH] OR adiposity OR adiposity[MeSH]

2) Diabetes OR diabetes [MeSH] OR glucose OR glucose [MeSH] OR "medical history" OR medical history[MeSH]

3) “blood pressure” OR blood pressure [MeSH] OR hypertension OR hypertension[MeSH])

4) smoking OR smoking[MeSH] OR cigarette OR cigarette[MeSH] OR cigarettes OR cigarettes[MeSH] smoke OR smoke[MeSH] OR tobacco OR tobacco[MeSH] OR snus OR snuff OR snuff[MeSH] OR "environmental tobacco smoke” OR environmental tobacco smoke[MeSH] OR “passive smoking” OR passive smoking[MeSH] OR “smoking cessation" OR smoking cessation[MeSH]

5) COVID-19 OR SARS-CoV-2) AND (MORTALITY OR DEATH)

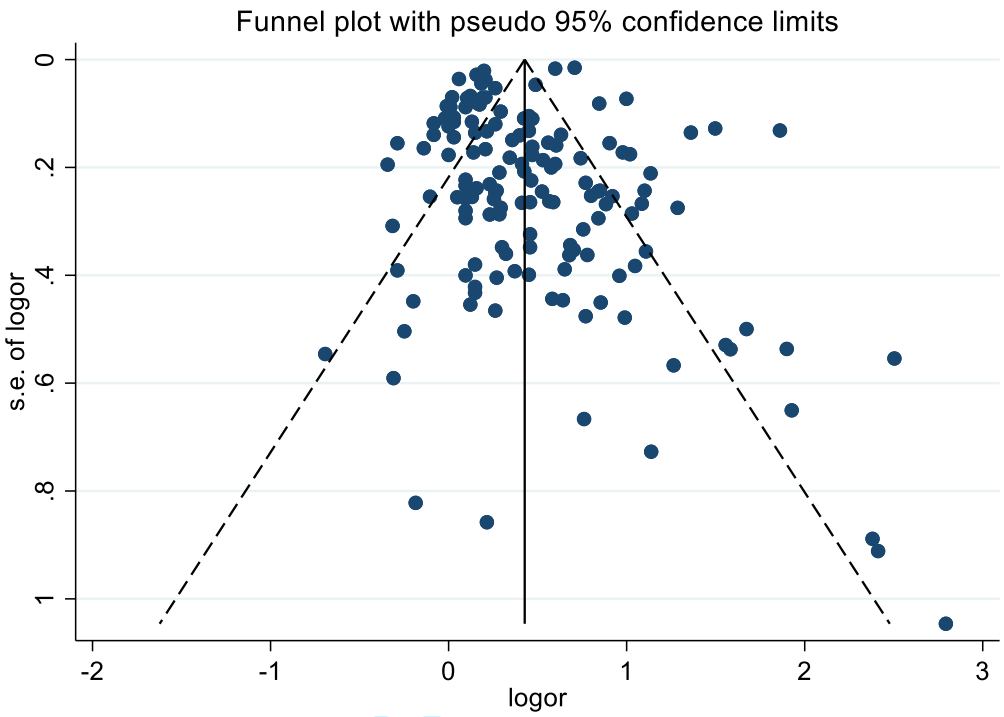
6) 1 OR 2 OR 3 OR 4

8) 6 AND 5

Embase search terms:

☐	1	(body mass index or bmi or overweight or obesity or anthropometry or fatness or body fatness or abdominal fatness or abdominal obesity or adiposity or diabetes or blood glucose or medical history or blood pressure or hypertension or smoking or cigarette or cigarettes or smoke or tobacco or snus or snuff or environmental tobacco smoke or passive smoking or smoking cessation).ab,ti.	2489416	Advanced
☐	2	body mass index/ or bmi/ or overweight/ or obesity/ or anthropometry/ or fatness/ or body fatness/ or abdominal fatness/ or abdominal obesity/ or adiposity/ or diabetes/ or blood glucose/ or medical history/ or blood pressure/ or hypertension/ or smoking/ or cigarette/ or cigarettes/ or smoke/ or tobacco/ or snus/ or snuff/ or environmental tobacco smoke/ or passive smoking/ or smoking cessation/	2393322	Advanced
☐	3	(COVID-19 or SARS-CoV-2).ab,ti.	66858	Advanced
☐	4	COVID-19/ or SARS-CoV-2/	95	Advanced
☐	5	(mortality or death).ab,ti.	2047722	Advanced
☐	6	mortality/ or death/	1077889	Advanced
☐	7	1 or 2	3256028	Advanced
☐	8	3 or 4	66898	Advanced
☐	9	5 or 6	2342175	Advanced
☐	10	7 and 8 and 9	2095	Advanced

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Supplementary Figure 1: Funnel plot of diabetes and mortality risk in patients with COVID-19

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Study omitted	Estimate	[95% Conf. Interval]
Albitar, 2020	1.5293853	1.4358113 1.6290579
Al-Salameh, 2020	1.5442724	1.4494652 1.645281
Hashemi, 2020	1.5388012	1.444257 1.6395345
Atkins, 2020	1.5295212	1.435613 1.6295722
Barbu, 2020	1.5139627	1.4230996 1.6106272
Barron, 2020	1.5295643	1.4380455 1.6269073
Boulle, 2020	1.5274702	1.4342617 1.626736
Bousquet, 2020	1.537884	1.4434227 1.638527
Calmes, 2020	1.5364693	1.4419811 1.6371492
Cao, 2020	1.5384178	1.4440672 1.6389329
Carrasco-Sánchez, 2020	1.5381721	1.4424546 1.6402413
Castelnuovo, 2020	1.5441017	1.4491032 1.645328
Chachkhiani, 2020	1.5411512	1.4464148 1.6420925
Chang, 2020	1.5396796	1.4451315 1.6404138
Chen, 2020	1.5313251	1.4371679 1.631651
Chen, 2020	1.5329227	1.4387097 1.6333052
Chilimuri, 2020	1.5377694	1.4431939 1.6385427
Ciardullo, 2020	1.5410501	1.4461583 1.6421683
Coppelli, 2020	1.5411114	1.446546 1.6418588
Crouse, 2020	1.5296242	1.435763 1.6296216
Cummings, 2020	1.5397375	1.4450227 1.6406603
Dennis, 2020	1.5427595	1.4469318 1.6449338
Desai, 2020	1.5401301	1.4455459 1.6409032
DF de Souza, 2020	1.5309588	1.4368759 1.6312019
Escalera-Antezana, 2020	1.5346913	1.4406917 1.6348242
Esme, 2020	1.5427371	1.4472408 1.6445348
Fang, 2020	1.5347261	1.4405845 1.63502
Fernandez-Cruz. , 2020	1.5323932	1.4382639 1.632683
Fried, 2020	1.5448198	1.4499104 1.6459421
Galloway, 2020	1.5412358	1.4462699 1.6424375
Ghany, 2020	1.5436468	1.448668 1.6448528
Grasselli, 2020	1.5424429	1.4473172 1.6438209
Gu, 2020	1.541069	1.4463067 1.6420403
Gu, 2020	1.5362892	1.4419804 1.6367661
Guisado-Vasco, 2020	1.5346212	1.4403979 1.6350082
Gupta, 2020	1.5425971	1.4475741 1.6438575
Gutierrez, 2020	1.5402496	1.441107 1.6462128
Haase, 2020	1.541796	1.4469641 1.6428429
Harrison, 2020	1.5433826	1.4483094 1.6446967
Narain, 2020	1.5444776	1.4496069 1.645557

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Hu, 2020	1.5414069	1.4467993	1.6422008
Imam, 2020	1.5233965	1.4304692	1.6223607
Kim, 2020	1.5359648	1.4414166	1.6367149
Kim, 2020	1.5423032	1.4471788	1.6436802
Klang, 2020	1.5443149	1.449327	1.6455281
Iaccarino, 2020	1.5368001	1.4421782	1.6376301
Lampasona, 2020	1.5336277	1.4393409	1.6340909
Lee, 2020	1.5330603	1.4390559	1.6332053
Li, 2020	1.5316471	1.4377936	1.6316268
Li, 2020	1.543546	1.4486066	1.6447074
Li, 2020	1.5410111	1.4462956	1.6419294
Liu, 2020	1.537012	1.4427184	1.6374683
Liu, 2020	1.5344501	1.4404455	1.6345894
Lombardi, 2020	1.5391879	1.444409	1.640186
Lunski, 2020	1.5452052	1.4502395	1.6463894
Maddaloni, 2020	1.5388812	1.4444	1.6395427
Mallow, 2020	1.5415205	1.4459832	1.64337
McCarty, 2020	1.5362325	1.4418561	1.6367861
Mccullough, 2020	1.5416657	1.4468532	1.6426914
Mendy, 2020	1.5371674	1.4427798	1.63773
Menezes-Soares, 2020	1.535192	1.4406074	1.6359866
Mikami, 2020	1.5455133	1.4505748	1.6466656
Mirani, 2020	1.5397638	1.4450005	1.6407416
Moon, 2020	1.530614	1.436552	1.6308349
Moon, 2020	1.5351865	1.4409643	1.6355697
Munblit, 2020	1.5340241	1.4396216	1.634617
Nachega, 2020	1.5413967	1.446624	1.6423783
Nakeshbandi, 2020	1.5433714	1.4483217	1.644659
Anzola, 2020	1.5396569	1.445129	1.640368
Nicholson, 2020	1.5342927	1.4400049	1.6347542
Nogueira, 2020	1.5475264	1.4526211	1.6486322
Palaiodimos, 2020	1.539917	1.4453287	1.6406955
Pan, 2020	1.5393479	1.444692	1.6402055
Perkin, 2020	1.5366874	1.4421818	1.637386
Petrilli, 2020	1.5445431	1.4495956	1.6457098
Pettit, 2020	1.542518	1.4479332	1.6432815
Poblador-Plou, 2020	1.538321	1.4433994	1.6394849
Portoles, 2020	1.535696	1.4411083	1.636492
Rastad, 2020	1.5370438	1.4424915	1.6377939
Recinella, 2020	1.5426378	1.4479485	1.6435194
Reilev, 2020	1.5376048	1.4427335	1.6387147

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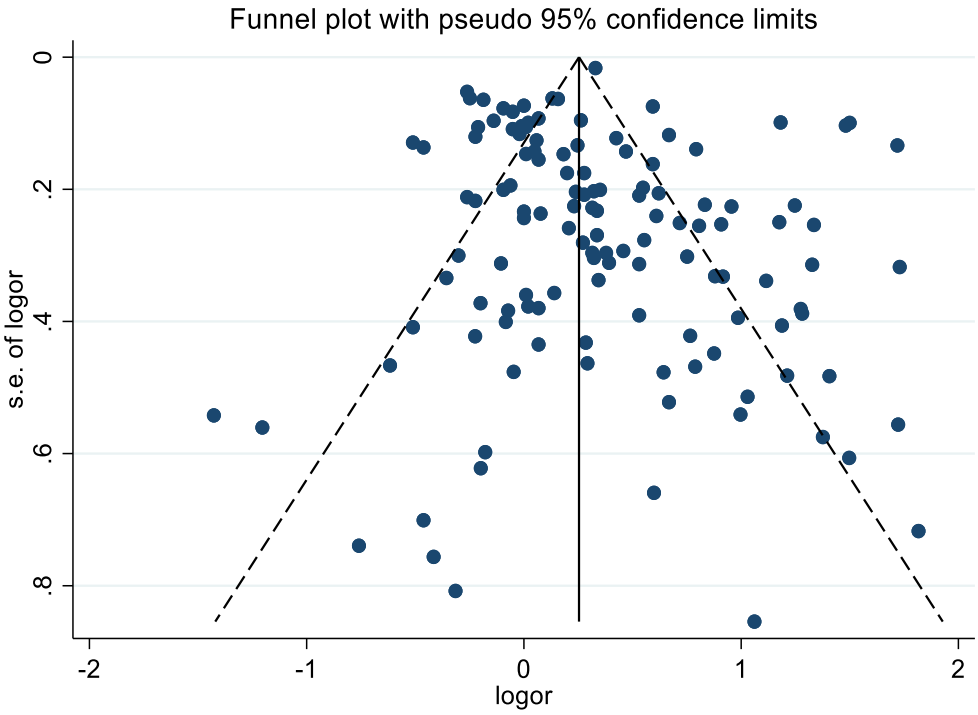
Rivera-Izquierdo, 2020	1.5335327 1.4392818 1.6339558
Rodrıguez-Molinero, 2020	1.5314742 1.4374141 1.6316895
Rossi, 2020	1.5375675 1.4428757 1.6384736
Rottoli, 2020	1.5418841 1.4470898 1.6428881
Rozaliyani, 2020	1.54023 1.4454752 1.6411963
Salacup, 2020	1.5383996 1.4439025 1.6390811
Salazar, 2020	1.5320424 1.4381227 1.6320957
Sands, 2020	1.5378478 1.4430354 1.6388896
Santos, 2020	1.5433739 1.4477333 1.6453327
Seiglie, 2020	1.5359882 1.441622 1.6365315
Shah, 2020	1.5382174 1.4436157 1.6390185
Shang, 2020	1.5354923 1.4411708 1.6359869
Shi, 2020	1.5378044 1.443289 1.6385093
Smith, 2020	1.5391345 1.4442916 1.6402056
Sousa, 2020	1.5397989 1.445085 1.6407204
Stony Brook COVID-19, 2020	Research Consortium 1.5362756 1.441714 1.6370395
Talavera, 2020	1.5393621 1.444719 1.6402053
Tartof, 2020	1.5421256 1.447147 1.6433378
Turcotte, 2020	1.5361555 1.4419419 1.6365249
Wollenstein-Betech, 2020	1.5441164 1.4472077 1.6475143
Wu, 2020	1.5398521 1.4451839 1.6407217
Wu, 2020	1.5378178 1.4433225 1.6384996
Xie, 2020	1.5365462 1.442167 1.6371019
Yahyavi, 2020	1.5406708 1.4456849 1.6418976
Yan, 2020	1.5409604 1.4461825 1.6419497
Yan, 2020	1.5381222 1.4434472 1.6390067
Yang, 2020	1.5376757 1.4430538 1.6385021
Yazdanpanah, 2020	1.5360827 1.4417021 1.636642
Yu, 2020	1.5335183 1.43924 1.6339724
Zandkarimi, 2020	1.5342196 1.4398625 1.6347601
Zhang, 2020	1.5332291 1.4392681 1.6333241
Zhou, 2020	1.53381 1.4396418 1.634138
Zhu, 2020	1.5385867 1.4437524 1.6396505
Zimering, 2020	1.5406566 1.4461459 1.6413438
An, 2020	1.5363024 1.441656 1.6371624
Almazeed, 2020	1.5392737 1.4448762 1.6398386
Bepouka, 2020	1.5390179 1.4444937 1.6397277
Biscarini, 2020	1.5398027 1.4451042 1.6407068
Abohamr, 2020	1.5307534 1.4367411 1.6309172
Czernichow, 2020	1.5404149 1.4453428 1.6417407
Docherty, 2020	1.5435574 1.4491569 1.6441072

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Foy, 2020	1.545491	1.4505337 1.6466644
Harmouch, 2020	1.5434103	1.4485663 1.6444643
Hewitt, 2020	1.544011	1.4490118 1.6452384
Ioannou, 2020	1.5431746	1.4480621 1.6445346
Kabarriti, 2020	1.5422004	1.4469972 1.6436675
Kolhe, 2020	1.5418558	1.4469479 1.6429888
Li, 2020	1.5343466	1.4403477 1.6344799
Meng, 2020	1.5472357	1.452306 1.6483704
Namendys-Silva, 2020	1.5329634	1.4388483 1.6332345
Neumann-Podczaska, 2020	1.5356869	1.4414098 1.6361301
Nie, 2020	1.5352678	1.4409221 1.6357909
Núñez-Gil, 2020	1.5301256	1.4361231 1.6302811
Park, 2020	1.5341833	1.4398587 1.6346871
Parker, 2020	1.5368429	1.4424789 1.6373801
Rosenberg, 2020	1.5379175	1.4430023 1.6390758
Shi, 2020	1.5396239	1.445085 1.6403475
Tehrani, 2020	1.5389986	1.4444335 1.6397549
Wang, 2020	1.5375615	1.4428405 1.6385009
Arshad, 2020	1.544768	1.4498014 1.6459552
Berenguer, 2020	1.5419745	1.4467498 1.6434668
Ciceri, 2020	1.5382968	1.4435889 1.6392183
Deng, 2020	1.5205251	1.428224 1.6187912
Wang, 2020	1.5330503	1.4390513 1.6331892
Combined	1.5379736	1.4437299 1.6383693

Supplementary Figure 2: Influence analysis of diabetes and mortality risk in patients with COVID-19

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Supplementary Figure 3: Funnel plot of hypertension and mortality risk in patients with COVID-19

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Study omitted	Estimate	[95% Conf. Interval]
Albitar, 2020	1.409848	1.2944931 1.5354825
Almazeedi, 2020	1.420794	1.3044698 1.5474913
An, 2020	1.4202995	1.3034456 1.5476295
Anzola, 2020	1.4250737	1.3083675 1.5521902
Arshad, 2020	1.42682	1.3096998 1.5544138
Atkins, 2020	1.4181341	1.3015437 1.5451684
Barbu, 2020	1.4032818	1.2905699 1.5258374
Bepouka, 2020	1.4205649	1.3040783 1.5474566
Berenguer, 2020	1.4144012	1.2976918 1.5416069
Biscarini, 2020	1.4260321	1.3090302 1.5534917
Boulle, 2020	1.4235396	1.3060689 1.551576
Bousquet, 2020	1.4184818	1.3022381 1.545102
Brouns, 2020	1.4248693	1.3082318 1.5519058
Calmes, 2020	1.4182589	1.3018024 1.5451334
Cao, 2020	1.4202471	1.3040493 1.5467987
Carrasco-Sánchez, 2020	1.4225355	1.3039615 1.5518919
Castelnuovo, 2020	1.4258839	1.3085872 1.5536945
Chachkhiani, 2020	1.4165195	1.3003861 1.5430244
Chen, 2020	1.4063388	1.2914803 1.5314124
Chen, 2020	1.4216918	1.3051524 1.5486373
Chen, 2020	1.4113356	1.2955974 1.5374128
Chilimuri, 2020	1.4177847	1.3014145 1.5445606
Ciardullo, 2020	1.4191939	1.3024282 1.5464277
Ciceri, 2020	1.4102607	1.2947223 1.5361094
Coppelli, 2020	1.4210031	1.3044831 1.5479311
Crouse, 2020	1.4099303	1.2945702 1.5355703
Cummings, 2020	1.416917	1.3006244 1.5436076
Czernichow, 2020	1.4236165	1.3062171 1.5515677
DE VITO, 2020	1.4222373	1.3059056 1.548932
DF de Souza, 2020	1.4171228	1.3003109 1.5444287
Deng, 2020	1.3963305	1.2868365 1.515141
Desai, 2020	1.4228125	1.3060952 1.54996
Du, 2020	1.4107752	1.295385 1.5364441
Escalera-Antezana, 2020	1.4110782	1.2955806 1.5368721
Esme, 2020	1.4221164	1.3035688 1.5514448
Fang, 2020	1.4226052	1.3060334 1.5495815
Foy, 2020	1.422677	1.3055518 1.55031
Fried, 2020	1.4274895	1.3108877 1.554463
Galloway, 2020	1.4198248	1.3028002 1.5473611

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Gao, 2020	1.4137607	1.2978215	1.5400571
Ghany, 2020	1.4183894	1.3019762	1.5452114
Giacomelli, 2020	1.4181474	1.3018019	1.5448911
Grasselli, 2020	1.423923	1.3065236	1.5518717
Gu, 2020	1.4185693	1.3020047	1.5455697
Guisado-Vasco, 2020	1.4154416	1.2991362	1.5421594
Gupta, 2020	1.4226934	1.3054665	1.5504467
Gutierrez, 2020	1.4256142	1.2931122	1.5716934
Haase, 2020	1.4236912	1.3067129	1.5511414
Harmouch, 2020	1.4136345	1.2976446	1.5399922
Hashemi, 2020	1.4149497	1.2990468	1.5411937
Hewitt, 2020	1.4244795	1.3071433	1.5523484
Hu, 2020	1.420043	1.3035676	1.5469255
Iaccarino, 2020	1.4224738	1.3056414	1.5497606
Imam, 2020	1.3971469	1.2870134	1.5167049
Ioannou, 2020	1.4247838	1.3071853	1.5529618
Kabarriti, 2020	1.4241887	1.3062947	1.5527228
Kim, 2020	1.4190946	1.3024367	1.5462012
Kim, 2020	1.4222981	1.3052446	1.5498487
Klang, 2020	1.4207023	1.303695	1.5482111
Lee, 2020	1.4160867	1.3001062	1.5424136
Li, 2020	1.4208537	1.304601	1.5474657
Li, 2020	1.4118376	1.2960577	1.5379604
Liu, 2020	1.4229921	1.3063434	1.5500568
Liu, 2020	1.4283701	1.3117267	1.5553858
Lombardi, 2020	1.4196392	1.3029786	1.5467449
Lopez-Mendez, 2020	1.4123951	1.2968653	1.5382168
Lunski, 2020	1.4306874	1.3140447	1.5576842
MCCULLOUGH, 2020	1.4113107	1.2956346	1.5373144
Maddaloni, 2020	1.4137343	1.2980008	1.5397871
Magleby, 2020	1.4250162	1.308177	1.5522907
Mallow, 2020	1.4274716	1.3114311	1.55378
Matsuzawa, 2020	1.4213893	1.3050824	1.5480613
McCarty, 2020	1.4147518	1.2988299	1.5410198
Meng, 2020	1.4232006	1.3060794	1.5508245
Mikami, 2020	1.4254555	1.3079046	1.5535716
Mirani, 2020	1.4148802	1.2987224	1.541427
Moon, 2020	1.4160312	1.3000233	1.5423911
Mowla, 2020	1.4159008	1.3000853	1.5420334
Munblit, 2020	1.4157315	1.2994165	1.5424583
Munoz, 2020	1.4121697	1.2966897	1.5379342

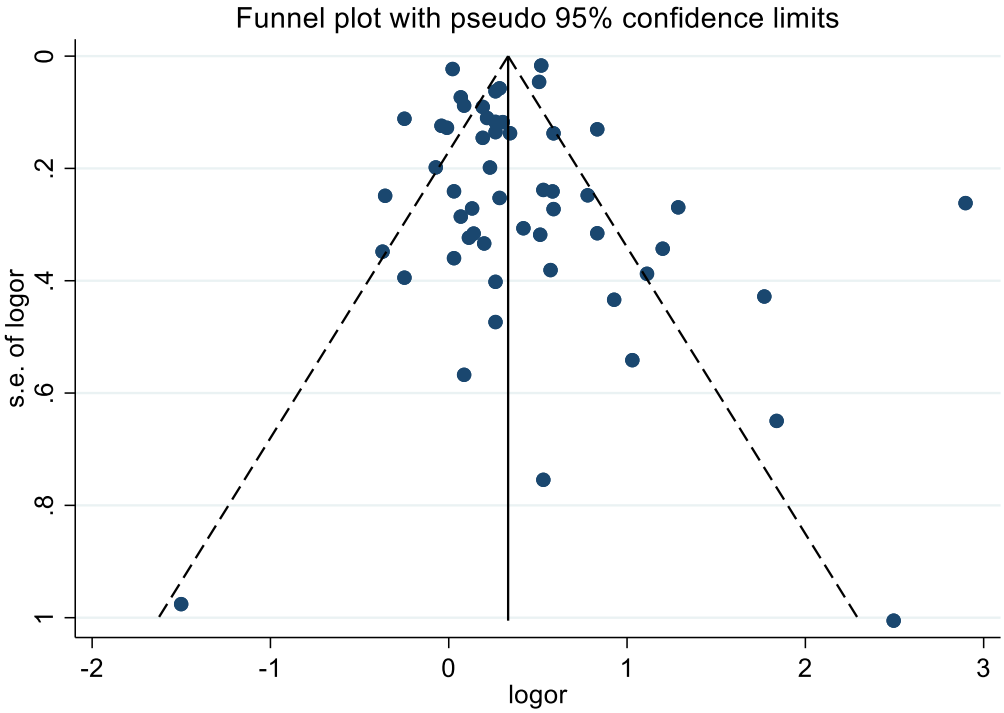
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Nachega, 2020	1.4223779 1.3055705 1.5496361
Nakeshbandi, 2020	1.4267973 1.3096594 1.5544121
Namendys-Silva, 2020	1.4127643 1.2969795 1.5388856
Narain, 2020	1.4267741 1.3096164 1.5544127
Nicholson, 2020	1.4146136 1.2983321 1.5413096
Nie, 2020	1.406029 1.2912035 1.5310657
Núñez-Gil, 2020	1.4157708 1.2995708 1.5423607
Palaiodimos, 2020	1.4217563 1.3051939 1.5487283
Pan, 2020	1.4125925 1.2967428 1.5387923
Parker, 2020	1.4141377 1.2983069 1.5403026
Perkin, 2020	1.4182899 1.3017601 1.5452511
Petrilli, 2020	1.4239464 1.3066469 1.5517758
Pettit, 2020	1.4267213 1.3101194 1.553701
Poblador-Plou, 2020	1.4229058 1.3053201 1.5510838
Recinella, 2020	1.4212246 1.3046689 1.548193
Reilev, 2020	1.4198674 1.3024342 1.5478889
Rivera-Izquierdo, 2020	1.4176491 1.301313 1.5443856
Rodrıguez-Molinero, 2020	1.4035665 1.2891709 1.5281131
Rosenberg, 2020	1.4131894 1.2970421 1.5397373
Rossi, 2020	1.4164165 1.2998153 1.5434775
Rottoli, 2020	1.4197984 1.3031938 1.5468363
Rozaliyani, 2020	1.4145004 1.2983296 1.5410658
Russo, 2020	1.4240398 1.3070652 1.5514829
Salacup, 2020	1.4208176 1.3044074 1.5476167
Salazar, 2020	1.4101171 1.2948774 1.5356129
Sands, 2020	1.42993 1.3131154 1.5571364
Seiglie, 2020	1.4185754 1.3022965 1.5452365
Shah, 2020	1.4121155 1.2965516 1.5379798
Shang, 2020	1.4250045 1.3082117 1.5522242
Shi, 2020	1.4124031 1.2966639 1.5384731
Shi, 2020	1.4202402 1.3038338 1.5470392
Stony Brook COVID-19	Research Consortium, 2020 1.40791881.2927667 1.5333279
Sun, 2020	1.4207453 1.3044379 1.547423
Talavera, 2020	1.4189814 1.3024817 1.5459015
Tartof, 2020	1.4185152 1.3018947 1.5455824
Tehrani, 2020	1.4184757 1.3020438 1.5453192
Wang, 2020	1.3947774 1.2842842 1.5147767
Wang, 2020	1.4169366 1.3009491 1.5432652
Wang, 2020	1.4196205 1.3029077 1.5467883
Wu, 2020	1.4162004 1.3000072 1.5427787
Wu, 2020	1.4254589 1.3084775 1.5528989

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Xie, 2020	1.4142019	1.2984338	1.5402918
Yang, 2020	1.4126209	1.2969178	1.5386463
Yazdanpanah, 2020	1.4077736	1.2926683	1.5331283
Yu, 2020	1.4214995	1.3047308	1.5487185
Zangrillo, 2020	1.412156	1.2967449	1.5378386
Zhou, 2020	1.4105095	1.2950244	1.536293
Combined	1.4180375	1.3021785	1.5442047

Supplementary Figure 4: Influence analysis of hypertension and mortality risk in patients with COVID-19



Supplementary Figure 5: Funnel plot of obesity and mortality risk in patients with COVID-19

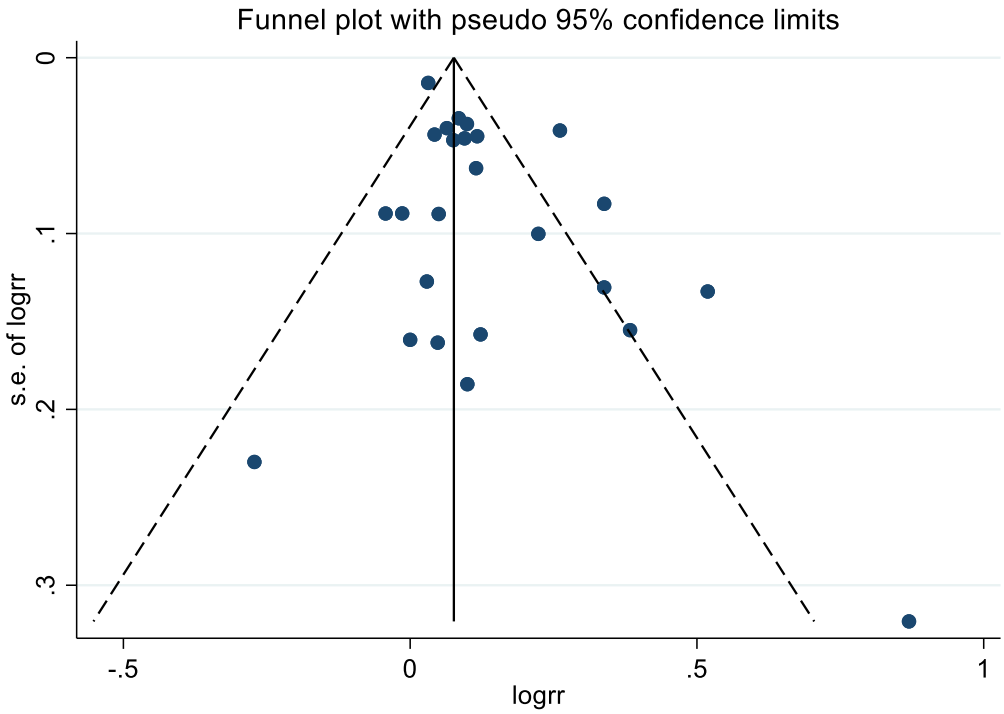
Study omitted	Estimate	[95% Conf.	Interval]
Almazeedi, 2020	1.4554933	1.3114676	1.6153359
McCarty, 2020	1.4626863	1.3171904	1.6242536
Mccullough, 2020	1.4678108	1.3216922	1.6300832
Salazar, 2020	1.4587231	1.3135937	1.6198866
Arshad, 2020	1.4721547	1.3261496	1.6342347
Stony Brook COVID-19, 2020	1.4625834	1.3163379	1.6250767
Rosenberg, 2020	1.4645751	1.317825	1.6276668
Price-Haywood, 2020	1.4633511	1.3166041	1.6264545
Salacup, 2020	1.4560728	1.3184568	1.6080528
Hashemi, 2020	1.454875	1.3098872	1.6159111
Biscarini, 2020	1.4580365	1.3122839	1.6199776
Fried, 2020	1.4623791	1.3147643	1.6265674
Tehrani, 2020	1.4557939	1.3104337	1.6172782
Kim, 2020	1.4611641	1.3138261	1.6250254
Mendy, 2020	1.4512935	1.3070762	1.6114231
Rodrıguez-Molinero, 2020	1.4539628	1.3089103	1.61509
Crouse, 2020	1.4545934	1.3092545	1.6160662
Hajifathalian, 2020	1.4535341	1.3084871	1.6146597
Berenguer, 2020	1.4570931	1.3097397	1.6210245
Castelnuovo, 2020	1.4555653	1.3092505	1.6182317
Portoles, 2020	1.4520395	1.3071913	1.6129383
Ramos-Rincon, 2020	1.4555131	1.3086935	1.618804
Lombardi, 2020	1.4531474	1.3074876	1.6150345
Lunski, 2020	1.4534943	1.3069357	1.616488
Mallow, 2020	1.4557445	1.3068982	1.6215434
Rossi, 2020	1.4502716	1.3058093	1.6107157
Nakeshbandi, 2020	1.4531265	1.3068702	1.615751
Parker, 2020	1.4498988	1.3056446	1.610091
Docherty, 2020	1.4553313	1.3059231	1.6218332
Mirani, 2020	1.450893	1.3057793	1.6121336
Zhang, 2020	1.4518838	1.3054672	1.6147219
Sands, 2020	1.4500164	1.3040735	1.6122923
Núñez-Gil, 2020	1.4474576	1.3029506	1.6079915
Wollenstein-Betech, 2020	1.4460288	1.2970564	1.6121114
Xie, 2020	1.4454393	1.3012034	1.6056634
Gutierrez, 2020	1.4410919	1.2967353	1.6015188
Sousa, 2020	1.447451	1.3037943	1.6069363
Pettit, 2020	1.4440517	1.2996014	1.6045576
DF de Souza, 2020	1.4449612	1.3009996	1.6048528
Shah, 2020	1.4426155	1.2983586	1.6029004

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Imam, 2020	1.4404271	1.295764	1.6012409
Calmes, 2020	1.4430385	1.2988892	1.6031855
Rottoli, 2020	1.4371952	1.2937157	1.5965873
Klang, 2020	1.4298403	1.2871754	1.5883176
Nachega, 2020	1.4382833	1.2949491	1.5974829
Alamdari, 2020	1.4401046	1.2969043	1.5991166
Coppelli, 2020	1.4412719	1.2981442	1.6001803
Giacomelli, 2020	1.4355083	1.2928371	1.5939237
Yazdanpanah, 2020	1.4316087	1.2893935	1.5895095
Gazzaruso, 2020	1.4244035	1.2831527	1.5812036
Mather, 2020	1.4267011	1.2856641	1.5832098
Steinberg, 2020	1.4363017	1.2941382	1.5940822
Escalera-Antezana, 2020	1.4401749	1.2977738	1.5982014
Abohamr, 2020	1.3768033	1.2478951	1.5190276
Combined	1.4483488	1.3049973	1.6074472

Supplementary Figure 6: Influence analysis of obesity and mortality risk in patients with COVID-19

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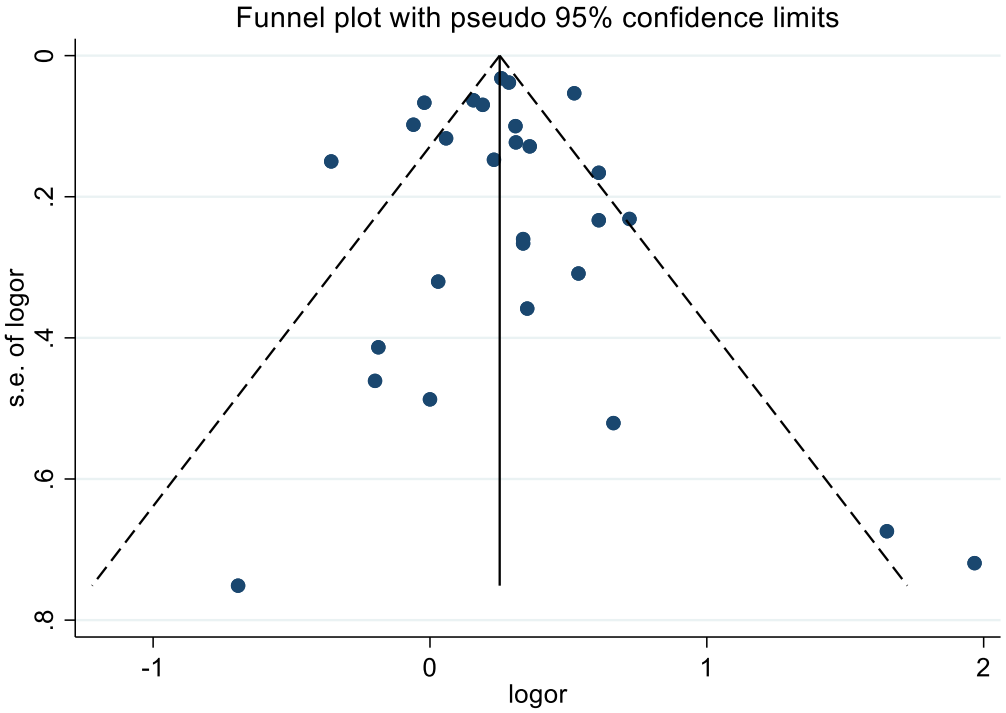


Supplementary Figure 7: Funnel plot of BMI per 5kg/km^2 and mortality risk in patients with COVID-19

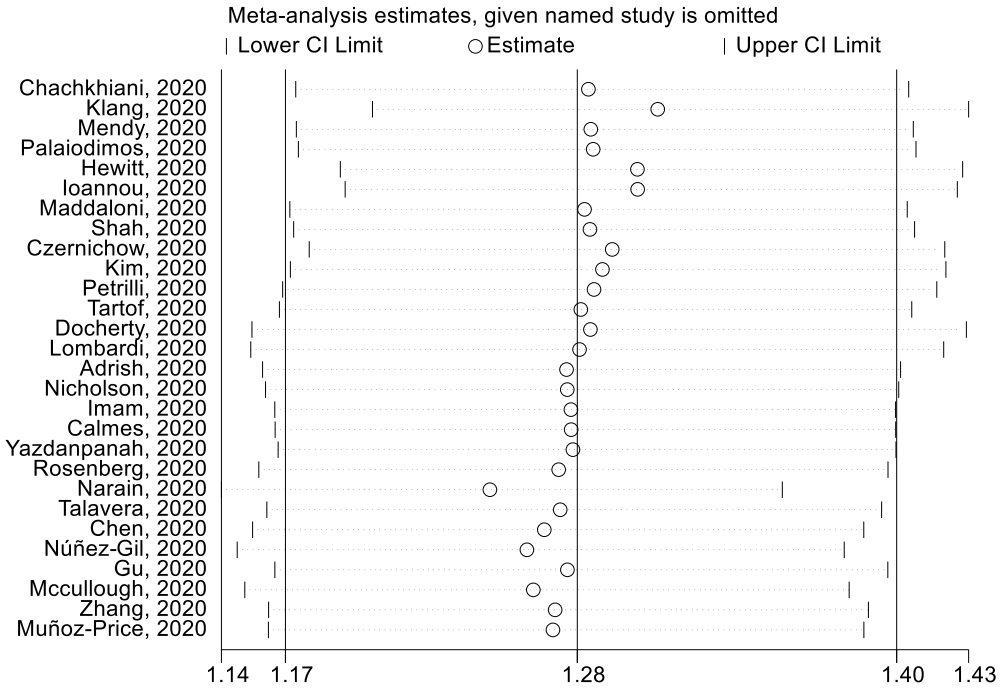
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Study omitted	Estimate	[95% Conf.	Interval]
Salazar, 2020	1.1262932	1.0773913	1.1774148
Haase, 2020	1.1295877	1.0794542	1.1820494
Li, 2020	1.1284989	1.0781709	1.1811761
Calmes, 2020	1.1250597	1.0753614	1.177055
Hajifathalian, 2020	1.1252736	1.0753136	1.1775547
Ferrando-Vivas, 2020	1.1313956	1.0795149	1.1857696
Anderson, 2020	1.1291318	1.0770124	1.1837733
Seiglie, 2020	1.1241773	1.0744675	1.176187
Lombardi, 2020	1.1259074	1.0754082	1.1787779
Petrilli, 2020	1.1278753	1.0753478	1.1829686
Kabarriti, 2020	1.126538	1.0745648	1.1810251
Santos, 2020	1.1266743	1.0735707	1.1824048
Gupta, 2020	1.1251055	1.0731854	1.1795374
Wang, 2020	1.1251912	1.0726496	1.1803064
Palaiodimos, 2020	1.1231542	1.073586	1.175011
Nakeshbandi, 2020	1.1232269	1.0722753	1.1765996
Klang, 2020 (b)	1.1233695	1.0716629	1.1775709
Gu, 2020	1.1227893	1.0731335	1.1747427
Shah, 2020	1.1186447	1.0692952	1.170272
Czernichow, 2020	1.1065391	1.0623916	1.1525211
Rottoli, 2020	1.116652	1.0682974	1.1671954
Gayam, 2020	1.1111181	1.0643468	1.1599447
Baronio, 2020	1.1171073	1.0688614	1.1675309
Klang, 2020 (a)	1.1115534	1.0657233	1.1593543
Muñoz-Price, 2020	1.1179643	1.0707144	1.1672993
Chachkhiani, 2020	1.1227052	1.0738231	1.1738126
Salazar, 2020	1.1227052	1.0738231	1.1738126
Ghany, 2020	1.1227052	1.0738231	1.1738126
Zimering, 2020	1.1227052	1.0738231	1.1738126
Tartof, 2020	1.1227052	1.0738231	1.1738126
Ioannou, 2020	1.1227052	1.0738231	1.1738126
Recinella	1.1227052	1.0738231	1.1738126
Harmouch, 2020	1.1227052	1.0738231	1.1738126
Combined	1.1227053	1.0738231	1.1738127

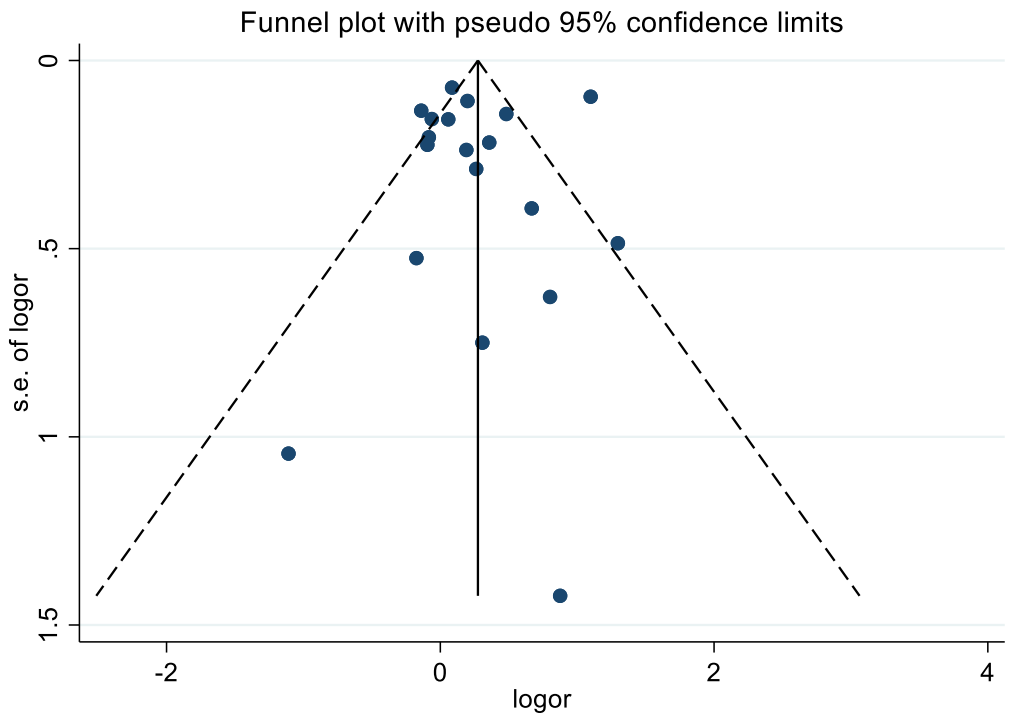
Supplementary Figure 8: Influence analysis of BMI per 5kg/km^2 and mortality risk in patients with COVID-19



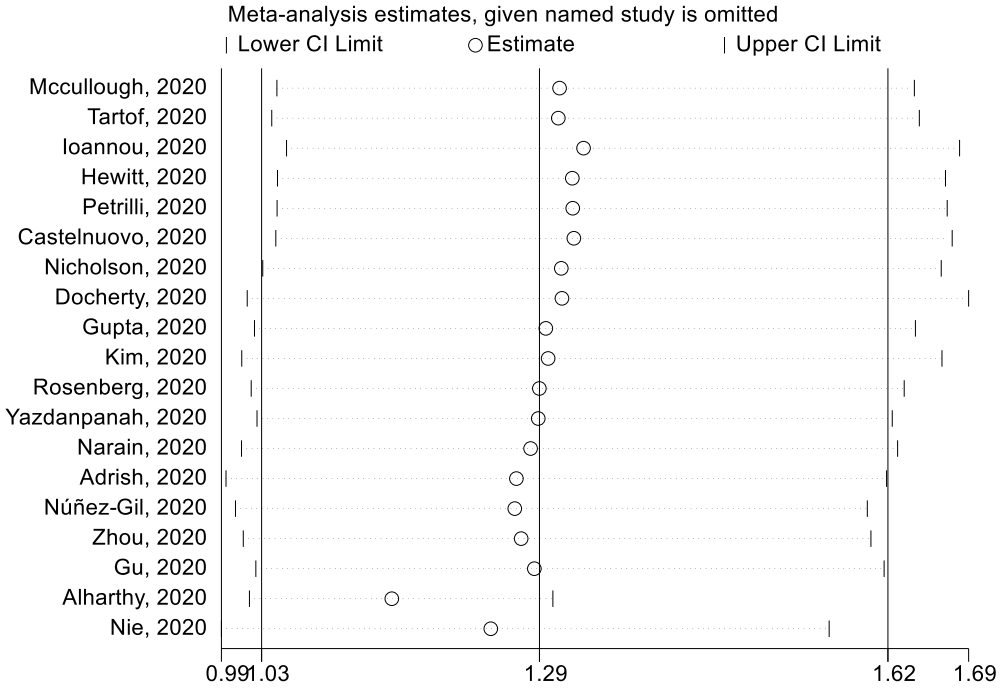
Supplementary Figure 9: Funnel plot of ever smoking vs never smoking and mortality risk in patients with COVID-19



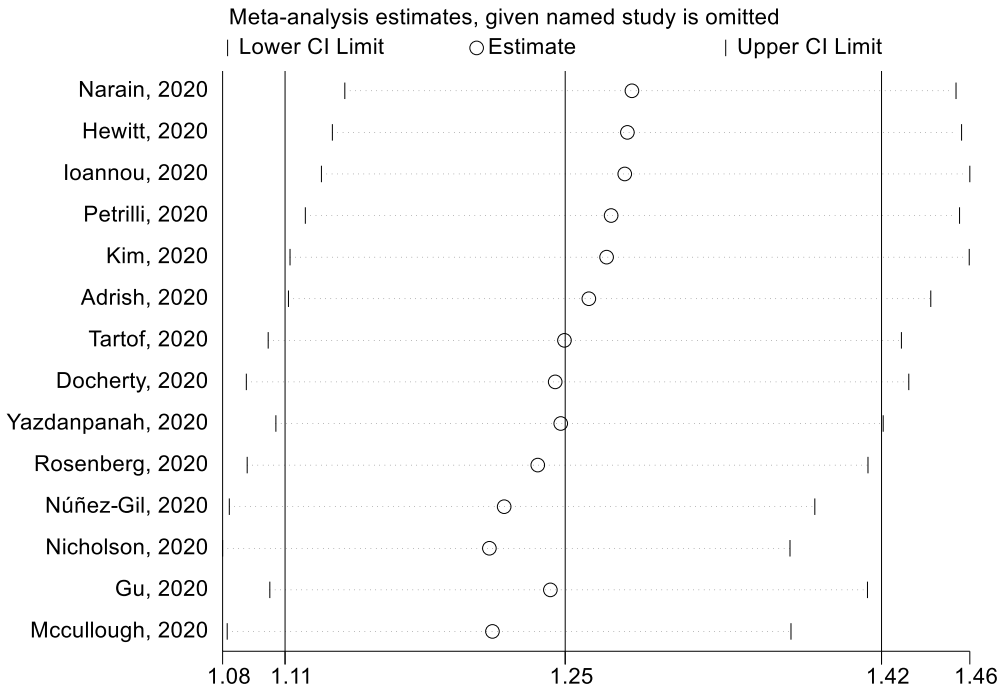
Supplementary Figure 10: Influence analysis of ever smoking vs never smoking and mortality risk in patients with COVID-19



Supplementary Figure 11: Funnel plot of current smoking vs never smoking and mortality risk in patients with COVID-19

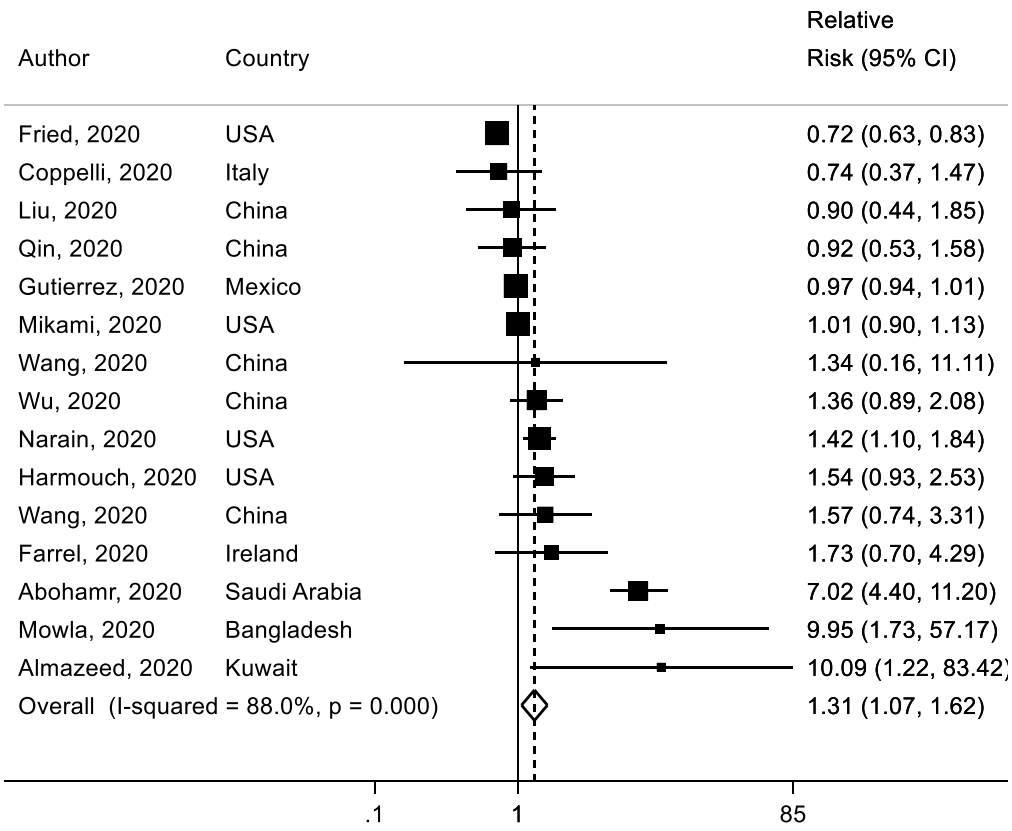


Supplementary Figure 12: Influence analysis of current smoking vs never smoking and mortality risk in patients with COVID-19



Supplementary Figure 13: Influence analysis of former smoking vs never smoking and mortality risk in patients with COVID-19

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Supplementary Figure 14: Association between undefined smoking status (versus never smoking) and mortality risk in patients with COVID-19

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Supplementary Table 1: Characteristics of studies included in this meta-analysis

Author, years, country	Design	Study description	Study periods	Study size, no. of death	Exposure	Comparison	RR (95 % CI)	Adjustment factors
Abohamr et al 2020, Saudi Arabia	Retrospective cohort study	King Saud Medical City, Riyadh	March 23 and June 15, 2020	768 hospitalized patients with COVID-19; 89 died	Diabetes Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no	3.01 (1.867-4.84) 18.143 (10.862-30.30) 7.018 (4.398-11.199)	None
Adrish et al, 2020, USA	Retrospective cohort study	Data from the Bronx Care Health System, the largest voluntary, not-for- profit health and teaching hospital system, serving the south and central Bronx in the New York City	March 9 to May 18, 2020	1173 COVID-19 patients; 390 died	Smoking	Ever smokers vs. never Current smokers vs. never Past smokers vs. never	1.34 (1.08-1.66) 1.62 (1.22-2.13) 1.15 (0.87-1.51)	Age
Alamdari et al, 2020, Iran	Retrospective cohort study	Patients admitted Shahid Modarres Hospital in Tehran,	April 2 and May 3	157 COVID-19 patients; 18 died	BMI	>30 kg/m2	2.53 (1.08-5.92)	None
Albitar et al, 2020, China	cross-sectional study	Data from national, provincial, and municipal health reported from Hubei and outside of Hubei	December 1, 2019 to April 21, 2020	828 COVID-19 patients; 219 died	Diabetes Hypertension	Yes vs. no Yes vs. no	12.23 (4.13-36.27) 3.58 (1.69-7.55)	Age, sex, geographic location and chronic disease (hypertension, diabetes, chronic lung diseases, chronic kidney and cardiovascular disease)
Alharthy et al, 2020, Saudi Arabia	Retrospective cohort study	Patients from King Saud Medical City (KSMC), Riyadh (adult > 18 years old, and admitted to ICU)	March 20 and May 31, 2020.	352 COVID-19 patients; 114 died	Smoking	Active smokers vs. never	3.00 (2.51–3.66)	Age, ICU length of stay, Spo_2/FiO_2 ratio, white blood cells, lymphocytes, d-dimers, lactate and pulmonary embolism
Almazeedi et al 2020, Kuwait	Retrospective cohort study	Data from JaberAl-Ahmad Al-Sabah hospitalisa (1240 bed tertiary hospital based in South Surra)	February 24 and April 20, 2020	1096 hospitalized patients with COVID-19; 19 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Yes vs. no	0.83 (0.17-4.16) 0.84 (0.46-4.81) 0.22 (0.03-1.51) 10.09 (1.22-83.4)	age, sex, hypertension, smoking, diabetes, asthma, chronic renal disease, qSOFA score, elevated CRP and elevated procalcitonin
Al-Salameh et al, 2020, France	Observational cohort study	Amiens University Hospital	case confirmed until April 21 2020	433 COVID-19 patients; 21 died	Diabetes	Yes vs. no	0.73 (0.40-1.34)	sex, age, the presence/absence of hypertension, the presence/absence of

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								cardiovascular or cardiac disease, the presence/absence of obesity, the presence/absence of chronic obstructive pulmonary disease (COPD) or sleep apnoea syndrome, a history of chronic kidney disease (CKD), a history of cancer, treatment with ACEIs or ARBs, glycaemia on admission, the white blood cell (WBC) count on admission, the glomerular filtration rate on admission, the serum C-reactive protein (CRP) on admission, and abnormal liver function test results
An et al 2020, South Korea	Retrospective cohort study	Data provided by the Korean National Health Insurance Service (KNHIS): the database of beneficiaries of national health insurance	January 23 andApril 2, 2020	10,237 hospitalized patients with COVID-19; 228 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.75 (1.29-2.36) 1.22 (0.87-1.73)	Age group, sex, income level, residence, household type, disablility, symptom and infection route.
Anderson et al, 2020, USA	Retrospective cohort study	Patients admitted from the emergency department (ED) to NewYork-Presbyterian/Columbia University Irving Medical Center	March 10 and 24 April 2020	2466 COVID-19 patients; 627 died	BMI	<18.5 kg/m2 18.5-24.9 kg/m2 25-29.9 kg/m2 30-34.9 kg/m2 35-39.9 kg/m2 ≥40 kg/m2	2.4 (1.1–5.3) 1.5 (1.02–2.2) 1 (reference) 1.4 (0.97–2.0) 1.0 (0.6–1.7) 1.6 (0.9–2.7)	Age, sex, race/ethnicity, hypertension, asthma or chronic obstructive pulmonary disease, chronic kidney disease, pulmonary hypertension, smoking, cancer, and diabetes
Anzola et al, 2020, Italy	Observational cohort study	Data from the Hospital of Gavardo, located in the neighborhood of Brescia in	March 12, 2020 to April 11, 2020	431 COVID-19 patients; 72 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.13 (0.46-2.73) 0.60 (0.27-1.34)	Age, creatinine and CRP

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		Lombardia						
Arshad et al 2020, USA	Retrospective cohort study	Data from the Henry Ford Health System (HFHS) in Southeast Michigan, a large six hospital integrated health system; the largest of hospital sisan802-bed quaternary academic teaching hospital in urban Detroit, Michigan.	March 10 and May 2, 2020	2,541 hospitalized patients with COVID-19; 460 died	Diabetes Hypertension Obesity	Yes vs no Yes vs no Yes vs no	0.98 (0.79-1.211) 0.80 (0.63-1.01) 0.78 (0.62-0.96)	Age, sex, white race, lung disease, BMI, COPD, hypertension, cancer, diabetes, admitted to ICU, ventilator and admission factors
Atkins et al, 2020, UK	Observational cohort study	UK Biobank Community Cohort; aged 65–86 years	March 16 to April 26, 2020.	507 COVID-19 patients; 41 died	Diabetes Hypertension	Yes vs. no Yes vs. no	3.11 (2.06-4.71) 1.42 (0.96-2.11)	Age group, sex, ethnicity, education, baseline assessment center, hypertension, diabetes and others chronic diseases
Barbu et al, 2020, Romania	Retrospective cohort study	Data from ministry of Health; 61.4% of males and 38.6% females; 68.2 years	March 22 to May 8, 2020	814 died	Diabetes Hypertension	Yes vs. no Yes vs. no	6.43 (4.97-8.32) 3.26 (2.69-3.96)	Age
Baronio et al 2020, Italy	Retrospective cohort study	Patients admitted to the “Fondazione Poliambulanza di Brescia” hospital (Brescia, Lombardy, Italy)	March 1, and April 11, 2020	191 hospitalized patients with COVID-19; 78 died	Obesity	18.5-<25 25-<30 30+	1 1.36 (0.70-2.65) 2.23 (1.15-4.35)	Age, sex, cardiovascular plus hypertension and temperature
Barron et al, 2020, England	Retrospective cohort study	Data from whole-population study (NHS England and NHS Improvement created a bespoke hub of relevant datasets in the National Commissioning Data Repository) covering all individuals registered with a general practice in England; 40.9 years	March 1 to May 11 2020	23 698 in-hospital COVID-19-related deaths	Diabetes	Yes vs. no	2.03 (1.97-2.09)	Age, sex, ethnicity, deprivation quintile and region.
Bepouka et al 2020,	Retrospective cohort study	Data from Kinshasa University Hospital; the tertiary level	March 23 and June 15, 2020	141 hospitalized patients with COVID-19; 41 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.31 (0.59-2.88) 1.07 (0.51-2.26)	Age, sex, civil status, admission procedure,

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Democratic Republic of the Congo		hospital in the city of Kinshasa with a capacity of 1000 beds						comorbidities, all admission signs, blood oxygen saturation, severity of illness, conventional treatment and biologic variables
Berenguer et al 2020, Spain	Retrospective observational study	Data from the COVID-19-Spain, a study of 127 Spanish centres until 17 March 2020		4035 hospitalized patients with COVID-19; 1131 died	Diabetes Hypertension Obesity	Yes vs no Yes vs no Yes vs no	1.23 (1.07-1.41) 1.81 (1.56-2.09) 1.21 (1.01-1.44)	Age, sex, hypertension, diabetes, obesity, cancer, others chronic diseases, admission signs and symptoms and vital signs.
Biscarini et al 2020, Italy	Retrospective cohort study	Data fro the SMAtteo Covid 19 Registry (SMACORE) cohort (Hospital, Fondazione IRCCS Policlinico San Matteo in Pavia)	February 21 and March 31, 2020	427 hospitalized patients with COVID-19; 139 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.29 (0.78-2.15) 0.77 (0.51-1.17) 1.03 (0.65-1.67)	Sex, age, presence of interstitial pneumonia at admission, CRP levels and PF ratio at admission, antiviral and antibiotic treatment, and main comorbidities (tumor, heart disease, hypertension, diabetes)
Boulle et al, 2020, South Africa	Cohort study	Data from the Western Cape Provincial Health Data Centre (WCPHDC) of public sector patients; aged ≥20	before March 1, 2020	15,203 COVID-19 patients; 625 died	Diabetes Hypertension Obesity	diabetes HbA1c <7% vs. no diabetes HbA1c 7 - 8.9% vs. no diabetes HbA1c ≥9% vs. no diabetes no HbA1c measurement vs. no Yes vs. no	2.21 (1.57-3.12) 3.41 (2.59-4.51) 3.62 (2.8- 4.59) 2.02 (1.47-2.76) 1.02 (0.84-1.24)	Age, sex, diabetes, hypertension, HIV, chronic kidney disease, chronic pulmonary diseases and tuberculosis.
Bousquet et al, 2020	Retrospective cohort study	The university hospital of Paris-Seine Saint Denis	March to April 2020	108 COVID-19 patients; 28 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.57 (0.72-3.44) 1.34 (0.54-3.32)	Age

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France		(Avicenne hospital; nearly 200 hospital beds for Covid-19 patients); aged 65			BMI	BMI < 21 kg/m2	0.89 (0.34-2.35)	
Brouns et al 2020, Netherlands	Retrospective cohort study	Data from 14 NH facilities from the NH organization Envida, Maastricht,	March 20 and May 1, 2020	67 hospitalized patients with COVID-19; 35 died	Hypertension	Yes vs. no	0.54 (0.22-1.37)	Age, sex, COVID-19 rRT-PCR result, CCI, hypertension, medication use, and OAT
Calmes et al 2020, Belgium	Retrospective cohort study	Patients hospitalized in University Hospital of Liege	March 18 and April 17, 2020	595 hospitalized patients with COVID-19; 17 died	Diabetes Hypertension Smoking BMI Obesity	Yes vs. no Yes vs. no smoker or ex-smoker vs never BMI per Yes vs. no	1.80 (1.10-3.10) 1.40 (0.80-2.30) 1.40 (0.81-2.30) 1.00 (0.97-1.10) 1.80 (1.10-3.20)	Age and gender
Cao et al, 2020 China	Retrospective cohort study	Wuhan Jin Yin-tan hospital and Wuhan Union hospital	January 5, and February 22, 2020	101 hospitalized patients with COVID-19; 66 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.24 (0.23-6.64) 0.73 (0.15-3.56)	Age, respiratory rate, acute respiratory distress syndrome, cardiovascular disease, hypertension, diabetes, chronic pulmonary disease and bacterial infection
Carrasco-Sanchez et al, 2020 Spain	Retrospective multi-center study	Data from the Spanish SEMI-COVID-19 Registry	March 1 to May 31, 2020	11312 hospitalized patients with COVID-19; 2289 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.63 (1.49-1.79) 1.14 (1.01-1.29)	Age, gender, hypertension, diabetes, COPD, dependency, lymphopenia, anemia (hemoglobin<10 g/dl), serum creatinine, CRP>60 mg/L, LDH>400 U/L, and Ddimer> 1000 ng/ml, elevated admission BG levels
Castelnuovo et al 2020, Italy	Retrospective cohort study	CORIST Collaboration Project (median age 67 years)	February 19 to May 23 2021	3894 hospitalized patients with COVID-19; 712 died	Diabetes Hypertension Smoking Obesity	Yes vs. no Yes vs. no Current smoker vs never ≥30 vs <30 kg/m2	1.03 (0.84-1.28) 0.87 (0.72-1.05) 0.94 (0.69-1.27) 1.21 (0.91-1.61)	Gender, age, diabetes, hypertension, history of myocardial infarction, heart failure, chronic pulmonary disease, chronic kidney disease, CRP, obesity, smoking habit and anti-COVID19

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								drugs during hospitalization as fixed effects and repeated measures
Chachkhiani et al 2020 USA	Retrospective cohort study	University Medical Center New Orleans (UMCNO); aged (60 mean)	March 1, through March 31	250 hospitalized patients with COVID-19; 56 died	Diabetes Hypertension Smoking BMI	Yes vs. no Yes vs. no Smoker vs. never per 1 kg/m2 increase	1.10 (0.60-1.80) 1.70 (0.80-3.70) 0.50 (0.10-1.90) 1.001 (1.000-1.002)	None
Chang et al 2020 South Korea	Retrospective cohort study	Kyungpook National University Chilgok Hospital and Kyungpook National University Hospital in Daegu; aged 67 (mean)	February 1 to April 10, 2020	106 hospitalized patients with COVID-19; 30 died	Diabetes	Yes vs. no	1.16 (0.51-2.65)	Age, chronic lung disease, fasting plasma glucose and white blood cell
Chen et al 2020 China	Retrospective cohort study	Data from Union Hospital (main part, Union West Hospital and Union Tumor Hospital), Wuhan Central Hospital, General Hospital of Central Theater Command, Wuhan Third Hospital and Wuhan Jin-Yin-Tan Hospital; ≥18 years patients	January 20 to April 4, 2020	1859 hospitalized patients with COVID-19; 208 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Smokers (current or former smoker) vs. never	2.47 (1.82-3.34) 2.21 (1.68–2.90) 1.84 (1.17–2.92)	Age
Chen et al 2020 China	Retrospective cohort study	Admitted to the Central Hospital of Wuhan (age : 15–99 years)	1 January 2020 to March, 3 2020	904 hospitalized patients with COVID-19; 92 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.51 (1.53-4.13) 3.48 (2.24-5.40)	None
Chen et al 2020, China	Retrospective cohort study	Data from patients admitted to The Central Hospital ofWuhan	1 January 2020 to 15 February 2020	660 hospitalized patients with COVID-19; 82 died	Hypertension	Yes vs. no	0.92 (0.42-2.02)	Age, cerebral infraction, SOFA, CRP and LDH
Chilimuri et al 2020 USA	Retrospective cohort study	The Bronxcare Health System (BCHS) hospital in the South Bronx, New York	March 9 to April 9, 2020	375 hospitalized patients with COVID-19; 160 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.58 (0.94-2.65) 1.46 (0.82-2.62)	several variables (age, sex, hypertension, diabetes, CVD, chronic kidney disease and biological measurement)
Ciardullo et al 2020 Italy	Retrospective cohort study	Policlinico di Monza hospital in Monza (aged ≥ 18 years)	February 22 and May 15, 2020	373 hospitalized patients with COVID-19; 142 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.23 (0.85-1.63) 1.32 (0.91-1.81)	Age, sex, chronic kidney disease, hypertension diabetes, COPD, chronic obstructive and

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								pulmonary disease
Ciceri et al 2020, Spain	Retrospective cohort study	Data from North-East of Milan's province and admitted at San Raffaele Hospital; aged ≥18 years	February 25th and March 24th	410 hospitalized patients with COVID-19; 95 died	Diabetes Hypertension BMI (not precised)	Yes vs no Yes vs no	1.51 (0.96–2.05) 2.60 (1.67–4.05) 0.96 (0.90–1.02)	Age
Coppelli et al 2020 Italy	Retrospective cohort study	The Pisa COVID-19 Study (University Uospital, Pisa)	20 March to 30 April 2020	271 hospitalized patients with COVID-19; 67 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no BMI >30 kg/m2 Smokers vs non-smokers	0.78 (0.29-2.09) 1.02 (0.49-2.15) 2.80 (0.97-8.10) 0.74 (0.37-1.47)	Age, sex, hypertension, cerebrovascular disease, chronic obstructive pulmonary disease, chronic kidney disease, cognitive impairment and several biomarkers
Crouse et al 2020 USA	Retrospective cohort study	University of Alabama at Birmingham Hospital (a tertiary health care center in the racially diverse Southern U.S.)	25 February to June 22, 2020	604 hospitalized patients with COVID-19; 45 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	3.62 (2.11-6.20) 3.60 (1.68-7.69) 1.14 (0.67-1.94)	Age, race, sex, obese status and hypertension status
Cummings et al 2020 USA	Prospective cohort study	Data from two NewYork-Presbyterian hospitals affiliated with Columbia University Irving Medical Center in northern Manhattan (aged ≥18 years)	March 2 to April 1, 2020	257 hospitalized patients with COVID-19; 101 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.31 (0.81-2.10) 1.58 (0.89-2.81) 0.76 (0.40-1.47)	Nine independent variables (age, sex, hypertension, COPD, chronic kidney disease, diabetes, symptom duration before hospital, chronic cardiac disease and BMI)
Czernichow et al 2020, France	Prospective cohort study	Data from the Paris hospitals, France	February 1 and 30 April 2020	5795 hospitalized patients with COVID-19; 891 died	Diabetes Hypertension Smoking Obesity	Yes vs. no Yes vs. no Yes vs. no <18.5 18.5-24.9 25-29.99 30-34.99 35-39.99 40+	1.34 (1.11-1.62) 1.01 (0.82-1.24) 1.06 (0.84-1.33) 1.24 (0.77-1.93) 1 1.41 (1.21-1.77) 1.89 (1.45-2.47) 2.79 (1.95-3.97) 2.55 (1.62-3.95)	Comorbidities, smoking status, age and sex using logistic regressions, both including and excluding patients with missing BMI and with stratification on age class
De Souza et al 2020 Brazil	Cross-sectional study	Data from the state of Alagoas in the northeastern of Brazil (older adults (aged ≥60 years)	until 01 August, 2020	9807 hospitalized patients with COVID-19; 1171 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	2.33 (1.99-2.74) 1.53 (1.20-1.94) 1.77 (0.84–3.74)	Age, sex, initial symptoms reporte and comorbidities
De Vito et al	Retrospective	Patients admitted to the Unit of	March 8 to April	87 hospitalized patients	Hypertension	Yes vs. no	0.47 (0.11-2.00)	Age, moderate ARDS,

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2020 Italy	cohort study	Infectious Diseases of the University Hospital of Sassari (North-West Sardinia, Italy)	8, 2020	with COVID-19; 57 died				not compliance patients lymphocytes and others comorbidities
Deng et al 2020, China	Retrospective cohort study	Data reported publicly by the Chinese center for disease control and prevention	updated through February 11, 2020	44,672 hospitalized patients with COVID-19; 1023 died	Diabetes Hypertension	Yes vs no Yes vs no	4.47 (3.49–5.61) 4.48 (3.69–5.45)	None
Dennis et al 2020 England	Retrospective cohort study	Patients admitted to hospital in England with COVID-19 requiring admission to a high dependency unit (HDU) or intensive care unit (ICU); aged 18–99	1 March 2020 and 27 July 2020	19256 hospitalized patients with COVID-19; 5077 died	Diabetes	Yes vs. no	1.23 (1.14-1.32)	Age, sex, ethnicity, obesity and comorbidity including hypertension and others
Desai et al 2020 Italy	Retrospective cohort study	Data from emergency Department (ED) of Humanitas Clinical and Research Center; aged 64.8 years (mean)	February 21 and April 14, 2020	372 hospitalized patients with COVID-19; 120 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.10 (0.50-2.40) 0.90 (0.50-1.70)	Age, gender, chronic use of ACE-I and ARBs), time interval between onset of symptoms and ED admission.
Docherty et al 2020, UK	Prosepctive cohort study	Data from the ISARIC WHO CCP-UK study which is an ongoing study in 208 acute care hospitals in England, Scotland, and Wales	February and 19 April 2020	15 194 hospitalized patients with COVID-19; 3911 died	Diabetes Obesity Smoking	Yes vs. no Yes vs. no current vs. never Former vs. never	1.06 (0.99-1.14) 1.33 (1.19-1.49) 1.09 (0.95-1.26) 1.35 (1.26-1.45	Age, sex, and major comorbidities (diabetes, hypertension, obesity and others)
Du et al 2020, China	Retrospective cohort study	Data from Wuhan Pulmonary Hospital	December 25th, 2019 and February 7, 2020	179 hospitalized patients with COVID-19; 21 died	Hypertension	Yes vs no	4.081 (1.584–10.510)	None
Escalera-Antezana et al 2020 Bolivia	Cross-sectional study	Data from the Hospitals and the Ministry of Health of Bolivia, obtaining the clinical and epidemiological data of the COVID-19 cases that were laboratory-diagnosed	March 2 to 29, 2020	107 hospitalized patients with COVID-19; 6 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	16.33 (2.10-126.82) 3.28 (1.28-6.29) 12.13 (1.69-86.95)	Age, sex and comorbidties for only hypertension association
Esme et al 2020 Turkey	Retrospective cohort study	Data from Turkish Ministry of Health National COVID-19; aged ≥ 60 years	March 11 to May 27, 2020	16942 hospitalized patients with COVID-19; 3141 died	Diabetes Hypertension	Yes vs. no (60-79 age) Yes vs. no (80+ age)	1.18 (1.06-1.30) 1.26 (1.07-1.49) 1.17 (1.03-1.32) for	Gender, presence of hypertension, diabetes, chronic obstructive pulmonary disease (COPD), coronary artery

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						Yes vs. no (60-79 age)	60-79 only	disease (CAD), atrial fibrillation (AF), chronic kidney disease (CKD), dementia, depression, malnutrition, and hyperlipidemia (HL)
Fang et al 2020 China	Retrospective cohort study	Data from admitted to the infectious ward of Wuhan Third Hospital; 615 men and 665 women, aged 7–101 years	January 27 to March 15, 2020	1280 hospitalized patients with COVID-19; 69 died	Diabetes Hypertension	Yes vs. no Yes vs. no	3.54 (1.17-10.8) 0.80 (0.35-1.83)	Sex, age, hypertension, COPD, diabetes, CRF, CHD, Arrhythmia
Farrel et al 2020, Ireland	Retrospective cohort study	Data collected from the Microbiology Department of hospital's laboratory (connolly Hospital, Ireland and Royal College of Surgeons)	March 13 and May 1, 2020	257 hospitalized patients with COVID-19; 39 died	Smoking Obesity	Yes vs. no Overweight/obese vs. no	1.73 (0.70-4.31) 3.09 (1.32-7.23)	Age
Fernandez-Cruz et al 2020, Spain	Retrospective cohort study	University hospital in Madrid, Spain	March of 2020	463 hospitalized patients with COVID-19; 71 died	Diabetes	Yes vs. no	2.80 (1.60-4.90)	None
Ferrando-Vivas et al 2020 Bolivia	Observational cohort study.	Adults from a critical care units across England, Wales, and Northern Ireland	March 1 and June 22, 2020	9,990 hospitalized patients with COVID-19; 3,933 died	BMI	per 5 kg/m2 increase	1.032 (1.003–1.061)	Several variables including age, sex, ethinicity, deprivation and others factors)
Foy et al 2020, USA	Retrospective cohort study	Data from one of 4 hospitals in the Boston, Massachusetts area (Massachusetts General Hospital, Brigham andWomen's Hospital, North Shore Medical Center, and Newton-Wellesley Hospital)	March 4 and April 28, 2020.	1641 hospitalized patients with COVID-19; 276 died	Diabetes Hypertension	Yes vs. no Yes vs. no	0.87 (0.63-1.20) 1.05 (0.80-1.4)	Age, ethnicity, D-dimer (dimerized plasmin fragment D) level, absolute lymphocyte count, hypertension, diabetes, COPD, CKD and CAD.
Fried et al 2020, USA	Retrospective cohort study	data from the hospital chargemaster, in 250 hospitals, across 38 states (patients ≥ 18 years)	February 15 and April 20, 2020	11721 hospitalized patients with COVID-19; 2514 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Yes vs. no	0.99 (0.77-1.08) 0.78 (0.69-0.88) 1.07 (0.93-1.24) 0.72 (0.63-0.83)	Age, sex, insurance status, chronic kidney disease, liver disease obesity and smoking
Galloway et al 2020, England	Prospective cohort study	Data from two London hospitals (King's College Hospital and Princess Royal University Hospital) ; aged 71	March 1 to April 17 2020.	1157 hospitalized patients with COVID-19; 244 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.24 (0.95-1.60) 1.28 (0.99-1.67)	age and sex

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		years (mean)						
Gao et al 2020, China	Retrospective cohort study	Huo Shen Shan Hospital (Wuhan)	February 5 to March 15, 2020	2877 hospitalized patients with COVID-19; 56 died	Hypertension	Yes vs. no	2.12 (1.17-3.82)	Age, sex, medical history of diabetes, insulin-treated diabetes, myocardial infarction, underwent PCI/CABG, renal failure, stroke, heart failure, and COPD
Gayam et al 2020, USA	Retrospective cohort study	Data from an inner-city teaching community hospital in New York City		408 hospitalized patients with COVID-19; 132 died	BMI	BMI per 1 unit increase	1.07 (1.04-1.11)	Age, BMI, myalgia, D-dimers, Ferritin, CRP and other biological samples
Gazzaruso et al 2020, Italy	Retrospective cohort study	Not mentionned (Italy)	15 to 30 Aril 2020	192 hospitalized patients with COVID-19; 62 died	Obesity	Yes vs no	3.62 (2.14–6.14)	Age, d-dimer, history of hypertension, eGFR, lactate dehydrogenase, history of a lung disease and C-reactive protein
Ghany et al 2020, USA	Observational cohort study	Elderly Medicare COVID-19 patients across eight states;	January 1 and May 4, 2020	400 hospitalized patients with COVID-19; 72 died	Diabetes Hypertension BMI	Yes vs. no Yes vs. no BMI per 1 unit increase	1.03 (0.58-1.02) 1.38 (0.76-2.50) 1.00 (0.96–1.05)	Several variables (not mentionned)
Giacomelli et al 2020, Italy	Prospective cohort study	Adult COVID-19 patients admitted to Luigi Sacco Hospital in Milan,	February 21 to March 19, 2020	233 hospitalized patients with COVID-19; 48 died	Being treated with at least one anti-hypertensive agent Obesity	Yes vs. no Yes vs. no	1.41 (0.73−2.74) 3.04 (1.42−6.49)	Age, sex, comorbidities, BMI, hypertension drugs, and biological parameters
Grasselli et al 2020, Italy	Retrospective cohort study	Patients requiring ICU admission in Lombardy	February 20 to April 22, 2020.	3988 hospitalized patients with COVID-19; 926 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.18 (1.01-1.39) 0.99 (0.81-1.22)	Demographics, comorbidities, drugs, and respiratory parameters
Gu et al 2020, China	Nested case–control study	Data from mainland China outside of Hubei Province	18 December 2019 to 8 March 2020	275 hospitalized patients with COVID-19; 94 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.14 (0.70-1.90) 1.37 (0.90-2.20)	Age
Gu et al 2020, USA	Retrospective cohort study	University of Michigan Medicine Health System,	March 10 to April 22, 2020	1119 hospitalized patients with COVID-19; 43 died	Diabetes BMI Past-smoker Current-Smoker	Yes vs. no per one SD (8.1 kg/m2) Yes vs. no Yes vs. no	2.16 (0.85-5.49) 1.22 (0.74, 2.01) 2.06 (0.73, 5.77) 2.40 (0.15, 39.6)	Race, age, and sex

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Guisado-Vasco et al 2020, Spain	Retrospective observational study	Data from adult patients admitted to Hospital Universitario Quironsalud Madrid, in Pozuelo de Alarcon, Madrid	March 10 to April 15, 2020	607 hospitalized patients with COVID-19; 141 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.61 (1.19-5.73) 1.73 (1.17-2.54)	Age, sex, baseline comorbidities, laboratories parameters and available therapies
Gupta et al 2020, USA	Prospective cohort study	data from multicenter cohort study of 65 hospitals, adult patients (≥18 years of age)	March 4 to April 4, 2020	2215 hospitalized patients with COVID-19; 784 died	Diabetes Hypertension current smoker BMI	Yes vs. no Yes vs. no Yes vs. no <25 25-29.9 30-34.9 35-39.9 ≥40	1.14 (0.91-1.43) 1.06 (0.83-1.36) 1.21 (0.76-1.93) 1 1.01 (0.73-1.39) 0.97 (0.69-1.37) 1.24 (0.81-1.89) 1.51 (1.01-2.25)	Age, sex, race, hypertension, BMI, diabetes, smoking and others comorbidities
Gutierrez et al 2020, Mexico	Retrospective cohort study	Data from directorate General of Epidemiology (Ministry of Health)		654,858 hospitalized patients with COVID-19; 71,706 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Yes vs. no	1.82 (1.76-1.88) 1.39 (1.35-1.44) 1.68 (1.62–1.73) 0.97 (0.94–1.01)	Sex, age, comorbidities, health provider, whether the individual speaks an indigenous language, quintiles of the proportion of the population living in poverty in the municipality (county) of residence
Haase et al 2020, Denmark	Retrospective observational study	Data from the Danish nationwide, including 29 ICUs in 27 hospitals	10 March to 19 May 2020 and	323 hospitalized patients with COVID-19; 118 died	Diabetes Hypertension BMI	Yes vs. no Yes vs. no <18 18-24.9 25-29.9 30-34.9 35-39.9 ≥40	1.10 (0.71-1.70) 0.94 (0.64-1.37) 0.95 (0.51-1.76) 1 0.77 (0.49-1.22) 0.71 (0.39-1.27) 0.80 (0.34-1.89) 0.86 (0.38-1.93)	Age and gender
Hajifathalian et al 2020, USA	Retrospective observational study	Data from two hospitals in New York City (an academic tertiary care referral center and a smaller community hospital) (age ≥ 18)	March 4 and April 9, 2020	770 hospitalized patients with COVID-19; 88 died	BMI	<18.5 18.5-30 kg/m2 ≥30 kg/m2	1.64 (0.84-3.19) 1 1.15 (0.62-2.14)	Age, race/ethnicity, and troponin I level

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Harmouch et al 2020, USA	Retrospective cohort study	Data from St. Luke's University Hospital, Bethlehem, Pennsylvania, which is part of the St. Luke's University Health Network	March and April 15, 2020	563 hospitalized patients with COVID-19; 81 died	Diabetes Hypertension Smoking BMI Obesity	Yes vs. no Yes vs. no Yes vs. no BMI per Morbid obesity vs. no	0.901 (0.547-1.482) 2.049 (1.253-3.351) 1.538 (0.935-2.530) 0.981 (0.951-1.013) 1.036 (0.587-1.829)	None
Harrison et al 2020, USA	Retrospective cohort study	Data form the TriNetX network (24 healthcare organizations in the US); aged up to 90 years	January 20 to May 26, 2020	31461 hospitalized patients with COVID-19; 1296 died	Diabetes	Yes vs. no	1.11 (0.96-1.27)	age, sex, ethnicity and comorbidities
Hashemi et al 2020, USA	Retrospective cohort study	data from nine hospitals (two large tertiary centres and seven community hospitals) and a single healthcare system in Massachusetts; aged of 63.4 years	11 March to 2 April 2020	363 hospitalized patients with COVID-19; 55 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.38 (0.68-2.79) 2.20 (0.88-5.52) 1.03 (0.51-2.09)	Age, gender, obesity, cardiac diseases, hypertension, hyperlipidaemia, and pulmonary disorders, CLD and NAFLD
Hewitt et al 2020, EUROPE	Prospective cohort study	Ten hospitals in the United Kingdom (Patients ≥ 18 years old) and one Italian Hospital (University Hospital of Modena Policlinico)	Feb 27, and April 28, 2020,	1564 hospitalized patients with COVID-19; 425 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Ex-smokers vs. never Current smokers vs. never	1.03 (0.82–1.29) 0.95 (0.77–1·18) 0.95 (0.77–1.18) 0.91 (0.59–1.42)	Age group, sex, smoking, C-reactive protein, diabetes, coronary artery disease, hypertension, renal function, and the clinical frailty scale
Hu et al 2020, China	Retrospective cohort study	Data from Renmin Hospital of Wuhan University	December 2019 to March 2020	821 hospitalized patients with COVID-19; 180 died	Diabetes Hypertension	Yes vs. no Yes vs. no	0.82 (0.34-1.97) 1.15 (0.57-2.31)	Several variables (including age, sex and comorbidities)
Iaccarino et al 2020, Italy	cross-sectional study	Data from 26 hospitals in Italy, a nationwide survey (study of the Italian Society of Hypertension)	March 9 to April 9, 2020.	1591 hospitalized patients with COVID-19; 188 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.70 (1.30-2.70) 1.00 (0.60-1.50)	Age, sex, hypertension, obesity, diabetes, HF, COPD, CKD and charlon index
Imam et al 2020, USA	Retrospective cohort study	Data from largest healthcare system in Southeast Michigan (8 hospitals);	April 1. 2020,	2040 hospitalized patients with COVID-19; 200 died	Diabetes Hypertension Smoking history Obesity	Yes vs. no Yes vs. no Yes vs. no BMI > 30 vs BMI < 30	3.90 (3.00-5.10) 4.40 (3.60-5.40) 1.40 (0.83-2.30) 1.80 (1.40-2.40)	None (multivariable for BMI only)

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Ioannou et al 2020, USA	Retrospective cohort study	Data from the Department of Veterans Affairs (VA) national health care system	February 28 and May 14, 2020	10 131 hospitalized patients with COVID-19; 1094 died	Diabetes BMI Hypertension Smoking	Yes vs. no <18.5 18.5-24.9 25.0-29.9 30.0-34.9 ≥35 Yes vs. no Former vs. Never Current vs. never	1.13 (0.99-1.29) 1.29 (0.98-1.70) 1 0.90 (0.77-1.06) 0.84 (0.69-1.01) 0.97 (0.77-1.21) 0.95 (0.81-1.12) 1.02 (0.88-1.19) 0.87 (0.67-1.13)	Age, sex, race/ethnicity, sociodemographic characteristics, comorbid conditions, and symptoms
Kabarriti et al 2020, USA	Retrospective cohort study	Patients from to the Montefiore Medical Center, a large urban academic medical center in the Bronx, New York,	March 14 and April 15, 2020,	5902 hospitalized patients with COVID-19; 918 died	Diabetes Hypertension BMI	Yes vs. no Yes vs. no <18.5 18.5-35 >35	1.21 (1.06-1.40) 1.00 (0.87-1.16) 0.97 (0.66-1.41) 1 1.42 (1.19-1.70)	sex, age, socioeconomic status, ethinicity and other comorbidities
Kim et al 2020, South Korea	retrospective cohort study	Korean National Health Insurance Service Big Data linked with COVID-19 registry data of Korea Centers for Disease Control & Prevention;	confirmed COVID-19 up to March 26, 2020	9148 hospitalized patients with COVID-19; 130 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.82 (1.25-2.67) 1.32 (0.88-1.99)	age, sex, race/ethnicity, smoker, hypertension, obesity, diabetes, chronic lung disease, cardiovascular disease, neurologic disease, renal disease, immunosuppression, hematologic disorders, and rheumatologic or autoimmune disease
Kim et al 2020, USA	Retrospective cohort study	Data from 74 counties in 13 states Surveillance Network (COVID-NET); aged 62 years (median)	March 1, 2020 and May 30, 2020.	2491 hospitalized patients with COVID-19; 420 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Former smokers vs. never Current smokers	1.19 (1.01-1.40) 1.07 (0.79-1.45) 1.14 (0.98-1.33) 1.22 (0.99-1.51)	Sex, age, type of distiricts, high epidemic region and socio-economic status

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					Obesity	BMI≥30 vs BMI<30	1.09 (0.92-1.30)	
Klang et al 2020, USA	Retrospective cohort study	Data from five hospital campuses serving different geographic populations (Mount Sinai Hospital, Mount Sinai Brooklyn, Mount Sinai Queens, Mount Sinai Morningside, and Mount Sinai West), in NY	March 1 and 5/9/2020	6760 hospitalized patients with COVID-19; 1320 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Smokers (past or current) vs. never	1.00 (0.80-1.30) 1.20 (0.90-1.60) 2.30 (1.80-3.00) 0.70 (0.50–0.90)	Age decile, CAD, DM, CHF, CKD, COPD, obesity, cancer, and smoking
Klang et al 2020, USA	Retrospective cohort study	Data of COVID-19 patients hospitalized to a large academic hospital system in New York City	March 1 and May 17, 2020	3406 hospitalized patients with COVID-19; 1136 died	Age ≤ 50 Diabetes Hypertension Smoking BMI Age > 50 Diabetes Hypertension Smoking BMI	Yes vs. no Yes vs. no past or present vs. never 30 30-40 >40 Yes vs. no Yes vs. no Smokers (past or present) vs. never 30 30-40 >40	1.30 (0.70-2.60) 0.50 (0.20-1.10) 1.70 (0.80-3.80) 1 1.10 (0.50-2.30) 5.10 (2.30-11.10) 1.4 (1.2-1.7) 1.1 (0.9-1.3) 1.0 (0.8-1.2) 1 1.1 (0.9-1.3) 1.6 (1.2-2.3)	Age, sex, CAD, CHF, hypertension, CKD, smoking, BMI, race, cancer and hyperlipidemia
Kolhe et al 2020, UK	Retrospective cohort study	Data from the Department of University Hospitals of Derby and Burton (UHDB) National Health Service (NHS), which comprised 5 hospitals in the UK.	March 5 and May 12, 2020	1161 hospitalized patients with COVID-19; 419 died	Diabetes with complications	Yes vs. no	1.15 (0.82, 1.61)	Age, gender, ethnicity, residence in care home, use of ACEI/ARB, need for mechanical ventilation, and comorbidities
Izzampasona et al 2020, Italy	Prospective cohort study	Data from the Emergency or Clinical departments of the Istituto di Ricovero e Cura a Carattere Scientifico;	480 February and 19 April 2020	509 hospitalized patients with COVID-19; 93 died	Diabetes	Yes vs. no	2.32 (1.44-3.75)	Age, sex, BMI and other relevant comorbidities
Lee et al 2020,	Retrospective	Patients in the	February 18 and	98 hospitalized patients	Diabetes	Yes vs. no	4.74 (1.68-13.38)	None

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South Korea	cohort study	Daegu/Gyeongsangbuk-do province; patients aged ≥ 65 years	March 4, 2020	with COVID-19;20 died	Hypertension	Yes vs. no	1.95 (0.70-5.42)	
Li et al 2020, China	Retrospective cohort study	Wuhan Union Hospital Affiliated with Tongji Medical College of Huazhong University of Science	22 January 2020 to 17 March 2020.	453 hospitalized patients with COVID-19; 39 died	Diabetes Hypertension	known diabetes vs. normal glucose Newly diagnosed diabetes vs. normal glucose	4.63 (1.02-21.00) 9.42 (2.18-40.7)	Age, sex, smoking, systolic blood pressure, total cholesterol, anti-hypertensive drugs and admission to ICU.
Li et al 2020, China	Retrospective cohort study	Patients admitted to Renmin Hospital of Wuhan University	January 31 to February 20, 2020	204 hospitalized patients with COVID-19; 76 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.10 (0.60-1.90) 2.30 (1.50-3.60)	Age
Li et al 2020, China	Retrospective cohort study	Patients from two critical and two general designated hospitals, in Wuhan	January 2 and February 15, 2020	523 hospitalized patients with COVID-19; 95 died	BMI	per 2.9 kg/m2 increase	0.99 (0.90-1.10)	age, gender, Hypertension, diabetes mellitus, coronary artery disease, þCr, PCT, ALC, cTnI, hs-CRP, LDL-c and FBG
Li et al 2020, China	Retrospective cohort study	Patients admitted to Wuhan Red Cross Hospital	January 24 and March 15, 2020	199 hospitalized patients with COVID-19; 18 died	Diabetes Hypertension	Yes vs. no Yes vs. no	10.816 (1.895-61.741) 0.660 (0.150-2.908)	Multivariate model but variable are not mentionned
Li et al 2020, USA	Retrospective cohort study	Yale New Haven Health System, a 5-hospital system in the Northeast of the United States	February 15 and April 25, 2020	1685 hospitalized patients with COVID-19; 318 died	Diabetes wit complications	Yes vs. no	1.00 (0.70-1.40)	Age, sex, race/ethnicity, medical comorbidities and hospital locations
Liu et al 2020, China	Retrospective cohort study	Leishenshan Hospital	NA	934 hospitalized patients with COVID-19; 13 died	Diabetes	Yes vs. no	2.14 (0.58-7.91)	Age, the history of cardiovascular disease, D-dimer, WBC, and lymphocyte count
Liu et al 2020, China	Retrospective cohort study	Wuhan Infectious Disease Hospital (18–94 years)	December 29, 2019 to February 28, 2020	1190 hospitalized patients with COVID-19; 157 died	Diabetes	Yes vs. no	11.16 (1.87-66.57)	Age, sex, diabetes, hypertension, CKD and some symptoms and signs
Liu et al 2020, China	Retrospective cohort study	Weibo posts and a telephone survey	January 20 and February 15, 2020,	599 hospitalized patients with COVID-19; 83 died	Hypertension Smoking	Yes vs. no Yes vs. no	0.24 (0.08-0.67) 0.90 (0.44-1.85)	Age and sex
Liu et al 2020, China	Retrospective cohort study	Patients of Eastern Campus of Renmin Hospital, Wuhan University	January 30 to February 20, 2020	214 hospitalized patients with COVID-19; 94 died	Hypertension	Yes vs. no	0.82 (0.40-1.70)	Age, lymphocytopenia, serum LDH, PCT, and BUN levels
Lombardi et al	Cross-sectional	13 Italian cardiology units	March 1 to April	614 hospitalized patients	Diabetes		1.41 (0.99-2.02)	For hypertension :

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2020, Italy	study		9, 2020	with COVID-19; 148 died	Hypertension Obesity BMI Smoking	Yes vs. no Yes vs. no Yes vs. no +1 kg/m2 Ever vs. never smoker	1.26 (0.81-1.96) 1.26 (0.85-1.85) 1.01 (0.97-1.04) 1.33 (1.24-1.44)	ajusted for comorbidities
Lopez-Mendez et al 2020, Mexico	Retrospective cohort study	Patients admitted to Medica Sur Clinic & Foundation	March 14 and June 5, 2020	155 hospitalized patients with COVID-19; 11 died	Hypertension	Yes vs. no	3.96 (1.28−12.19)	None
Lunski et al 2020, USA	Retrospective cohort study	Data from ochsner Health System in the state of Louisiana,	March 1 through April 30, 2020	4760 hospitalized patients with COVID-19; 484 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	0.92 (0.70-1.21) 0.60 (0.47-0.78) 1.30 (1.03-1.63)	Age, sex, race, comorbidities (COPD, diabetes, BMI, hypertension, CKD) and other variables
Maddaloni et al 2020, Italy	Retrospective cohort study	Data from four academic hospitals located in the Lazio region of Italy (the Covid-19 & Diabetes (CoViDiab))	March 1 to May 15th, 2020	354 hospitalized patients with COVID-19; 77 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Ever vs. never smoker	1.30 (0.50-3.10) 2.80 (1.00-7.50) 1.00 (0.40-2.70)	Age and sex
Magleby et al 2020, USA	Retrospective cohort study	Hospitalized patients in NewYork-Presbyterian Hospital/Weill Cornell Medical Center and affiliated Lower Manhattan Hospital	March 30 to April 30, 2020	678 hospitalized patients with COVID-19; 127 died	Hypertension	Yes vs. no	0.74 (0.41-1.33)	Age, race, coronary artery disease, congestive heart failure, cerebrovascular disease, hypertension, COPD, days of symptoms prior to admission, symptoms upon presentation, initial chest x-ray findings, and level of oxygen support within three hours of arrival to the ED
Mallow et al 2020, USA	Retrospective cohort study	Data were obtained retrospectively from a commercially-available, allpayer administrative database of inpatient and hospital-based outpatient (276 hospitals in the database	March 15 and April 30, 2020	21 676 hospitalized patients with COVID-19; 4936 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.30 (1.17-1.44) 0.77 (0.70-0.86) 1.30 (1.15 1.47)	Geographic region, teaching status and bed size
Mather et al	Retrospective	Data from Hartford Hospital,	February 24 and	878 hospitalized patients	Obesity	Yes vs no	5.87 (2.54-13.6)	Age, history of atrial

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2020, USA	cohort study	an 890-bed tertiary care medical center in Hartford, Connecticut	May 13, 2020	with COVID-19; 191 died				fibrillation, heart failure, CKD, obesity, neutrophil-lymphocyte ratio, white blood cell count, and asthma
Matsuzawa et al 2020, Japan	Retrospective cohort study	Patients from six hospitals in Kanagawa Prefecture	February 1 and May 1, 2020	151 hospitalized patients with COVID-19; 14 died	Hypertension	Yes vs. no	0.63 (0.15-2.34)	Age, sex, and diabetes mellitus
McCarty et al 2020, USA	Retrospective cohort study	Data from two tertiary medical centers and the seven community hospitals which comprise Mass General Brigham	March 22, through April 2, 2020	379 hospitalized patients with COVID-19; 57 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.97 (0.97-4.02) 2.15 (0.94-4.91) 0.69 (0.35-1.37)	Age, gender, obesity, cardiopulmonary comorbidities, hypertension, and diabetes
MCCULLOUGH et al 2020, USA	Retrospective observational cohort	Data from weill Cornell Medicine/New York-Presbyterian Hospital, a quaternary referral center and 862-bed teaching hospital	March 3 through April 9, 2020	756 hospitalized patients with COVID-19; 90 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Former vs. never Active vs. never	1.10 (0.68-1.70) 2.48 (1.51-4.07) 0.70 (0.43-1.14) 2.26 (1.42-3.61) 0.33 (0.04-2.47)	Age, clinical characteristics, and ECG variables, CAD, an immunosuppressed state, and hypoxemia
Mendy et al 2020, USA	Retrospective observational cohort	Data from university of Cincinnati health system, consisted of 4 hospitals in the states of Ohio. Kentucky. and Indiana	March 13 to May 31, 2020	689 hospitalized patients with COVID-19; 26 died	Diabetes Smoking Obesity	Yes vs. no Ever vs. never smokers Yes vs. no	1.79 (0.75-4.27) 0.82 (0.33-2.01) 1.09 (0.36-3.33)	Age, gender, race/ethnicity, and smoking
Menezes-Soares et al 2020, Brazil	Retrospective cohort study	Data from State Health Secretariat of Espırito Santo; Brazil	February 29 to June 11th, 2020	1,152 hospitalized patients with COVID-19; 456 died	Diabetes	Yes vs. no	1.88 (1.43-2.47)	None
Meng et al 2020, China	Retrospective cohort study	Data from Tongji Hospital in Wuhan	January 18 and March 27, 2020	2665 hospitalized patients with COVID-19; 299 died	Diabetes Hypertension	Yes vs. no Yes vs. no	0.71 (0.48-1.03) 1.01 (0.76-1.35)	Age, sex and comorbidities (diabetes, hypertension, COPD, CKD, CVD and cancer)
Mikami et al 2020, USA	Retrospective cohort study	Data from the medical records of the Mount Sinai Health System (8 hospitals and/or over 400 ambulatory practices)	March 13 and April 17, 2020	3708 hospitalized patients with COVID-19; 806 died	Diabetes Hypertension Cigarettes use history	Yes vs. no Yes vs. no Yes vs. no	0.92 (0.73-1.16) 0.91 (0.79-1.07) 1.01 (0.90–1.13)	Age, sex, race, cigarette use history, past medical history of asthma, hypertension, diabetes, or

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		in the New York City metropolitan area						cancer, systolic BP, RR, SpO2, BMI, initial laboratory values (lymphocyte proportion, Ddimer IL-6), and hydroxychloroquine use
Mirani et al 2020, Italy	large case series	Patients from single center in the region of Lombardy	February 20 and 9 April 2020	385 hospitalized patients with COVID-19; 102 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no BMI>30 kg/m2	1.33 (0.88-2.00) 1.84 (1.15–2.95) 1.33 (0.81–2.18)	Age and sex
Moon et al 2020, South Korea	Retrospective cohort Study	Data identified from the Korean Health Insurance Review and Assessment Service	prior to May 15. 2020.	5307 hospitalized patients with COVID-19; 211 died	Diabetes	Yes vs. no	2.66 (1.90-3.73	Age, sex, hypertension, dyslipidemia, chronic kidney disease, chronic obstructive lung disease, cardiovascular disease, atrial fibrillation, end-stage renal disease and cancer
Moon et al 2020, South Korea	Retrospective cohort study	Data from Dongguk University Gyeongju Hospital (DUGH) and the Andong Medical Center (AMC)	February 18 and June 30, 2020	352 hospitalized patients with COVID-19; 24 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.69 (1.05-6.85) 1.90 (0.75-4.84)	Age
MOWLA et al 2020, Bangladesh	Cross-sectional Study	Patients admitted in Dhaka Medical College Hospital (DMCH)	May 2 and 15, 2020	100 hospitalized patients with COVID-19; 10 died	Hypertension Smoking	Yes vs no Yes vs no	2.89 (0.54-15.37) 9.95 (1.73-57.12)	Multiple variables (hypertension, heart disease and renal disease)
Munblit et al 2020, Russia	Retrospective cohort study	Patients from the Sechenov University hospital network in Moscow city	April 22 and June 6, 2020	3480 hospitalized patients with COVID-19; 191 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.10 (1.46-2.99) 1.70 (1.14-2.59)	Age, sex, hypertension, diabetes, dementia, neoplasm, COPD and CKD
Munoz et al 2020, Spain	Retrospective cohort study	Patients a large hospital in Madrid,	NA	100 hospitalized patients with COVID-19; 23 died	Hypertension	Yes vs. no	4.47 (1.36-14.66)	Age, BMI, diabetes and antiobiotic treatment
Munoz-price et al 2020, USA	cross-sectional study	Froedtert Health and Medical College of Wisconsin (Milwaukee), the largest academic system in Wisconsin,	March 12 to March 31, 2020	369 hospitalized patients with COVID-19; 20 died	Smoking Obesity	History of smoking or current Yes vs. no	7.15 (0.55-9.22) 1.19 (1.05-1.35)	clustering at the zip code–level of patients' residence
Nachega et al 2020, Congo	Retrospective cohort study	Patients from seven hospitals in Kinshasa (one private, two	March 10, and July 31 2020	766 hospitalized patients with COVID-19; 101	Diabetes Hypertension	Yes vs. no Yes vs. no	1.10 (0.66-1.81) 1.00 (0.62-1.61)	age-group, the presence of comorbidities

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		faith-based Catholic, and four public)		died	Obesity	Yes vs. no	2.30 (1.24–4.27)	(hypertension, heart disease, diabetes, obesity, CKD, and cancer) and treatment received
Nakeshbandi et al 2020, USA	Retrospective cohort study	Patients tested for COVID-19 at SUNY Downstate Health Sciences University, hospital in New York	March 10 and April 13, 2020	504 hospitalized patients with COVID-19; 219 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Overweight Obese	1.10 (0.92-1.30) 0.81 (0.66–1.0) 1.4 (1.1–1.9) 1.3 (1.0–1.7)	Age, sex, smoking, obesity and hypertension
Namendys-Silva et al 2020, Mexico	Retrospective cohort study	Data from 10 hospitals located in Mexico	April 1 and April 30, 2020	164 hospitalized patients with COVID-19; 85 died	Diabetes Hypertension	Yes vs. no Yes vs. no	3.03 (1.51-6.09) 2.41 (1.26-4.62)	None
Narain et al 2020, USA	Retrospective cohort study	Data from electronic health records across the Northwell Health system	March 1 and April 24, 2020	5,776 hospitalized patients with COVID-19; 282 died	Diabetes Hypertension Smoking BMI	Yes vs. no Yes vs. no Never Active Former Smoker (unknown active/former) Unknown < 18.5 18.5-24.9 25-29.9 >30	1.02 (0.89-1.17) 0.83 (0.73-0.94) 1 1.43 (0.94-2.21) 0.93 (0.78-1.11) 1.42 (1.09-1.83) 3.02 (2.58-3.56) 1.15 (0.93-1.44) 1 0.98 (0.83-1.42) 1.07 (0.90-1.27)	Several covariables not mentionned
Neumann-Podczaska et al 2020, Poland	Retrospective cohort study	Data from the Jozef Strus Hospital, Poznan, Poland	March 12 and 5 April 2020	50 hospitalized patients with COVID-19; 20 died	Diabetes	Yes vs. no	2.35 (0.97–5.67)	Age
Nicholsoni et al 2020, USA	Retrospective cohort study	Patients admitted in five hospitals in the Mass General Brigham health care system, Boston	COVID-19 before May 19, 2020.	1042 hospitalized patients with COVID-19; 211 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Former vs. no Current vs. no	2.32 (1.30-4.12) 1.81 (1.32-2.49) 2.05 (1.39-3.03) 1.06 (0.78-1.44)	Age, sex, statin, C-reactive protein, albumin, C-reactive protein, diabetes, neutrophil-lymphocyte ratio, mean corpuscular volume, platelet count, and procalcitonin

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Nie et al 2020, China	Retrospective cohort study	Patients admitted to Union Hospital (West Campus), Huazhong University of Science and Technology	January 12 and March 12, 2020	311 hospitalized patients with COVID-19; 111 died	Diabetes Hypertension smoking	Yes vs. no Yes vs. no Current vs. never	2.13 (1.15–3.95) 3.80 (2.31–6.25) 3.66 (1.41–9.46)	None
Nogueira et al 2020, Portugal	Retrospective cohort study	Data from the General Health Directorate of Portugal (DGS)	January to April, 21 2020	20293 hospitalized patients with COVID-19; 502 died	Diabetes	Yes vs. no	0.75 (0.55-1.01)	Age, sex and pregnancy
Núñez-Gil et al 2020, Spain	Retrospective cohort study	Data from the HOPE COVID-19 registry (Health Outcome Predictive Evaluation for COVID-19)	February 8 and April 1, 2020	908 hospitalized patients with COVID-19; 301 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Yes vs. exsmoker No vs. exsmoker	2.77 (1.96-3.90) 1.74 (1.01-2.99) 1.52 (0.83-2.76) 1.07 (0.54-2.11) 0.55 (0.39-0.80)	None
Palaiodimos et al 2020, USA	Retrospective cohort study	Conducted at the Montefiore Medical Center, a tertiary academic institution in the Bronx, New York	43899	200 hospitalized patients with COVID-19; 48 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no <25 BMI ≥35 BMI Current or former smoker vs. never	1.16 (0.55-2.44) 0.93 (0.44–1.98) 1.57 (0.68–3.66) 3.35 (1.43–7.87) 0.83 (0.37–1.87)	Age, BMI, smoking and diabetes
Pan et al 2020, China	Retrospective cohort study	Renmin Hospital of Wuhan University	January 4, to February 14, 2020	996 hospitalized patients with COVID-19; 89 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.34 (0.78-2.29) 2.24 (1.36–3.70)	Age, sex, diabetes mellitus, matched with hypertensive
Park et al 2020, South Korea	Retrospective cohort study	Patients from five tertiary hospitals of Daegu, South Korea; aged >19 years	February 20 and April 14, 2020	289 hospitalized patients with COVID-19; 70 died	Diabetes	Yes vs. no	2.226 (1.357 to 3.652)	Age, sex, comorbidities (diabetes, hypertension, COPD, CKD and CLD) and other biological parameters
Parker et al 2020, South Africa	Retrospective cohort study	Patients admitted to the Tygerberg Hospital, Cape Town	25 March 25 and May 11, 2020	113 hospitalized patients with COVID-19; 28 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.90 (0.80-4.60) 2.40 (1.00-5.80) 1.30 (0.50-3.20)	None

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Perkin et al 2020, England	Retrospective cohort study	Data for deaths occurring at St George's HOSPITAL (Black, Asian and minority ethnic (BAME) communities), London	6 weeks commencing 12th of March 2020	243 hospitalized patients with COVID-19 and died with it	Diabetes Hypertension	Yes vs. no Yes vs. no	1.76 (1.05-2.94) 1.40 (0.80-1.99)	Age
Petrilli et al 2020, USA	Prospective cohort study.	Data from 260 outpatient office sites and four acute care hospitals (Tisch Hospital and NYU Langone Orthopedic Hospital in Manhattan. NYU Langone Hospital – Brooklyn in Brooklyn. and NYU Winthrop on Long Island).	March 1 to April 8 2020	2737 hospitalized patients with COVID-19; 391 died	Diabetes Hypertension BMI Smoking	Yes vs. no Yes vs. no <25 25.0-29.9 30.0-39.9 ≥40 Former vs no Current vs no Unknown vs no	1.01 (0.85-1.21) 0.98 (0.78-1.23) 1 1.01 (0.82-1.25) 1.08 (0.87-1.36) 1.45 (0.99-2.13) 1.07 (0.88-1.31) 0.92 (0.62-1.38) 1.50 (1.21-1.86)	Age, sex, race/ethnicity, smoking status, BMI, and chronic diseases
Pettit et al 2020, USA	Retrospective cohort study	Patients admitted to the University of Chicago Medical Center, an 811-bed academic medical center on the south side of Chicago, who had completed their hospital course	March 1 and April 18, 2020	238 hospitalized patients with COVID-19; 24 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	0.50 (0.20-1.70) 0.30 (0.10-0.90) 1.70 (1.10-2.80)	Demographics variable (age, sex, race/ethnicity) and comorbidities (diabetes, hypertension, pulmonary diseases, CVD, cancer, HIV and stroke)
Poblador-Plou et al 2020, Spain	Retrospective cohort study	The PRECOVID Study; in the Spanish region of Aragon	March 4 and April 17, 2020	4412 hospitalized patients with COVID-19; 771 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.65 (1.23-2.21), men 1.40 (1.02-1.91), women 0.96 (0.75-1.23), men 1.20 (0.92-1.56), women	Age
Portoles et al 2020, Spain	Prospective cohort study.	Patients form Puerta de Hierro University Hospital	February 25 to 24 April, 2020	1603 hospitalized patients with COVID-19; 197 died	Diabetes Obesity	Yes vs. no Yes vs. no	1.83 (1.34-2.50) 1.22 (0.63–2.33)	Age
Price-Haywood et al 2020, USA	Retrospective cohort study	Data from patients seen within an integrated-delivery health system (Ochsner Health) in Louisiana	March 1 and April 11, 2020	3481 hospitalized patients with COVID-19; 326 died	Obesity	Yes vs. no	0.99 (0.77–1.27)	Age, sex, race, Charlson Comorbidity Index score, residence in a low-income area, and indicators for baseline vital signs and laboratory measures (respiratory

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								rate; levels of aspartate aminotransferase, venous lactate, creatinine, bilirubin, procalcitonin, and C-reactive protein; and counts of lymphocytes and platelets).
Qin et al 2020, China	Retrospective cohort study	Patients admitted to the Sino-French New City Branch of Tongji Hospital, Tongji Medical college, Huazhong University of Science and Technology	January 26 and February 5, 2020	548 hospitalized patients with COVID-19; 90 died	Smoking	Smoking history vs. no	0.92 (0.53-1.57)	Age, sex, and comorbidities
Ramos-Rincon et al 2020, Spain	Retrospective cohort study	Data from the Spanish Society of Internal Medicine's SEMI-COVID-19 Registry, in which 150 hospitals in Spain participate (patients ≥18 years-old of age)	March 1 and May 29, 2020	2772 hospitalized patients with COVID-19; 1301 died	Obesity	BMI ≥ 30 vs. BMI < 30	1.24 (1.00-1.54)	None
Rapp et al 2020, USA	Retrospective cohort study	Data from the Mount Sinai Hospital System (MSHS) COVID-19 registry	February 29 and May 19, 2020	4,062 hospitalized patients with COVID-19; 1,190 died	Obesity	BMI ≥ 35 vs. BMI < 35	1.53 (1.21-1.94)	Sex, age and history of CKD
Rastad et al 2020, Iran	Retrospective cohort study	Alborz province, Iran	February 20 to March 25, 2020	2957 hospitalized patients with COVID-19; 301 died	Diabetes	Yes vs. no	1.69 (1.05-2.74)	Gender, age and laboratory tests
Recinella et al 2020, Italy	Retrospective cohort study	Patients admitted to two COVID-19 units of Sant'Orsola-Malpighi University Hospital in Bologna; age ≥ 65	March 30 and May 15, 2020	109 hospitalized patients with COVID-19; 43 died	Diabetes Hypertension BMI	Yes vs. no Yes vs. no BMI per	0.751 (0.347–1.622) 1.006 (0.495–2.047) 0.875 (0.782–0.979)	Age
Reilev et al 2020, Denmark	Retrospective cohort study	Data from the Danish Microbiology Database	February 27 until 19 May 2020	11 122 hospitalized patients with COVID-19; 577 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no overweight and obesity vs. no	1.60 (1.30–2.00) 1.30 (1.10-1.60) 1.50 (1.10-2.00)	Age and sex
Rivera-Izquierdo et al 2020, Spain	Retrospective cohort study	Patients from Hospital Universitario Clı´nico San Cecilio (Granada, Spain); aged 24 to 97	16 March and 10 April 2020	238 hospitalized patients with COVID-19; 38 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.42 (1.43-4.09) 1.48 (0.80-2.71)	Age, sex, comorbidities and medication on admission, symptoms, physical exam on

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								admission
Rodrıguez-Molinero et al 2020, Spain	Retrospective cohort study	Patients from the hospitals of the Consorci Sanitari de l'Alt Penedès i Garraf (CSAPG).	March 12 and May 2 2020	418 hospitalized patients with COVID-19;79 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Yes vs. no	2.96 (1.75-4.99) 5.64 (3.13-10.87) 1.12 (0.58–2.06) 0.38 (0.09–1.11)	None
Rosenberg et al 2020, USA	Retrospective cohort study	Patients admitted to hospitals in the New York City (NYC) metropolitan region between	March 15 and 28, 2020.	1438 hospitalized patients with COVID-19; 264 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no BMI >30 vs BMI <30 kg/m2 Current vs. never Farmer vs. never	1.57 (1.28-1.93) 1.95 (1.55-2.46) 0.96 (0.75-1.22) 1.30 (0.74-2.29) 1.47 (1.11-1.95)	Age
Rossi et al 2020, Italy	Prospective cohort	Patients from the Reggio Emilia province (Hospital, outpatient, primary, and preventive care to all the resident population)	February 27 to April 2. 2020	2653 hospitalized patients with COVID-19; 217 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.60 (1.10-2.20) 1.60 (1.20-2.10) 1.30 (0.60-2.90)	Age and sex and Charlson Comorbidity Index
Rottoli et al 2020, Italy	Retrospective cohort study	Data of the ongoing PREDI-CO multicentric study, from Sant'Orsola Hospital in Bologna (≥18 years)	March 1 and April 20. 2020.	482 hospitalized patients with COVID-19; 31 died	Diabetes Hypertension Obesity BMI	Yes vs. no Yes vs. no Yes vs. no per 1 unit increase <30 30-35 ≥35	1.05 (0.64-1.74) 1.23 (0.74-2.04) 2.18 (1.34–3.54) 1.07 (1.02–1.13) 1 1.21 (0.64–2.27) 1.72 (1.00–2.99)	Age, gender, hypertension and type 2 diabetes
Rozaliyani et al 2020, Indonesia	Retrospective cohort	Data from the ongoing recapitulation of Epidemiological Surveillance (ES) by the Provincial Health Office of Jakarta	March 2 and to April 27th 2020	4052 hospitalized patients with COVID-19; 381 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.26 (0.80-1.98) 1.86 (1.24-2.78)	Age, sex, symptoms, and other comorbidities
Russo et al 2020, Italy	Retrospective cohort study	Patients admitted to the Policlinico San Martino Hospital, Genoa and aged ≥ 18 years	February 25 and April 13, 2020	777 hospitalized patients with COVID-19; 275 died	Hypertension	Yes vs. no	0.91 (0.61–1.34)	Age, sex, Charlson morbidity and others biological informations
Salacup et al 2020, USA	Retrospective cohort study	A Single Tertiary Center Cohort in an underserved-Inner City Population with minority population in the United States	March 1 to April 24,2020	242 hospitalized patients with COVID-19; 52 died	Diabetes Hypertension BMI	Yes vs. no Yes vs. no BMI	1.45 (0.67-3.13) 0.95 (0.38-2.43) 1.02 (0.98-1.07)	Age, BMI, race/ethinicty and comorbidities

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Salazar et al 2020, USA	Retrospective cohort study	Data from the eight Houston Methodist hospitals	March 28 to July 6, 2020	316 hospitalized patients with COVID-19; 24 died	Diabetes Hypertension Obesity BMI	Yes vs. no Yes vs. no <30 ≥30 <18.5 18.5-24.9 25-29.9 ≥30 BMI (unit)	5.33 (2.00-14.18) 5.60 (1.88-16.63) 1 0.78 (0.36, 1.69) 201.94 (28.85, 1413.70) 1 0.40 (0.11, 1.39) 0.49 (0.18, 1.33) 1.02 (0.95, 1.09)	Age
Sands et al 2020, USA	Retrospective cohort study	Data from HCA Healthcare, a large healthcare system with >180 affiliated acute-care facilities among its >2,000 associated sites of care in 21 US states and the United Kingdom.	March 1 and discharge dates as of May 12. 2020.	6,180 hospitalized patients with COVID-19; 1,181 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.57 (1.21-2.03) 0.63 (0.48-0.82) 1.41 (1.08-1.85)	Age, sex, all comorbidities, vital signs at admission, and other patient characteristics
Santos et al 2020, Brazil	Retrospective cohort study	Data registered in the public national epidemiological surveillance system and obtained from the open access database (3103 health units and 1238 municipalities)	20 February 2020 to 2 June 2020	46 285 hospitalized patients with COVID-19; 21408 died	Diabetes BMI	Yes vs. no 18.5-24.9 25-29.9 30-34.9 35-40 >40	1.19 (1.14–1.23) 1 1.18 (0.53–2.61) 1.00 (0.47–2.12) 1.16 (0.54–2.45) 1.27 (0.59–2.70)	Age, sex, race/ethnicity,, education level, symptoms and signs and comorbidities
Seiglie et al 2020, USA	Retrospective cohort study	Data from the Massachusetts General Hospital, registry of patients hospitalized	March 11 and April 30, 2020	417 hospitalized patients with COVID-19; 49 died	Diabetes Hypertension BMI	Yes vs. no Yes vs. no 25-29.9 ≥30	2.02 (1.01-4.03) 1.33 (0.57-3.10) 0.60 (0.25-1.44) 1.10 (0.50-2.45)	Age, sex, race/ethnicity, and comorbidities
Shah et al 2020, USA	Retrospective cohort study	Patients from Phoebe Putney Health System, the largest community health system in Southwest Georgia	March 2 and May 6, 2020.	522 hospitalized patients with COVID-19; 92 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no	1.51 (0.90-2.55) 3.36 (1.30-8.60) 1.03 (0.55-1.93)	Age, BMI, sex, race and comorbidities

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					BMI	Former/current vs. never smoker <=30 30-40 >=40	1 1.49 (0.79-2.77) 2.29 (1.11-4.69)	
Shang et al 2020, China	Retrospective cohort study	Data from the Wuhan No.7 Hospital; temporarily taken over by Zhongnan Hospital of Wuhan University.	December 25, 2019 to March 20, 2020	584 hospitalized patients with COVID-19; 57 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.18 1.07-4.44) 0.70 (0.36-1.35)	Age, sex, hypertension, cardiovascular diseases, diabetes, chronic respiratory diseases, chronic kidney diseases, chronic liver diseases, acute kidney injury, acute liver injury, respiratory failure, and acute cardiac injury
Shi et al 2020, China	Retrospective cohort study	Patients from two tertiary hospitals in Wuhan (Renmin Hospital of Wuhan University and Zhongnan Hospital of Wuhan University)	January 1 and March 8, 2020	1561 hospitalized patients with COVID-19; 31 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.58 (0.84-2.99) 2.50 (1.30-4.78)	Age and sex
Shi et al 2020, China	Retrospective cohort study	Patients admitted to the Renmin Hospital of Wuhan University with laboratory-confirmed COVID-19	January 1 and February 23, 2020	671 hospitalized patients with COVID-19; 62 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.16 (0.47-2.56) 1.07 (0.46-2.53)	Age, preexisting cardiovascular disease (hypertension and other), diabetes, cerebrovascular disease, COPD and ARDS
Smith et al 2020, USA	retrospective cohort study	Data from four Trinity Health of New England hospitals (THONE) in Connecticut and Massachusetts	March 1 to April 22, 2020	346 hospitalized patients with COVID-19; 117 died	Diabetes	Yes vs. no	1.43 (1.07-1.92)	Age, COPD and hyperlipidemia
Sousa et al 2020, Brazil	Historical cohort	Datat from Fortaleza (Ceará's capital city), a state of the Brazilian Northeast	until 14 April 2020	2070 hospitalized patients with COVID-19; 131 died	Diabetes Obesity	Yes vs. no Yes vs. no	1.30 (0.80-2.10) 1.70 (0.40-7.70)	Age, cardiovascular disease, diabetes, BMI, Neurologic disease, pneumopathis and renal disease.
Steinberg et al 2020, USA	retrospective cohort study	Patients two sites: a high-volume, urban, academic,	March 8–April 4, 2020	210 hospitalized patients with COVID-19; 18 died	Obesity	Yes vs no	6.29 (1.76-22.46)	None

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		tertiary-care medical center and an affiliated suburban community hospital; 18-45 years old						
Stony Brook COVID-19 Research Consortium et al 2020, USA	Retrospective cohort study	Data of 1325 sequentially hospitalized patients with COVID-19 in New York;177 died; aged 49 to 75	March 2 to May 11, 2020	1325 hospitalized patients with COVID-19; 177 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no BMI >30 vs BMI <30 kg/m2	1.78 (1.21-2.65) 3.24 (1.98-5.27) 0.93 (0.63-1.37)	None
Sun et al 2020, China	Retrospective cohort study	Data of the Sino-French New City Branch of Tongji Hospital,	January 29 and March 5, 2020	244 hospitalized patients with COVID-19; 121 died	Hypertension	Yes vs no	0.82 (0.24-2.75)	Age, sex, biological parameter, symptoms and signs
Talavera et al 2020, Spain	Retrospective cohort study	Patients from Hospital Clínico Universitario de Valladolid	March 8 to April 11, 2020	576 hospitalized patients with COVID-19; 127 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Current or recent vs. never	1.33 (0.76-2.34) 1.31 (0.76-2.28) 1.71 (0.93-3.12)	Age, sex, hypertention, anosmia, cardiological disorders, cancer, smoking and Prior mRS ≥ 3
Tartof et al 2020, USA	Retrospective cohort study	Data from Kaiser Permanente Southern California members, a large integrated health care organization	February 13 to May 2 2020	6916 hospitalized patients with COVID-19; 206 died	Diabetes Smoking Hypertension BMI	DM with hemoglobin A1c level <7.5% vs. no DM with hemoglobin A1c level >7.5% vs. no Current Former Never Yes vs. no <18.5 18.5–24 25–29 30–34 35–39 40–44 ≥45	1.12 (0.79-1.59) 1.22 (0.81-1.84) 0.84 (0.30–2.35) 1.30 (0.96–1.75) 1 1.38 (0.93–2.06) 1.81 (0.99–3.30) 1 0.91 (0.62–1.35) 1.26 (0.82–1.95) 1.16 (0.63–2.17) 2.68 (1.43–5.04) 4.18 (2.12–8.26)	Sex, race/ethnicity, age, tobacco use, myocardial infarction, congestive heart failure, peripheral vascular disease, cerebrovascular disease, chronic pulmonary disease, renal disease, metastatic tumor or malignancy, other immune disease, hyperlipidemia, hypertension, asthma, organ transplant, and diabetes status and hemoglobin A1c level.
Tehrani et al 2020, Sweden	Retrospective cohort study	Patients admitted to the Danderyd University Hospital	March 5 and April 28, 2020	255 hospitalized patients with COVID-19; 70 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.35 (0.68–2.66) 1.37 (0.77–2.46) 1.07 (0.61–1.87)	None

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Turcotte et al 2020, USA	Retrospective cohort study	Data from anne Arundel Medical Center, a 350-bed acute care hospital	March 1 to April 12, 2020	117 hospitalized patients with COVID-19; 29 died	Diabetes	Yes vs. no	3.12 (0.75-12.97)	Temperature at admission, supplemental O2, sputum production and chronic kidney diseases
Wang et al 2020, China	Retrospective cohort study	Data from the intensive care unit of the Caidian Branch of Tongji Hospital	February 2 AND March 1, 2020	59 hospitalized patients with COVID-19; 41 died	Smoking	Smoking history vs. no	1.57 (0.74–3.31)	Age
Wang et al 2020, China	Retrospective cohort study	Patients admitted to Wuhan Union West Hospital and Wuhan Red Cross Hospital;	January 24 to February 10, 2020	605 hospitalized patients with COVID-19; 114 died	Hypertension	Yes vs. no	1.27 (0.85-1.89)	Age
Wang et al 2020, China	Retrospective cohort study	Data from Several wuhan hospitals	February 23; 2020	7283 hospitalized patients with COVID-19; 649 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.67 (1.22-2.30) 5.58 (4.30-7.26)	age, sex, location, occupation, dichotomous self-reported co-morbidities, number of presenting symptoms
Wang et al 2020, China	Retrospective cohort study	Patients hospitalized in the First People's Hospital of Jiangxia District in Wuhan, China	January 7, to February 11, 2020	296 hospitalized patients with COVID-19; 19 died	Diabetes Hypertension Smoking	Yes vs no Yes vs no Yes vs no	4.87 (1.70, 13.96) 1.82 (0.50, 6.63) 1.34 (0.16, 10.99)	Several variables not mentionned
Wang et al 2020, USA	Retrospective cohort study	Data from five member hospitals from the Mount Sinai Health System in New York City (NYC)	February 24 and April 15, 2020	3273 hospitalized patients with COVID-19; 742 died	BMI	per one unit increase	1.02 (1.00-1.03)	Age, sex and race
Wollenstein-Betech et al 2020, Brazil	Retrospective cohort study	Data from the nationwide surveilance program for acute respiratory distress syndrome (ARDS) in the state of São Paulo a	March 1 and June 30, 2020	113,214 hospitalized patients with COVID-19; 50,387 died	Diabetes Obesity	Yes vs. no Yes vs. no	1.22 (1.17-1.27) 1.66 (1.52-1.82)	None
Wu et al 2020, China	Retrospective cohort study	Data from Wuhan Hankou Hospital and No Six Hospital of Wuhan	December 26, 2019 and March 15 2020	1690 hospitalized patients with COVID-19; 556 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Yes vs. no	1.26 (0.72-2.22) 0.80 (0.52-1.22) 1.36 (0.89-2.08)	Age, gender, smoking, diabetes, hypertension, others comorbidites and severall biological measurements
Wu et al 2020, China	Retrospective cohort study	Data from to Wuhan Jinyintan Hospital	December 25, 2019 and January 26, 2020	201 hospitalized patients with COVID-19; 44 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.58 (0.80-3.13) 1.70 (0.92-3.14)	Age

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Xie et al 2020, USA	Retrospective cohort study	Black patients from two hospitals in New Orleans	March 30 to 5 April 2020	287 hospitalized patients with COVID-19; 58 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.92 (0.98-4.50) 2.71 (0.99-8.25) 1.67 (0.98-3.41)	Age, sex, race/ethnicity, hospital site, and the Charlson Comorbidity Index
Yahyavi et al 2020, Iran	Retrospective cohort study	Data from 17 referral hospitals affiliated to Iran University of medical sciences (IUMS)	February 20 and April 4, 2020	2553 hospitalized patients with COVID-19; 478 died	Diabetes	Yes vs. no	1.30 (1.00-1.60)	Angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, cardiovascular disease, chronic kidney disease, chronic pulmonary disease, diabetes mellitus, intensive care unit, diuretics, beta-blockers, and calcium channel blocker
Yan et al 2020, China	Retrospective cohort study	Patients from main tertiary teaching hospitals (Tongji Hospital, Optical Valley Branch and Sino-French New City Branch)	January 27 and February 17, 2020	882 hospitalized patients with COVID-19; 128 died	Diabetes	Yes vs. no	1.17 (0.73-1.86)	Age, sex, and comorbidities (cardiovascular disease, chronic respiratory disease, chronic kidney disease, cerebrovascular disease, diabetes, and malignancy), lymphopenia, d-dimer, lactate dehydrogenase, interleukin-6, myocardia injury, and liver injury
Yan et al 2020, China	Retrospective cohort study	Patients from Tongji Hospital (Wuhan)	January 10 to February 24, 2020	193 hospitalized patients with COVID-19; 108 died	Diabetes	Yes vs. no	1.53 (1.02-2.30)	Age, sex, hypertension, cardiovascular disease and cerebrovascular disease,
Yang et al 2020, China	Retrospective cohort study	Patients from the Central Hospital of Wuhan, a tertiary teaching hospital	January 1 to February 29, 2020	226 hospitalized patients with COVID-19; 50 died	Hypertension	Yes vs. no	2.68 (1.24-5.81)	Propensity score matching (multiple)
Yang et al 2020, China	Retrospective cohort study	Adult patients with COVID-19 hospitalized in Tongji Hospital (Wuhan)	February 4 to March 10, 2020	263 hospitalized patients with COVID-19; 107 died	Diabetes	Yes vs. no	1.59 (1.03-2.48)	Age

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Yazdanpanah et al 2020, France	Cohort	French COVID cohort; patients from 25 hospitals participating in the cohort throughout France	January 24 and March 15, 2020	264 hospitalized patients with COVID-19; 42 died	Diabetes Obesity Hypertension Smoking	Yes vs. no Yes vs. no Yes vs. no Former vs. no Current vs. no	1.98 (1.01– 3.89) 3.32 (1.70-6.52) 3.77 (2.04 -6.99) 1.44 (0.65-3.22) 1.36 (0.31-5.86)	Age, sex, ethnicity, smoking and comorbidites
Yu et al 2020, China	Retrospective cohort study	Data from Tongji Hospital, Wuhan	January 14 and February 28, 2020	1663 hospitalized patients with COVID-19; 212 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.34 (1.45-3.76) 1.08 (0.68-1.72)	Age, sex, history of hypertension and diabetes, lymphopenia, increased alanine aminotransferase, increased lactate dehydrogenase, increased D-dimer, and increased PCT
Zandkarimi et al 2020, Iran	Cross-sectional Study	Kurdistan province COVID-19 patients, western Iran, (identified by 16-hour centers) admitted to hospitals (16 hospitals)	February 22 and May 18, 2020	1831 hospitalized patients with COVID-19; 142 died	Diabetes	Yes vs. no	2.16 (1.38-3.38)	Age, sex, CHD, CLD, CKD and cancer
Zangrillo et al 2020, Italy	Retrospective cohort study	Patients from the COVID-BioB study held in a 1350-bed university hospital in Milan and performed at Istituto di Ricovero e Cura a Carattere Scientifico (IRCCS) San Raffaele Scientific Institute	February 20 and April 2, 2020	73 hospitalized patients with COVID-19; 17 died	Hypertension	Yes vs. no	6.15 (1.75-29.11)	Age, sex and other variable not mentionned
Zhang et al 2020, China	Retrospective cohort study	multicenter hospitals in Wuhan (Department of Infectious Disease and the Department of Oncology of the Union Hospital of Tongji Medical College, Huazhong University of Science and Departments of Endocrinology in the following four hospitals; aged 38 to 66	Jan 1 to Mar 17, 2020	312 hospitalized patients with COVID-19; 33 died	Diabetes	Yes vs. no	6.87 (1.92-24.58)	Age, sex, hospital, aspartate aminotransferase, estimated glomerular filtration rate, prothrombin time, and procalcitonin
Zhang et al 2020, China	Retrospective cohort study	Data from Tongji Hospital and Wuhan Pulmonary Hospital	February 7 and March 27, 2020,	53 hospitalized patients with COVID-19; 13 died	BMI	BMI per 1 unit increase	1.35 (1.08-1.70)	None
Zhang et al	Retrospective	Patients admitted to Zhongnan	December 29,	289 hospitalized patients	Smoking	Yes vs. no	5.21 (1.39-19.52)	Age, chest

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2020, China	cohort study	hospital of Wuhan Universityand No.7 hospital of Wuhan	2019 and February 16, 2020)	with COVID-19; 49 died				tightness/dyspnea, CRP and others biological parameters
Zhou et al 2020, China	Retrospective cohort study	Patients from Jinyintan Hospital and Wuhan Pulmonary Hospital (≥18 years old)		191 hospitalized patients with COVID-19; 54 died	Diabetes Hypertension Current smoker	Yes vs. no Yes vs. no Current smoker vs no	2.85 (1.35-6.05) 3.05 (1.57–5.92) 2.23 (0.65–7.63)	None
Zhu et al 2020, China	Retrospective cohort study	Data obtained from patient cohorts admitted in the 19 hospitals in Hubei Province, Chin	December 30, 2019 and March 20, 2020	7,337 hospitalized patients with COVID-19; 248 died	Diabetes	Yes vs. no	1.49 (1.13-1.96)	Age, gender, indicators of the severity of COVID-19 and hospital sites
Zimering et al 2020, USA	Retrospective cohort study	US military veterans hospitalized for COVID-19 infection during the peak of the pandemic in New Jersey	March 2020 and June 2020	55 hospitalized patients with COVID-19; 17 died	Diabetes BMI	Yes vs. no BMI per one unit increase	0.73 (0.23-2.35) 0.99 (0.92–1.07)	None

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Supplementary Table 2: Risk of bias judgements for each studies from the ROBINS-I

Author, years, country	**Bias due to confounding**	**Bias due to selection of participants**	**Bias due to exposure assessment**	**Bias due to misclassification during follow-up**	**Bias due to missing data**	**Bias due to measurement of the outcome**	**Bias due to selective reporting of the results**	**Overall judgement**
Abohamr et al 2020, Saudi Arabia	Critical	Moderate	Low	Low	No information	Low	Moderate	Critical
Adrish et al, 2020, USA	Serious	Low	Low	Low	No information	Low	Low	Serious
Alamdari et al, 2020, Iran	Critical	Moderate	Moderate	Low	No information	Low	Low	Critical
Albitar et al, 2020, China	Low	Low	Low	Moderate	Low	Low	Moderate	Moderate
Alharthy et al, 2020, Saudi Arabia	Low	Moderate	Moderate	Low	Low	Low	Low	Moderate
Almazeedi et al 2020, Kuwait	No information	Low	Low	Low	No information	Low	Low	Low
Al-Salameh et al, 2020, France	Low	Low	Low	Low	Low	Low	Low	Low
An et al 2020, South Korea	Low	Low	Low	Low	Low	Low	Low	Low
Anderson et al, 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Anzola et al, 2020, Italy	Low	Low	Low	Low	Low	Low	Low	Low
Arshad et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Atkins et al, 2020, UK	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Barbu et al, 2020, Romania	Serious	Low	Low	Low	No information	Low	Low	Serious
Baronio et al 2020, Italy	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Barron et al, 2020, England	Low	Low	Low	Low	Low	Low	Low	Low
Bepouka et al 2020, Democratic Republic of the Congo	Low	Low	Low	Low	Low	Low	Low	Low
Berenguer et al 2020, Spain	Low	Low	Low	Low	Low	Low	Low	Low
Biscarini et al 2020, Italy	Low	Low	Low	Low	Low	Low	Low	Low
Boulle et al, 2020, South Africa	Low	Low	Low	Low	No information	Low	Low	Low
Bousquet et al, 2020 France	Serious	Moderate	Low	Low	No information	Low	Low	Serious
Brouns et al 2020, Netherlands	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Calmes et al 2020, Belgium	Moderate	Low	Low	Low	Moderate	Low	Low	Moderate
Cao et al, 2020, China	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Carrasco-Sanchez et al, 2020, Spain	Low	Low	Low	Low	Low	Low	Low	Low
Castelnuovo et al 2020, Italy	Low	Low	Low	Low	Low	Low	Low	Low
Chachkhiani et al 2020, USA	Critical	Moderate	Low	Low	Moderate	Low	Low	Critical
Chang et al 2020, South Korea	Low	Low	Low	Low	No information	Low	Low	Low
Chen et al 2020, China	Serious	Low	Low	Low	Moderate	Low	Low	Serious
Chen et al 2020, China	Critical	Low	Low	Low	Low	Low	Low	Critical
Chen et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low

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Chilimuri et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Ciardullo et al 2020 Italy	Low	Low	Low	Low	Low	Low	Low	Low
Ciceri et al 2020, Spain	Serious	Low	Low	Low	Low	Low	Low	Serious
Coppelli et al 2020, Italy	Low	Low	Low	Low	No information	Low	Low	Low
Crouse et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Cummings et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Czernichow et al 2020, France	Low	Low	Low	Low	Low	Low	Low	Low
De Souza et al 2020, Brazil	No information	Moderate	Low	Moderate	No information	Low	Low	Moderate
De Vito et al 2020, Italy	Low	Low	Low	Low	No information	Low	Low	Low
Deng et al 2020, China	Critical	Low	Low	Low	No information	Low	Low	Critical
Dennis et al 2020, England	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Desai et al 2020, Italy	Low	Low	Low	Low	Low	Low	Low	Low
Docherty et al 2020, UK	Low	Low	Low	Low	Low	Low	Low	Low
Du et al 2020, China	Critical	Low	Low	Low	No information	Low	Low	Critical
Escalera-Antezana et al 2020, Bolivia	Low	Moderate	Low	Moderate	No information	Low	Low	Moderate
Esme et al 2020, Turkey	Low	Moderate	Low	Low	Moderate	Low	Low	Moderate
Fang et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Farrel et al 2020, Ireland	Serious	Low	Low	Low	Low	Low	Low	Serious
Fernandez-Cruz et al 2020, Spain	Critical	Moderate	Low	Moderate	No information	Low	Low	Critical
Ferrando-Vivas et al 2020, Bolivia	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Foy et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Fried et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Galloway et al 2020, England	Moderate	Moderate	Moderate	Low	No information	Low	Low	Moderate
Gao et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Gayam et al 2020, USA	Low	Moderate	Moderate	Moderate	No information	Low	Moderate	Moderate
Gazzaruso et al 2020, Italy	Low	Low	Low	Low	No information	Low	Low	Low
Ghany et al 2020, USA	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Giacomelli et al 2020, Italy	Low	Low	Low	Low	No information	Low	Low	Low
Grasselli et al 2020, Italy	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Gu et al 2020, China	Serious	Low	Low	Low	Low	Low	Low	Serious
Gu et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Guisado-Vasco et al 2020, Spain	Low	Low	Low	Low	Low	Low	Low	Low
Gupta et al 2020, USA	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Gutierrez et al 2020, Mexico	Low	Low	Low	Low	Low	Low	Low	Low

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Haase et al 2020, Denmark	Moderate	Low	Low	Low	Low	Low	Low	Moderate
Hajifathalian et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Harmouch et al 2020, USA	Critical	Low	Low	Low	Low	Low	Low	Critical
Harrison et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Hashemi et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Hewitt et al 2020, EUROPE	Low	Low	Low	Low	Low	Low	Low	Low
Hu et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Iaccarino et al 2020, Italy	No information	Low	Low	Low	No information	Low	Low	Low
Imam et al 2020, USA	Critical	Low	Low	Low	No information	Low	Low	Critical
Ioannou et al 2020, USA	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Kabarriti et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Kim et al 2020, South Korea	Low	Low	Low	Low	No information	Low	Low	Low
Kim et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Klang et al 2020, USA	Low	Low	Low	Low	Moderate	Low	Low	Moderate
Klang et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Kolhe et al 2020, UK	Low	Low	Low	Low	Low	Low	Low	Low
Lampasona et al 2020, Italy	Low	Low	Low	Low	Low	Low	Low	Low
Lee et al 2020, South Korea	Critical	Moderate	Low	Low	No information	Low	Low	Critical
Li et al 2020, China	Low	Low	Low	Low	Low	Low	Low	Low
Li et al 2020, China	Serious	Moderate	Low	Low	No information	Low	Low	Serious
Li et al 2020, China	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Li et al 2020, China	No information	Low	Low	Low	No information	Low	Moderate	Moderate
Li et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Liu et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Liu et al 2020, China	Low	Low	Low	Low	Low	Low	Low	Low
Liu et al 2020, China	Moderate	Low	Low	Low	No information	Low	Low	Moderate
Liu et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Lombardi et al 2020, Italy	Moderate	Low	Low	Low	No information	Low	Low	Moderate
Lopez-Mendez et al 2020, Mexico	Critical	Low	Low	Low	No information	Low	Low	Critical
Lunski et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Maddaloni et al 2020, Italy	Moderate	Low	Low	Low	No information	Low	Low	Moderate
Magleby et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Mallow et al 2020, USA	Low	Moderate	Low	Low	Moderate	Low	Low	Moderate
Mather et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Matsuzawa et al 2020, Japan	Low	Low	Low	Low	No information	Low	Low	Low

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McCarty et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
MCCULLOUGH et al 2020, USA	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Mendy et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Menezes-Soares et al 2020, Brazil	Critical	Low	Low	Low	No information	Low	Low	Critical
Meng et al 2020, China	No information	Low	Low	Low	Low	Low	Low	Low
Mikami et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Mirani et al 2020, Italy	Moderate	Low	Low	Low	No information	Low	Low	Moderate
Moon et al 2020, South Korea	Low	Low	Low	Low	Low	Low	Low	Low
Moon et al 2020, South Korea	Serious	Low	Low	Low	No information	Low	Low	Serious
MOWLA et al 2020, Bangladesh	Low	Low	Low	Moderate	No information	Low	Low	Moderate
Munblit et al 2020, Russia	Low	Low	Low	Low	No information	Low	Low	Low
Munoz et al 2020, Spain	Low	Low	Low	Low	No information	Low	Low	Low
Munoz-price et al 2020, USA	Low	Low	Low	Moderate	Moderate	Low	Low	Moderate
Nachega et al 2020, Congo	Low	Moderate	Low	Low	Moderate	Low	Low	Moderate
Nakeshbandi et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Namendys-Silva et al 2020, Mexico	Critical	Moderate	Low	Low	Moderate	Low	Low	Critical
Narain et al 2020, USA	Low	Moderate	Low	Low	Moderate	Low	Low	Moderate
Neumann-Podczaska et al 2020, Poland	Serious	Moderate	Low	Low	No information	Low	Low	Serious
Nicholsoni et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Nie et al 2020, China	Critical	Low	Low	Low	No information	Low	Low	Critical
Nogueira et al 2020, Portugal	Low	Low	Low	Low	Moderate	Low	Low	Moderate
Núñez-Gil et al 2020, Spain	Critical	Low	Low	Low	No information	Low	Moderate	Critical
Palaiodimos et al 2020, USA	Moderate	Low	Low	Low	Low	Low	Low	Moderate
Pan et al 2020, China	Low	Low	Low	Low	Low	Low	Low	Low
Park et al 2020, South Korea	Low	Low	Low	Low	No information	Low	Low	Low
Parker et al 2020, South Africa	Critical	Low	Low	Low	No information	Low	Moderate	Critical
Perkin et al 2020, England	Serious	Low	Low	Low	No information	Low	Low	Serious
Petrilli et al 2020, USA	Low	Low	Low	Low	Moderate	Low	Low	Moderate
Pettit et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Poblador-Plou et al 2020, Spain	Serious	Low	Low	Low	No information	Low	Low	Serious
Portoles et al 2020, Spain	Serious	Low	Low	Low	No information	Low	Low	Serious
Price-Haywood et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Qin et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Ramos-Rincon et al 2020, Spain	Critical	Moderate	Low	Low	No information	Low	Low	Critical
Rapp et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low

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Rastad et al 2020, Iran	Low	Low	Low	Low	Moderate	Low	Low	Low
Recinella et al 2020, Italy	Serious	Moderate	Low	Low	Low	Low	Low	Serious
Reilev et al 2020, Denmark	Moderate	Low	Low	Low	No information	Low	Low	Moderate
Rivera-Izquierdo et al 2020, Spain	Low	Low	Low	Low	No information	Low	Low	Low
Rodrıguez-Molinero et al 2020, Spain	Critical	Low	Low	Low	Low	Low	Low	Critical
Rosenberg et al 2020, USA	Serious	Low	Low	Low	No information	Low	Low	Serious
Rossi et al 2020, Italy	Low	Low	Low	Low	No information	Low	Low	Low
Rottoli et al 2020, Italy	Low	Low	Low	Low	Low	Low	Low	Low
Rozaliyani et al 2020, Indonesia	Low	Low	Low	Low	No information	Low	Low	Low
Russo et al 2020, Italy	Low	Low	Low	Low	No information	Low	Low	Low
Salacup et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Salazar et al 2020, USA	Serious	Moderate	Low	Low	No information	Low	Low	Serious
Sands et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Santos et al 2020, Brazil	Low	Low	Low	Low	Low	Low	Low	Low
Seiglie et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Shah et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Shang et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Shi et al 2020, China	Moderate	Low	Low	Low	Moderate	Low	Low	Moderate
Shi et al 2020, China	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Smith et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Sousa et al 2020,Brazil	No information	Low	Low	Low	Moderate	Low	Low	Moderate
Steinberg et al 2020, USA	Critical	Low	Low	Low	No information	Low	Low	Critical
Stony Brook COVID-19 Research Consortium et al 2020, USA	Critical	Low	Moderate	Low	Low	Low	Low	Critical
Sun et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Talavera et al 2020, Spain	Low	Low	Low	Low	Low	Low	Low	Low
Tartof et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Tehrani et al 2020, Sweden	Critical	Low	Low	Low	No information	Low	Low	Critical
Turcotte et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Wang et al 2020, China	Serious	Moderate	Low	Low	No information	Low	Low	Serious
Wang et al 2020, China	Serious	Low	Low	Low	Moderate	Low	Low	Serious
Wang et al 2020, China	Low	Moderate	Low	Low	Moderate	Low	Low	Moderate
Wang et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Wang et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Wollenstein-Betech et al 2020, Brazil	Critical	Low	Low	Low	Moderate	Low	Low	Critical

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Wu et al 2020, China	Low	Moderate	Low	Low	Low	Moderate	Low	Moderate
Wu et al 2020, China	Serious	Low	Low	Low	No information	Low	Low	Serious
Xie et al 2020, USA	Low	Low	Low	Low	No information	Low	Moderate	Moderate
Yahyavi et al 2020, Iran	Low	Low	Low	Low	No information	Low	Low	Low
Yan et al 2020, China	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Yan et al 2020, China	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Yang et al 2020, China	Low	Low	Low	Low	Low	Low	Low	Low
Yang et al 2020, China	Serious	Low	Low	Low	Moderate	Low	Low	Serious
Yazdanpanah et al 2020, France	Low	Low	Low	Low	Low	Low	Low	Low
Yu et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Zandkarimi et al 2020, Iran	No information	Low	Low	Moderate	No information	Low	Low	Moderate
Zangrillo et al 2020, Italy	No information	Moderate	Low	Low	No information	Low	Moderate	Moderate
Zhang et al 2020, China	Low	Low	Low	Low	Low	Low	Low	Low
Zhang et al 2020, China	Critical	Moderate	Moderate	Low	No information	Low	Moderate	Critical
Zhang et al 2020, China	No information	Moderate	Low	Low	Moderate	Low	Low	Moderate
Zhou et al 2020, China	Serious	Low	Low	Low	No information	Low	Low	Serious
Zhu et al 2020, China	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Zimering et al 2020, USA	Critical	Low	Low	Low	No information	Low	Low	Critical

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Supplementary Table 3. Subgroup analyses of association between BMI and mortality risk in patients with COVID-19										
	Obesity (obese vs non-obese)					**BMI per 5 kg/m^2**				
	n	RR (95% CI)	I^2 (%)	P_{within} [a]	$P_{between}$ [b]	n	RR (95% CI)	I^2 (%)	P_{within} [a]	$P_{between}$ [b]
All studies	54	1.45 (1.31-1.61)	91.0	<0.0001		25	1.12 (1.07-1.17)	68.2	<0.0001	
Study design										
Retrospective	47	1.44 (1.29-1.61)	92.1	<0.0001		20	1.11 (1.06-1.16)	59.8	<0.0001	
Prospective	4	1.45 (1.07-1.97)	34.1	0.208		3	1.15 (1.02-1.30)	84.8	0.001	
Cross-sectional	3	1.88 (0.87-4.02)	61.8	0.07	0.85	2	1.50 (0.68-3.31)	83.5		0.89
Geographical location										
North America	26	1.26 (1.09-1.46)	94.2	<0.0001		16	1.13 (1.07-1.19)	52.3	<0.0001	
Europe	17	1.40 (1.23-1.60)	46.2	0.02		7	1.14 (1.01-1.29)	84.2	<0.0001	
Asia	5	2.50 (0.72-8.68)	95.5	<0.0001		1	0.99 (0.83-1.17)			
South America	4	1.80 (1.24-2.62)	23.7	0.27		1	1.09 (1.02-1.16)			0.81
Africa	2	1.93 (1.15-3.23)	0.5	0.32	0.03					
Number of patients										
<1000	31	1.76 (1.42-2.18)	87.2	<0.0001	0.0001	13	1.19 (1.07-1.31)	62.6	0.001	
≥1000	23	1.27 (1.13-1.42)	88.3	<0.0001		11	1.10 (1.05-1.15)	70.1	<0.0001	0.001
Risk of bias										
Low	28	1.39 (1.20-1.60)	93.3	<0.0001		12	1.12 (1.06-1.18)	70.3	<0.0001	
Moderate	11	1.52 (1.19-1.95)	68.9	<0.0001		12	1.15 (1.06-1.25)	66.7	0.001	
Serious	3	0.97 (0.78-1.21)	0.0	0.675		1	0.76 (0.49-1.20)			0.36
Critical	12	1.75 (1.32-2.32)	91.9	<0.0001	0.42					
Adjustment for confounders										
No	12	1.69 (1.30-2.21)	91.3	<0.0001		1	1.05 (0.88-1.25)	-		
Age only	3	0.97 (0.78-1.21)	0.0	0.675	0.31	1	0.76 (0.49-1.20)	-		
Multiple	36	1.43 (1.26-1.61)	91.6	<0.0001		23	1.13 (1.08-1.18)	69.8	<0.0001	0.35

Footnote: RR summary relative risk; CI confidence interval

I^2 (%) is a measure of the proportion of the heterogeneity attributed to between study variation rather than due to chance. I^2-valuesof 25%, 50% and 75% indicates low, moderate and high between study heterogeneity, respectively.

[a]P-value for heterogeneity among the studies within each cancer type

[b]P-value for between subgroup or category heterogeneity generated from meta-regression analysis

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Supplementary Table 4. Subgroup analyses of association between smoking status and mortality risk in patients with COVID-19

	Smoking (ever versus never smoker)					Current smoker (yes vs no)					Ex-smoker (yes vs no)				
	n	RR (95% CI)	I^2 (%)	P_{within} [a]	$P_{between}$ [b]	n	RR (95% CI)	I^2 (%)	P_{within} [a]	$P_{between}$ [b]	n	RR (95% CI)	I^2 (%)	P_{within} [a]	$P_{between}$ [b]
All studies	28	1.28 (1.17-1.40)	74.0	<0.0001		19	1.29 (1.03-1.62)	84.2	<0.0001		14	1.25 (1.11-1.42)	74.8	<0.0001	
Study design															
Retrospective	21	1.32 (1.15-1.52)	75.9	<0.0001		14	1.41 (1.04-1.91)	86.2	<0.0001		10	1.33 (1.12-1.57)	74.6	<0.0001	
Prospective	5	1.16 (1.01-1.32)	65.8	0.020		5	1.07 (0.94-1.21)	0.0	0.83		4	1.15 (0.93-1.42)	76.8	0.005	
Cross-sectional	2	2.65 (0.52-13.38)	81.7	0.020	0.45	-	-	-		0.01		-	-	-	
Geographical location															0.007
North America	17	1.25 (1.08-1.45)	79.4	<0.0001		11	1.16 (0.99-1.35)	34.3	0.125		9	1.23 (1.06-1.43)	71.9	<0.0001	
Europe	9	1.26 (1.13-1.40)	59.2	0.012		5	1.07 (0.95-1.21)	0.0	0.446		4	1.30 (1.00-1.69)	76.3	<0.0001	
Asia	2	2.56 (0.99-6.59)	53.0	0.15		3	3.00 (2.50-3.61)	0.0	0.82		1	2.06 (0.73-5.79)			0.71
South America	_	_	_	_		-	-	-				-	-	-	
Africa	_	_	_	_	0.21	-	-	-		<0.0001		-	-	-	
Number of patients															
<1000	13	1.47 (1.19-1.80)	46.1	0.035		6	2.47 (1.68-3.62)	27.1	0.23	<0.0001	4	1.91 (1.48-2.48)	0.0	0.784	
≥1000	15	1.22 (1.10-1.37)	82.1	<0.0001	0.001	13	1.11 (1.00-1.24)	27.1	0.17		10	1.17 (1.04-1.33)	76.9	<0.0001	0.002
Risk of bias															
Low	12	1.20 (1.08-1.34)	46.4	0.039		8	1.09 (0.98-1.20)	0.0	0.854		6	1.26 (1.07-1.49)	71.5	0.004	
Moderate	10	1.25 (1.04-1.51)	87.3	<0.0001		6	1.23 (0.69-2.22)	93.3	<0.0001		4	1.12 (0.90-1.39)	75.8	0.006	
Serious	3	1.43 (1.24-1.66)	0.0	0.495		3	1.57 (1.23-2.01)	0.0	0.674		3	1.32 (1.07-1.64)	10.4	0.327	
Critical	3	1.51 (0.99-2.30)	40.3	0.19	0.60	2	2.50 (1.37-4.58)	1.6%	0.313	0.18	1	1.82 (1.27-2.60)			0.53
Adjustment for confounders															
No	4	1.43 (1.13-1.80)	44.4	0.15		3	2.45 (1.43-4.19)	0.0	0.594		1	1.82 (1.27-2.60)			
Age only	3	1.43 (1.24-1.66)	0.0	0.495	0.40	2	1.55 (1.21-1.99)	0.0	0.493		3	1.32 (1.07-1.64)	10.4	0.327	
Multiple	21	1.23 (1.09-1.38)	78.3	<0.0001		14	1.16 (0.88-1.52)	87.6	<0.0001	0.13	10	1.20 (1.05-1.39)	79.6	<0.0001	0.41

Footnote: RR summary relative risk; CI confidence interval

I^2 (%) is a measure of the proportion of the heterogeneity attributed to between study variation rather than due to chance. I^2-valuesof 25%, 50% and 75% indicates low, moderate and high between study heterogeneity, respectively.

[a]P-value for heterogeneity among the studies within each cancer type

[b]P-value for between subgroup or category heterogeneity generated from meta-regression analysis

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Supplementary Table 5: GRADE Evidence profile

		Downgrading factors					Upgrading factors					
Prognostic factors	Initial rate of confidence	Risk of bias	Inconsistency	Indirectness	Imprecision (ratio of the upper to lower 95%CI)	Publication bias	Large magnitude of association	Dose response	Residual confounding	Consistency	Overall quality	Level of evidence for the health effect
Diabetes	Low (observational studies)	Not likely	Not likely	Not likely	Not likely (ratio: 0.97)	Not likely	No upgrade	No upgrade	No upgrade	No upgrade	⊕⊕oo Low	Low
Hypertension	Low (observational studies)	Not likely	Not likely	Not likely	Not likely (ratio : 1.18)	Not likely	No upgrade	No upgrade	No upgrade	No upgrade	⊕⊕oo Low	Low
Obesity	Low (observational studies)	Not likely	Not likely	Not likely	Not likely (ratio : 1.23)	Not likely	Upgrade (SRR>2 for BMI from 40 to 50)	Upgrade (evidence of nonlinear relation)	No upgrade	No upgrade	⊕⊕⊕⊕ High	High
Smoking	Low (observational studies)	Not likely	Not likely	Not likely	Not likely (ratio : 1.20)	Not likely	No upgrade	No upgrade	No upgrade	No upgrade	⊕⊕oo Low	Low

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Supplementary method for GRADE

GRADE (Grading of Recommendations, Assessment, Development and Evaluations) is a transparent framework for developing and presenting summaries of evidence and provides a systematic approach for making clinical practice recommendations. The aim is to rate the certainty of evidence (or known as quality of evidence) for each outcome across studies (i.e. a body of evidence) using four levels of evidence: very low, low, moderate, and high. Guidelines are described in a serie of articles published in the Journal of Clinical Epidemiology: https://www.jclinepi.com/action/showPdf?pii=S0895-4356%2810%2900330-6

GRADE does not mean rating each study as a single unit. Rather, rating is made for each outcome, and quality may differ indeed, is likely to differ from one outcome to another within a single study and across a body of evidence.

Quality of evidence	Interpretation
High	We are very confident that the true effect lies close to that of the estimate of the effect
Moderate	We are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect but there is a possibility that it is substantially different
Low	Our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect
Very Low	We have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect

In the GRADE framework, observational study initially starts as low-quality evidence. Then, there are 5 criteria which can be used to downgrade the level of evidence:

- Risk of bias in individual studies – e.g. methodological issues in included studies such as inadequate blinding (participants knew they were in control/treatment group)
- Inconsistency of results between studies
- Indirectness of evidence – e.g. participants were children although the systematic review was about adults
- Imprecision – results were not statistically significant, or the effect was clinically important once the studies were meta-analysed
- Publication bias – result was biased due to a file-drawering effect, as studies not showing a statistically significant effect are less likely to be published.

Additionally, observational studies starting at low can be upgraded based on 3 criteria: large effect, dose-response effect and "Effect of all plausible confounding factors would be to reduce the effect (where an effect is observed) or suggest a spurious effect (when no effect is observed)".

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Guidance to rate the outcome is briefly described below:

1- Risk of bias across studies (internal validity)

Risk of bias was assessed using ROBIN-I rating the overall risk of bias into four categories: low, moderate, serious and critical risk of bias.

Downgrade	Guidance
Not likely	Most information is at low risk of bias
Serious	Does not clearly meet guidance for “not serious” or “very serious”
Very serious	Most information is at serious or critical risk of bias

2- Inconsistency

Magnitude of intervention effects differs across studies, it may be explained by the difference in population, intervention, outcomes or study methods. If large variability (heterogeneity) in magnitude of effect remains unexplained, the quality of evidence decreases.

Downgrade	Guidance
Not likely	Point estimates similar Confidence intervals overlap Statistical heterogeneity is non-significant ($p \geq 0.1$) I^2 of $\leq 50\%$
Serious	Point estimates vary Confidence intervals show minimal overlap Statistical heterogeneity has low p-value ($p \leq 0.1$) I^2 of >50% to 75%
Very serious	Point estimates vary widely Confidence intervals show minimal or no overlap Statistical heterogeneity has low p-value ($p \leq 0.1$) I^2 of >75%

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3- Directness

Certainty can be rated down if the patients studied are different from those for whom the recommendation applies. Decisions regarding indirectness of patients and interventions depend on an understanding of whether biological or social factors are sufficiently different that one might expect substantial differences in the magnitude of effect. Indirectness occurs when the population, intervention, or outcomes differ from those in which we are interested

4- Precision is the degree of certainty surrounding an effect estimate with respect to a given outcome (AHRQ 2012a). A precise estimate enables the evaluator to determine whether there is an effect (i.e., it is different from the comparison group). We used 95% confidence intervals as the primary method to assess imprecision (Guyatt et al. 2011b).

Downgrade	Guidance
Not likely	No or minimal indications of large standard deviations (i.e., SD > mean) For ratio measures (e.g., odds ratio, OR) the ratio of the upper to lower 95% CI for most studies (or meta-estimate) is < 10; or for absolute measures (e.g., percent control response) the absolute difference between the upper and lower 95% CI for most studies (or meta-estimate) is < 100
Serious	Does not clearly meet guidance for "not serious" or "very serious"
Very serious	Large standard deviations (i.e., SD > mean) For ratio measures (e.g., OR) the ratio of the upper to lower 95% CI for most studies (or meta-estimate) is ≥ 10; or for absolute measures (e.g., percent control response) the absolute difference between the upper and lower 95% CI for most studies (or meta-estimate) is ≥ 100

5- Publication bias

We characterized publication bias as "undetected" (no downgrade) or "strongly suspected" as recommended by GRADE (Guyatt et al. 2011e). In general, studies with statistically significant results are more likely to be published than studies without statistically significant results (''negative studies'') (Guyatt et al. 2011e). Thus, some degree of publication bias is likely on any topic; however, downgrading is reserved for cases where the concern is serious enough to significantly reduce confidence in the body of evidence.

Publication bias is suspected when studies are uniformly small, particularly when sponsored by industries, NGOs or authors with conflicts of interests. Funnel plots and Egge's regression can help to visualize publication bias.

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6- Domains that can increase confidence

Upgrade	Guidance
Large magnitude of association or effect	GRADE Working Group has previously suggested guidelines for rating quality of evidence up by one category (typically from low to moderate) for associations greater than 2, and up by two categories for associations greater than 5 (Guyatt et al.2011g).
Dose response	Upgrade for evidence of a monotonic dose-response gradient (Guyatt et al. 2011g) and for evidence of a non-monotonic dose response when data fit the expected pattern, i.e., prior knowledge leads to expectation for non-monotonic dose response, and/or non-monotonic dose response is consistently observed in the evidence base.
Residual Confounding or Other Related Factors	Confidence in the results is increased when the body of evidence is potentially biased by factors counter to the observed effect. Upgrading should be considered when there are indications that residual confounding or bias would underestimate an apparent association or treatment effect (i.e., bias towards the null), or suggest a spurious effect when results suggest no effect.
Cross-Species/Population/Study Consistency	Three types of consistency in the body of evidence can be used to support an increase in confidence in the results: •across animal studies–consistent results reported in multiple experimental animal models or species •across dissimilar populations–consistent results reported across populations (human or wildlife) that differ in factors such as time, location, and/or exposure •across study types–consistent results reported from studies with different design features, e.g., between prospective cohort and case-control human studies or between chronic and multigenerational animal studies

Translation of confidence rating into level of evidence		
Confidence in the body of evidence	Direction of the effect (Health effect)	Level of evidence for the health effect
High	⇨	High
Moderate	⇨	Moderate
Low	⇨	Low
Very low or no evidence	⇨	Inadequate
Confidence in the body of evidence	Direction of the effect (No Health effect)	Level of evidence for the health effect
High	⇨	Evidence of no health effect
Moderate	⇨	Inadequate
Low	⇨	Inadequate
Very low or no evidence	⇨	Inadequate

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Page 1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Page 3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Pages 5-6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Page 6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Page 6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Pages 6-7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Page 6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Page 6, supplement text
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Pages 6-7, Figure 1
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Page 7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Page 6-7, Table S1
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Page è
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Pages 7-9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Pages 7-9

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Page 8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Pages 7-9
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Page 9, Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Page 9, Table S1
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Page 9, Table S2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Table S1, Figures 2-3, 5-7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Pages 9-12
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Table S2, Pages 9-12
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Pages 9-13, Table 1-2, Figures S1-14
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Pages 13, 16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Page 16
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Page 16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Page 17

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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BMJ Open

Diabetes, hypertension, body mass index, smoking and COVID-19-related mortality: A systematic review and meta-analysis of observational studies

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-052777.R1
Article Type:	Original research
Date Submitted by the Author:	26-Aug-2021
Complete List of Authors:	Mahamat-Saleh, Yahya; INSERM U1018 Equipe 11, Inserm, Gustave Roussy, "Exposome and Heredity" team, CESP; Gustave Roussy, Gustave Roussy Fiolet, Thibault; Université Paris-Saclay, Espace Maurice-Tubiana Rebeaud , Mathieu Edouard; University of Lausanne Faculty of Biology and Medicine, Department of Plant Molecular Biology Mulot , Matthieu; University of Neuchatel Faculty of Sciences, Laboratory of Soil Biodiversity Guihur , Anthony; University of Lausanne Faculty of Biology and Medicine El Fatouhi , Douae; Paris-Saclay University Faculty of Medicine, Paris-Saclay University, UVSQ, Inserm, Gustave Roussy, "Exposome and Heredity" team, CESP Laouali , Nasser; Paris-Saclay University Faculty of Science Orsay, Paris-Saclay University, UVSQ, Inserm, Gustave Roussy, "Exposome and Heredity" team, CESP Peiffer-Smadja, Nathan; Hospital Bichat - Claude-Bernard, Infectious and Tropical Diseases Department Aune, Dagfinn; Imperial College London, Department of Epidemiology and Biostatistics; Imperial College London Severi, Gianluca; INSERM
Primary Subject Heading:	Epidemiology
Secondary Subject Heading:	Diabetes and endocrinology, Cardiovascular medicine, Smoking and tobacco, Infectious diseases
Keywords:	COVID-19, Hypertension < CARDIOLOGY, General diabetes < DIABETES & ENDOCRINOLOGY, EPIDEMIOLOGY

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Diabetes, hypertension, body mass index, smoking and COVID-19-related mortality: A systematic review and meta-analysis of observational studies

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Word: Abstract: 314; Full text: 4263

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Abstract

Objectives: We conducted a systematic literature review and meta-analysis of observational studies to investigate the association between diabetes, hypertension, body mass index (BMI) or smoking with the risk of death in patients with COVID-19 and to estimate the proportion of deaths attributable to these conditions.

Methods: Relevant observational studies were identified by searches in the PubMed, Cochrane library and Embase databases through November 14, 2020. Random-effects models were used to estimate summary relative risks (SRRs) and 95% confidence intervals (CIs). Certainty of evidence was assessed using the Cochrane methods and the GRADE framework.

Results: A total of 186 studies representing 210,447 deaths among 1,304,587 patients with COVID-19 were included in this analysis. The SRR for death in COVID-19 patients was 1.54 (95% CI=1.44-1.64, I^2=92%, n=145, low certainty) for diabetes and 1.42 (95% CI=1.30-1.54, I^2=90%, n=127, low certainty) for hypertension compared to patients without each of these comorbidities. Regarding obesity, the SSR was 1.45 (95% CI=1.31-1.61, I^2=91%, n=54, high certainty) for patients with BMI ≥30kg/m^2 compared to those with BMI<30kg/m^2 and 1.12 (95%CI=1.07-1.17, I^2=68%, n=25) per 5 kg/m^2 increase in BMI. There was evidence of a J-shaped non-linear dose-response relationship between BMI and mortality from COVID-19, with the nadir of the curve at a BMI of around 22-24, and a 1.5-2 fold increase in COVID-19 mortality with extreme obesity (BMI of 40-45). The SRR was 1.28 (95% CI=1.29-1.50, I^2=74%, n=28, low certainty) for ever, 1.29 (95% CI=1.03-1.62, I^2=84%) for current and 1.26 (95% CI=1.11-1.42, I^2=84%) for former smokers compared to never smokers. The absolute risk of COVID-19 death was increased by 14%, 11%, 12% and 7 % for diabetes, hypertension, obesity, and smoking, respectively. The proportion of deaths attributable to diabetes, hypertension, obesity, and smoking was 8%, 7%, 11%, and 2%, respectively.

Conclusion: Our findings suggest that diabetes, hypertension, obesity and smoking were associated with higher COVID-19 mortality, contributing to nearly 30% of COVID-19 deaths.

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Systematic review registration This study is registered with PROSPERO, CRD42020218115.

Keywords: diabetes; hypertension; obesity; smoking; COVID-19; mortality; meta-analysis

Strengths and limitations of this study

- We did a systematic review and meta-analysis of 186 observational studies (210,447 deaths among 1,304,587 patients with COVID-19) to investigate the association between diabetes, hypertension, obesity and smoking and risk of death in patients with COVID-19 and to estimate the proportion of deaths attributable to these four conditions.
- We performed several subgroup analyses, and assessed certainty of evidence using the Cochrane methods and the GRADE framework. A non-linear dose-response relation between BMI and COVID-19 mortality was also explored.
- However, we did not investigate the association between presence of two or more coexisting comorbidities and risk of death in patients with COVID-19.

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Introduction

Coronavirus disease 2019 (COVID-19) is a viral infectious disease caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), which was first reported in Wuhan City, China, in December 2019[1]. SARS-CoV-2 has since spread to all countries worldwide and COVID-19 has been declared a pandemic by the World Health Organization (WHO)[2]. As of 24 August 2021, over 212,3 million cases and 4,4 million deaths have been reported globally since the start of the pandemic[3]. Age is the main risk factor for poor outcome in people with COVID-19 infection[4,5], as it is correlated with more comorbidities. About 70 to 87% of COVID-19 deaths are among people aged 70 years or older[6,7]. Patients with comorbidities, including diabetes, cardiovascular disease, respiratory disease, chronic kidney disease and others chronic diseases are at increased risk of developing severe or critical COVID-19[8,9], which may partly explain a greater mortality in hospital[10–14]. Studies suggest that about 20 to 51% of patients hospitalized with COVID-19 have at least one comorbidity[10,15].

Previous meta-analyses reported a higher mortality rate from COVID-19 in patients with comorbidities[16–22]. Ssentongo et al, based on 25 studies published from December 2019 to July 9th, 2020, suggested that diabetes and hypertension were respectively associated with a 1.48-fold and 1.82-fold greater risk of COVID-19 death compared to those without these comorbidities[16]. Based on studies published during the same period, Luo et al reported similar results[17]. Du et al found that patients with obesity had a 2.68-fold risk of dying from COVID-19 compared to non-obese patients[18]. Most of the published meta-analyses did not investigate the shape of the dose-response relationship between Body-Mass Index (BMI) and risk of death in order to clarify whether the association is dose-dependent or if there are threshold effects[20,23–25]. In addition, evidence suggests that smoking may increase risk of severe disease and death from COVID-19[19]. However, it is not clear whether such an increase in COVID-19 mortality is different in current and past smokers since previous meta-analyses have not performed separate analyses[19,26,27].

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However, since the publication of these meta-analyses, several observational studies have been published on diabetes, hypertension, obesity or smoking and risk of death in patients with COVID-19[4,28–122] and the strength of the associations differed greatly between studies. Moreover, the proportion of deaths attributable to diabetes, hypertension, obesity or smoking habits has not been estimated. This last aspect may help adapting public health measures and vaccination strategies to populations at risk of severe COVID-19.

Given the rapidly increasing death from COVID-19 globally, and since diabetes, hypertension, obesity and tobacco smoking represent the most important public health problems worldwide, which contributed to higher risk of death globally; we thus conducted a systematic review and meta-analysis of published observational studies to investigate the association between diabetes, hypertension, smoking and obesity and risk of death in patient with COVID-19 and to clarify the strength of these associations. We further estimated the proportion of deaths attributable to these conditions.

Materials and methods

Search strategy and selection criteria

The meta-analysis was reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement[123] and Meta-analysis of Observational Studies in Epidemiology (MOOSE)[124]. This study was registered and accepted in the International Prospective Register of PROSPERO (registration number CRD42020218115) in October 2020. PubMed (MEDLINE), Cochrane library and Embase databases were searched to identify relevant articles published in English from December 2019 to 14th November 2020. The search terms that we used are provided in the **supplementary text**. In addition, we searched the reference lists of the relevant publications, reviews and meta-analyses to identify additional potentially relevant studies. We only included observational studies (cohort studies and cross-sectional) that reported relative risk estimates (such as hazard ratios (HR), risk ratios (RR) or odds ratios (OR)) with the 95% confidence intervals (CIs) with or without adjustment for

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potential confounders. The search was independently screened by two researchers (Y.MS and T.F) and discrepancies were resolved by discussion with a third researcher (M.ER).

Data collection

From each included publication, we extracted results and study characteristics which included first author's last name, publication year, country where the research was conducted in, study design, study description or name, study period, sample size with number of deaths, exposure, categories, risk estimate and 95% CIs, and adjustment factors. Data was extracted by Y.MS and extractions were checked for accuracy by T.F. Discrepancies were resolved through discussion with a third researcher (M.ER).

Quality assessment and risk of bias

The quality of individual studies was assessed independently by two researchers (Y.MS and T.F) using the Cochrane risk of bias tool ROBINS-I, which grades studies on a scale from critical risk of bias to low risk of bias considering bias due to confounding, selection of study participants, exposure measurement, misclassification of exposure during follow-up, missing data, measurement of outcomes, and bias due to selection of reported results[125]. Following the assessment of risk of bias, the body of evidence for each comorbidity and risk of death was rated independently using the GRADE (Grading of Recommendations, Assessment, Development and Evaluations) approach[126]. Discrepancies were resolved through discussion with a third researcher (D.A).

Data analysis

We used random effects models that consider both within study and between-study variation to calculate summary RRs (SRR) (95% CIs) of COVID-19 mortality for patients with diabetes compared to those without diabetes, for patients with hypertension versus those without hypertension, for obese versus non-obese and for current, former and ever smoker compared to never smokers. The natural logarithm of the RRs was weighted using random effects weights[127]. Statistical heterogeneity between studies was assessed by the Cochrane Q test and the I^2 statistic[128]. We calculated the absolute Risk difference (RD) from the

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baseline risk of mortality (BR) from Docherty et al[129], large cohort and Relative Risk (RR) using the formula RD=BR×(RR−1)[130].

We further performed a dose-response analysis for the associations between BMI and COVID-19 mortality using the method described by Greenland and Longnecker to compute the linear trend from the natural logs of the RRs and CIs across categories of BMI[131]. We calculated SRRs and 95% CIs for a 5-unit increment in BMI using random effects models. This method required mean or median of BMI, RRs and 95% CIs for at least three categories. The mean or median BMI level per category was used if provided in the publication, and if not, the midpoint of the upper and lower boundaries was estimated as a range in each category. When the highest and lowest categories were open-ended, we used the width of the adjacent interval to estimate the upper and lower boundaries for the category. For studies that reported results separately for young and adults, for current and former smoker, but not overall, we pooled the results using a fixed-effects model as reported by the Hamling procedure to obtain an overall estimate to be used in the meta-analysis[132].

To explore the potential nonlinear dose-response relation between BMI and mortality among patients with COVID-19, we used fractional polynomial models[133]. We determined the best fitting second order fractional polynomial regression model, defined as the one with the lowest deviance. Only studies which presented more than two categories of BMI were included in the nonlinear analysis. Subgroup and meta-regression analyses were conducted to investigate potential sources of heterogeneity. Small-study effects, such as publication bias, were visually assessed by examining funnel plots for asymmetry, and with Egger's test[134], and the results were considered to indicate potential small-study bias when p-values were<0.10. We conducted sensitivity analyses excluding one study at a time to clarify whether the results were driven by one large study or a study with an extreme result.

We finally calculated the population attributable fraction (PAF) of mortality among patients with COVID-19 due to diabetes, hypertension, obesity and smoking, worldwide using the following formula[135]:

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$$PAF = \frac{p \times (RR - 1)}{[p \times (RR - 1)] + 1}$$

Where RR was the relative risk, p was the prevalence of the exposure in patient with COVID-19. The prevalence of diabetes (11.5%), hypertension (22.9 %), obesity (29%) and smoking (9%) were obtained from previous meta-analyses[136–138].

Patient and public involvement

Patients or the public were not involved in any aspect of the study design, conduct or in the development of the research question or outcome measures.

Results

A total of 6,007 records were identified in MEDLINE, Cochrane library and in EMBASE (**Figure 1)**. A total of 4,665 publications were excluded after reading title and abstract or because of duplicates. Among 1,342 full-text articles retrieved, 994 were excluded as not meeting the inclusion criteria, leaving a total of 348 publications. Of these, 162 articles were not eligible because they lacked sufficient data[139,140], reported no risk estimate or irrelevant data[139,140], because they had identical populations[141–151] or were retracted[152]. Finally, a total of 186 observational studies were included in this meta-analysis. Of the included studies, 58 were from Europe, 58 from North America, 60 from Asia, 6 from South America and 4 from Africa (**Table S1)**. From the 186 publications assessed using the ROBINS-I tool, 92 were evaluated as being at low risk of bias, 49 at moderate risk of bias, 21 at serious risk of bias and 24 at critical risk of bias because of insufficient adjustment of relevant confounders (**Table S2**).

Diabetes and mortality in patient with COVID-19

A total of 145 studies[4,28–119,121,122,129,153–201] were included in the analysis of the association between diabetes and mortality, including a total of 198,491 deaths among 1,165,897 patients with COVID-19. The SRR for diabetes patients compared to those without diabetes was 1.54 (95% CI=1.44-1.64) and there was high heterogeneity among studies (I^2=92%). We found that diabetes increased the absolute risk of death

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by 14% (**Table 1**). The funnel plot did not provide evidence of publication bias (**Figure S1**) and we found no evidence of small study effects (Egger's test, P-value=0.54).

The positive association persisted across all subgroups analyses stratified by study design, number of patients, geographic location, and adjustment for some confounding factors (**Table 2**). However, meta-regression analyses suggested some indication of heterogeneity between studies that adjusted for multiple factors such as age, sex and other comorbidities versus studies without such adjustment or adjusted for age only ($P_{heterogeneity}$=0.003), with a stronger association for the studies with no adjustment factors or those adjusted for age only. The positive association was also stronger among studies with critical or serious risk of bias compared to those with moderate or low risk of bias, with heterogeneity detected across studies ($P_{heterogeneity=}$0.001).

The influence analysis showed no substantial influence of any of the included studies on the global estimate of diabetes and mortality. The SRR ranged from 1.51 (95% CI=1.42-1.61) when excluding the Barbu et al. study[35] to 1.55 (95% CI=1.45-1.64) when excluding the Meng et al. study[114] (**Figure S2)**.

Hypertension and mortality in patient with COVID-19

We included 127 observational studies[4,29,30,32,33,35–39,42–45,47–50,52–56,58,59,61–63,65,66,69,71–76,78,81–84,86,88–90,92–95,97,103,105,105,106,106,108,110–112,114–118,120–122,153,154,156,156,156–159,161–167,169,173–180,182–188,193–195,195–200,202–208,208–221] in the analysis of the association between hypertension and mortality, including a total of 113,243 deaths among 934,958 patients with COVID-19. The SRR for hypertension patients versus those without hypertension was 1.42 (95% CI=1.30-1.54) with high evidence of heterogeneity (I^2=90%) (**Table 1**). The absolute risk of death for COVID-19 patients with hypertension compared to without hypertension was increased by 11%. Small study effects, such as publication bias, were not indicated with the funnel plot (**Figure S3**) or Egger's test (P-value=0.26).

Here again, the positive association persisted in most subgroup analyses (**Table 2**). Our meta-regression analysis showed that the study design and the geographic location did not significantly influence the magnitude of the overall association. However, heterogeneity between subgroup analyses

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was observed in analyses stratified by risk of bias with higher association for studies with critical compared to those with low, moderate or serious risk of bias ($P_{heterogeneity}$=<0.0001) and by adjustment for confounding factors with higher association for studies without adjustment compared to those that adjusted for age only or for multiple factors ($P_{heterogeneity}$=<0.0001).

When excluding the most influential studies, we found no substantial influence of any of the included studies (**Figure S4**).

BMI and mortality in patient with COVID-19

Fifty-four studies [29,32,38,44,45,48,49,52,61,62,64,65,67,69,73,74,79,84–86,88–90,93,99,103,107,118,120–122,129,154,164,165,178,181,182,184,192,196,197,202,207,222–230] investigated BMI and mortality risk including 145,605 deaths among 858,374 patients with COVID-19 were identified. The analysis yielded a SRR of 1.45 (95% CI=1.31-1.61) for obese (BMI≥30kg/m^2) versus non-obese (BMI<30kg/m^2) patients, with high heterogeneity detected between studies (I^2=91%) (**Figure 2**) (**Table 1**). We found that obesity increased the absolute risk of death by 12%. There was no statistically significant evidence of publication bias (P-value Egger's test=0.92) (**Figure S5**).The meta-regression analysis showed that study characteristics did not significantly alter the overall estimate (**Table S3**). When excluding the most influential studies one by one, the global estimate did not substantially change (**Figure S6**).

Twenty-five studies (32,072 deaths among 95,852 patients with COVID-19)[38,56,61,74,86,89,91–93,97,108,111,112,161,162,175,177,223,224,231–237] were included in the dose-response meta-analysis of BMI and mortality risk. The summary RR for a 5 kg/m^2 increment in BMI was 1.12 (95% CI=1.07-1.17, I^2=68%) (**Figure 3**) and no statistically significant evidence of publication bias (P-value=0.11) or by inspection of the funnel plot was observed (**Figure S7**). However, evidence of a J-shaped nonlinear relation between BMI and mortality risk was observed ($P_{nonlinearity}$=<0.0001), suggesting a flat dose–response curve at a BMI around 22-24 kg/m^2 with a slight increase in risk of death below that range and a 1.5-2 fold increase in risk of death with a BMI of 40-45 vs. 22-24 kg/m^2 (**Figure 4**).

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In sensitivity analyses excluding one study at a time from the analysis, the summary for a 5 kg/m^2 increment in BMI ranged from 1.11 (95% CI=1.06-1.15) when excluding the Czernichow et al. Study[108] to 1.13 (95% CI=1.08–1.19) when excluding the Ferrando-Vivas et al. study[232] (**Figure S8**).

Smoking and mortality in patient with COVID-19

Twenty-eight studies[38,58,61,63,66,67,76,93,97,103,108,110,111,118,121,129,161,164–166,175,177,183,204,234,238–240] were included in the analysis of ever smoking versus never smoking and mortality with a total of 11,333 deaths among 47,096 COVID-19 patients. The SRR for hospital death in patient with COVID-19 was 1.28 (95% CI=1.17-1.40, I^2=74%) for ever smokers versus never smokers (**Figure 5**). The absolute risk of death for smoking was increased by 7% (**Table 1**). There was no evidence of publication or small study bias (P-value=0.91) (**Figure S9**). In sensitivity analyses, the results persisted when excluding one study at a time (**Figure S10**).

For current smoking versus never smoking, nineteen studies [4,58,66,76,97,103,110,111,117,118,121,129,161,162,165,177,203,238,241], which included 9,845 deaths among 33,147 patients with COVID-19, were identified. The SRR of current smoking was 1.29 (95%CI=1.03-1.62, I^2=84%) (**Figure 6**) and no statistically significant evidence of publication bias was observed (P-value=0.86) (**Figure S11**).

A total of 14 studies[58,66,76,97,103,110,111,118,121,129,161,165,177,238] were included in the analysis of former smoking versus never smoking and mortality risk, including 8,121 deaths among 25,340 patients with COVID-19. The SRR was 1.25 (95% CI=1.11-1.42) with moderate to high heterogeneity (I^2=75%) (**Figure 7**). There was no evidence of publication bias with Egger's test (P-value=0.70). In sensitivity analyses excluding one study at a time from the analyses of current and former smoking, the results were not materially altered (**Figures S12-13**). Globally, results did not change in nearly all subgroup analyses (**Table S4**).

Finally, a total of 15 studies[29,44,52,58,58,107,174,186,192,209,220,242–245] did not provide a definition of the smoking variable (>76,400 deaths, 682,310 patients) and the SRR was 1.31 (95% CI=1.07-1.62, I^2=88%) (**Figure S14**) which seemed to be similar to current and ever smoking.

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Population attributable fraction of deaths and assessment of certainty of the body of evidence

The estimated PAF was 10.6 %, 11.8 %, 11.5 % and 2.5 % for diabetes, hypertension, obesity and ever smoking, respectively when considering all studies included in this meta-analysis (**Table 1)**. Based on studies that adjusted for multiple risk factors, attributable death was 8% for diabetes, 7% for hypertension, 11% for obesity and 2% for ever smoking.

Regarding the body of evidence, certainty of evidence was rated "low" for diabetes, hypertension and smoking and "high" for obesity (**Table S5**).

Discussion

- **Principal findings**

To our knowledge, this study is one of the largest meta-analyses of association between diabetes, hypertension, obesity, smoking and COVID-19 mortality. We found that patients with diabetes had a 54% higher risk of death from COVID-19 compared to patients without diabetes; those with hypertension had a 42% increase in the relative risk of death from COVID-19 compared to patients without hypertension and those with obesity have a 45% greater in the relative risk of COVID-19 death compared to non-obese patients. In addition, we found that ever, current and former smoking was associated with 28%, 29% and 25% increases in the relative risk of death in COVID-19 patients. Our linear dose-response meta-analysis suggested that each 5 kg/km^2 increment in BMI was associated with a 12% greater risk of COVID-19 death. However, evidence of nonlinearity was observed in the analysis of BMI and risk of COVID-19 death, with a J-shaped dose-response relation with flattening of the dose–response curve between 22 and 24 of BMI level and a slight increase below that range and a 1.5-2 fold increase in risk with a BMI of 40-45. While there was no publication bias, study heterogeneity was high for all exposure and this persisted in most of the subgroup analyses. However, the heterogeneity appeared to be driven to a larger extent by differences in the strength of the associations, than differences in the direction of the effect, as the vast majority of studies reported significant or non-significant positive associations between these exposures

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and increased mortality, and relatively few studies reported risk estimates in the direction of an inverse association. Given this meta-analysis included more studies than a typical meta-analysis, I^2 and heterogeneity were high as the likelihood of divergent findings increases with increasing number of studies.

- **Comparisons to findings from previous epidemiologic studies and biological mechanisms**

Since the first reports of COVID-19, several studies have shown that COVID-19 patients with comorbidities have a higher risk of death[4,246–248]. However, these studies have differed greatly in term of sample size, and results are conflicting and heterogeneous. Previous meta-analyses have shown that patients with diabetes, hypertension and obesity had an increased risk of mortality[16,18,21,248–250]. Unfortunately, these previous meta-analyses were limited by the lack of subgroup analyses, which is crucial to evaluate heterogeneity and no previous meta-analysis has estimated the number of deaths attributed to these comorbidities or conditions.

This meta-analysis summarizes the results of 186 observational studies published up to November 2020, including 210,447 deaths among 1,304,587 million patients with COVID-19. Our findings are similar to results from previously published systematic reviews, suggesting a higher mortality rate of COVID-19 in patients with cardiovascular or chronic condition[16,21,22,248]. We found a 1.54-fold greater mortality from COVID-19 among patient with diabetes compared with those without (n=145 studies), which is similar to those yielded in previous meta-analyses[16], whereas the estimate magnitude is weaker than those in Almeida-Pititto et al., 2020[21], Kumar et al., 2020[249], Shang et al.[251] and Luo et al., 2020[17]. In our subgroup analyses, we found that the positive association was stronger in studies without any adjustment or adjusted for age only compared to studies that adjusted for multiple factors. Thus, the higher magnitude observed in the previous meta-analyses may be in part due to the important number of studies without adjustment for confounding factors such as age and comorbidities, which are mostly with critical or serious risk of bias. In addition, we estimated that 8% of deaths by COVID-19 were attributed to diabetes; this aspect has to our knowledge not been investigated previously. The absolute risk of death

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associated with diabetes was increased by 14%. The mechanism underlying the increased mortality from COVID-19 in patients with diabetes may be explained by chronic inflammatory conditions. COVID-19 patients with diabetes have a significantly higher inflammatory markers such as C-reactive protein, interleukin-6 (IL-6) compared to patients without diabetes[252]. Inflammatory markers such as IL-6, CRP, IL-10, LDH and TNF-α, which are indicative of different aspects of COVID-19 severity, requirements of intensive care support including dialysis and ventilation, are associated with higher risk of death[253,254].

Regarding hypertension, our findings yielded a 1.42-fold higher risk of death from COVID-19 in patient with hypertension (n=127 studies), which also is slightly weaker than the results from previous meta-analyses[16,22,250]. Nearly 7% of death in patients with COVID-19 could be attributed to hypertension and we found that hypertension increased the absolute risk of death by 11%. Previous clinical studies showed that hypertension is a major risk factor for worse outcome in patients infected with SARS and Middle East respiratory syndrome[255,256]. Although, the exact mechanism by which hypertension increase mortality rate remains unclear, chronic inflammation may play an active role in increasing risk of death.

Du et al. suggested that patients with obesity had a 2.68-fold risk for COVID-19 mortality compared with non-obese patients (n=7 studies)[18]. In dose-response analysis, they showed that for each 1 kg/m^2 increase in BMI, the risk of death increased by 6%. Partly consistent with this study, we found that a 5 kg/m^2 increase in BMI level was associated with a 12% increase in the risk of death in patient with COVID-19 (n=25 studies). In addition, we found evidence of a nonlinear J-shaped association between BMI and mortality from COVID-19, with a flattening of the dose–response curve for BMI values between 22 and 24 kg/m^2 and with a slightly higher risk below that range and a moderate to strong increase in mortality with severe obesity (BMI 40-45). Obesity is associated with a low-grade systemic inflammation, which plays a major role in the pathogenesis of respiratory conditions. Patients with COVID-19 and with preexisting obesity may have an overactivated inflammation response, which may induce excessive inflammatory response[257]. Obesity is also strongly associated with increased risk of diabetes, hypertension, and several other chronic diseases[258] that increases risk of COVID-19 mortality[259].

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In addition, our finding showed that ever, current and former smoking was associated with 28%, 29% and 26% increases in the relative risk of death in patients with COVID-19 compared to never smokers. Tobacco smoking is known to alter the function of the immune system; therefore smokers are more likely to get a severe infection of COVID-19 due to their poor mucociliary clearance, which could lead to the release of pro-inflammatory markers and oxidative stress and thereby contribute to higher mortality rates. The risk of death in former smokers was higher than in never smokers and only slightly lower than risk in current smoker. However, more studies are required to clarify the impact of longer durations of smoking cessation in former smokers on risk of death in COVID-19 patients.

Although we found that diabetes, hypertension, BMI and smoking were associated with greater COVID-19 mortality, a recent meta-analysis suggested that mortality was more frequently observed in COVID-19 patients with cardiovascular disease, cerebrovascular accident, and chronic kidney disease[260]. The authors observed that COVID-19 mortality among all comorbidities was high in European and Latin American patients compared to the US patients. It is possible that geographical differences in therapeutic practice of COVID-19 such as the use of antibiotics, antivirals and others drugs may partly explain the greater COVID-19 death in some regions[261], while there was no evidence of heterogeneity in findings across geographic location in our study. The review also suggested that COVID-19 mortality among those with underlying medical diseases was high in mostly elderly patients[261]. However, we did not perform subgroup analysis by age because this information was lacking in most of the included studies.

- **Strengths, limitations and public health implications**

This present meta-analysis of observational studies on diabetes, hypertension, obesity and smoking and risk of death in patients with COVID-19 has several strengths, including the large sample size and number of COVID-19 deaths, the detailed subgroup and sensitivity analyses, as well as the linear and nonlinear dose-response analyses, which clarified the strength and shape of the dose-response relationship. Original aspects of our study included the estimation of the number of deaths attributable to these conditions. This is, to our knowledge, the first meta-analysis that perform a separate analysis of ever,

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current and former smoking versus never smoking in relation to COVID-19 mortality. As any previous published meta-analyses, the current analysis has some limitations that should be considered in the interpretation of our findings. First, we did not investigate the association between presence of two or more coexisting comorbidities and risk of death in patients with COVID-19. Finally, subgroup analyses stratified by clinical or lifestyle factors such as medications, diabetes type and duration, adherence to specific diet, or physical activity were not possible because of the lack of such data from the studies included.

Despite these limitations, our findings may have important public health implications in the context of increasing numbers of severe COVID-19 cases, overburdened hospitals and leading to higher hospital death due to COVID-19 and suggest that people with cardiovascular risk factors, in particular those with diabetes, hypertension and obesity, should be considered as a high priority to get vaccinated. In addition, since smoking is a risk factor for several chronic diseases, including cancer and cardiovascular disease, our finding lend support to the importance of smoking prevention and smoking cessation and support policies and public health efforts to reduce the prevalence smoking in the general population.

- **Conclusion**

Our finding suggests that presence of diabetes, hypertension, obesity, and smoking in patients with COVID-19 is associated with a 1.54-fold, 1.42 fold, 1.45 fold and 1.28 fold greater risk of mortality, respectively. We have found that the proportion of death attributable to diabetes, hypertension, obesity, and smoking was 8%, 7%, 11%, and 2% respectively. These findings support that people with diabetes, hypertension, obesity should be prioritized for vaccination in order to limit the higher death rates in hospital. Public policies should promote a healthier lifestyle including healthier diets and regular physical activity to reduce patient risk factors and comorbidities.

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Author Contributions: Y.MS conceived and designed the research. Y.MS performed statistical analysis and wrote the first draft of the paper. YMS, TF and MR performed the literature search and literature screening. Y.MS and TF assessed the risk of bias of the studies and assessed the certainty of evidence of the associations. MM assisted with the statistical analysis. YMS, TF, MR, MM, AG, DEF, NL, NPS, DA and GS contributed to data interpretation and revision of the manuscript for intellectual content. Y.MS takes responsibility for the integrity of the data and the accuracy of the data analysis.

Conflict of Interest Disclosures: We declare no competing interests

Funding: There was no funding source for this study

Data sharing statement: All data relevant to the study are included in the article or uploaded as supplementary information.

Provenance and peer review: Not commissioned; externally peer reviewed.

Patient consent for publication: Not required

Ethics approval statement: We adhered to the Preferred Reporting Items for Systemic Reviews and Meta-Analyses (PRISMA) guidelines. Ethical committee approval was not required for this study.

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Table 1: Summary relative risk, risk difference and fraction of deaths attributable to diabetes, hypertension, obesity and smoking in patients with COVID-19

	Number of studies	SRR (95% CI)	I^2 (%)	Risk difference (95% CI) (%)	PAF (95% CI) (all included studies) (%)	PAF (95% CI) (for fully adjusted studies) (%)	Number of patients	Number of deaths
Prognostic factors								
Diabetes	145	1.54 (1.44-1.64)	92	+14 (+11 to +17)	10.6 (8.8 to 12.4)	8.1 (6.2 to 9.7)	1,165,897	198,491
Hypertension	127	1.42 (1.30-1.54)	90	+11 (+8 to +14)	11.8 (8.7 to 14.7)	6.9 (4.0 to 9.6)	934,958	113,243
Obesity	54	1.45 (1.31-1.61)	91	+12 (+8 to +16)	11.5 (8.2 to 15.0)	11.1 (7.1 to 15.0)	858,374	145,605
Ever smoking	28	1.28 (1.17-1.40)	74	+7 (+4 to +10)	2.45 (1.5 to 3.5)	2.0 (0.8 to 3.3)	47,096	11,333

Footnote: CI confidence interval, PAF population attributable fraction of mortality; SRR summary relative risk

I^2 (%) is a measure of the proportion of the heterogeneity attributed to between study variation rather than due to chance. I^2-values of 25%, 50% and 75% indicates low, moderate and high between study heterogeneity, respectively.

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Table 2. Subgroup analyses of association between diabetes and hypertension and mortality risk in patients with COVID-19

	Diabetes					Hypertension				
	n	SRR (95% CI)	I^2 (%)	P_{within}[a]	$P_{between}$[b]	n	SRR (95% CI)	I^2 (%)	P_{within}[a]	$P_{between}$[b]
All studies	145	1.54 (1.44-1.64)	91.7	<0.0001		127	1.42 (1.30-1.54)	90.1	<0.0001	
Study design										
Retrospective	129	1.54 (1.44-1.64)	91.7	<0.0001		112	1.42 (1.30-1.56)	90.9	<0.0001	
Prospective	11	1.28 (1.12-1.48)	70.0	<0.0001		9	1.21 (0.98-1.50)	73.4	<0.0001	
Cross-sectional	5	2.47 (1.56-3.94)	81.4	<0.0001	0.05	6	1.70 (1.18-2.44)	61.8	0.02	0.50
Geographical location										
North America	45	1.33 (1.19-1.48)	90.3	<0.0001		40	1.30 (1.12-1.50)	93.4	<0.0001	
South America	6	1.54 (1.27-1.87)	93.7	<0.0001		2	2.03 (0.99-4.18)	69.1	0.072	
Europe	46	1.53 (1.36-1.72)	93.8	<0.0001		43	1.36 (1.20-1.54)	82.8	<0.0001	
Asia	43	1.94 (1.65-2.29)	75.1	<0.0001		38	1.62 (1.28-2.04)	86.1	<0.0001	
Africa	5	1.56 (0.92-2.62)	87.3	<0.0001	0.10	4	1.08 (0.86-1.35)	14.9	0.318	0.33
Number of patients										
<1000	79	1.73 (1.52-1.97)	74.8	<0.0001		78	1.56 (1.36-1.79)	74.8	<0.0001	
≥1000	66	1.43 (1.32-1.55)	95.4	<0.0001	0.001	49	1.28 (1.14-1.44)	94.8	<0.0001	<0.0001
Patients admission unit										
Non-ICU admitted	142	1.55 (1.45-1.66)	91.6	<0.0001	0.24	125	1.43 (1.31-1.56)	90.2	<0.0001	0.28
ICU admitted	3	1.22 (1.14-1.30)	0.0	0.81		2	0.98 (0.82-1.17)	0.0	0.81	
Risk of bias										
Low	72	1.46 (1.34-1.59)	92.4	<0.0001		63	1.17 (1.08-1.29)	80.2	<0.0001	
Moderate	37	1.28 (1.19-1.39)	64.9	<0.0001		34	1.34 (1.13-1.59)	88.9	<0.0001	
Serious	17	1.90 (1.44-2.51)	86.4	<0.0001		15	1.83 (1.42-2.37)	84.4	<0.0001	
Critical	19	2.11 (1.61-2.77)	93.4	<0.0001	0.001	15	2.64 (1.61-4.34)	96.3	<0.0001	<0.0001
Adjustment for confounders										
No	21	2.09 (1.62-2.69)	89.4	<0.0001		17	2.54 (1.62-3.99)	95.9	<0.0001	
Age only	18	1.90 (1.56-2.31)	85.4	<0.0001		14	1.78 (1.36-2.32)	85.1	<0.0001	
Multiple	106	1.40 (1.30-1.49)	87.5	<0.0001	0.003	96	1.23 (1.13-1.33)	85.4	<0.0001	<0.0001

Footnote: CI confidence interval; SRR summary relative risk

I^2 (%) is a measure of the proportion of the heterogeneity attributed to between study variation rather than due to chance. I^2-values of 25%, 50% and 75% indicates low, moderate and high between study heterogeneity, respectively.

[a]P-value for heterogeneity within each subgroup

[b]P-value for heterogeneity between subgroups with meta-regression analysis

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Figure legends:

Figure 1: Flow-chart of study selection

Figure 2: Association between obesity and mortality risk in patients with COVID-19

Figure 3: Association between BMI and mortality risk in patients with COVID-19, linear dose-response analysis, per 5kg/m^2 increment of BMI

Figure 4: Association between BMI and mortality risk in patients with COVID-19: nonlinear dose-response analysis

Figure 5: Association between ever smoking (versus never smoking) and mortality risk in patients with COVID-19

Figure 6: Association between current smoking (versus never smoking) and mortality risk in patients with COVID-19

Figure 7: Association between former smoking (versus never smoking) and mortality risk in patients with COVID-19

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Figure 1: Flow-chart of study selection

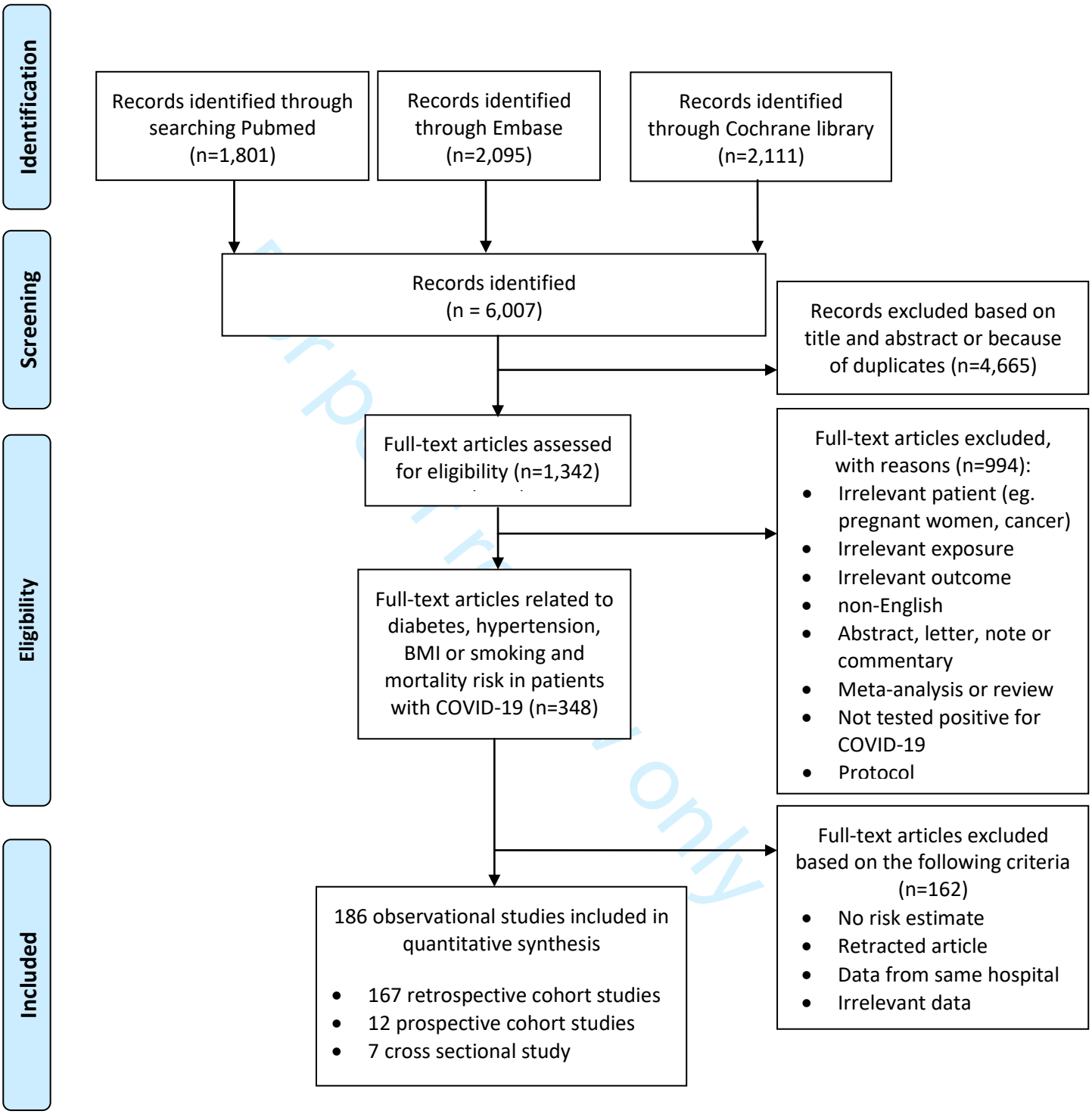


Figure 2: Association between obesity and mortality risk in patients with COVID-19

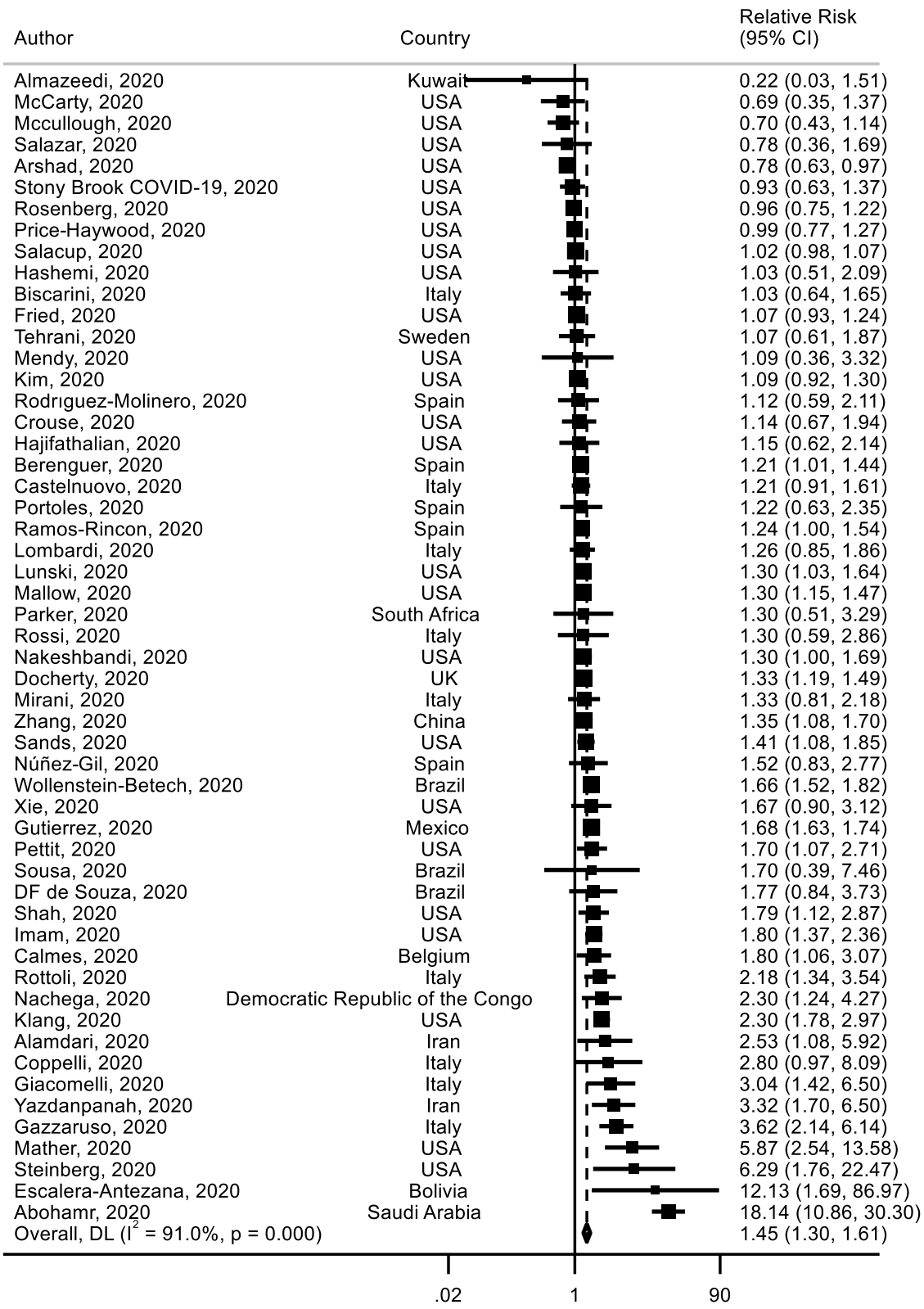


Figure 3: Association between BMI and mortality risk in patients with COVID-19, linear dose-response analysis, per 5kg/m^2 increment of BMI

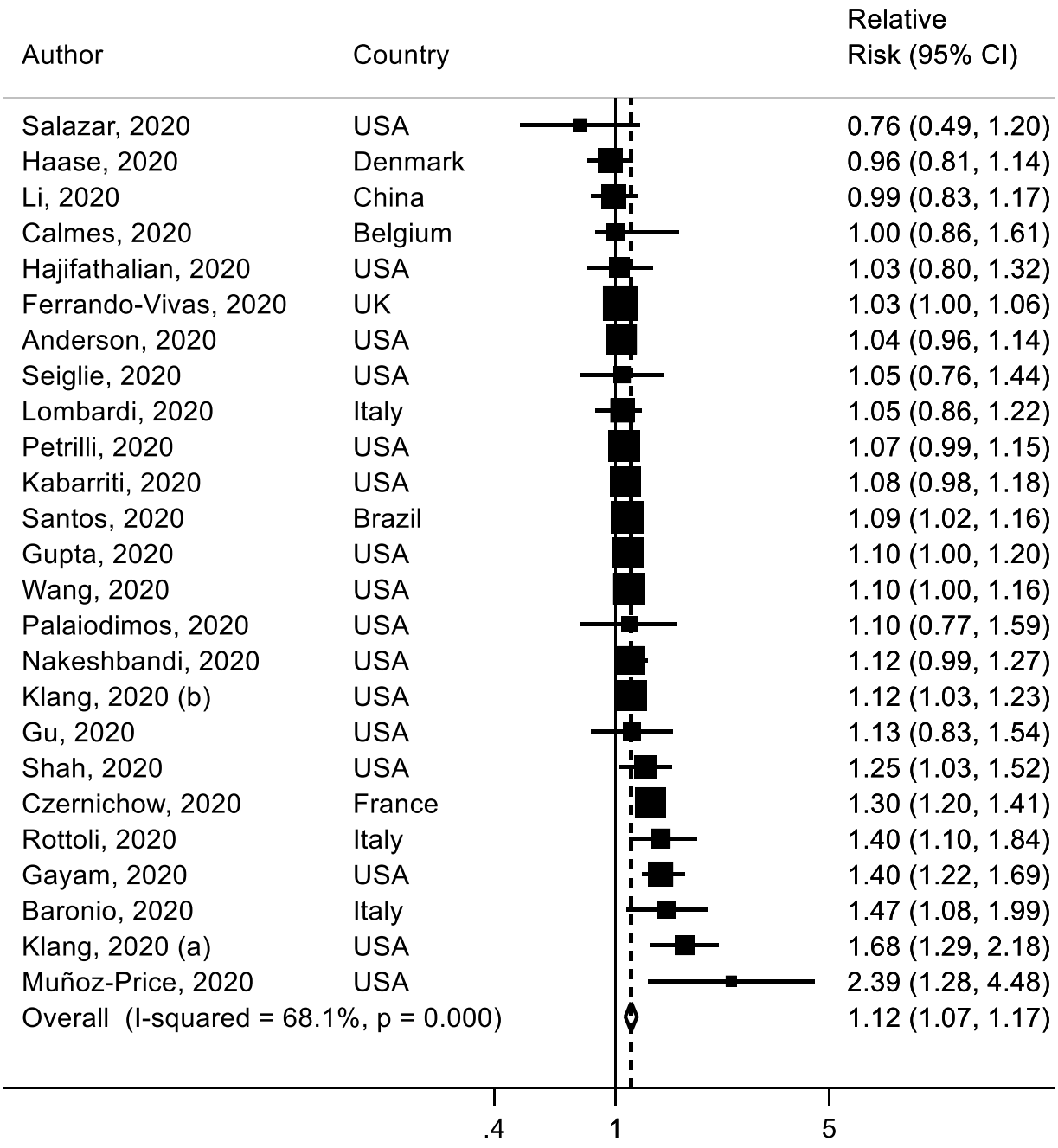


Figure 4: Association between BMI and mortality risk in patients with COVID-19: nonlinear dose-response analysis

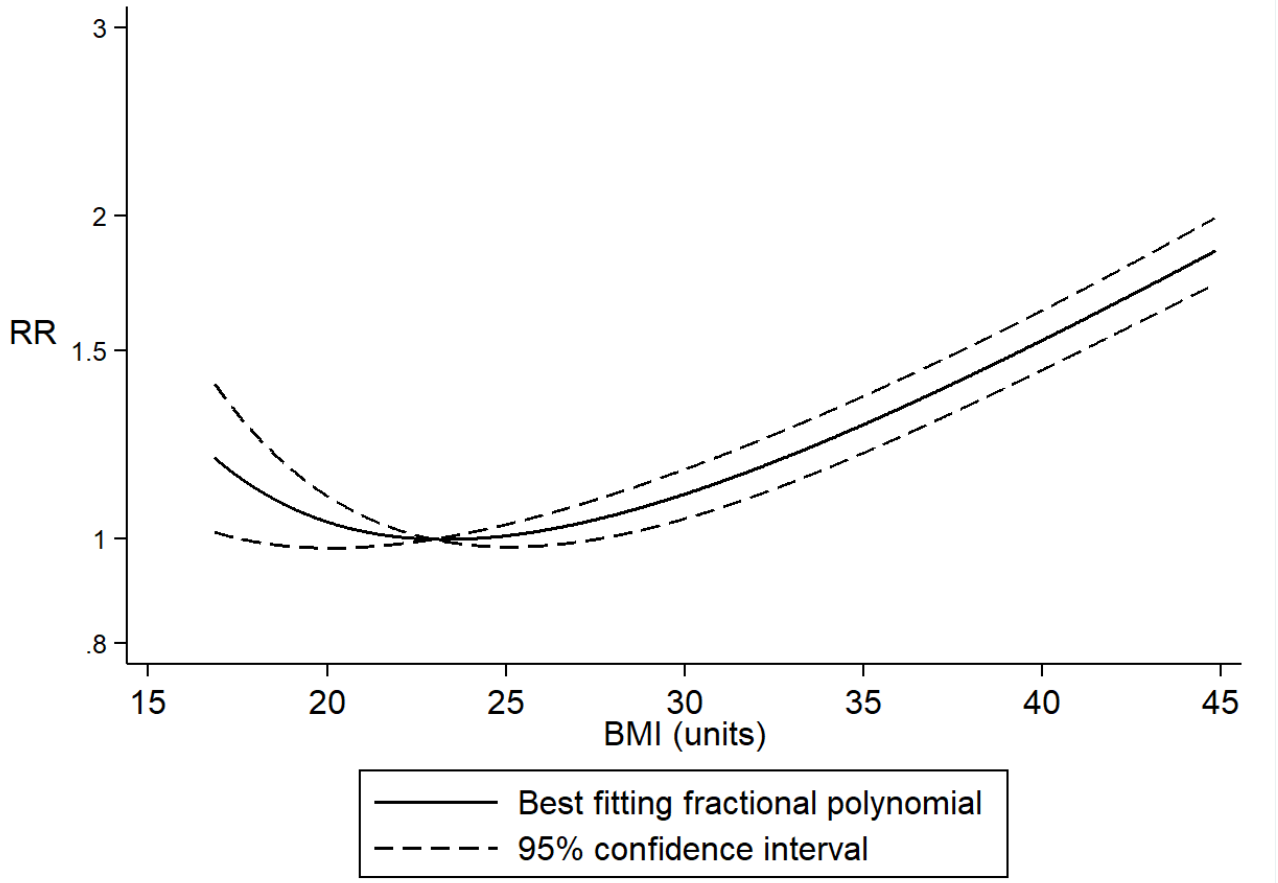


Figure 5: Association between ever smoking (versus never smoking) and mortality risk in patients with COVID-19

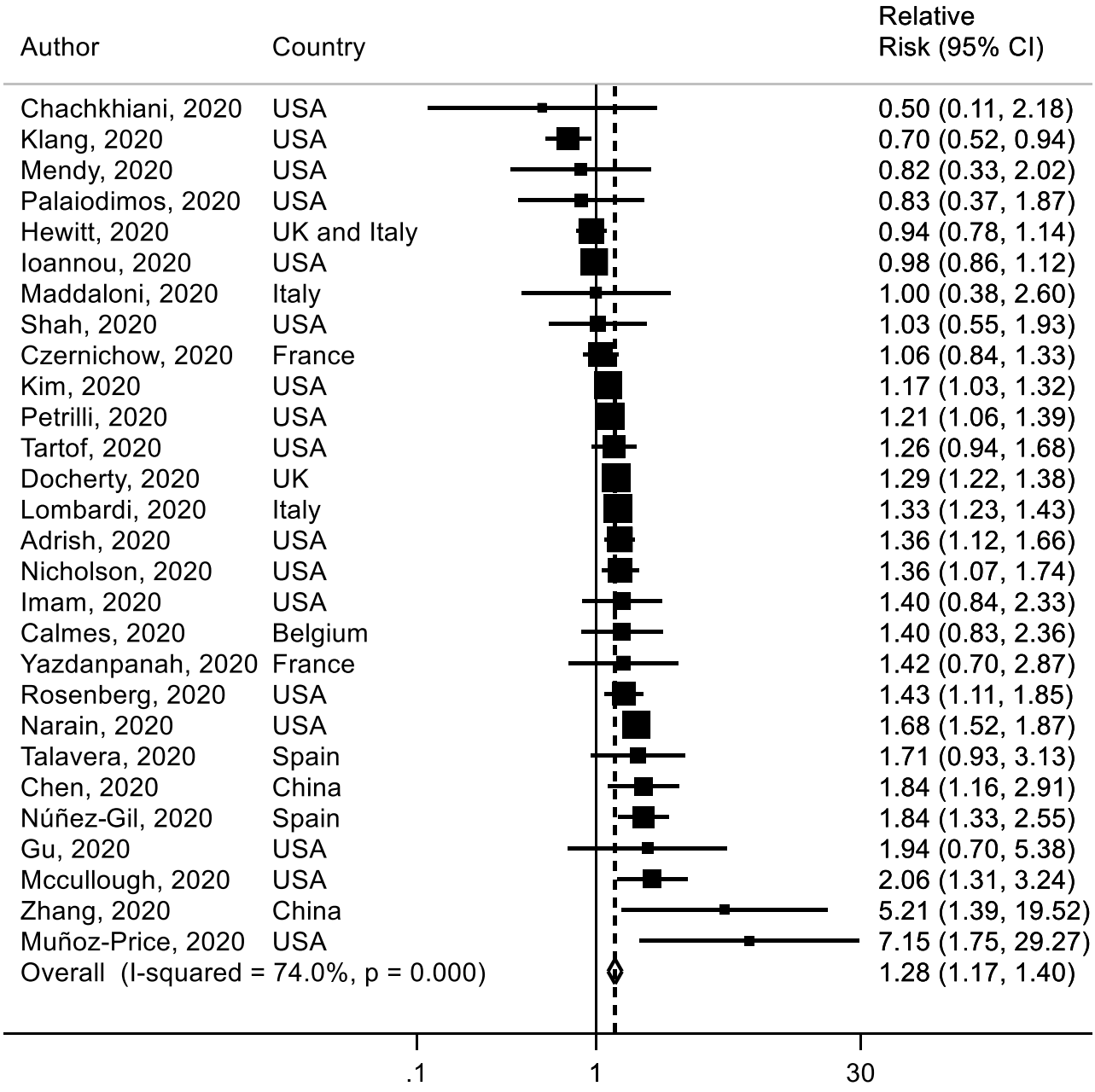


Figure 6: Association between current smoking (versus never smoking) and mortality risk in patients with COVID-19

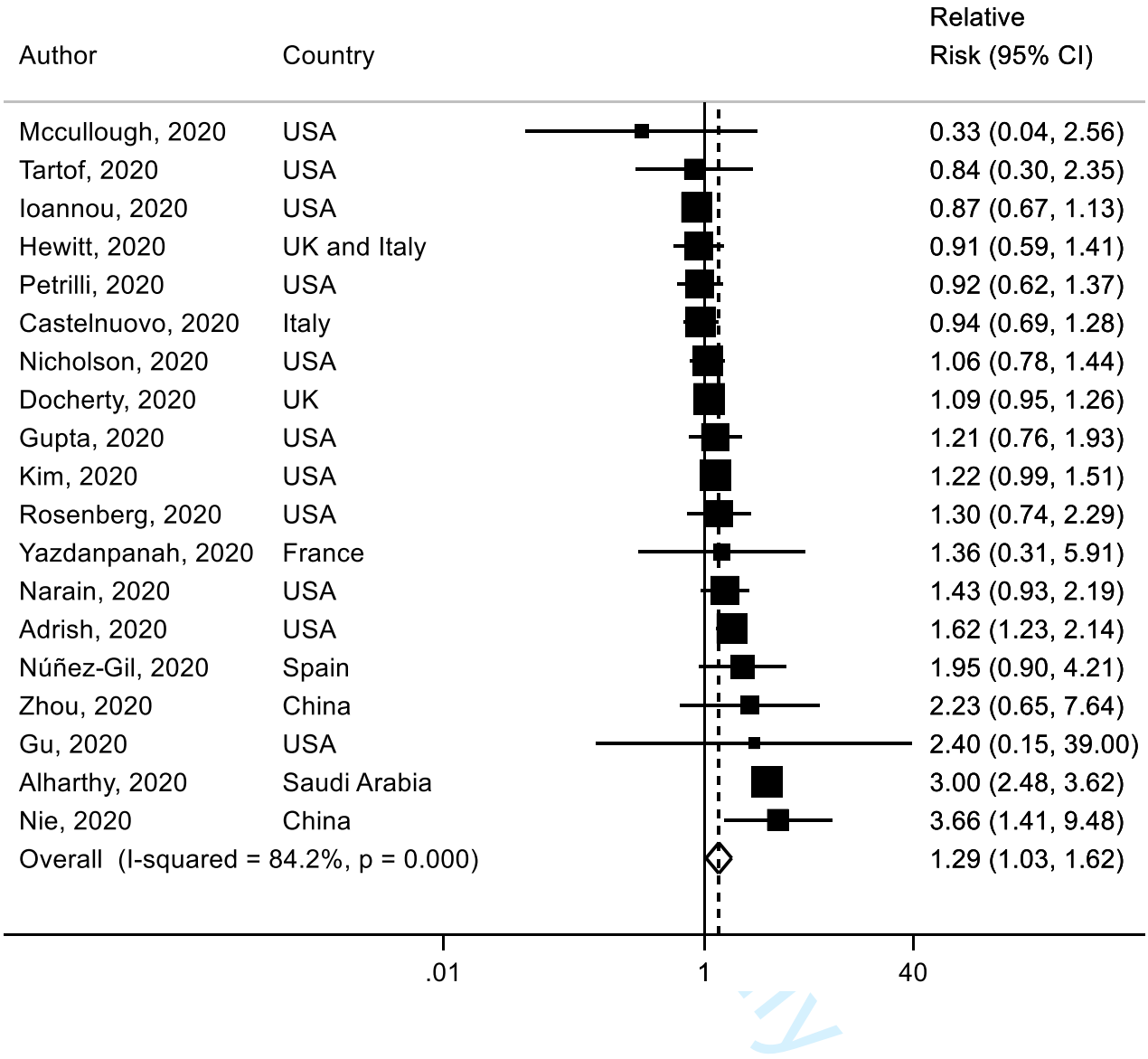
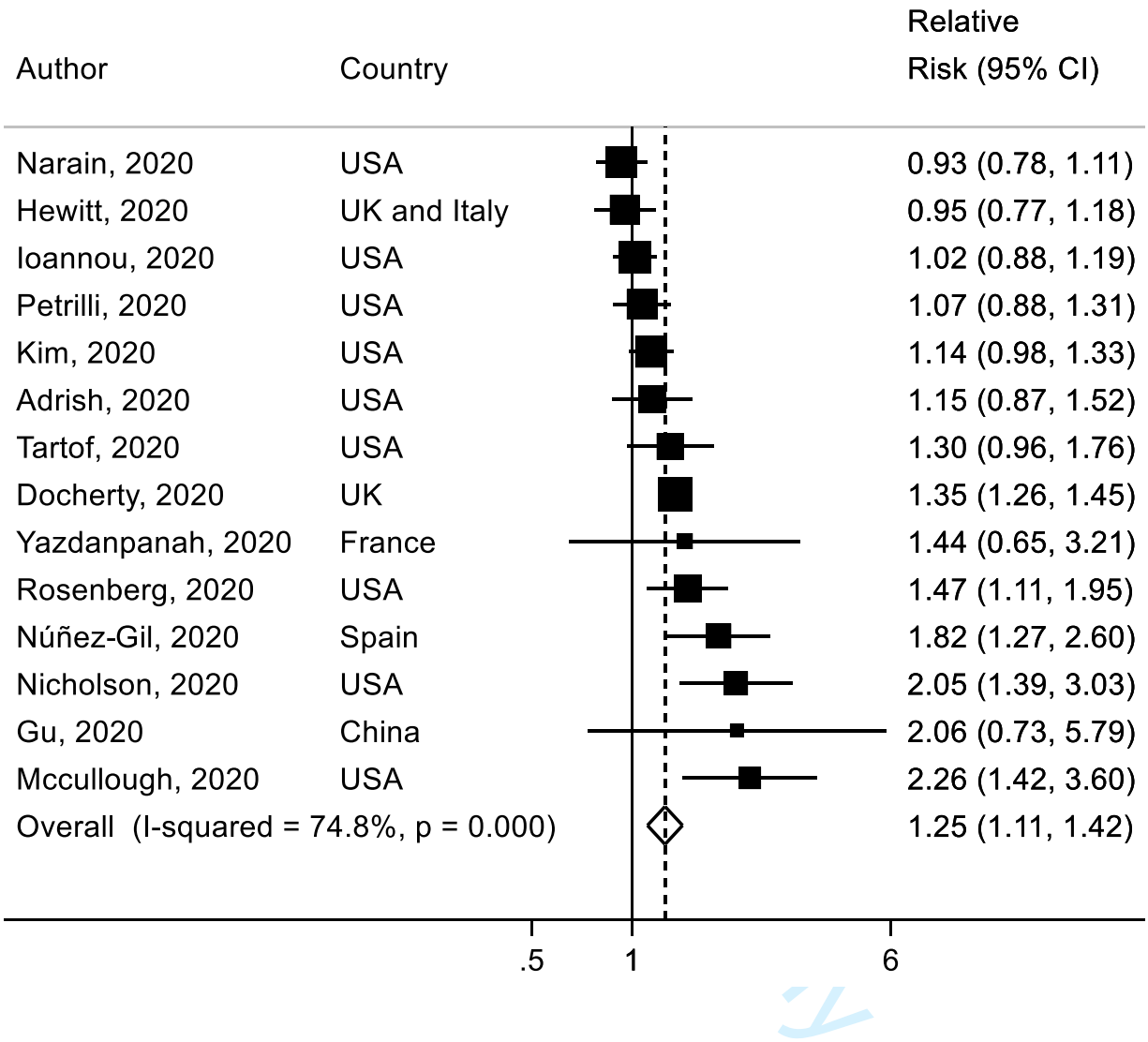


Figure 7: Association between former smoking (versus never smoking) and mortality risk in patients with COVID-19



Supplementary Table 1: Characteristics of studies included in this meta-analysis

Author, years, country	Design	Study description	Study periods	Study size, no. of death	Exposure	Comparison	RR (95 % CI)	Adjustment factors
Abohamr et al 2020, Saudi Arabia	Retrospective cohort study	King Saud Medical City, Riyadh	March 23 and June 15, 2020	768 hospitalized patients with COVID-19; 89 died	Diabetes Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no	3.01 (1.867-4.84) 18.143 (10.862-30.30) 7.018 (4.398-11.199)	None
Adrish et al, 2020, USA	Retrospective cohort study	Data from the Bronx Care Health System, the largest voluntary, not-for- profit health and teaching hospital system, serving the south and central Bronx in the New York City	March 9 to May 18, 2020	1173 COVID-19 patients; 390 died	Smoking	Ever smokers vs. never Current smokers vs. never Past smokers vs. never	1.34 (1.08-1.66) 1.62 (1.22-2.13) 1.15 (0.87-1.51)	Age
Alamdari et al, 2020, Iran	Retrospective cohort study	Patients admitted Shahid Modarres Hospital in Tehran,	April 2 and May 3	157 COVID-19 patients; 18 died	BMI	>30 kg/m2	2.53 (1.08-5.92)	None
Albitar et al, 2020, China	cross-sectional study	Data from national, provincial, and municipal health reported from Hubei and outside of Hubei	December 1, 2019 to April 21, 2020	828 COVID-19 patients; 219 died	Diabetes Hypertension	Yes vs. no Yes vs. no	12.23 (4.13-36.27) 3.58 (1.69-7.55)	Age, sex, geographic location and chronic disease (hypertension, diabetes, chronic lung diseases, chronic kidney and cardiovascular disease)
Alharthy et al, 2020, Saudi Arabia	Retrospective cohort study	Patients from King Saud Medical City (KSMC), Riyadh (adult > 18 years old, and admitted to ICU)	March 20 and May 31, 2020.	352 COVID-19 patients; 114 died	Smoking	Active smokers vs. never	3.00 (2.51–3.66)	Age, ICU length of stay, Spo_2/FiO_2 ratio, white blood cells, lymphocytes, d-dimers, lactate and pulmonary embolism
Almazeedi et al 2020, Kuwait	Retrospective cohort study	Data from JaberAl-Ahmad Al-Sabah hospitalisa (1240 bed tertiary hospital based in South Surra)	February 24 and April 20, 2020	1096 hospitalized patients with COVID-19; 19 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Yes vs. no	0.83 (0.17-4.16) 0.84 (0.46-4.81) 0.22 (0.03-1.51) 10.09 (1.22-83.4)	age, sex, hypertension, smoking, diabetes, asthma, chronic renal disease, qSOFA score, elevated CRP and elevated procalcitonin
Al-Salameh et al, 2020, France	Observational cohort study	Amiens University Hospital	case confirmed until April 21 2020	433 COVID-19 patients; 21 died	Diabetes	Yes vs. no	0.73 (0.40-1.34)	sex, age, the presence/absence of hypertension, the presence/absence of

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								cardiovascular or cardiac disease, the presence/absence of obesity, the presence/absence of chronic obstructive pulmonary disease (COPD) or sleep apnoea syndrome, a history of chronic kidney disease (CKD), a history of cancer, treatment with ACEIs or ARBs, glycaemia on admission, the white blood cell (WBC) count on admission, the glomerular filtration rate on admission, the serum C-reactive protein (CRP) on admission, and abnormal liver function test results
An et al 2020, South Korea	Retrospective cohort study	Data provided by the Korean National Health Insurance Service (KNHIS): the database of beneficiaries of national health insurance	January 23 andApril 2, 2020	10,237 hospitalized patients with COVID-19; 228 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.75 (1.29-2.36) 1.22 (0.87-1.73)	Age group, sex, income level, residence, household type, disablility, symptom and infection route.
Anderson et al, 2020, USA	Retrospective cohort study	Patients admitted from the emergency department (ED) to NewYork-Presbyterian/Columbia University Irving Medical Center	March 10 and 24 April 2020	2466 COVID-19 patients; 627 died	BMI	<18.5 kg/m2 18.5-24.9 kg/m2 25-29.9 kg/m2 30-34.9 kg/m2 35-39.9 kg/m2 ≥40 kg/m2	2.4 (1.1–5.3) 1.5 (1.02–2.2) 1 (reference) 1.4 (0.97–2.0) 1.0 (0.6–1.7) 1.6 (0.9–2.7)	Age, sex, race/ethnicity, hypertension, asthma or chronic obstructive pulmonary disease, chronic kidney disease, pulmonary hypertension, smoking, cancer, and diabetes
Anzola et al, 2020, Italy	Observational cohort study	Data from the Hospital of Gavardo, located in the neighborhood of Brescia in	March 12, 2020 to April 11, 2020	431 COVID-19 patients; 72 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.13 (0.46-2.73) 0.60 (0.27-1.34)	Age, creatinine and CRP

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		Lombardia						
Arshad et al 2020, USA	Retrospective cohort study	Data from the Henry Ford Health System (HFHS) in Southeast Michigan, a large six hospital integrated health system; the largest of hospital sisan802-bed quaternary academic teaching hospital in urban Detroit, Michigan.	March 10 and May 2, 2020	2,541 hospitalized patients with COVID-19; 460 died	Diabetes Hypertension Obesity	Yes vs no Yes vs no Yes vs no	0.98 (0.79-1.211) 0.80 (0.63-1.01) 0.78 (0.62-0.96)	Age, sex, white race, lung disease, BMI, COPD, hypertension, cancer, diabetes, admitted to ICU, ventilator and admission factors
Atkins et al, 2020, UK	Observational cohort study	UK Biobank Community Cohort; aged 65–86 years	March 16 to April 26, 2020.	507 COVID-19 patients; 41 died	Diabetes Hypertension	Yes vs. no Yes vs. no	3.11 (2.06-4.71) 1.42 (0.96-2.11)	Age group, sex, ethnicity, education, baseline assessment center, hypertension, diabetes and others chronic diseases
Barbu et al, 2020, Romania	Retrospective cohort study	Data from ministry of Health; 61.4% of males and 38.6% females; 68.2 years	March 22 to May 8, 2020	814 died	Diabetes Hypertension	Yes vs. no Yes vs. no	6.43 (4.97-8.32) 3.26 (2.69-3.96)	Age
Baronio et al 2020, Italy	Retrospective cohort study	Patients admitted to the “Fondazione Poliambulanza di Brescia” hospital (Brescia, Lombardy, Italy)	March 1, and April 11, 2020	191 hospitalized patients with COVID-19; 78 died	Obesity	18.5-<25 25-<30 30+	1 1.36 (0.70-2.65) 2.23 (1.15-4.35)	Age, sex, cardiovascular plus hypertension and temperature
Barron et al, 2020, England	Retrospective cohort study	Data from whole-population study (NHS England and NHS Improvement created a bespoke hub of relevant datasets in the National Commissioning Data Repository) covering all individuals registered with a general practice in England; 40.9 years	March 1 to May 11 2020	23 698 in-hospital COVID-19-related deaths	Diabetes	Yes vs. no	2.03 (1.97-2.09)	Age, sex, ethnicity, deprivation quintile and region.
Bepouka et al 2020,	Retrospective cohort study	Data from Kinshasa University Hospital; the tertiary level	March 23 and June 15, 2020	141 hospitalized patients with COVID-19; 41 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.31 (0.59-2.88) 1.07 (0.51-2.26)	Age, sex, civil status, admission procedure,

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Democratic Republic of the Congo		hospital in the city of Kinshasa with a capacity of 1000 beds						comorbidities, all admission signs, blood oxygen saturation, severity of illness, conventional treatment and biologic variables
Berenguer et al 2020, Spain	Retrospective observational study	Data from the COVID-19-Spain, a study of 127 Spanish centres until 17 March 2020		4035 hospitalized patients with COVID-19; 1131 died	Diabetes Hypertension Obesity	Yes vs no Yes vs no Yes vs no	1.23 (1.07-1.41) 1.81 (1.56-2.09) 1.21 (1.01-1.44)	Age, sex, hypertension, diabetes, obesity, cancer, others chronic diseases, admission signs and symptoms and vital signs.
Biscarini et al 2020, Italy	Retrospective cohort study	Data fro the SMAtteo Covid 19 Registry (SMACORE) cohort (Hospital, Fondazione IRCCS Policlinico San Matteo in Pavia)	February 21 and March 31, 2020	427 hospitalized patients with COVID-19; 139 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.29 (0.78-2.15) 0.77 (0.51-1.17) 1.03 (0.65-1.67)	Sex, age, presence of interstitial pneumonia at admission, CRP levels and PF ratio at admission, antiviral and antibiotic treatment, and main comorbidities (tumor, heart disease, hypertension, diabetes)
Boulle et al, 2020, South Africa	Cohort study	Data from the Western Cape Provincial Health Data Centre (WCPHDC) of public sector patients; aged ≥20	before March 1, 2020	15,203 COVID-19 patients; 625 died	Diabetes Hypertension Obesity	diabetes HbA1c <7% vs. no diabetes HbA1c 7 - 8.9% vs. no diabetes HbA1c ≥9% vs. no diabetes no HbA1c measurement vs. no Yes vs. no	2.21 (1.57-3.12) 3.41 (2.59-4.51) 3.62 (2.8- 4.59) 2.02 (1.47-2.76) 1.02 (0.84-1.24)	Age, sex, diabetes, hypertension, HIV, chronic kidney disease, chronic pulmonary diseases and tuberculosis.
Bousquet et al, 2020	Retrospective cohort study	The university hospital of Paris-Seine Saint Denis	March to April 2020	108 COVID-19 patients; 28 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.57 (0.72-3.44) 1.34 (0.54-3.32)	Age

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France		(Avicenne hospital; nearly 200 hospital beds for Covid-19 patients); aged 65			BMI	BMI < 21 kg/m2	0.89 (0.34-2.35)	
Brouns et al 2020, Netherlands	Retrospective cohort study	Data from 14 NH facilities from the NH organization Envida, Maastricht,	March 20 and May 1, 2020	67 hospitalized patients with COVID-19; 35 died	Hypertension	Yes vs. no	0.54 (0.22-1.37)	Age, sex, COVID-19 rRT-PCR result, CCI, hypertension, medication use, and OAT
Calmes et al 2020, Belgium	Retrospective cohort study	Patients hospitalized in University Hospital of Liege	March 18 and April 17, 2020	595 hospitalized patients with COVID-19; 17 died	Diabetes Hypertension Smoking BMI Obesity	Yes vs. no Yes vs. no smoker or ex-smoker vs never BMI per Yes vs. no	1.80 (1.10-3.10) 1.40 (0.80-2.30) 1.40 (0.81-2.30) 1.00 (0.97-1.10) 1.80 (1.10-3.20)	Age and gender
Cao et al, 2020 China	Retrospective cohort study	Wuhan Jin Yin-tan hospital and Wuhan Union hospital	January 5, and February 22, 2020	101 hospitalized patients with COVID-19; 66 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.24 (0.23-6.64) 0.73 (0.15-3.56)	Age, respiratory rate, acute respiratory distress syndrome, cardiovascular disease, hypertension, diabetes, chronic pulmonary disease and bacterial infection
Carrasco-Sanchez et al, 2020 Spain	Retrospective multi-center study	Data from the Spanish SEMI-COVID-19 Registry	March 1 to May 31, 2020	11312 hospitalized patients with COVID-19; 2289 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.63 (1.49-1.79) 1.14 (1.01-1.29)	Age, gender, hypertension, diabetes, COPD, dependency, lymphopenia, anemia (hemoglobin<10 g/dl), serum creatinine, CRP>60 mg/L, LDH>400 U/L, and Ddimer> 1000 ng/ml, elevated admission BG levels
Castelnuovo et al 2020, Italy	Retrospective cohort study	CORIST Collaboration Project (median age 67 years)	February 19 to May 23 2021	3894 hospitalized patients with COVID-19; 712 died	Diabetes Hypertension Smoking Obesity	Yes vs. no Yes vs. no Current smoker vs never ≥30 vs <30 kg/m2	1.03 (0.84-1.28) 0.87 (0.72-1.05) 0.94 (0.69-1.27) 1.21 (0.91-1.61)	Gender, age, diabetes, hypertension, history of myocardial infarction, heart failure, chronic pulmonary disease, chronic kidney disease, CRP, obesity, smoking habit and anti-COVID19

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								drugs during hospitalization as fixed effects and repeated measures
Chachkhiani et al 2020 USA	Retrospective cohort study	University Medical Center New Orleans (UMCNO); aged (60 mean)	March 1, through March 31	250 hospitalized patients with COVID-19; 56 died	Diabetes Hypertension Smoking BMI	Yes vs. no Yes vs. no Smoker vs. never per 1 kg/m2 increase	1.10 (0.60-1.80) 1.70 (0.80-3.70) 0.50 (0.10-1.90) 1.001 (1.000-1.002)	None
Chang et al 2020 South Korea	Retrospective cohort study	Kyungpook National University Chilgok Hospital and Kyungpook National University Hospital in Daegu; aged 67 (mean)	February 1 to April 10, 2020	106 hospitalized patients with COVID-19; 30 died	Diabetes	Yes vs. no	1.16 (0.51-2.65)	Age, chronic lung disease, fasting plasma glucose and white blood cell
Chen et al 2020 China	Retrospective cohort study	Data from Union Hospital (main part, Union West Hospital and Union Tumor Hospital), Wuhan Central Hospital, General Hospital of Central Theater Command, Wuhan Third Hospital and Wuhan Jin-Yin-Tan Hospital; ≥18 years patients	January 20 to April 4, 2020	1859 hospitalized patients with COVID-19; 208 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Smokers (current or former smoker) vs. never	2.47 (1.82-3.34) 2.21 (1.68–2.90) 1.84 (1.17–2.92)	Age
Chen et al 2020 China	Retrospective cohort study	Admitted to the Central Hospital of Wuhan (age : 15–99 years)	1 January 2020 to March, 3 2020	904 hospitalized patients with COVID-19; 92 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.51 (1.53-4.13) 3.48 (2.24-5.40)	None
Chen et al 2020, China	Retrospective cohort study	Data from patients admitted to The Central Hospital ofWuhan	1 January 2020 to 15 February 2020	660 hospitalized patients with COVID-19; 82 died	Hypertension	Yes vs. no	0.92 (0.42-2.02)	Age, cerebral infraction, SOFA, CRP and LDH
Chilimuri et al 2020 USA	Retrospective cohort study	The Bronxcare Health System (BCHS) hospital in the South Bronx, New York	March 9 to April 9, 2020	375 hospitalized patients with COVID-19; 160 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.58 (0.94-2.65) 1.46 (0.82-2.62)	several variables (age, sex, hypertension, diabetes, CVD, chronic kidney disease and biological measurement)
Ciardullo et al 2020 Italy	Retrospective cohort study	Policlinico di Monza hospital in Monza (aged ≥ 18 years)	February 22 and May 15, 2020	373 hospitalized patients with COVID-19; 142 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.23 (0.85-1.63) 1.32 (0.91-1.81)	Age, sex, chronic kidney disease, hypertension diabetes, COPD, chronic obstructive and

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								pulmonary disease
Ciceri et al 2020, Spain	Retrospective cohort study	Data from North-East of Milan's province and admitted at San Raffaele Hospital; aged ≥18 years	February 25th and March 24th	410 hospitalized patients with COVID-19; 95 died	Diabetes Hypertension BMI (not precised)	Yes vs no Yes vs no	1.51 (0.96–2.05) 2.60 (1.67–4.05) 0.96 (0.90–1.02)	Age
Coppelli et al 2020 Italy	Retrospective cohort study	The Pisa COVID-19 Study (University Uospital, Pisa)	20 March to 30 April 2020	271 hospitalized patients with COVID-19; 67 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no BMI >30 kg/m2 Smokers vs non-smokers	0.78 (0.29-2.09) 1.02 (0.49-2.15) 2.80 (0.97-8.10) 0.74 (0.37-1.47)	Age, sex, hypertension, cerebrovascular disease, chronic obstructive pulmonary disease, chronic kidney disease, cognitive impairment and several biomarkers
Crouse et al 2020 USA	Retrospective cohort study	University of Alabama at Birmingham Hospital (a tertiary health care center in the racially diverse Southern U.S.)	25 February to June 22, 2020	604 hospitalized patients with COVID-19; 45 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	3.62 (2.11-6.20) 3.60 (1.68-7.69) 1.14 (0.67-1.94)	Age, race, sex, obese status and hypertension status
Cummings et al 2020 USA	Prospective cohort study	Data from two NewYork-Presbyterian hospitals affiliated with Columbia University Irving Medical Center in northern Manhattan (aged ≥18 years)	March 2 to April 1, 2020	257 hospitalized patients with COVID-19; 101 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.31 (0.81-2.10) 1.58 (0.89-2.81) 0.76 (0.40-1.47)	Nine independent variables (age, sex, hypertension, COPD, chronic kidney disease, diabetes, symptom duration before hospital, chronic cardiac disease and BMI)
Czernichow et al 2020, France	Prospective cohort study	Data from the Paris hospitals, France	February 1 and 30 April 2020	5795 hospitalized patients with COVID-19; 891 died	Diabetes Hypertension Smoking Obesity	Yes vs. no Yes vs. no Yes vs. no <18.5 18.5-24.9 25-29.99 30-34.99 35-39.99 40+	1.34 (1.11-1.62) 1.01 (0.82-1.24) 1.06 (0.84-1.33) 1.24 (0.77-1.93) 1 1.41 (1.21-1.77) 1.89 (1.45-2.47) 2.79 (1.95-3.97) 2.55 (1.62-3.95)	Comorbidities, smoking status, age and sex using logistic regressions, both including and excluding patients with missing BMI and with stratification on age class
De Souza et al 2020 Brazil	Cross-sectional study	Data from the state of Alagoas in the northeastern of Brazil (older adults (aged ≥60 years)	until 01 August, 2020	9807 hospitalized patients with COVID-19; 1171 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	2.33 (1.99-2.74) 1.53 (1.20-1.94) 1.77 (0.84–3.74)	Age, sex, initial symptoms reporte and comorbidities
De Vito et al	Retrospective	Patients admitted to the Unit of	March 8 to April	87 hospitalized patients	Hypertension	Yes vs. no	0.47 (0.11-2.00)	Age, moderate ARDS,

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2020 Italy	cohort study	Infectious Diseases of the University Hospital of Sassari (North-West Sardinia, Italy)	8, 2020	with COVID-19; 57 died				not compliance patients lymphocytes and others comorbidities
Deng et al 2020, China	Retrospective cohort study	Data reported publicly by the Chinese center for disease control and prevention	updated through February 11, 2020	44,672 hospitalized patients with COVID-19; 1023 died	Diabetes Hypertension	Yes vs no Yes vs no	4.47 (3.49–5.61) 4.48 (3.69–5.45)	None
Dennis et al 2020 England	Retrospective cohort study	Patients admitted to hospital in England with COVID-19 requiring admission to a high dependency unit (HDU) or intensive care unit (ICU); aged 18–99	1 March 2020 and 27 July 2020	19256 hospitalized patients with COVID-19; 5077 died	Diabetes	Yes vs. no	1.23 (1.14-1.32)	Age, sex, ethnicity, obesity and comorbidity including hypertension and others
Desai et al 2020 Italy	Retrospective cohort study	Data from emergency Department (ED) of Humanitas Clinical and Research Center; aged 64.8 years (mean)	February 21 and April 14, 2020	372 hospitalized patients with COVID-19; 120 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.10 (0.50-2.40) 0.90 (0.50-1.70)	Age, gender, chronic use of ACE-I and ARBs), time interval between onset of symptoms and ED admission.
Docherty et al 2020, UK	Prosepctive cohort study	Data from the ISARIC WHO CCP-UK study which is an ongoing study in 208 acute care hospitals in England, Scotland, and Wales	February and 19 April 2020	15 194 hospitalized patients with COVID-19; 3911 died	Diabetes Obesity Smoking	Yes vs. no Yes vs. no current vs. never Former vs. never	1.06 (0.99-1.14) 1.33 (1.19-1.49) 1.09 (0.95-1.26) 1.35 (1.26-1.45	Age, sex, and major comorbidities (diabetes, hypertension, obesity and others)
Du et al 2020, China	Retrospective cohort study	Data from Wuhan Pulmonary Hospital	December 25th, 2019 and February 7, 2020	179 hospitalized patients with COVID-19; 21 died	Hypertension	Yes vs no	4.081 (1.584–10.510)	None
Escalera-Antezana et al 2020 Bolivia	Cross-sectional study	Data from the Hospitals and the Ministry of Health of Bolivia, obtaining the clinical and epidemiological data of the COVID-19 cases that were laboratory-diagnosed	March 2 to 29, 2020	107 hospitalized patients with COVID-19; 6 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	16.33 (2.10-126.82) 3.28 (1.28-6.29) 12.13 (1.69-86.95)	Age, sex and comorbidties for only hypertension association
Esme et al 2020 Turkey	Retrospective cohort study	Data from Turkish Ministry of Health National COVID-19; aged ≥ 60 years	March 11 to May 27, 2020	16942 hospitalized patients with COVID-19; 3141 died	Diabetes Hypertension	Yes vs. no (60-79 age) Yes vs. no (80+ age)	1.18 (1.06-1.30) 1.26 (1.07-1.49) 1.17 (1.03-1.32) for	Gender, presence of hypertension, diabetes, chronic obstructive pulmonary disease (COPD), coronary artery

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						Yes vs. no (60-79 age)	60-79 only	disease (CAD), atrial fibrillation (AF), chronic kidney disease (CKD), dementia, depression, malnutrition, and hyperlipidemia (HL)
Fang et al 2020 China	Retrospective cohort study	Data from admitted to the infectious ward of Wuhan Third Hospital; 615 men and 665 women, aged 7–101 years	January 27 to March 15, 2020	1280 hospitalized patients with COVID-19; 69 died	Diabetes Hypertension	Yes vs. no Yes vs. no	3.54 (1.17-10.8) 0.80 (0.35-1.83)	Sex, age, hypertension, COPD, diabetes, CRF, CHD, Arrhythmia
Farrel et al 2020, Ireland	Retrospective cohort study	Data collected from the Microbiology Department of hospital's laboratory (connolly Hospital, Ireland and Royal College of Surgeons)	March 13 and May 1, 2020	257 hospitalized patients with COVID-19; 39 died	Smoking Obesity	Yes vs. no Overweight/obese vs. no	1.73 (0.70-4.31) 3.09 (1.32-7.23)	Age
Fernandez-Cruz et al 2020, Spain	Retrospective cohort study	University hospital in Madrid, Spain	March of 2020	463 hospitalized patients with COVID-19; 71 died	Diabetes	Yes vs. no	2.80 (1.60-4.90)	None
Ferrando-Vivas et al 2020 Bolivia	Observational cohort study.	Adults from a critical care units across England, Wales, and Northern Ireland	March 1 and June 22, 2020	9,990 hospitalized patients with COVID-19; 3,933 died	BMI	per 5 kg/m2 increase	1.032 (1.003–1.061)	Several variables including age, sex, ethinicity, deprivation and others factors)
Foy et al 2020, USA	Retrospective cohort study	Data from one of 4 hospitals in the Boston, Massachusetts area (Massachusetts General Hospital, Brigham andWomen's Hospital, North Shore Medical Center, and Newton-Wellesley Hospital)	March 4 and April 28, 2020.	1641 hospitalized patients with COVID-19; 276 died	Diabetes Hypertension	Yes vs. no Yes vs. no	0.87 (0.63-1.20) 1.05 (0.80-1.4)	Age, ethnicity, D-dimer (dimerized plasmin fragment D) level, absolute lymphocyte count, hypertension, diabetes, COPD, CKD and CAD.
Fried et al 2020, USA	Retrospective cohort study	data from the hospital chargemaster, in 250 hospitals, across 38 states (patients ≥ 18 years)	February 15 and April 20, 2020	11721 hospitalized patients with COVID-19; 2514 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Yes vs. no	0.99 (0.77-1.08) 0.78 (0.69-0.88) 1.07 (0.93-1.24) 0.72 (0.63-0.83)	Age, sex, insurance status, chronic kidney disease, liver disease obesity and smoking
Galloway et al 2020, England	Prospective cohort study	Data from two London hospitals (King's College Hospital and Princess Royal University Hospital) ; aged 71	March 1 to April 17 2020.	1157 hospitalized patients with COVID-19; 244 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.24 (0.95-1.60) 1.28 (0.99-1.67)	age and sex

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		years (mean)						
Gao et al 2020, China	Retrospective cohort study	Huo Shen Shan Hospital (Wuhan)	February 5 to March 15, 2020	2877 hospitalized patients with COVID-19; 56 died	Hypertension	Yes vs. no	2.12 (1.17-3.82)	Age, sex, medical history of diabetes, insulin-treated diabetes, myocardial infarction, underwent PCI/CABG, renal failure, stroke, heart failure, and COPD
Gayam et al 2020, USA	Retrospective cohort study	Data from an inner-city teaching community hospital in New York City		408 hospitalized patients with COVID-19; 132 died	BMI	BMI per 1 unit increase	1.07 (1.04-1.11)	Age, BMI, myalgia, D-dimers, Ferritin, CRP and other biological samples
Gazzaruso et al 2020, Italy	Retrospective cohort study	Not mentionned (Italy)	15 to 30 Aril 2020	192 hospitalized patients with COVID-19; 62 died	Obesity	Yes vs no	3.62 (2.14–6.14)	Age, d-dimer, history of hypertension, eGFR, lactate dehydrogenase, history of a lung disease and C-reactive protein
Ghany et al 2020, USA	Observational cohort study	Elderly Medicare COVID-19 patients across eight states;	January 1 and May 4, 2020	400 hospitalized patients with COVID-19; 72 died	Diabetes Hypertension BMI	Yes vs. no Yes vs. no BMI per 1 unit increase	1.03 (0.58-1.02) 1.38 (0.76-2.50) 1.00 (0.96–1.05)	Several variables (not mentionned)
Giacomelli et al 2020, Italy	Prospective cohort study	Adult COVID-19 patients admitted to Luigi Sacco Hospital in Milan,	February 21 to March 19, 2020	233 hospitalized patients with COVID-19; 48 died	Being treated with at least one anti-hypertensive agent Obesity	Yes vs. no Yes vs. no	1.41 (0.73−2.74) 3.04 (1.42−6.49)	Age, sex, comorbidities, BMI, hypertension drugs, and biological parameters
Grasselli et al 2020, Italy	Retrospective cohort study	Patients requiring ICU admission in Lombardy	February 20 to April 22, 2020.	3988 hospitalized patients with COVID-19; 926 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.18 (1.01-1.39) 0.99 (0.81-1.22)	Demographics, comorbidities, drugs, and respiratory parameters
Gu et al 2020, China	Nested case–control study	Data from mainland China outside of Hubei Province	18 December 2019 to 8 March 2020	275 hospitalized patients with COVID-19; 94 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.14 (0.70-1.90) 1.37 (0.90-2.20)	Age
Gu et al 2020, USA	Retrospective cohort study	University of Michigan Medicine Health System,	March 10 to April 22, 2020	1119 hospitalized patients with COVID-19; 43 died	Diabetes BMI Past-smoker Current-Smoker	Yes vs. no per one SD (8.1 kg/m2) Yes vs. no Yes vs. no	2.16 (0.85-5.49) 1.22 (0.74, 2.01) 2.06 (0.73, 5.77) 2.40 (0.15, 39.6)	Race, age, and sex

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Guisado-Vasco et al 2020, Spain	Retrospective observational study	Data from adult patients admitted to Hospital Universitario Quironsalud Madrid, in Pozuelo de Alarcon, Madrid	March 10 to April 15, 2020	607 hospitalized patients with COVID-19; 141 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.61 (1.19-5.73) 1.73 (1.17-2.54)	Age, sex, baseline comorbidities, laboratories parameters and available therapies
Gupta et al 2020, USA	Prospective cohort study	data from multicenter cohort study of 65 hospitals, adult patients (≥18 years of age)	March 4 to April 4, 2020	2215 hospitalized patients with COVID-19; 784 died	Diabetes Hypertension current smoker BMI	Yes vs. no Yes vs. no Yes vs. no <25 25-29.9 30-34.9 35-39.9 ≥40	1.14 (0.91-1.43) 1.06 (0.83-1.36) 1.21 (0.76-1.93) 1 1.01 (0.73-1.39) 0.97 (0.69-1.37) 1.24 (0.81-1.89) 1.51 (1.01-2.25)	Age, sex, race, hypertension, BMI, diabetes, smoking and others comorbidities
Gutierrez et al 2020, Mexico	Retrospective cohort study	Data from directorate General of Epidemiology (Ministry of Health)		654,858 hospitalized patients with COVID-19; 71,706 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Yes vs. no	1.82 (1.76-1.88) 1.39 (1.35-1.44) 1.68 (1.62–1.73) 0.97 (0.94–1.01)	Sex, age, comorbidities, health provider, whether the individual speaks an indigenous language, quintiles of the proportion of the population living in poverty in the municipality (county) of residence
Haase et al 2020, Denmark	Retrospective observational study	Data from the Danish nationwide, including 29 ICUs in 27 hospitals	10 March to 19 May 2020 and	323 hospitalized patients with COVID-19; 118 died	Diabetes Hypertension BMI	Yes vs. no Yes vs. no <18 18-24.9 25-29.9 30-34.9 35-39.9 ≥40	1.10 (0.71-1.70) 0.94 (0.64-1.37) 0.95 (0.51-1.76) 1 0.77 (0.49-1.22) 0.71 (0.39-1.27) 0.80 (0.34-1.89) 0.86 (0.38-1.93)	Age and gender
Hajifathalian et al 2020, USA	Retrospective observational study	Data from two hospitals in New York City (an academic tertiary care referral center and a smaller community hospital) (age ≥ 18)	March 4 and April 9, 2020	770 hospitalized patients with COVID-19; 88 died	BMI	<18.5 18.5-30 kg/m2 ≥30 kg/m2	1.64 (0.84-3.19) 1 1.15 (0.62-2.14)	Age, race/ethnicity, and troponin I level

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Harmouch et al 2020, USA	Retrospective cohort study	Data from St. Luke's University Hospital, Bethlehem, Pennsylvania, which is part of the St. Luke's University Health Network	March and April 15, 2020	563 hospitalized patients with COVID-19; 81 died	Diabetes Hypertension Smoking BMI Obesity	Yes vs. no Yes vs. no Yes vs. no BMI per Morbid obesity vs. no	0.901 (0.547-1.482) 2.049 (1.253-3.351) 1.538 (0.935-2.530) 0.981 (0.951-1.013) 1.036 (0.587-1.829)	None
Harrison et al 2020, USA	Retrospective cohort study	Data form the TriNetX network (24 healthcare organizations in the US); aged up to 90 years	January 20 to May 26, 2020	31461 hospitalized patients with COVID-19; 1296 died	Diabetes	Yes vs. no	1.11 (0.96-1.27)	age, sex, ethnicity and comorbidities
Hashemi et al 2020, USA	Retrospective cohort study	data from nine hospitals (two large tertiary centres and seven community hospitals) and a single healthcare system in Massachusetts; aged of 63.4 years	11 March to 2 April 2020	363 hospitalized patients with COVID-19; 55 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.38 (0.68-2.79) 2.20 (0.88-5.52) 1.03 (0.51-2.09)	Age, gender, obesity, cardiac diseases, hypertension, hyperlipidaemia, and pulmonary disorders, CLD and NAFLD
Hewitt et al 2020, EUROPE	Prospective cohort study	Ten hospitals in the United Kingdom (Patients ≥ 18 years old) and one Italian Hospital (University Hospital of Modena Policlinico)	Feb 27, and April 28, 2020,	1564 hospitalized patients with COVID-19; 425 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Ex-smokers vs. never Current smokers vs. never	1.03 (0.82–1.29) 0.95 (0.77–1·18) 0.95 (0.77–1.18) 0.91 (0.59–1.42)	Age group, sex, smoking, C-reactive protein, diabetes, coronary artery disease, hypertension, renal function, and the clinical frailty scale
Hu et al 2020, China	Retrospective cohort study	Data from Renmin Hospital of Wuhan University	December 2019 to March 2020	821 hospitalized patients with COVID-19; 180 died	Diabetes Hypertension	Yes vs. no Yes vs. no	0.82 (0.34-1.97) 1.15 (0.57-2.31)	Several variables (including age, sex and comorbidities)
Iaccarino et al 2020, Italy	cross-sectional study	Data from 26 hospitals in Italy, a nationwide survey (study of the Italian Society of Hypertension)	March 9 to April 9, 2020.	1591 hospitalized patients with COVID-19; 188 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.70 (1.30-2.70) 1.00 (0.60-1.50)	Age, sex, hypertension, obesity, diabetes, HF, COPD, CKD and charlon index
Imam et al 2020, USA	Retrospective cohort study	Data from largest healthcare system in Southeast Michigan (8 hospitals);	April 1. 2020,	2040 hospitalized patients with COVID-19; 200 died	Diabetes Hypertension Smoking history Obesity	Yes vs. no Yes vs. no Yes vs. no BMI > 30 vs BMI < 30	3.90 (3.00-5.10) 4.40 (3.60-5.40) 1.40 (0.83-2.30) 1.80 (1.40-2.40)	None (multivariable for BMI only)

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Ioannou et al 2020, USA	Retrospective cohort study	Data from the Department of Veterans Affairs (VA) national health care system	February 28 and May 14, 2020	10 131 hospitalized patients with COVID-19; 1094 died	Diabetes BMI Hypertension Smoking	Yes vs. no <18.5 18.5-24.9 25.0-29.9 30.0-34.9 ≥35 Yes vs. no Former vs. Never Current vs. never	1.13 (0.99-1.29) 1.29 (0.98-1.70) 1 0.90 (0.77-1.06) 0.84 (0.69-1.01) 0.97 (0.77-1.21) 0.95 (0.81-1.12) 1.02 (0.88-1.19) 0.87 (0.67-1.13)	Age, sex, race/ethnicity, sociodemographic characteristics, comorbid conditions, and symptoms
Kabarriti et al 2020, USA	Retrospective cohort study	Patients from to the Montefiore Medical Center, a large urban academic medical center in the Bronx, New York,	March 14 and April 15, 2020,	5902 hospitalized patients with COVID-19; 918 died	Diabetes Hypertension BMI	Yes vs. no Yes vs. no <18.5 18.5-35 >35	1.21 (1.06-1.40) 1.00 (0.87-1.16) 0.97 (0.66-1.41) 1 1.42 (1.19-1.70)	sex, age, socioeconomic status, ethinicity and other comorbidities
Kim et al 2020, South Korea	retrospective cohort study	Korean National Health Insurance Service Big Data linked with COVID-19 registry data of Korea Centers for Disease Control & Prevention;	confirmed COVID-19 up to March 26, 2020	9148 hospitalized patients with COVID-19; 130 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.82 (1.25-2.67) 1.32 (0.88-1.99)	age, sex, race/ethnicity, smoker, hypertension, obesity, diabetes, chronic lung disease, cardiovascular disease, neurologic disease, renal disease, immunosuppression, hematologic disorders, and rheumatologic or autoimmune disease
Kim et al 2020, USA	Retrospective cohort study	Data from 74 counties in 13 states Surveillance Network (COVID-NET); aged 62 years (median)	March 1, 2020 and May 30, 2020.	2491 hospitalized patients with COVID-19; 420 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Former smokers vs. never Current smokers	1.19 (1.01-1.40) 1.07 (0.79-1.45) 1.14 (0.98-1.33) 1.22 (0.99-1.51)	Sex, age, type of distiricts, high epidemic region and socio-economic status

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					Obesity	BMI≥30 vs BMI<30	1.09 (0.92-1.30)	
Klang et al 2020, USA	Retrospective cohort study	Data from five hospital campuses serving different geographic populations (Mount Sinai Hospital, Mount Sinai Brooklyn, Mount Sinai Queens, Mount Sinai Morningside, and Mount Sinai West), in NY	March 1 and 5/9/2020	6760 hospitalized patients with COVID-19; 1320 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Smokers (past or current) vs. never	1.00 (0.80-1.30) 1.20 (0.90-1.60) 2.30 (1.80-3.00) 0.70 (0.50–0.90)	Age decile, CAD, DM, CHF, CKD, COPD, obesity, cancer, and smoking
Klang et al 2020, USA	Retrospective cohort study	Data of COVID-19 patients hospitalized to a large academic hospital system in New York City	March 1 and May 17, 2020	3406 hospitalized patients with COVID-19; 1136 died	Age ≤ 50 Diabetes Hypertension Smoking BMI Age > 50 Diabetes Hypertension Smoking BMI	Yes vs. no Yes vs. no past or present vs. never 30 30-40 >40 Yes vs. no Yes vs. no Smokers (past or present) vs. never 30 30-40 >40	1.30 (0.70-2.60) 0.50 (0.20-1.10) 1.70 (0.80-3.80) 1 1.10 (0.50-2.30) 5.10 (2.30-11.10) 1.4 (1.2-1.7) 1.1 (0.9-1.3) 1.0 (0.8-1.2) 1 1.1 (0.9-1.3) 1.6 (1.2-2.3)	Age, sex, CAD, CHF, hypertension, CKD, smoking, BMI, race, cancer and hyperlipidemia
Kolhe et al 2020, UK	Retrospective cohort study	Data from the Department of University Hospitals of Derby and Burton (UHDB) National Health Service (NHS), which comprised 5 hospitals in the UK.	March 5 and May 12, 2020	1161 hospitalized patients with COVID-19; 419 died	Diabetes with complications	Yes vs. no	1.15 (0.82, 1.61)	Age, gender, ethnicity, residence in care home, use of ACEI/ARB, need for mechanical ventilation, and comorbidities
Izzampasona et al 2020, Italy	Prospective cohort study	Data from the Emergency or Clinical departments of the Istituto di Ricovero e Cura a Carattere Scientifico;	480 February and 19 April 2020	509 hospitalized patients with COVID-19; 93 died	Diabetes	Yes vs. no	2.32 (1.44-3.75)	Age, sex, BMI and other relevant comorbidities
Lee et al 2020,	Retrospective	Patients in the	February 18 and	98 hospitalized patients	Diabetes	Yes vs. no	4.74 (1.68-13.38)	None

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South Korea	cohort study	Daegu/Gyeongsangbuk-do province; patients aged ≥ 65 years	March 4, 2020	with COVID-19;20 died	Hypertension	Yes vs. no	1.95 (0.70-5.42)	
Li et al 2020, China	Retrospective cohort study	Wuhan Union Hospital Affiliated with Tongji Medical College of Huazhong University of Science	22 January 2020 to 17 March 2020.	453 hospitalized patients with COVID-19; 39 died	Diabetes Hypertension	known diabetes vs. normal glucose Newly diagnosed diabetes vs. normal glucose	4.63 (1.02-21.00) 9.42 (2.18-40.7)	Age, sex, smoking, systolic blood pressure, total cholesterol, anti-hypertensive drugs and admission to ICU.
Li et al 2020, China	Retrospective cohort study	Patients admitted to Renmin Hospital of Wuhan University	January 31 to February 20, 2020	204 hospitalized patients with COVID-19; 76 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.10 (0.60-1.90) 2.30 (1.50-3.60)	Age
Li et al 2020, China	Retrospective cohort study	Patients from two critical and two general designated hospitals, in Wuhan	January 2 and February 15, 2020	523 hospitalized patients with COVID-19; 95 died	BMI	per 2.9 kg/m2 increase	0.99 (0.90-1.10)	age, gender, Hypertension, diabetes mellitus, coronary artery disease, þCr, PCT, ALC, cTnI, hs-CRP, LDL-c and FBG
Li et al 2020, China	Retrospective cohort study	Patients admitted to Wuhan Red Cross Hospital	January 24 and March 15, 2020	199 hospitalized patients with COVID-19; 18 died	Diabetes Hypertension	Yes vs. no Yes vs. no	10.816 (1.895-61.741) 0.660 (0.150-2.908)	Multivariate model but variable are not mentionned
Li et al 2020, USA	Retrospective cohort study	Yale New Haven Health System, a 5-hospital system in the Northeast of the United States	February 15 and April 25, 2020	1685 hospitalized patients with COVID-19; 318 died	Diabetes wit complications	Yes vs. no	1.00 (0.70-1.40)	Age, sex, race/ethnicity, medical comorbidities and hospital locations
Liu et al 2020, China	Retrospective cohort study	Leishenshan Hospital	NA	934 hospitalized patients with COVID-19; 13 died	Diabetes	Yes vs. no	2.14 (0.58-7.91)	Age, the history of cardiovascular disease, D-dimer, WBC, and lymphocyte count
Liu et al 2020, China	Retrospective cohort study	Wuhan Infectious Disease Hospital (18–94 years)	December 29, 2019 to February 28, 2020	1190 hospitalized patients with COVID-19; 157 died	Diabetes	Yes vs. no	11.16 (1.87-66.57)	Age, sex, diabetes, hypertension, CKD and some symptoms and signs
Liu et al 2020, China	Retrospective cohort study	Weibo posts and a telephone survey	January 20 and February 15, 2020,	599 hospitalized patients with COVID-19; 83 died	Hypertension Smoking	Yes vs. no Yes vs. no	0.24 (0.08-0.67) 0.90 (0.44-1.85)	Age and sex
Liu et al 2020, China	Retrospective cohort study	Patients of Eastern Campus of Renmin Hospital, Wuhan University	January 30 to February 20, 2020	214 hospitalized patients with COVID-19; 94 died	Hypertension	Yes vs. no	0.82 (0.40-1.70)	Age, lymphocytopenia, serum LDH, PCT, and BUN levels
Lombardi et al	Cross-sectional	13 Italian cardiology units	March 1 to April	614 hospitalized patients	Diabetes		1.41 (0.99-2.02)	For hypertension :

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2020, Italy	study		9, 2020	with COVID-19; 148 died	Hypertension Obesity BMI Smoking	Yes vs. no Yes vs. no Yes vs. no +1 kg/m2 Ever vs. never smoker	1.26 (0.81-1.96) 1.26 (0.85-1.85) 1.01 (0.97-1.04) 1.33 (1.24-1.44)	ajusted for comorbidities
Lopez-Mendez et al 2020, Mexico	Retrospective cohort study	Patients admitted to Medica Sur Clinic & Foundation	March 14 and June 5, 2020	155 hospitalized patients with COVID-19; 11 died	Hypertension	Yes vs. no	3.96 (1.28−12.19)	None
Lunski et al 2020, USA	Retrospective cohort study	Data from ochsner Health System in the state of Louisiana,	March 1 through April 30, 2020	4760 hospitalized patients with COVID-19; 484 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	0.92 (0.70-1.21) 0.60 (0.47-0.78) 1.30 (1.03-1.63)	Age, sex, race, comorbidities (COPD, diabetes, BMI, hypertension, CKD) and other variables
Maddaloni et al 2020, Italy	Retrospective cohort study	Data from four academic hospitals located in the Lazio region of Italy (the Covid-19 & Diabetes (CoViDiab))	March 1 to May 15th, 2020	354 hospitalized patients with COVID-19; 77 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Ever vs. never smoker	1.30 (0.50-3.10) 2.80 (1.00-7.50) 1.00 (0.40-2.70)	Age and sex
Magleby et al 2020, USA	Retrospective cohort study	Hospitalized patients in NewYork-Presbyterian Hospital/Weill Cornell Medical Center and affiliated Lower Manhattan Hospital	March 30 to April 30, 2020	678 hospitalized patients with COVID-19; 127 died	Hypertension	Yes vs. no	0.74 (0.41-1.33)	Age, race, coronary artery disease, congestive heart failure, cerebrovascular disease, hypertension, COPD, days of symptoms prior to admission, symptoms upon presentation, initial chest x-ray findings, and level of oxygen support within three hours of arrival to the ED
Mallow et al 2020, USA	Retrospective cohort study	Data were obtained retrospectively from a commercially-available, allpayer administrative database of inpatient and hospital-based outpatient (276 hospitals in the database	March 15 and April 30, 2020	21 676 hospitalized patients with COVID-19; 4936 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.30 (1.17-1.44) 0.77 (0.70-0.86) 1.30 (1.15 1.47)	Geographic region, teaching status and bed size
Mather et al	Retrospective	Data from Hartford Hospital,	February 24 and	878 hospitalized patients	Obesity	Yes vs no	5.87 (2.54-13.6)	Age, history of atrial

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2020, USA	cohort study	an 890-bed tertiary care medical center in Hartford, Connecticut	May 13, 2020	with COVID-19; 191 died				fibrillation, heart failure, CKD, obesity, neutrophil-lymphocyte ratio, white blood cell count, and asthma
Matsuzawa et al 2020, Japan	Retrospective cohort study	Patients from six hospitals in Kanagawa Prefecture	February 1 and May 1, 2020	151 hospitalized patients with COVID-19; 14 died	Hypertension	Yes vs. no	0.63 (0.15-2.34)	Age, sex, and diabetes mellitus
McCarty et al 2020, USA	Retrospective cohort study	Data from two tertiary medical centers and the seven community hospitals which comprise Mass General Brigham	March 22, through April 2, 2020	379 hospitalized patients with COVID-19; 57 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.97 (0.97-4.02) 2.15 (0.94-4.91) 0.69 (0.35-1.37)	Age, gender, obesity, cardiopulmonary comorbidities, hypertension, and diabetes
MCCULLOUGH et al 2020, USA	Retrospective observational cohort	Data from weill Cornell Medicine/New York-Presbyterian Hospital, a quaternary referral center and 862-bed teaching hospital	March 3 through April 9, 2020	756 hospitalized patients with COVID-19; 90 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Former vs. never Active vs. never	1.10 (0.68-1.70) 2.48 (1.51-4.07) 0.70 (0.43-1.14) 2.26 (1.42-3.61) 0.33 (0.04-2.47)	Age, clinical characteristics, and ECG variables, CAD, an immunosuppressed state, and hypoxemia
Mendy et al 2020, USA	Retrospective observational cohort	Data from university of Cincinnati health system, consisted of 4 hospitals in the states of Ohio. Kentucky. and Indiana	March 13 to May 31, 2020	689 hospitalized patients with COVID-19; 26 died	Diabetes Smoking Obesity	Yes vs. no Ever vs. never smokers Yes vs. no	1.79 (0.75-4.27) 0.82 (0.33-2.01) 1.09 (0.36-3.33)	Age, gender, race/ethnicity, and smoking
Menezes-Soares et al 2020, Brazil	Retrospective cohort study	Data from State Health Secretariat of Espırito Santo; Brazil	February 29 to June 11th, 2020	1,152 hospitalized patients with COVID-19; 456 died	Diabetes	Yes vs. no	1.88 (1.43-2.47)	None
Meng et al 2020, China	Retrospective cohort study	Data from Tongji Hospital in Wuhan	January 18 and March 27, 2020	2665 hospitalized patients with COVID-19; 299 died	Diabetes Hypertension	Yes vs. no Yes vs. no	0.71 (0.48-1.03) 1.01 (0.76-1.35)	Age, sex and comorbidities (diabetes, hypertension, COPD, CKD, CVD and cancer)
Mikami et al 2020, USA	Retrospective cohort study	Data from the medical records of the Mount Sinai Health System (8 hospitals and/or over 400 ambulatory practices)	March 13 and April 17, 2020	3708 hospitalized patients with COVID-19; 806 died	Diabetes Hypertension Cigarettes use history	Yes vs. no Yes vs. no Yes vs. no	0.92 (0.73-1.16) 0.91 (0.79-1.07) 1.01 (0.90–1.13)	Age, sex, race, cigarette use history, past medical history of asthma, hypertension, diabetes, or

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		in the New York City metropolitan area						cancer, systolic BP, RR, SpO2, BMI, initial laboratory values (lymphocyte proportion, Ddimer IL-6), and hydroxychloroquine use
Mirani et al 2020, Italy	large case series	Patients from single center in the region of Lombardy	February 20 and 9 April 2020	385 hospitalized patients with COVID-19; 102 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no BMI>30 kg/m2	1.33 (0.88-2.00) 1.84 (1.15–2.95) 1.33 (0.81–2.18)	Age and sex
Moon et al 2020, South Korea	Retrospective cohort Study	Data identified from the Korean Health Insurance Review and Assessment Service	prior to May 15. 2020.	5307 hospitalized patients with COVID-19; 211 died	Diabetes	Yes vs. no	2.66 (1.90-3.73	Age, sex, hypertension, dyslipidemia, chronic kidney disease, chronic obstructive lung disease, cardiovascular disease, atrial fibrillation, end-stage renal disease and cancer
Moon et al 2020, South Korea	Retrospective cohort study	Data from Dongguk University Gyeongju Hospital (DUGH) and the Andong Medical Center (AMC)	February 18 and June 30, 2020	352 hospitalized patients with COVID-19; 24 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.69 (1.05-6.85) 1.90 (0.75-4.84)	Age
MOWLA et al 2020, Bangladesh	Cross-sectional Study	Patients admitted in Dhaka Medical College Hospital (DMCH)	May 2 and 15, 2020	100 hospitalized patients with COVID-19; 10 died	Hypertension Smoking	Yes vs no Yes vs no	2.89 (0.54-15.37) 9.95 (1.73-57.12)	Multiple variables (hypertension, heart disease and renal disease)
Munblit et al 2020, Russia	Retrospective cohort study	Patients from the Sechenov University hospital network in Moscow city	April 22 and June 6, 2020	3480 hospitalized patients with COVID-19; 191 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.10 (1.46-2.99) 1.70 (1.14-2.59)	Age, sex, hypertension, diabetes, dementia, neoplasm, COPD and CKD
Munoz et al 2020, Spain	Retrospective cohort study	Patients a large hospital in Madrid,	NA	100 hospitalized patients with COVID-19; 23 died	Hypertension	Yes vs. no	4.47 (1.36-14.66)	Age, BMI, diabetes and antiobiotic treatment
Munoz-price et al 2020, USA	cross-sectional study	Froedtert Health and Medical College of Wisconsin (Milwaukee), the largest academic system in Wisconsin,	March 12 to March 31, 2020	369 hospitalized patients with COVID-19; 20 died	Smoking Obesity	History of smoking or current Yes vs. no	7.15 (0.55-9.22) 1.19 (1.05-1.35)	clustering at the zip code–level of patients' residence
Nachega et al 2020, Congo	Retrospective cohort study	Patients from seven hospitals in Kinshasa (one private, two	March 10, and July 31 2020	766 hospitalized patients with COVID-19; 101	Diabetes Hypertension	Yes vs. no Yes vs. no	1.10 (0.66-1.81) 1.00 (0.62-1.61)	age-group, the presence of comorbidities

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		faith-based Catholic, and four public)		died	Obesity	Yes vs. no	2.30 (1.24–4.27)	(hypertension, heart disease, diabetes, obesity, CKD, and cancer) and treatment received
Nakeshbandi et al 2020, USA	Retrospective cohort study	Patients tested for COVID-19 at SUNY Downstate Health Sciences University, hospital in New York	March 10 and April 13, 2020	504 hospitalized patients with COVID-19; 219 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Overweight Obese	1.10 (0.92-1.30) 0.81 (0.66–1.0) 1.4 (1.1–1.9) 1.3 (1.0–1.7)	Age, sex, smoking, obesity and hypertension
Namendys-Silva et al 2020, Mexico	Retrospective cohort study	Data from 10 hospitals located in Mexico	April 1 and April 30, 2020	164 hospitalized patients with COVID-19; 85 died	Diabetes Hypertension	Yes vs. no Yes vs. no	3.03 (1.51-6.09) 2.41 (1.26-4.62)	None
Narain et al 2020, USA	Retrospective cohort study	Data from electronic health records across the Northwell Health system	March 1 and April 24, 2020	5,776 hospitalized patients with COVID-19; 282 died	Diabetes Hypertension Smoking BMI	Yes vs. no Yes vs. no Never Active Former Smoker (unknown active/former) Unknown < 18.5 18.5-24.9 25-29.9 >30	1.02 (0.89-1.17) 0.83 (0.73-0.94) 1 1.43 (0.94-2.21) 0.93 (0.78-1.11) 1.42 (1.09-1.83) 3.02 (2.58-3.56) 1.15 (0.93-1.44) 1 0.98 (0.83-1.42) 1.07 (0.90-1.27)	Several covariables not mentionned
Neumann-Podczaska et al 2020, Poland	Retrospective cohort study	Data from the Jozef Strus Hospital, Poznan, Poland	March 12 and 5 April 2020	50 hospitalized patients with COVID-19; 20 died	Diabetes	Yes vs. no	2.35 (0.97–5.67)	Age
Nicholsoni et al 2020, USA	Retrospective cohort study	Patients admitted in five hospitals in the Mass General Brigham health care system, Boston	COVID-19 before May 19, 2020.	1042 hospitalized patients with COVID-19; 211 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Former vs. no Current vs. no	2.32 (1.30-4.12) 1.81 (1.32-2.49) 2.05 (1.39-3.03) 1.06 (0.78-1.44)	Age, sex, statin, C-reactive protein, albumin, C-reactive protein, diabetes, neutrophil-lymphocyte ratio, mean corpuscular volume, platelet count, and procalcitonin

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Nie et al 2020, China	Retrospective cohort study	Patients admitted to Union Hospital (West Campus), Huazhong University of Science and Technology	January 12 and March 12, 2020	311 hospitalized patients with COVID-19; 111 died	Diabetes Hypertension smoking	Yes vs. no Yes vs. no Current vs. never	2.13 (1.15–3.95) 3.80 (2.31–6.25) 3.66 (1.41–9.46)	None
Nogueira et al 2020, Portugal	Retrospective cohort study	Data from the General Health Directorate of Portugal (DGS)	January to April, 21 2020	20293 hospitalized patients with COVID-19; 502 died	Diabetes	Yes vs. no	0.75 (0.55-1.01)	Age, sex and pregnancy
Núñez-Gil et al 2020, Spain	Retrospective cohort study	Data from the HOPE COVID-19 registry (Health Outcome Predictive Evaluation for COVID-19)	February 8 and April 1, 2020	908 hospitalized patients with COVID-19; 301 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Yes vs. exsmoker No vs. exsmoker	2.77 (1.96-3.90) 1.74 (1.01-2.99) 1.52 (0.83-2.76) 1.07 (0.54-2.11) 0.55 (0.39-0.80)	None
Palaiodimos et al 2020, USA	Retrospective cohort study	Conducted at the Montefiore Medical Center, a tertiary academic institution in the Bronx, New York	43899	200 hospitalized patients with COVID-19; 48 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no <25 BMI ≥35 BMI Current or former smoker vs. never	1.16 (0.55-2.44) 0.93 (0.44–1.98) 1.57 (0.68–3.66) 3.35 (1.43–7.87) 0.83 (0.37–1.87)	Age, BMI, smoking and diabetes
Pan et al 2020, China	Retrospective cohort study	Renmin Hospital of Wuhan University	January 4, to February 14, 2020	996 hospitalized patients with COVID-19; 89 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.34 (0.78-2.29) 2.24 (1.36–3.70)	Age, sex, diabetes mellitus, matched with hypertensive
Park et al 2020, South Korea	Retrospective cohort study	Patients from five tertiary hospitals of Daegu, South Korea; aged >19 years	February 20 and April 14, 2020	289 hospitalized patients with COVID-19; 70 died	Diabetes	Yes vs. no	2.226 (1.357 to 3.652)	Age, sex, comorbidities (diabetes, hypertension, COPD, CKD and CLD) and other biological parameters
Parker et al 2020, South Africa	Retrospective cohort study	Patients admitted to the Tygerberg Hospital, Cape Town	25 March 25 and May 11, 2020	113 hospitalized patients with COVID-19; 28 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.90 (0.80-4.60) 2.40 (1.00-5.80) 1.30 (0.50-3.20)	None

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Perkin et al 2020, England	Retrospective cohort study	Data for deaths occurring at St George's HOSPITAL (Black, Asian and minority ethnic (BAME) communities), London	6 weeks commencing 12th of March 2020	243 hospitalized patients with COVID-19 and died with it	Diabetes Hypertension	Yes vs. no Yes vs. no	1.76 (1.05-2.94) 1.40 (0.80-1.99)	Age
Petrilli et al 2020, USA	Prospective cohort study.	Data from 260 outpatient office sites and four acute care hospitals (Tisch Hospital and NYU Langone Orthopedic Hospital in Manhattan. NYU Langone Hospital – Brooklyn in Brooklyn. and NYU Winthrop on Long Island).	March 1 to April 8 2020	2737 hospitalized patients with COVID-19; 391 died	Diabetes Hypertension BMI Smoking	Yes vs. no Yes vs. no <25 25.0-29.9 30.0-39.9 ≥40 Former vs no Current vs no Unknown vs no	1.01 (0.85-1.21) 0.98 (0.78-1.23) 1 1.01 (0.82-1.25) 1.08 (0.87-1.36) 1.45 (0.99-2.13) 1.07 (0.88-1.31) 0.92 (0.62-1.38) 1.50 (1.21-1.86)	Age, sex, race/ethnicity, smoking status, BMI, and chronic diseases
Pettit et al 2020, USA	Retrospective cohort study	Patients admitted to the University of Chicago Medical Center, an 811-bed academic medical center on the south side of Chicago, who had completed their hospital course	March 1 and April 18, 2020	238 hospitalized patients with COVID-19; 24 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	0.50 (0.20-1.70) 0.30 (0.10-0.90) 1.70 (1.10-2.80)	Demographics variable (age, sex, race/ethnicity) and comorbidities (diabetes, hypertension, pulmonary diseases, CVD, cancer, HIV and stroke)
Poblador-Plou et al 2020, Spain	Retrospective cohort study	The PRECOVID Study; in the Spanish region of Aragon	March 4 and April 17, 2020	4412 hospitalized patients with COVID-19; 771 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.65 (1.23-2.21), men 1.40 (1.02-1.91), women 0.96 (0.75-1.23), men 1.20 (0.92-1.56), women	Age
Portoles et al 2020, Spain	Prospective cohort study.	Patients form Puerta de Hierro University Hospital	February 25 to 24 April, 2020	1603 hospitalized patients with COVID-19; 197 died	Diabetes Obesity	Yes vs. no Yes vs. no	1.83 (1.34-2.50) 1.22 (0.63–2.33)	Age
Price-Haywood et al 2020, USA	Retrospective cohort study	Data from patients seen within an integrated-delivery health system (Ochsner Health) in Louisiana	March 1 and April 11, 2020	3481 hospitalized patients with COVID-19; 326 died	Obesity	Yes vs. no	0.99 (0.77–1.27)	Age, sex, race, Charlson Comorbidity Index score, residence in a low-income area, and indicators for baseline vital signs and laboratory measures (respiratory

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								rate; levels of aspartate aminotransferase, venous lactate, creatinine, bilirubin, procalcitonin, and C-reactive protein; and counts of lymphocytes and platelets).
Qin et al 2020, China	Retrospective cohort study	Patients admitted to the Sino-French New City Branch of Tongji Hospital, Tongji Medical college, Huazhong University of Science and Technology	January 26 and February 5, 2020	548 hospitalized patients with COVID-19; 90 died	Smoking	Smoking history vs. no	0.92 (0.53-1.57)	Age, sex, and comorbidities
Ramos-Rincon et al 2020, Spain	Retrospective cohort study	Data from the Spanish Society of Internal Medicine's SEMI-COVID-19 Registry, in which 150 hospitals in Spain participate (patients ≥18 years-old of age)	March 1 and May 29, 2020	2772 hospitalized patients with COVID-19; 1301 died	Obesity	BMI ≥ 30 vs. BMI < 30	1.24 (1.00-1.54)	None
Rapp et al 2020, USA	Retrospective cohort study	Data from the Mount Sinai Hospital System (MSHS) COVID-19 registry	February 29 and May 19, 2020	4,062 hospitalized patients with COVID-19; 1,190 died	Obesity	BMI ≥ 35 vs. BMI < 35	1.53 (1.21-1.94)	Sex, age and history of CKD
Rastad et al 2020, Iran	Retrospective cohort study	Alborz province, Iran	February 20 to March 25, 2020	2957 hospitalized patients with COVID-19; 301 died	Diabetes	Yes vs. no	1.69 (1.05-2.74)	Gender, age and laboratory tests
Recinella et al 2020, Italy	Retrospective cohort study	Patients admitted to two COVID-19 units of Sant'Orsola-Malpighi University Hospital in Bologna; age ≥ 65	March 30 and May 15, 2020	109 hospitalized patients with COVID-19; 43 died	Diabetes Hypertension BMI	Yes vs. no Yes vs. no BMI per	0.751 (0.347–1.622) 1.006 (0.495–2.047) 0.875 (0.782–0.979)	Age
Reilev et al 2020, Denmark	Retrospective cohort study	Data from the Danish Microbiology Database	February 27 until 19 May 2020	11 122 hospitalized patients with COVID-19; 577 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no overweight and obesity vs. no	1.60 (1.30–2.00) 1.30 (1.10-1.60) 1.50 (1.10-2.00)	Age and sex
Rivera-Izquierdo et al 2020, Spain	Retrospective cohort study	Patients from Hospital Universitario Clı´nico San Cecilio (Granada, Spain); aged 24 to 97	16 March and 10 April 2020	238 hospitalized patients with COVID-19; 38 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.42 (1.43-4.09) 1.48 (0.80-2.71)	Age, sex, comorbidities and medication on admission, symptoms, physical exam on

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								admission
Rodrıguez-Molinero et al 2020, Spain	Retrospective cohort study	Patients from the hospitals of the Consorci Sanitari de l'Alt Penedès i Garraf (CSAPG).	March 12 and May 2 2020	418 hospitalized patients with COVID-19;79 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Yes vs. no	2.96 (1.75-4.99) 5.64 (3.13-10.87) 1.12 (0.58–2.06) 0.38 (0.09–1.11)	None
Rosenberg et al 2020, USA	Retrospective cohort study	Patients admitted to hospitals in the New York City (NYC) metropolitan region between	March 15 and 28, 2020.	1438 hospitalized patients with COVID-19; 264 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no BMI >30 vs BMI <30 kg/m2 Current vs. never Farmer vs. never	1.57 (1.28-1.93) 1.95 (1.55-2.46) 0.96 (0.75-1.22) 1.30 (0.74-2.29) 1.47 (1.11-1.95)	Age
Rossi et al 2020, Italy	Prospective cohort	Patients from the Reggio Emilia province (Hospital, outpatient, primary, and preventive care to all the resident population)	February 27 to April 2. 2020	2653 hospitalized patients with COVID-19; 217 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.60 (1.10-2.20) 1.60 (1.20-2.10) 1.30 (0.60-2.90)	Age and sex and Charlson Comorbidity Index
Rottoli et al 2020, Italy	Retrospective cohort study	Data of the ongoing PREDI-CO multicentric study, from Sant'Orsola Hospital in Bologna (≥18 years)	March 1 and April 20. 2020.	482 hospitalized patients with COVID-19; 31 died	Diabetes Hypertension Obesity BMI	Yes vs. no Yes vs. no Yes vs. no per 1 unit increase <30 30-35 ≥35	1.05 (0.64-1.74) 1.23 (0.74-2.04) 2.18 (1.34–3.54) 1.07 (1.02–1.13) 1 1.21 (0.64–2.27) 1.72 (1.00–2.99)	Age, gender, hypertension and type 2 diabetes
Rozaliyani et al 2020, Indonesia	Retrospective cohort	Data from the ongoing recapitulation of Epidemiological Surveillance (ES) by the Provincial Health Office of Jakarta	March 2 and to April 27th 2020	4052 hospitalized patients with COVID-19; 381 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.26 (0.80-1.98) 1.86 (1.24-2.78)	Age, sex, symptoms, and other comorbidities
Russo et al 2020, Italy	Retrospective cohort study	Patients admitted to the Policlinico San Martino Hospital, Genoa and aged ≥ 18 years	February 25 and April 13, 2020	777 hospitalized patients with COVID-19; 275 died	Hypertension	Yes vs. no	0.91 (0.61–1.34)	Age, sex, Charlson morbidity and others biological informations
Salacup et al 2020, USA	Retrospective cohort study	A Single Tertiary Center Cohort in an underserved-Inner City Population with minority population in the United States	March 1 to April 24,2020	242 hospitalized patients with COVID-19; 52 died	Diabetes Hypertension BMI	Yes vs. no Yes vs. no BMI	1.45 (0.67-3.13) 0.95 (0.38-2.43) 1.02 (0.98-1.07)	Age, BMI, race/ethinicty and comorbidities

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Salazar et al 2020, USA	Retrospective cohort study	Data from the eight Houston Methodist hospitals	March 28 to July 6, 2020	316 hospitalized patients with COVID-19; 24 died	Diabetes Hypertension Obesity BMI	Yes vs. no Yes vs. no <30 ≥30 <18.5 18.5-24.9 25-29.9 ≥30 BMI (unit)	5.33 (2.00-14.18) 5.60 (1.88-16.63) 1 0.78 (0.36, 1.69) 201.94 (28.85, 1413.70) 1 0.40 (0.11, 1.39) 0.49 (0.18, 1.33) 1.02 (0.95, 1.09)	Age
Sands et al 2020, USA	Retrospective cohort study	Data from HCA Healthcare, a large healthcare system with >180 affiliated acute-care facilities among its >2,000 associated sites of care in 21 US states and the United Kingdom.	March 1 and discharge dates as of May 12. 2020.	6,180 hospitalized patients with COVID-19; 1,181 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.57 (1.21-2.03) 0.63 (0.48-0.82) 1.41 (1.08-1.85)	Age, sex, all comorbidities, vital signs at admission, and other patient characteristics
Santos et al 2020, Brazil	Retrospective cohort study	Data registered in the public national epidemiological surveillance system and obtained from the open access database (3103 health units and 1238 municipalities)	20 February 2020 to 2 June 2020	46 285 hospitalized patients with COVID-19; 21408 died	Diabetes BMI	Yes vs. no 18.5-24.9 25-29.9 30-34.9 35-40 >40	1.19 (1.14–1.23) 1 1.18 (0.53–2.61) 1.00 (0.47–2.12) 1.16 (0.54–2.45) 1.27 (0.59–2.70)	Age, sex, race/ethnicity,, education level, symptoms and signs and comorbidities
Seiglie et al 2020, USA	Retrospective cohort study	Data from the Massachusetts General Hospital, registry of patients hospitalized	March 11 and April 30, 2020	417 hospitalized patients with COVID-19; 49 died	Diabetes Hypertension BMI	Yes vs. no Yes vs. no 25-29.9 ≥30	2.02 (1.01-4.03) 1.33 (0.57-3.10) 0.60 (0.25-1.44) 1.10 (0.50-2.45)	Age, sex, race/ethnicity, and comorbidities
Shah et al 2020, USA	Retrospective cohort study	Patients from Phoebe Putney Health System, the largest community health system in Southwest Georgia	March 2 and May 6, 2020.	522 hospitalized patients with COVID-19; 92 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no	1.51 (0.90-2.55) 3.36 (1.30-8.60) 1.03 (0.55-1.93)	Age, BMI, sex, race and comorbidities

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					BMI	Former/current vs. never smoker <=30 30-40 >=40	1 1.49 (0.79-2.77) 2.29 (1.11-4.69)	
Shang et al 2020, China	Retrospective cohort study	Data from the Wuhan No.7 Hospital; temporarily taken over by Zhongnan Hospital of Wuhan University.	December 25, 2019 to March 20, 2020	584 hospitalized patients with COVID-19; 57 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.18 1.07-4.44) 0.70 (0.36-1.35)	Age, sex, hypertension, cardiovascular diseases, diabetes, chronic respiratory diseases, chronic kidney diseases, chronic liver diseases, acute kidney injury, acute liver injury, respiratory failure, and acute cardiac injury
Shi et al 2020, China	Retrospective cohort study	Patients from two tertiary hospitals in Wuhan (Renmin Hospital of Wuhan University and Zhongnan Hospital of Wuhan University)	January 1 and March 8, 2020	1561 hospitalized patients with COVID-19; 31 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.58 (0.84-2.99) 2.50 (1.30-4.78)	Age and sex
Shi et al 2020, China	Retrospective cohort study	Patients admitted to the Renmin Hospital of Wuhan University with laboratory-confirmed COVID-19	January 1 and February 23, 2020	671 hospitalized patients with COVID-19; 62 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.16 (0.47-2.56) 1.07 (0.46-2.53)	Age, preexisting cardiovascular disease (hypertension and other), diabetes, cerebrovascular disease, COPD and ARDS
Smith et al 2020, USA	retrospective cohort study	Data from four Trinity Health of New England hospitals (THONE) in Connecticut and Massachusetts	March 1 to April 22, 2020	346 hospitalized patients with COVID-19; 117 died	Diabetes	Yes vs. no	1.43 (1.07-1.92)	Age, COPD and hyperlipidemia
Sousa et al 2020, Brazil	Historical cohort	Datat from Fortaleza (Ceará's capital city), a state of the Brazilian Northeast	until 14 April 2020	2070 hospitalized patients with COVID-19; 131 died	Diabetes Obesity	Yes vs. no Yes vs. no	1.30 (0.80-2.10) 1.70 (0.40-7.70)	Age, cardiovascular disease, diabetes, BMI, Neurologic disease, pneumopathis and renal disease.
Steinberg et al 2020, USA	retrospective cohort study	Patients two sites: a high-volume, urban, academic,	March 8–April 4, 2020	210 hospitalized patients with COVID-19; 18 died	Obesity	Yes vs no	6.29 (1.76-22.46)	None

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		tertiary-care medical center and an affiliated suburban community hospital; 18-45 years old						
Stony Brook COVID-19 Research Consortium et al 2020, USA	Retrospective cohort study	Data of 1325 sequentially hospitalized patients with COVID-19 in New York;177 died; aged 49 to 75	March 2 to May 11, 2020	1325 hospitalized patients with COVID-19; 177 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no BMI >30 vs BMI <30 kg/m2	1.78 (1.21-2.65) 3.24 (1.98-5.27) 0.93 (0.63-1.37)	None
Sun et al 2020, China	Retrospective cohort study	Data of the Sino-French New City Branch of Tongji Hospital,	January 29 and March 5, 2020	244 hospitalized patients with COVID-19; 121 died	Hypertension	Yes vs no	0.82 (0.24-2.75)	Age, sex, biological parameter, symptoms and signs
Talavera et al 2020, Spain	Retrospective cohort study	Patients from Hospital Clínico Universitario de Valladolid	March 8 to April 11, 2020	576 hospitalized patients with COVID-19; 127 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Current or recent vs. never	1.33 (0.76-2.34) 1.31 (0.76-2.28) 1.71 (0.93-3.12)	Age, sex, hypertention, anosmia, cardiological disorders, cancer, smoking and Prior mRS ≥ 3
Tartof et al 2020, USA	Retrospective cohort study	Data from Kaiser Permanente Southern California members, a large integrated health care organization	February 13 to May 2 2020	6916 hospitalized patients with COVID-19; 206 died	Diabetes Smoking Hypertension BMI	DM with hemoglobin A1c level <7.5% vs. no DM with hemoglobin A1c level >7.5% vs. no Current Former Never Yes vs. no <18.5 18.5–24 25–29 30–34 35–39 40–44 ≥45	1.12 (0.79-1.59) 1.22 (0.81-1.84) 0.84 (0.30–2.35) 1.30 (0.96–1.75) 1 1.38 (0.93–2.06) 1.81 (0.99–3.30) 1 0.91 (0.62–1.35) 1.26 (0.82–1.95) 1.16 (0.63–2.17) 2.68 (1.43–5.04) 4.18 (2.12–8.26)	Sex, race/ethnicity, age, tobacco use, myocardial infarction, congestive heart failure, peripheral vascular disease, cerebrovascular disease, chronic pulmonary disease, renal disease, metastatic tumor or malignancy, other immune disease, hyperlipidemia, hypertension, asthma, organ transplant, and diabetes status and hemoglobin A1c level.
Tehrani et al 2020, Sweden	Retrospective cohort study	Patients admitted to the Danderyd University Hospital	March 5 and April 28, 2020	255 hospitalized patients with COVID-19; 70 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.35 (0.68–2.66) 1.37 (0.77–2.46) 1.07 (0.61–1.87)	None

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Turcotte et al 2020, USA	Retrospective cohort study	Data from anne Arundel Medical Center, a 350-bed acute care hospital	March 1 to April 12, 2020	117 hospitalized patients with COVID-19; 29 died	Diabetes	Yes vs. no	3.12 (0.75-12.97)	Temperature at admission, supplemental O2, sputum production and chronic kidney diseases
Wang et al 2020, China	Retrospective cohort study	Data from the intensive care unit of the Caidian Branch of Tongji Hospital	February 2 AND March 1, 2020	59 hospitalized patients with COVID-19; 41 died	Smoking	Smoking history vs. no	1.57 (0.74–3.31)	Age
Wang et al 2020, China	Retrospective cohort study	Patients admitted to Wuhan Union West Hospital and Wuhan Red Cross Hospital;	January 24 to February 10, 2020	605 hospitalized patients with COVID-19; 114 died	Hypertension	Yes vs. no	1.27 (0.85-1.89)	Age
Wang et al 2020, China	Retrospective cohort study	Data from Several wuhan hospitals	February 23; 2020	7283 hospitalized patients with COVID-19; 649 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.67 (1.22-2.30) 5.58 (4.30-7.26)	age, sex, location, occupation, dichotomous self-reported co-morbidities, number of presenting symptoms
Wang et al 2020, China	Retrospective cohort study	Patients hospitalized in the First People's Hospital of Jiangxia District in Wuhan, China	January 7, to February 11, 2020	296 hospitalized patients with COVID-19; 19 died	Diabetes Hypertension Smoking	Yes vs no Yes vs no Yes vs no	4.87 (1.70, 13.96) 1.82 (0.50, 6.63) 1.34 (0.16, 10.99)	Several variables not mentionned
Wang et al 2020, USA	Retrospective cohort study	Data from five member hospitals from the Mount Sinai Health System in New York City (NYC)	February 24 and April 15, 2020	3273 hospitalized patients with COVID-19; 742 died	BMI	per one unit increase	1.02 (1.00-1.03)	Age, sex and race
Wollenstein-Betech et al 2020, Brazil	Retrospective cohort study	Data from the nationwide surveilance program for acute respiratory distress syndrome (ARDS) in the state of São Paulo a	March 1 and June 30, 2020	113,214 hospitalized patients with COVID-19; 50,387 died	Diabetes Obesity	Yes vs. no Yes vs. no	1.22 (1.17-1.27) 1.66 (1.52-1.82)	None
Wu et al 2020, China	Retrospective cohort study	Data from Wuhan Hankou Hospital and No Six Hospital of Wuhan	December 26, 2019 and March 15 2020	1690 hospitalized patients with COVID-19; 556 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Yes vs. no	1.26 (0.72-2.22) 0.80 (0.52-1.22) 1.36 (0.89-2.08)	Age, gender, smoking, diabetes, hypertension, others comorbidites and severall biological measurements
Wu et al 2020, China	Retrospective cohort study	Data from to Wuhan Jinyintan Hospital	December 25, 2019 and January 26, 2020	201 hospitalized patients with COVID-19; 44 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.58 (0.80-3.13) 1.70 (0.92-3.14)	Age

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Xie et al 2020, USA	Retrospective cohort study	Black patients from two hospitals in New Orleans	March 30 to 5 April 2020	287 hospitalized patients with COVID-19; 58 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.92 (0.98-4.50) 2.71 (0.99-8.25) 1.67 (0.98-3.41)	Age, sex, race/ethnicity, hospital site, and the Charlson Comorbidity Index
Yahyavi et al 2020, Iran	Retrospective cohort study	Data from 17 referral hospitals affiliated to Iran University of medical sciences (IUMS)	February 20 and April 4, 2020	2553 hospitalized patients with COVID-19; 478 died	Diabetes	Yes vs. no	1.30 (1.00-1.60)	Angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, cardiovascular disease, chronic kidney disease, chronic pulmonary disease, diabetes mellitus, intensive care unit, diuretics, beta-blockers, and calcium channel blocker
Yan et al 2020, China	Retrospective cohort study	Patients from main tertiary teaching hospitals (Tongji Hospital, Optical Valley Branch and Sino-French New City Branch)	January 27 and February 17, 2020	882 hospitalized patients with COVID-19; 128 died	Diabetes	Yes vs. no	1.17 (0.73-1.86)	Age, sex, and comorbidities (cardiovascular disease, chronic respiratory disease, chronic kidney disease, cerebrovascular disease, diabetes, and malignancy), lymphopenia, d-dimer, lactate dehydrogenase, interleukin-6, myocardia injury, and liver injury
Yan et al 2020, China	Retrospective cohort study	Patients from Tongji Hospital (Wuhan)	January 10 to February 24, 2020	193 hospitalized patients with COVID-19; 108 died	Diabetes	Yes vs. no	1.53 (1.02-2.30)	Age, sex, hypertension, cardiovascular disease and cerebrovascular disease,
Yang et al 2020, China	Retrospective cohort study	Patients from the Central Hospital of Wuhan, a tertiary teaching hospital	January 1 to February 29, 2020	226 hospitalized patients with COVID-19; 50 died	Hypertension	Yes vs. no	2.68 (1.24-5.81)	Propensity score matching (multiple)
Yang et al 2020, China	Retrospective cohort study	Adult patients with COVID-19 hospitalized in Tongji Hospital (Wuhan)	February 4 to March 10, 2020	263 hospitalized patients with COVID-19; 107 died	Diabetes	Yes vs. no	1.59 (1.03-2.48)	Age

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Yazdanpanah et al 2020, France	Cohort	French COVID cohort; patients from 25 hospitals participating in the cohort throughout France	January 24 and March 15, 2020	264 hospitalized patients with COVID-19; 42 died	Diabetes Obesity Hypertension Smoking	Yes vs. no Yes vs. no Yes vs. no Former vs. no Current vs. no	1.98 (1.01– 3.89) 3.32 (1.70-6.52) 3.77 (2.04 -6.99) 1.44 (0.65-3.22) 1.36 (0.31-5.86)	Age, sex, ethnicity, smoking and comorbidites
Yu et al 2020, China	Retrospective cohort study	Data from Tongji Hospital, Wuhan	January 14 and February 28, 2020	1663 hospitalized patients with COVID-19; 212 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.34 (1.45-3.76) 1.08 (0.68-1.72)	Age, sex, history of hypertension and diabetes, lymphopenia, increased alanine aminotransferase, increased lactate dehydrogenase, increased D-dimer, and increased PCT
Zandkarimi et al 2020, Iran	Cross-sectional Study	Kurdistan province COVID-19 patients, western Iran, (identified by 16-hour centers) admitted to hospitals (16 hospitals)	February 22 and May 18, 2020	1831 hospitalized patients with COVID-19; 142 died	Diabetes	Yes vs. no	2.16 (1.38-3.38)	Age, sex, CHD, CLD, CKD and cancer
Zangrillo et al 2020, Italy	Retrospective cohort study	Patients from the COVID-BioB study held in a 1350-bed university hospital in Milan and performed at Istituto di Ricovero e Cura a Carattere Scientifico (IRCCS) San Raffaele Scientific Institute	February 20 and April 2, 2020	73 hospitalized patients with COVID-19; 17 died	Hypertension	Yes vs. no	6.15 (1.75-29.11)	Age, sex and other variable not mentionned
Zhang et al 2020, China	Retrospective cohort study	multicenter hospitals in Wuhan (Department of Infectious Disease and the Department of Oncology of the Union Hospital of Tongji Medical College, Huazhong University of Science and Departments of Endocrinology in the following four hospitals; aged 38 to 66	Jan 1 to Mar 17, 2020	312 hospitalized patients with COVID-19; 33 died	Diabetes	Yes vs. no	6.87 (1.92-24.58)	Age, sex, hospital, aspartate aminotransferase, estimated glomerular filtration rate, prothrombin time, and procalcitonin
Zhang et al 2020, China	Retrospective cohort study	Data from Tongji Hospital and Wuhan Pulmonary Hospital	February 7 and March 27, 2020,	53 hospitalized patients with COVID-19; 13 died	BMI	BMI per 1 unit increase	1.35 (1.08-1.70)	None
Zhang et al	Retrospective	Patients admitted to Zhongnan	December 29,	289 hospitalized patients	Smoking	Yes vs. no	5.21 (1.39-19.52)	Age, chest

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2020, China	cohort study	hospital of Wuhan Universityand No.7 hospital of Wuhan	2019 and February 16, 2020)	with COVID-19; 49 died				tightness/dyspnea, CRP and others biological parameters
Zhou et al 2020, China	Retrospective cohort study	Patients from Jinyintan Hospital and Wuhan Pulmonary Hospital (≥18 years old)		191 hospitalized patients with COVID-19; 54 died	Diabetes Hypertension Current smoker	Yes vs. no Yes vs. no Current smoker vs no	2.85 (1.35-6.05) 3.05 (1.57–5.92) 2.23 (0.65–7.63)	None
Zhu et al 2020, China	Retrospective cohort study	Data obtained from patient cohorts admitted in the 19 hospitals in Hubei Province, Chin	December 30, 2019 and March 20, 2020	7,337 hospitalized patients with COVID-19; 248 died	Diabetes	Yes vs. no	1.49 (1.13-1.96)	Age, gender, indicators of the severity of COVID-19 and hospital sites
Zimering et al 2020, USA	Retrospective cohort study	US military veterans hospitalized for COVID-19 infection during the peak of the pandemic in New Jersey	March 2020 and June 2020	55 hospitalized patients with COVID-19; 17 died	Diabetes BMI	Yes vs. no BMI per one unit increase	0.73 (0.23-2.35) 0.99 (0.92–1.07)	None

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Supplementary Table 2: Risk of bias judgements for each studies from the ROBINS-I

Author, years, country	**Bias due to confounding**	**Bias due to selection of participants**	**Bias due to exposure assessment**	**Bias due to misclassification during follow-up**	**Bias due to missing data**	**Bias due to measurement of the outcome**	**Bias due to selective reporting of the results**	**Overall judgement**
Abohamr et al 2020, Saudi Arabia	Critical	Moderate	Low	Low	No information	Low	Moderate	Critical
Adrish et al, 2020, USA	Serious	Low	Low	Low	No information	Low	Low	Serious
Alamdari et al, 2020, Iran	Critical	Moderate	Moderate	Low	No information	Low	Low	Critical
Albitar et al, 2020, China	Low	Low	Low	Moderate	Low	Low	Moderate	Moderate
Alharthy et al, 2020, Saudi Arabia	Low	Moderate	Moderate	Low	Low	Low	Low	Moderate
Almazeedi et al 2020, Kuwait	No information	Low	Low	Low	No information	Low	Low	Low
Al-Salameh et al, 2020, France	Low	Low	Low	Low	Low	Low	Low	Low
An et al 2020, South Korea	Low	Low	Low	Low	Low	Low	Low	Low
Anderson et al, 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Anzola et al, 2020, Italy	Low	Low	Low	Low	Low	Low	Low	Low
Arshad et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Atkins et al, 2020, UK	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Barbu et al, 2020, Romania	Serious	Low	Low	Low	No information	Low	Low	Serious
Baronio et al 2020, Italy	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Barron et al, 2020, England	Low	Low	Low	Low	Low	Low	Low	Low
Bepouka et al 2020, Democratic Republic of the Congo	Low	Low	Low	Low	Low	Low	Low	Low
Berenguer et al 2020, Spain	Low	Low	Low	Low	Low	Low	Low	Low
Biscarini et al 2020, Italy	Low	Low	Low	Low	Low	Low	Low	Low
Boulle et al, 2020, South Africa	Low	Low	Low	Low	No information	Low	Low	Low
Bousquet et al, 2020 France	Serious	Moderate	Low	Low	No information	Low	Low	Serious
Brouns et al 2020, Netherlands	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Calmes et al 2020, Belgium	Moderate	Low	Low	Low	Moderate	Low	Low	Moderate
Cao et al, 2020, China	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Carrasco-Sanchez et al, 2020, Spain	Low	Low	Low	Low	Low	Low	Low	Low
Castelnuovo et al 2020, Italy	Low	Low	Low	Low	Low	Low	Low	Low
Chachkhiani et al 2020, USA	Critical	Moderate	Low	Low	Moderate	Low	Low	Critical
Chang et al 2020, South Korea	Low	Low	Low	Low	No information	Low	Low	Low
Chen et al 2020, China	Serious	Low	Low	Low	Moderate	Low	Low	Serious
Chen et al 2020, China	Critical	Low	Low	Low	Low	Low	Low	Critical
Chen et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low

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Chilimuri et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Ciardullo et al 2020 Italy	Low	Low	Low	Low	Low	Low	Low	Low
Ciceri et al 2020, Spain	Serious	Low	Low	Low	Low	Low	Low	Serious
Coppelli et al 2020, Italy	Low	Low	Low	Low	No information	Low	Low	Low
Crouse et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Cummings et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Czernichow et al 2020, France	Low	Low	Low	Low	Low	Low	Low	Low
De Souza et al 2020, Brazil	No information	Moderate	Low	Moderate	No information	Low	Low	Moderate
De Vito et al 2020, Italy	Low	Low	Low	Low	No information	Low	Low	Low
Deng et al 2020, China	Critical	Low	Low	Low	No information	Low	Low	Critical
Dennis et al 2020, England	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Desai et al 2020, Italy	Low	Low	Low	Low	Low	Low	Low	Low
Docherty et al 2020, UK	Low	Low	Low	Low	Low	Low	Low	Low
Du et al 2020, China	Critical	Low	Low	Low	No information	Low	Low	Critical
Escalera-Antezana et al 2020, Bolivia	Low	Moderate	Low	Moderate	No information	Low	Low	Moderate
Esme et al 2020, Turkey	Low	Moderate	Low	Low	Moderate	Low	Low	Moderate
Fang et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Farrel et al 2020, Ireland	Serious	Low	Low	Low	Low	Low	Low	Serious
Fernandez-Cruz et al 2020, Spain	Critical	Moderate	Low	Moderate	No information	Low	Low	Critical
Ferrando-Vivas et al 2020, Bolivia	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Foy et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Fried et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Galloway et al 2020, England	Moderate	Moderate	Moderate	Low	No information	Low	Low	Moderate
Gao et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Gayam et al 2020, USA	Low	Moderate	Moderate	Moderate	No information	Low	Moderate	Moderate
Gazzaruso et al 2020, Italy	Low	Low	Low	Low	No information	Low	Low	Low
Ghany et al 2020, USA	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Giacomelli et al 2020, Italy	Low	Low	Low	Low	No information	Low	Low	Low
Grasselli et al 2020, Italy	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Gu et al 2020, China	Serious	Low	Low	Low	Low	Low	Low	Serious
Gu et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Guisado-Vasco et al 2020, Spain	Low	Low	Low	Low	Low	Low	Low	Low
Gupta et al 2020, USA	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Gutierrez et al 2020, Mexico	Low	Low	Low	Low	Low	Low	Low	Low

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Haase et al 2020, Denmark	Moderate	Low	Low	Low	Low	Low	Low	Moderate
Hajifathalian et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Harmouch et al 2020, USA	Critical	Low	Low	Low	Low	Low	Low	Critical
Harrison et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Hashemi et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Hewitt et al 2020, EUROPE	Low	Low	Low	Low	Low	Low	Low	Low
Hu et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Iaccarino et al 2020, Italy	No information	Low	Low	Low	No information	Low	Low	Low
Imam et al 2020, USA	Critical	Low	Low	Low	No information	Low	Low	Critical
Ioannou et al 2020, USA	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Kabarriti et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Kim et al 2020, South Korea	Low	Low	Low	Low	No information	Low	Low	Low
Kim et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Klang et al 2020, USA	Low	Low	Low	Low	Moderate	Low	Low	Moderate
Klang et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Kolhe et al 2020, UK	Low	Low	Low	Low	Low	Low	Low	Low
Lampasona et al 2020, Italy	Low	Low	Low	Low	Low	Low	Low	Low
Lee et al 2020, South Korea	Critical	Moderate	Low	Low	No information	Low	Low	Critical
Li et al 2020, China	Low	Low	Low	Low	Low	Low	Low	Low
Li et al 2020, China	Serious	Moderate	Low	Low	No information	Low	Low	Serious
Li et al 2020, China	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Li et al 2020, China	No information	Low	Low	Low	No information	Low	Moderate	Moderate
Li et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Liu et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Liu et al 2020, China	Low	Low	Low	Low	Low	Low	Low	Low
Liu et al 2020, China	Moderate	Low	Low	Low	No information	Low	Low	Moderate
Liu et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Lombardi et al 2020, Italy	Moderate	Low	Low	Low	No information	Low	Low	Moderate
Lopez-Mendez et al 2020, Mexico	Critical	Low	Low	Low	No information	Low	Low	Critical
Lunski et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Maddaloni et al 2020, Italy	Moderate	Low	Low	Low	No information	Low	Low	Moderate
Magleby et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Mallow et al 2020, USA	Low	Moderate	Low	Low	Moderate	Low	Low	Moderate
Mather et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Matsuzawa et al 2020, Japan	Low	Low	Low	Low	No information	Low	Low	Low

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McCarty et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
MCCULLOUGH et al 2020, USA	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Mendy et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Menezes-Soares et al 2020, Brazil	Critical	Low	Low	Low	No information	Low	Low	Critical
Meng et al 2020, China	No information	Low	Low	Low	Low	Low	Low	Low
Mikami et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Mirani et al 2020, Italy	Moderate	Low	Low	Low	No information	Low	Low	Moderate
Moon et al 2020, South Korea	Low	Low	Low	Low	Low	Low	Low	Low
Moon et al 2020, South Korea	Serious	Low	Low	Low	No information	Low	Low	Serious
MOWLA et al 2020, Bangladesh	Low	Low	Low	Moderate	No information	Low	Low	Moderate
Munblit et al 2020, Russia	Low	Low	Low	Low	No information	Low	Low	Low
Munoz et al 2020, Spain	Low	Low	Low	Low	No information	Low	Low	Low
Munoz-price et al 2020, USA	Low	Low	Low	Moderate	Moderate	Low	Low	Moderate
Nachega et al 2020, Congo	Low	Moderate	Low	Low	Moderate	Low	Low	Moderate
Nakeshbandi et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Namendys-Silva et al 2020, Mexico	Critical	Moderate	Low	Low	Moderate	Low	Low	Critical
Narain et al 2020, USA	Low	Moderate	Low	Low	Moderate	Low	Low	Moderate
Neumann-Podczaska et al 2020, Poland	Serious	Moderate	Low	Low	No information	Low	Low	Serious
Nicholsoni et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Nie et al 2020, China	Critical	Low	Low	Low	No information	Low	Low	Critical
Nogueira et al 2020, Portugal	Low	Low	Low	Low	Moderate	Low	Low	Moderate
Núñez-Gil et al 2020, Spain	Critical	Low	Low	Low	No information	Low	Moderate	Critical
Palaiodimos et al 2020, USA	Moderate	Low	Low	Low	Low	Low	Low	Moderate
Pan et al 2020, China	Low	Low	Low	Low	Low	Low	Low	Low
Park et al 2020, South Korea	Low	Low	Low	Low	No information	Low	Low	Low
Parker et al 2020, South Africa	Critical	Low	Low	Low	No information	Low	Moderate	Critical
Perkin et al 2020, England	Serious	Low	Low	Low	No information	Low	Low	Serious
Petrilli et al 2020, USA	Low	Low	Low	Low	Moderate	Low	Low	Moderate
Pettit et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Poblador-Plou et al 2020, Spain	Serious	Low	Low	Low	No information	Low	Low	Serious
Portoles et al 2020, Spain	Serious	Low	Low	Low	No information	Low	Low	Serious
Price-Haywood et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Qin et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Ramos-Rincon et al 2020, Spain	Critical	Moderate	Low	Low	No information	Low	Low	Critical
Rapp et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low

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Rastad et al 2020, Iran	Low	Low	Low	Low	Moderate	Low	Low	Low
Recinella et al 2020, Italy	Serious	Moderate	Low	Low	Low	Low	Low	Serious
Reilev et al 2020, Denmark	Moderate	Low	Low	Low	No information	Low	Low	Moderate
Rivera-Izquierdo et al 2020, Spain	Low	Low	Low	Low	No information	Low	Low	Low
Rodrıguez-Molinero et al 2020, Spain	Critical	Low	Low	Low	Low	Low	Low	Critical
Rosenberg et al 2020, USA	Serious	Low	Low	Low	No information	Low	Low	Serious
Rossi et al 2020, Italy	Low	Low	Low	Low	No information	Low	Low	Low
Rottoli et al 2020, Italy	Low	Low	Low	Low	Low	Low	Low	Low
Rozaliyani et al 2020, Indonesia	Low	Low	Low	Low	No information	Low	Low	Low
Russo et al 2020, Italy	Low	Low	Low	Low	No information	Low	Low	Low
Salacup et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Salazar et al 2020, USA	Serious	Moderate	Low	Low	No information	Low	Low	Serious
Sands et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Santos et al 2020, Brazil	Low	Low	Low	Low	Low	Low	Low	Low
Seiglie et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Shah et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Shang et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Shi et al 2020, China	Moderate	Low	Low	Low	Moderate	Low	Low	Moderate
Shi et al 2020, China	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Smith et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Sousa et al 2020,Brazil	No information	Low	Low	Low	Moderate	Low	Low	Moderate
Steinberg et al 2020, USA	Critical	Low	Low	Low	No information	Low	Low	Critical
Stony Brook COVID-19 Research Consortium et al 2020, USA	Critical	Low	Moderate	Low	Low	Low	Low	Critical
Sun et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Talavera et al 2020, Spain	Low	Low	Low	Low	Low	Low	Low	Low
Tartof et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Tehrani et al 2020, Sweden	Critical	Low	Low	Low	No information	Low	Low	Critical
Turcotte et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Wang et al 2020, China	Serious	Moderate	Low	Low	No information	Low	Low	Serious
Wang et al 2020, China	Serious	Low	Low	Low	Moderate	Low	Low	Serious
Wang et al 2020, China	Low	Moderate	Low	Low	Moderate	Low	Low	Moderate
Wang et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Wang et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Wollenstein-Betech et al 2020, Brazil	Critical	Low	Low	Low	Moderate	Low	Low	Critical

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Wu et al 2020, China	Low	Moderate	Low	Low	Low	Moderate	Low	Moderate
Wu et al 2020, China	Serious	Low	Low	Low	No information	Low	Low	Serious
Xie et al 2020, USA	Low	Low	Low	Low	No information	Low	Moderate	Moderate
Yahyavi et al 2020, Iran	Low	Low	Low	Low	No information	Low	Low	Low
Yan et al 2020, China	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Yan et al 2020, China	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Yang et al 2020, China	Low	Low	Low	Low	Low	Low	Low	Low
Yang et al 2020, China	Serious	Low	Low	Low	Moderate	Low	Low	Serious
Yazdanpanah et al 2020, France	Low	Low	Low	Low	Low	Low	Low	Low
Yu et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Zandkarimi et al 2020, Iran	No information	Low	Low	Moderate	No information	Low	Low	Moderate
Zangrillo et al 2020, Italy	No information	Moderate	Low	Low	No information	Low	Moderate	Moderate
Zhang et al 2020, China	Low	Low	Low	Low	Low	Low	Low	Low
Zhang et al 2020, China	Critical	Moderate	Moderate	Low	No information	Low	Moderate	Critical
Zhang et al 2020, China	No information	Moderate	Low	Low	Moderate	Low	Low	Moderate
Zhou et al 2020, China	Serious	Low	Low	Low	No information	Low	Low	Serious
Zhu et al 2020, China	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Zimering et al 2020, USA	Critical	Low	Low	Low	No information	Low	Low	Critical

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Supplementary Table 3. Subgroup analyses of association between BMI and mortality risk in patients with COVID-19										
	Obesity (obese vs non-obese)					**BMI per 5 kg/m^2**				
	n	SRR (95% CI)	I^2 (%)	P_{within} [a]	$P_{between}$ [b]	n	SRR (95% CI)	I^2 (%)	P_{within} [a]	$P_{between}$ [b]
All studies	54	1.45 (1.31-1.61)	91.0	<0.0001		25	1.12 (1.07-1.17)	68.2	<0.0001	
Study design										
Retrospective	47	1.44 (1.29-1.61)	92.1	<0.0001		20	1.11 (1.06-1.16)	59.8	<0.0001	
Prospective	4	1.45 (1.07-1.97)	34.1	0.208		3	1.15 (1.02-1.30)	84.8	0.001	
Cross-sectional	3	1.88 (0.87-4.02)	61.8	0.07	0.85	2	1.50 (0.68-3.31)	83.5		0.89
Geographical location										
North America	26	1.26 (1.09-1.46)	94.2	<0.0001		16	1.13 (1.07-1.19)	52.3	<0.0001	
South America	4	1.80 (1.24-2.62)	23.7	0.27		1	1.09 (1.02-1.16)			
Europe	17	1.40 (1.23-1.60)	46.2	0.02		7	1.14 (1.01-1.29)	84.2	<0.0001	
Asia	5	2.50 (0.72-8.68)	95.5	<0.0001		1	0.99 (0.83-1.17)			
Africa	2	1.93 (1.15-3.23)	0.5	0.32	0.07					0.81
Number of patients										
<1000	31	1.76 (1.42-2.18)	87.2	<0.0001	0.0001	13	1.19 (1.07-1.31)	62.6	0.001	
≥1000	23	1.27 (1.13-1.42)	88.3	<0.0001		11	1.10 (1.05-1.15)	70.1	<0.0001	0.001
Risk of bias										
Low	28	1.39 (1.20-1.60)	93.3	<0.0001		12	1.12 (1.06-1.18)	70.3	<0.0001	
Moderate	11	1.52 (1.19-1.95)	68.9	<0.0001		12	1.15 (1.06-1.25)	66.7	0.001	
Serious	3	0.97 (0.78-1.21)	0.0	0.675		1	0.76 (0.49-1.20)			0.36
Critical	12	1.75 (1.32-2.32)	91.9	<0.0001	0.42					
Adjustment for confounders										
No	12	1.69 (1.30-2.21)	91.3	<0.0001		1	1.05 (0.88-1.25)	-		
Age only	3	0.97 (0.78-1.21)	0.0	0.675	0.31	1	0.76 (0.49-1.20)	-		
Multiple	36	1.43 (1.26-1.61)	91.6	<0.0001		23	1.13 (1.08-1.18)	69.8	<0.0001	0.35

Footnote: CI confidence interval; SRR summary relative risk

I^2 (%) is a measure of the proportion of the heterogeneity attributed to between study variation rather than due to chance. I^2-values of 25%, 50% and 75% indicates low, moderate and high between study heterogeneity, respectively.

[a]P-value for heterogeneity within each subgroup

[b]P-value for heterogeneity between subgroups with meta-regression analysisbetween

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Supplementary Table 4. Subgroup analyses of association between smoking status and mortality risk in patients with COVID-19

	Smoking (ever versus never smoker)					Current smoker (yes vs no)					Ex-smoker (yes vs no)				
	n	SRR (95% CI)	I^2 (%)	P_{within} [a]	$P_{between}$ [b]	n	SRR (95% CI)	I^2 (%)	P_{within} [a]	$P_{between}$ [b]	n	SRR (95% CI)	I^2 (%)	P_{within} [a]	$P_{between}$ [b]
All studies	28	1.28 (1.17-1.40)	74.0	<0.0001		19	1.29 (1.03-1.62)	84.2	<0.0001		14	1.25 (1.11-1.42)	74.8	<0.0001	
Study design															
Retrospective	21	1.32 (1.15-1.52)	75.9	<0.0001		14	1.41 (1.04-1.91)	86.2	<0.0001		10	1.33 (1.12-1.57)	74.6	<0.0001	
Prospective	5	1.16 (1.01-1.32)	65.8	0.020		5	1.07 (0.94-1.21)	0.0	0.83		4	1.15 (0.93-1.42)	76.8	0.005	
Cross-sectional	2	2.65 (0.52-13.38)	81.7	0.020	0.45	-	-	-		0.01		-	-	-	
Geographical location															0.007
North America	17	1.25 (1.08-1.45)	79.4	<0.0001		11	1.16 (0.99-1.35)	34.3	0.125		9	1.23 (1.06-1.43)	71.9	<0.0001	
Europe	9	1.26 (1.13-1.40)	59.2	0.012		5	1.07 (0.95-1.21)	0.0	0.446		4	1.30 (1.00-1.69)	76.3	<0.0001	
Asia	2	2.56 (0.99-6.59)	53.0	0.15		3	3.00 (2.50-3.61)	0.0	0.82		1	2.06 (0.73-5.79)			0.71
South America	_	_	_	_		-	-	-				-	-	-	
Africa	_	_	_	_	0.21	-	-	-		<0.0001		-	-	-	
Number of patients															
<1000	13	1.47 (1.19-1.80)	46.1	0.035		6	2.47 (1.68-3.62)	27.1	0.23	<0.0001	4	1.91 (1.48-2.48)	0.0	0.784	
≥1000	15	1.22 (1.10-1.37)	82.1	<0.0001	0.001	13	1.11 (1.00-1.24)	27.1	0.17		10	1.17 (1.04-1.33)	76.9	<0.0001	0.002
Risk of bias															
Low	12	1.20 (1.08-1.34)	46.4	0.039		8	1.09 (0.98-1.20)	0.0	0.854		6	1.26 (1.07-1.49)	71.5	0.004	
Moderate	10	1.25 (1.04-1.51)	87.3	<0.0001		6	1.23 (0.69-2.22)	93.3	<0.0001		4	1.12 (0.90-1.39)	75.8	0.006	
Serious	3	1.43 (1.24-1.66)	0.0	0.495		3	1.57 (1.23-2.01)	0.0	0.674		3	1.32 (1.07-1.64)	10.4	0.327	
Critical	3	1.51 (0.99-2.30)	40.3	0.19	0.60	2	2.50 (1.37-4.58)	1.6%	0.313	0.18	1	1.82 (1.27-2.60)			0.53
Adjustment for confounders															
No	4	1.43 (1.13-1.80)	44.4	0.15		3	2.45 (1.43-4.19)	0.0	0.594		1	1.82 (1.27-2.60)			
Age only	3	1.43 (1.24-1.66)	0.0	0.495	0.40	2	1.55 (1.21-1.99)	0.0	0.493		3	1.32 (1.07-1.64)	10.4	0.327	
Multiple	21	1.23 (1.09-1.38)	78.3	<0.0001		14	1.16 (0.88-1.52)	87.6	<0.0001	0.13	10	1.20 (1.05-1.39)	79.6	<0.0001	0.41

Footnote: CI confidence interval; SRR summary relative risk

I^2 (%) is a measure of the proportion of the heterogeneity attributed to between study variation rather than due to chance. I^2-values of 25%, 50% and 75% indicates low, moderate and high between study heterogeneity, respectively.

[a]P-value for heterogeneity within each subgroup

[b]P-value for heterogeneity between subgroups with meta-regression analysisbetween

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Supplementary Table 5: GRADE Evidence profile

		Downgrading factors					Upgrading factors					
Prognostic factors	Initial rate of confidence	Risk of bias	Inconsistency	Indirectness	Imprecision (ratio of the upper to lower 95%CI)	Publication bias	Large magnitude of association	Dose response	Residual confounding	Consistency	Overall quality	Level of evidence for the health effect
Diabetes	Low (observational studies)	Not likely	Not likely	Not likely	Not likely (ratio: 0.97)	Not likely	No upgrade	No upgrade	No upgrade	No upgrade	⨁⨁◯◯ Low	Low
Hypertension	Low (observational studies)	Not likely	Not likely	Not likely	Not likely (ratio : 1.18)	Not likely	No upgrade	No upgrade	No upgrade	No upgrade	⨁⨁◯◯ Low	Low
Obesity	Low (observational studies)	Not likely	Not likely	Not likely	Not likely (ratio : 1.23)	Not likely	Upgrade (SRR>2 for BMI ~ 45)	Upgrade (evidence of nonlinear relation)	No upgrade	No upgrade	⨁⨁⨁⨁ High	High
Smoking	Low (observational studies)	Not likely	Not likely	Not likely	Not likely (ratio : 1.20)	Not likely	No upgrade	No upgrade	No upgrade	No upgrade	⨁⨁◯◯ Low	Low

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Supplementary method for GRADE

GRADE (Grading of Recommendations, Assessment, Development and Evaluations) is a transparent framework for developing and presenting summaries of evidence and provides a systematic approach for making clinical practice recommendations. The aim is to rate the certainty of evidence (or known as quality of evidence) for each outcome across studies (i.e. a body of evidence) using four levels of evidence: very low, low, moderate, and high. Guidelines are described in a serie of articles published in the Journal of Clinical Epidemiology: https://www.jclinepi.com/action/showPdf?pii=S0895-4356%2810%2900330-6

GRADE does not mean rating each study as a single unit. Rather, rating is made for each outcome, and quality may differ indeed, is likely to differ from one outcome to another within a single study and across a body of evidence.

Quality of evidence	Interpretation
High	We are very confident that the true effect lies close to that of the estimate of the effect
Moderate	We are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect but there is a possibility that it is substantially different
Low	Our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect
Very Low	We have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect

In the GRADE framework, observational study initially starts as low-quality evidence. Then, there are 5 criteria which can be used to downgrade the level of evidence:

- Risk of bias in individual studies – e.g. methodological issues in included studies such as inadequate blinding (participants knew they were in control/treatment group)
- Inconsistency of results between studies
- Indirectness of evidence – e.g. participants were children although the systematic review was about adults
- Imprecision – results were not statistically significant, or the effect was clinically important once the studies were meta-analysed
- Publication bias – result was biased due to a file-drawering effect, as studies not showing a statistically significant effect are less likely to be published.

Additionally, observational studies starting at low can be upgraded based on 3 criteria: large effect, dose-response effect and “Effect of all plausible confounding factors would be to reduce the effect (where an effect is observed) or suggest a spurious effect (when no effect is observed)”.

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Guidance to rate the outcome is briefly described below:

1- Risk of bias across studies (internal validity)

Risk of bias was assessed using ROBIN-I rating the overall risk of bias into four categories: low, moderate, serious and critical risk of bias.

Downgrade	Guidance
Not likely	Most information is at low risk of bias
Serious	Does not clearly meet guidance for “not serious” or “very serious”
Very serious	Most information is at serious or critical risk of bias

2- Inconsistency

Magnitude of intervention effects differs across studies, it may be explained by the difference in population, intervention, outcomes or study methods. If large variability (heterogeneity) in magnitude of effect remains unexplained, the quality of evidence decreases.

Downgrade	Guidance
Not likely	Point estimates similar Confidence intervals overlap Statistical heterogeneity is non-significant ($p \geq 0.1$) I^2 of $\leq 50\%$
Serious	Point estimates vary Confidence intervals show minimal overlap Statistical heterogeneity has low p-value ($p \leq 0.1$) I^2 of >50% to 75%
Very serious	Point estimates vary widely Confidence intervals show minimal or no overlap Statistical heterogeneity has low p-value ($p \leq 0.1$) I^2 of >75%

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3- Directness

Certainty can be rated down if the patients studied are different from those for whom the recommendation applies. Decisions regarding indirectness of patients and interventions depend on an understanding of whether biological or social factors are sufficiently different that one might expect substantial differences in the magnitude of effect. Indirectness occurs when the population, intervention, or outcomes differ from those in which we are interested

4- Precision is the degree of certainty surrounding an effect estimate with respect to a given outcome (AHRQ 2012a). A precise estimate enables the evaluator to determine whether there is an effect (i.e., it is different from the comparison group). We used 95% confidence intervals as the primary method to assess imprecision (Guyatt et al. 2011b).

Downgrade	Guidance
Not likely	No or minimal indications of large standard deviations (i.e., SD > mean) For ratio measures (e.g., odds ratio, OR) the ratio of the upper to lower 95% CI for most studies (or meta-estimate) is < 10; or for absolute measures (e.g., percent control response) the absolute difference between the upper and lower 95% CI for most studies (or meta-estimate) is < 100
Serious	Does not clearly meet guidance for "not serious" or "very serious"
Very serious	Large standard deviations (i.e., SD > mean) For ratio measures (e.g., OR) the ratio of the upper to lower 95% CI for most studies (or meta-estimate) is ≥ 10; or for absolute measures (e.g., percent control response) the absolute difference between the upper and lower 95% CI for most studies (or meta-estimate) is ≥ 100

5- Publication bias

We characterized publication bias as "undetected" (no downgrade) or "strongly suspected" as recommended by GRADE (Guyatt et al. 2011e). In general, studies with statistically significant results are more likely to be published than studies without statistically significant results (''negative studies'') (Guyatt et al. 2011e). Thus, some degree of publication bias is likely on any topic; however, downgrading is reserved for cases where the concern is serious enough to significantly reduce confidence in the body of evidence.

Publication bias is suspected when studies are uniformly small, particularly when sponsored by industries, NGOs or authors with conflicts of interests. Funnel plots and Egge's regression can help to visualize publication bias.

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6- Domains that can increase confidence

Upgrade	Guidance
Large magnitude of association or effect	GRADE Working Group has previously suggested guidelines for rating quality of evidence up by one category (typically from low to moderate) for associations greater than 2, and up by two categories for associations greater than 5 (Guyatt et al.2011g).
Dose response	Upgrade for evidence of a monotonic dose-response gradient (Guyatt et al. 2011g) and for evidence of a non-monotonic dose response when data fit the expected pattern, i.e., prior knowledge leads to expectation for non-monotonic dose response, and/or non-monotonic dose response is consistently observed in the evidence base.
Residual Confounding or Other Related Factors	Confidence in the results is increased when the body of evidence is potentially biased by factors counter to the observed effect. Upgrading should be considered when there are indications that residual confounding or bias would underestimate an apparent association or treatment effect (i.e., bias towards the null), or suggest a spurious effect when results suggest no effect.
Cross-Species/Population/Study Consistency	Three types of consistency in the body of evidence can be used to support an increase in confidence in the results: •across animal studies–consistent results reported in multiple experimental animal models or species •across dissimilar populations–consistent results reported across populations (human or wildlife) that differ in factors such as time, location, and/or exposure •across study types–consistent results reported from studies with different design features, e.g., between prospective cohort and case-control human studies or between chronic and multigenerational animal studies

Translation of confidence rating into level of evidence		
Confidence in the body of evidence	Direction of the effect (Health effect)	Level of evidence for the health effect
High	⇨	High
Moderate	⇨	Moderate
Low	⇨	Low
Very low or no evidence	⇨	Inadequate
Confidence in the body of evidence	Direction of the effect (No Health effect)	Level of evidence for the health effect
High	⇨	Evidence of no health effect
Moderate	⇨	Inadequate
Low	⇨	Inadequate
Very low or no evidence	⇨	Inadequate

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Supplementary text

PubMed search terms:

1) “Body mass index” OR body mass index [MeSH] OR BMI OR overweight OR overweight [MeSH] OR obesity OR obesity[MeSH] OR anthropometry OR anthropometry[MeSH] OR fatness OR “body fatness” OR “abdominal fatness” OR “abdominal obesity” OR abdominal obesity[MeSH] OR adiposity OR adiposity[MeSH]

2) Diabetes OR diabetes [MeSH] OR glucose OR glucose [MeSH] OR "medical history" OR medical history[MeSH]

3) “blood pressure” OR blood pressure [MeSH] OR hypertension OR hypertension[MeSH])

4) smoking OR smoking[MeSH] OR cigarette OR cigarette[MeSH] OR cigarettes OR cigarettes[MeSH] smoke OR smoke[MeSH] OR tobacco OR tobacco[MeSH] OR snus OR snuff OR snuff[MeSH] OR "environmental tobacco smoke” OR environmental tobacco smoke[MeSH] OR “passive smoking” OR passive smoking[MeSH] OR “smoking cessation" OR smoking cessation[MeSH]

5) COVID-19 OR SARS-CoV-2) AND (MORTALITY OR DEATH)

6) 1 OR 2 OR 3 OR 4

8) 6 AND 5

Embase search terms:

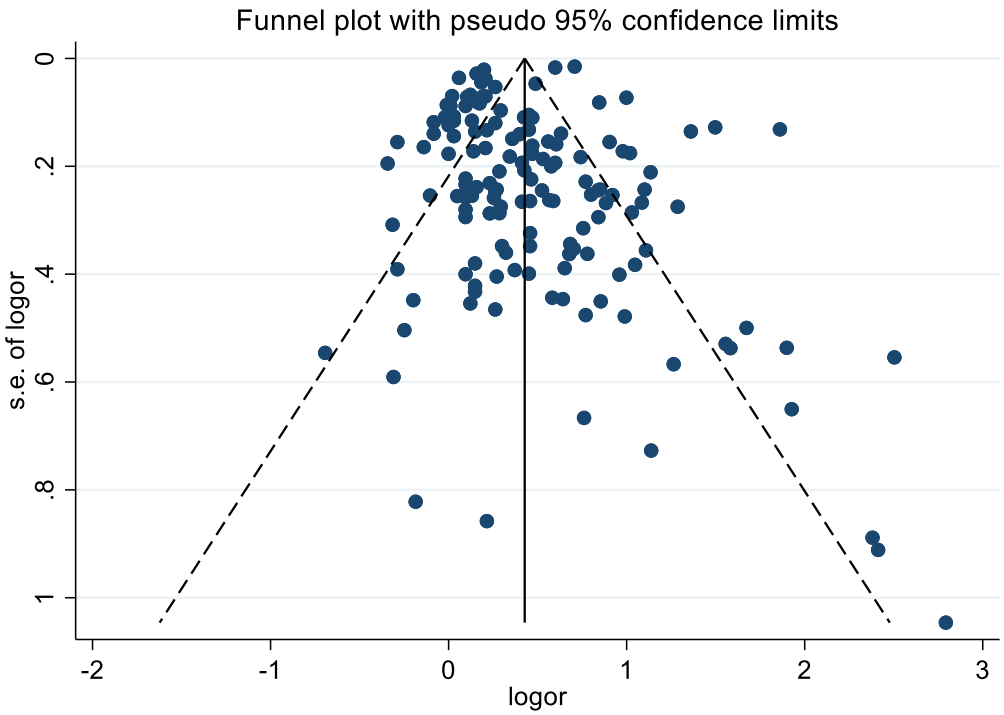
☐	1	(body mass index or bmi or overweight or obesity or anthropometry or fatness or body fatness or abdominal fatness or abdominal obesity or adiposity or diabetes or blood glucose or medical history or blood pressure or hypertension or smoking or cigarette or cigarettes or smoke or tobacco or snus or snuff or environmental tobacco smoke or passive smoking or smoking cessation).ab,ti.	2489416	Advanced
☐	2	body mass index/ or bmi/ or overweight/ or obesity/ or anthropometry/ or fatness/ or body fatness/ or abdominal fatness/ or abdominal obesity/ or adiposity/ or diabetes/ or blood glucose/ or medical history/ or blood pressure/ or hypertension/ or smoking/ or cigarette/ or cigarettes/ or smoke/ or tobacco/ or snus/ or snuff/ or environmental tobacco smoke/ or passive smoking/ or smoking cessation/	2393322	Advanced
☐	3	(COVID-19 or SARS-CoV-2).ab,ti.	66858	Advanced
☐	4	COVID-19/ or SARS-CoV-2/	95	Advanced
☐	5	(mortality or death).ab,ti.	2047722	Advanced
☐	6	mortality/ or death/	1077889	Advanced
☐	7	1 or 2	3256028	Advanced
☐	8	3 or 4	66898	Advanced
☐	9	5 or 6	2342175	Advanced
☐	10	7 and 8 and 9	2095	Advanced

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Cochrane library search terms:

#1	Body mass index
#2	anthropometry
#3	obesity
#4	overweight
#5	adiposity
#6	#1 OR #2 OR #3 OR #4 OR #5
#7	Diabetes
#8	glucose
#9	medical history
#10	#7 OR #8 OR #9
#11	blood pressure
#12	hypertension
#13	#11 OR #12
#14	smoking
#15	cigarette
#16	tobacco
#17	#14 OR #15 OR #16
#18	COVID-19
#19	SARS-CoV-2
#20	#18 OR #19
#21	MORTALITY
#22	DEATH
#23	#21 OR #22
#24	(#6 OR #10 OR #13 OR #20) AND #23

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Supplementary Figure 1: Funnel plot of diabetes and mortality risk in patients with COVID-19

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Study omitted	Estimate	[95% Conf. Interval]
Albitar, 2020	1.5293853	1.4358113 1.6290579
Al-Salameh, 2020	1.5442724	1.4494652 1.645281
Hashemi, 2020	1.5388012	1.444257 1.6395345
Atkins, 2020	1.5295212	1.435613 1.6295722
Barbu, 2020	1.5139627	1.4230996 1.6106272
Barron, 2020	1.5295643	1.4380455 1.6269073
Boulle, 2020	1.5274702	1.4342617 1.626736
Bousquet, 2020	1.537884	1.4434227 1.638527
Calmes, 2020	1.5364693	1.4419811 1.6371492
Cao, 2020	1.5384178	1.4440672 1.6389329
Carrasco-Sánchez, 2020	1.5381721	1.4424546 1.6402413
Castelnuovo, 2020	1.5441017	1.4491032 1.645328
Chachkhiani, 2020	1.5411512	1.4464148 1.6420925
Chang, 2020	1.5396796	1.4451315 1.6404138
Chen, 2020	1.5313251	1.4371679 1.631651
Chen, 2020	1.5329227	1.4387097 1.6333052
Chilimuri, 2020	1.5377694	1.4431939 1.6385427
Ciardullo, 2020	1.5410501	1.4461583 1.6421683
Coppelli, 2020	1.5411114	1.446546 1.6418588
Crouse, 2020	1.5296242	1.435763 1.6296216
Cummings, 2020	1.5397375	1.4450227 1.6406603
Dennis, 2020	1.5427595	1.4469318 1.6449338
Desai, 2020	1.5401301	1.4455459 1.6409032
DF de Souza, 2020	1.5309588	1.4368759 1.6312019
Escalera-Antezana, 2020	1.5346913	1.4406917 1.6348242
Esme, 2020	1.5427371	1.4472408 1.6445348
Fang, 2020	1.5347261	1.4405845 1.63502
Fernandez-Cruz. , 2020	1.5323932	1.4382639 1.632683
Fried, 2020	1.5448198	1.4499104 1.6459421
Galloway, 2020	1.5412358	1.4462699 1.6424375
Ghany, 2020	1.5436468	1.448668 1.6448528
Grasselli, 2020	1.5424429	1.4473172 1.6438209
Gu, 2020	1.541069	1.4463067 1.6420403
Gu, 2020	1.5362892	1.4419804 1.6367661
Guisado-Vasco, 2020	1.5346212	1.4403979 1.6350082
Gupta, 2020	1.5425971	1.4475741 1.6438575
Gutierrez, 2020	1.5402496	1.441107 1.6462128
Haase, 2020	1.541796	1.4469641 1.6428429
Harrison, 2020	1.5433826	1.4483094 1.6446967
Narain, 2020	1.5444776	1.4496069 1.645557

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Hu, 2020	1.5414069	1.4467993	1.6422008
Imam, 2020	1.5233965	1.4304692	1.6223607
Kim, 2020	1.5359648	1.4414166	1.6367149
Kim, 2020	1.5423032	1.4471788	1.6436802
Klang, 2020	1.5443149	1.449327	1.6455281
Iaccarino, 2020	1.5368001	1.4421782	1.6376301
Lampasona, 2020	1.5336277	1.4393409	1.6340909
Lee, 2020	1.5330603	1.4390559	1.6332053
Li, 2020	1.5316471	1.4377936	1.6316268
Li, 2020	1.543546	1.4486066	1.6447074
Li, 2020	1.5410111	1.4462956	1.6419294
Liu, 2020	1.537012	1.4427184	1.6374683
Liu, 2020	1.5344501	1.4404455	1.6345894
Lombardi, 2020	1.5391879	1.444409	1.640186
Lunski, 2020	1.5452052	1.4502395	1.6463894
Maddaloni, 2020	1.5388812	1.4444	1.6395427
Mallow, 2020	1.5415205	1.4459832	1.64337
McCarty, 2020	1.5362325	1.4418561	1.6367861
Mccullough, 2020	1.5416657	1.4468532	1.6426914
Mendy, 2020	1.5371674	1.4427798	1.63773
Menezes-Soares, 2020	1.535192	1.4406074	1.6359866
Mikami, 2020	1.5455133	1.4505748	1.6466656
Mirani, 2020	1.5397638	1.4450005	1.6407416
Moon, 2020	1.530614	1.436552	1.6308349
Moon, 2020	1.5351865	1.4409643	1.6355697
Munblit, 2020	1.5340241	1.4396216	1.634617
Nachega, 2020	1.5413967	1.446624	1.6423783
Nakeshbandi, 2020	1.5433714	1.4483217	1.644659
Anzola, 2020	1.5396569	1.445129	1.640368
Nicholson, 2020	1.5342927	1.4400049	1.6347542
Nogueira, 2020	1.5475264	1.4526211	1.6486322
Palaiodimos, 2020	1.539917	1.4453287	1.6406955
Pan, 2020	1.5393479	1.444692	1.6402055
Perkin, 2020	1.5366874	1.4421818	1.637386
Petrilli, 2020	1.5445431	1.4495956	1.6457098
Pettit, 2020	1.542518	1.4479332	1.6432815
Poblador-Plou, 2020	1.538321	1.4433994	1.6394849
Portoles, 2020	1.535696	1.4411083	1.636492
Rastad, 2020	1.5370438	1.4424915	1.6377939
Recinella, 2020	1.5426378	1.4479485	1.6435194
Reilev, 2020	1.5376048	1.4427335	1.6387147

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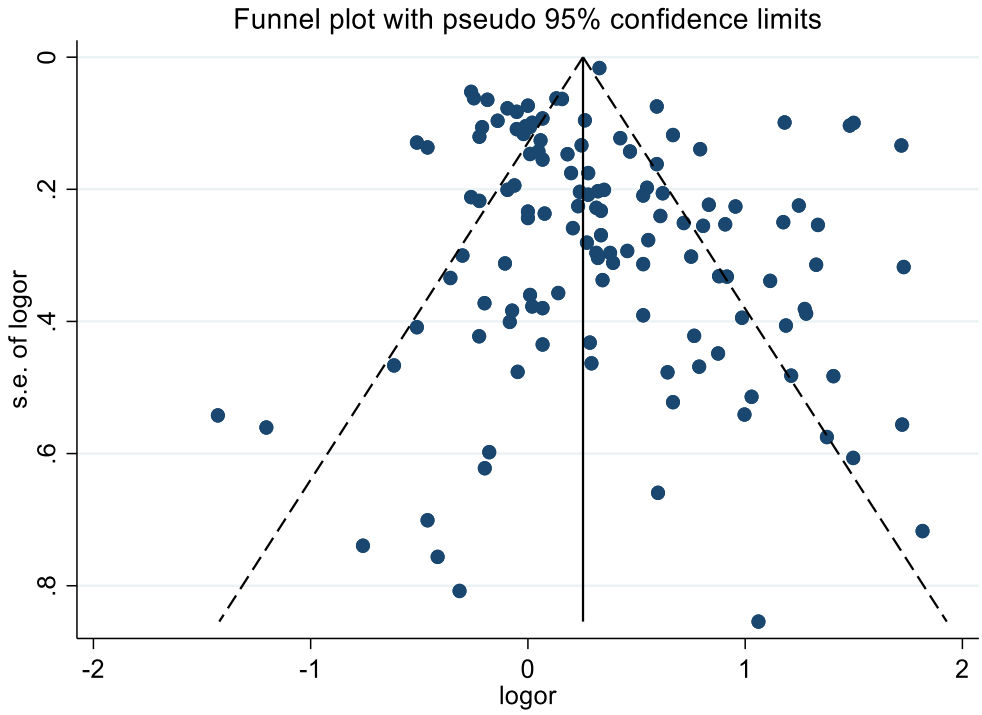
Rivera-Izquierdo, 2020	1.5335327 1.4392818 1.6339558
Rodrıguez-Molinero, 2020	1.5314742 1.4374141 1.6316895
Rossi, 2020	1.5375675 1.4428757 1.6384736
Rottoli, 2020	1.5418841 1.4470898 1.6428881
Rozaliyani, 2020	1.54023 1.4454752 1.6411963
Salacup, 2020	1.5383996 1.4439025 1.6390811
Salazar, 2020	1.5320424 1.4381227 1.6320957
Sands, 2020	1.5378478 1.4430354 1.6388896
Santos, 2020	1.5433739 1.4477333 1.6453327
Seiglie, 2020	1.5359882 1.441622 1.6365315
Shah, 2020	1.5382174 1.4436157 1.6390185
Shang, 2020	1.5354923 1.4411708 1.6359869
Shi, 2020	1.5378044 1.443289 1.6385093
Smith, 2020	1.5391345 1.4442916 1.6402056
Sousa, 2020	1.5397989 1.445085 1.6407204
Stony Brook COVID-19, 2020	Research Consortium 1.5362756 1.441714 1.6370395
Talavera, 2020	1.5393621 1.444719 1.6402053
Tartof, 2020	1.5421256 1.447147 1.6433378
Turcotte, 2020	1.5361555 1.4419419 1.6365249
Wollenstein-Betech, 2020	1.5441164 1.4472077 1.6475143
Wu, 2020	1.5398521 1.4451839 1.6407217
Wu, 2020	1.5378178 1.4433225 1.6384996
Xie, 2020	1.5365462 1.442167 1.6371019
Yahyavi, 2020	1.5406708 1.4456849 1.6418976
Yan, 2020	1.5409604 1.4461825 1.6419497
Yan, 2020	1.5381222 1.4434472 1.6390067
Yang, 2020	1.5376757 1.4430538 1.6385021
Yazdanpanah, 2020	1.5360827 1.4417021 1.636642
Yu, 2020	1.5335183 1.43924 1.6339724
Zandkarimi, 2020	1.5342196 1.4398625 1.6347601
Zhang, 2020	1.5332291 1.4392681 1.6333241
Zhou, 2020	1.53381 1.4396418 1.634138
Zhu, 2020	1.5385867 1.4437524 1.6396505
Zimering, 2020	1.5406566 1.4461459 1.6413438
An, 2020	1.5363024 1.441656 1.6371624
Almazeed, 2020	1.5392737 1.4448762 1.6398386
Bepouka, 2020	1.5390179 1.4444937 1.6397277
Biscarini, 2020	1.5398027 1.4451042 1.6407068
Abohamr, 2020	1.5307534 1.4367411 1.6309172
Czernichow, 2020	1.5404149 1.4453428 1.6417407
Docherty, 2020	1.5435574 1.4491569 1.6441072

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Foy, 2020	1.545491	1.4505337 1.6466644
Harmouch, 2020	1.5434103	1.4485663 1.6444643
Hewitt, 2020	1.544011	1.4490118 1.6452384
Ioannou, 2020	1.5431746	1.4480621 1.6445346
Kabarriti, 2020	1.5422004	1.4469972 1.6436675
Kolhe, 2020	1.5418558	1.4469479 1.6429888
Li, 2020	1.5343466	1.4403477 1.6344799
Meng, 2020	1.5472357	1.452306 1.6483704
Namendys-Silva, 2020	1.5329634	1.4388483 1.6332345
Neumann-Podczaska, 2020	1.5356869	1.4414098 1.6361301
Nie, 2020	1.5352678	1.4409221 1.6357909
Núñez-Gil, 2020	1.5301256	1.4361231 1.6302811
Park, 2020	1.5341833	1.4398587 1.6346871
Parker, 2020	1.5368429	1.4424789 1.6373801
Rosenberg, 2020	1.5379175	1.4430023 1.6390758
Shi, 2020	1.5396239	1.445085 1.6403475
Tehrani, 2020	1.5389986	1.4444335 1.6397549
Wang, 2020	1.5375615	1.4428405 1.6385009
Arshad, 2020	1.544768	1.4498014 1.6459552
Berenguer, 2020	1.5419745	1.4467498 1.6434668
Ciceri, 2020	1.5382968	1.4435889 1.6392183
Deng, 2020	1.5205251	1.428224 1.6187912
Wang, 2020	1.5330503	1.4390513 1.6331892
Combined	1.5379736	1.4437299 1.6383693

Supplementary Figure 2: Influence analysis of diabetes and mortality risk in patients with COVID-19

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Supplementary Figure 3: Funnel plot of hypertension and mortality risk in patients with COVID-19

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Study omitted	Estimate	[95% Conf. Interval]
Albitar, 2020	1.409848	1.2944931 1.5354825
Almazeedi, 2020	1.420794	1.3044698 1.5474913
An, 2020	1.4202995	1.3034456 1.5476295
Anzola, 2020	1.4250737	1.3083675 1.5521902
Arshad, 2020	1.42682	1.3096998 1.5544138
Atkins, 2020	1.4181341	1.3015437 1.5451684
Barbu, 2020	1.4032818	1.2905699 1.5258374
Bepouka, 2020	1.4205649	1.3040783 1.5474566
Berenguer, 2020	1.4144012	1.2976918 1.5416069
Biscarini, 2020	1.4260321	1.3090302 1.5534917
Boulle, 2020	1.4235396	1.3060689 1.551576
Bousquet, 2020	1.4184818	1.3022381 1.545102
Brouns, 2020	1.4248693	1.3082318 1.5519058
Calmes, 2020	1.4182589	1.3018024 1.5451334
Cao, 2020	1.4202471	1.3040493 1.5467987
Carrasco-Sánchez, 2020	1.4225355	1.3039615 1.5518919
Castelnuovo, 2020	1.4258839	1.3085872 1.5536945
Chachkhiani, 2020	1.4165195	1.3003861 1.5430244
Chen, 2020	1.4063388	1.2914803 1.5314124
Chen, 2020	1.4216918	1.3051524 1.5486373
Chen, 2020	1.4113356	1.2955974 1.5374128
Chilimuri, 2020	1.4177847	1.3014145 1.5445606
Ciardullo, 2020	1.4191939	1.3024282 1.5464277
Ciceri, 2020	1.4102607	1.2947223 1.5361094
Coppelli, 2020	1.4210031	1.3044831 1.5479311
Crouse, 2020	1.4099303	1.2945702 1.5355703
Cummings, 2020	1.416917	1.3006244 1.5436076
Czernichow, 2020	1.4236165	1.3062171 1.5515677
DE VITO, 2020	1.4222373	1.3059056 1.548932
DF de Souza, 2020	1.4171228	1.3003109 1.5444287
Deng, 2020	1.3963305	1.2868365 1.515141
Desai, 2020	1.4228125	1.3060952 1.54996
Du, 2020	1.4107752	1.295385 1.5364441
Escalera-Antezana, 2020	1.4110782	1.2955806 1.5368721
Esme, 2020	1.4221164	1.3035688 1.5514448
Fang, 2020	1.4226052	1.3060334 1.5495815
Foy, 2020	1.422677	1.3055518 1.55031
Fried, 2020	1.4274895	1.3108877 1.554463
Galloway, 2020	1.4198248	1.3028002 1.5473611

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Gao, 2020	1.4137607	1.2978215	1.5400571
Ghany, 2020	1.4183894	1.3019762	1.5452114
Giacomelli, 2020	1.4181474	1.3018019	1.5448911
Grasselli, 2020	1.423923	1.3065236	1.5518717
Gu, 2020	1.4185693	1.3020047	1.5455697
Guisado-Vasco, 2020	1.4154416	1.2991362	1.5421594
Gupta, 2020	1.4226934	1.3054665	1.5504467
Gutierrez, 2020	1.4256142	1.2931122	1.5716934
Haase, 2020	1.4236912	1.3067129	1.5511414
Harmouch, 2020	1.4136345	1.2976446	1.5399922
Hashemi, 2020	1.4149497	1.2990468	1.5411937
Hewitt, 2020	1.4244795	1.3071433	1.5523484
Hu, 2020	1.420043	1.3035676	1.5469255
Iaccarino, 2020	1.4224738	1.3056414	1.5497606
Imam, 2020	1.3971469	1.2870134	1.5167049
Ioannou, 2020	1.4247838	1.3071853	1.5529618
Kabarriti, 2020	1.4241887	1.3062947	1.5527228
Kim, 2020	1.4190946	1.3024367	1.5462012
Kim, 2020	1.4222981	1.3052446	1.5498487
Klang, 2020	1.4207023	1.303695	1.5482111
Lee, 2020	1.4160867	1.3001062	1.5424136
Li, 2020	1.4208537	1.304601	1.5474657
Li, 2020	1.4118376	1.2960577	1.5379604
Liu, 2020	1.4229921	1.3063434	1.5500568
Liu, 2020	1.4283701	1.3117267	1.5553858
Lombardi, 2020	1.4196392	1.3029786	1.5467449
Lopez-Mendez, 2020	1.4123951	1.2968653	1.5382168
Lunski, 2020	1.4306874	1.3140447	1.5576842
MCCULLOUGH, 2020	1.4113107	1.2956346	1.5373144
Maddaloni, 2020	1.4137343	1.2980008	1.5397871
Magleby, 2020	1.4250162	1.308177	1.5522907
Mallow, 2020	1.4274716	1.3114311	1.55378
Matsuzawa, 2020	1.4213893	1.3050824	1.5480613
McCarty, 2020	1.4147518	1.2988299	1.5410198
Meng, 2020	1.4232006	1.3060794	1.5508245
Mikami, 2020	1.4254555	1.3079046	1.5535716
Mirani, 2020	1.4148802	1.2987224	1.541427
Moon, 2020	1.4160312	1.3000233	1.5423911
Mowla, 2020	1.4159008	1.3000853	1.5420334
Munblit, 2020	1.4157315	1.2994165	1.5424583
Munoz, 2020	1.4121697	1.2966897	1.5379342

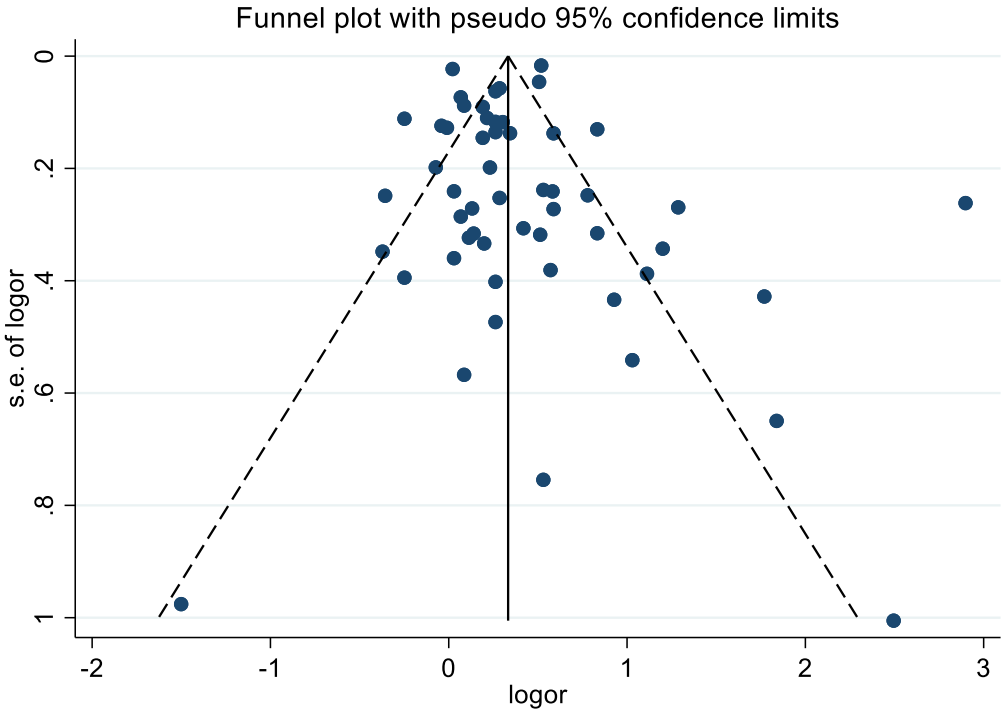
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Nachega, 2020	1.4223779 1.3055705 1.5496361
Nakeshbandi, 2020	1.4267973 1.3096594 1.5544121
Namendys-Silva, 2020	1.4127643 1.2969795 1.5388856
Narain, 2020	1.4267741 1.3096164 1.5544127
Nicholson, 2020	1.4146136 1.2983321 1.5413096
Nie, 2020	1.406029 1.2912035 1.5310657
Núñez-Gil, 2020	1.4157708 1.2995708 1.5423607
Palaiodimos, 2020	1.4217563 1.3051939 1.5487283
Pan, 2020	1.4125925 1.2967428 1.5387923
Parker, 2020	1.4141377 1.2983069 1.5403026
Perkin, 2020	1.4182899 1.3017601 1.5452511
Petrilli, 2020	1.4239464 1.3066469 1.5517758
Pettit, 2020	1.4267213 1.3101194 1.553701
Poblador-Plou, 2020	1.4229058 1.3053201 1.5510838
Recinella, 2020	1.4212246 1.3046689 1.548193
Reilev, 2020	1.4198674 1.3024342 1.5478889
Rivera-Izquierdo, 2020	1.4176491 1.301313 1.5443856
Rodrıguez-Molinero, 2020	1.4035665 1.2891709 1.5281131
Rosenberg, 2020	1.4131894 1.2970421 1.5397373
Rossi, 2020	1.4164165 1.2998153 1.5434775
Rottoli, 2020	1.4197984 1.3031938 1.5468363
Rozaliyani, 2020	1.4145004 1.2983296 1.5410658
Russo, 2020	1.4240398 1.3070652 1.5514829
Salacup, 2020	1.4208176 1.3044074 1.5476167
Salazar, 2020	1.4101171 1.2948774 1.5356129
Sands, 2020	1.42993 1.3131154 1.5571364
Seiglie, 2020	1.4185754 1.3022965 1.5452365
Shah, 2020	1.4121155 1.2965516 1.5379798
Shang, 2020	1.4250045 1.3082117 1.5522242
Shi, 2020	1.4124031 1.2966639 1.5384731
Shi, 2020	1.4202402 1.3038338 1.5470392
Stony Brook COVID-19	Research Consortium, 2020 1.40791881.2927667 1.5333279
Sun, 2020	1.4207453 1.3044379 1.547423
Talavera, 2020	1.4189814 1.3024817 1.5459015
Tartof, 2020	1.4185152 1.3018947 1.5455824
Tehrani, 2020	1.4184757 1.3020438 1.5453192
Wang, 2020	1.3947774 1.2842842 1.5147767
Wang, 2020	1.4169366 1.3009491 1.5432652
Wang, 2020	1.4196205 1.3029077 1.5467883
Wu, 2020	1.4162004 1.3000072 1.5427787
Wu, 2020	1.4254589 1.3084775 1.5528989

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Xie, 2020	1.4142019	1.2984338	1.5402918
Yang, 2020	1.4126209	1.2969178	1.5386463
Yazdanpanah, 2020	1.4077736	1.2926683	1.5331283
Yu, 2020	1.4214995	1.3047308	1.5487185
Zangrillo, 2020	1.412156	1.2967449	1.5378386
Zhou, 2020	1.4105095	1.2950244	1.536293
Combined	1.4180375	1.3021785	1.5442047

Supplementary Figure 4: Influence analysis of hypertension and mortality risk in patients with COVID-19



Supplementary Figure 5: Funnel plot of obesity and mortality risk in patients with COVID-19

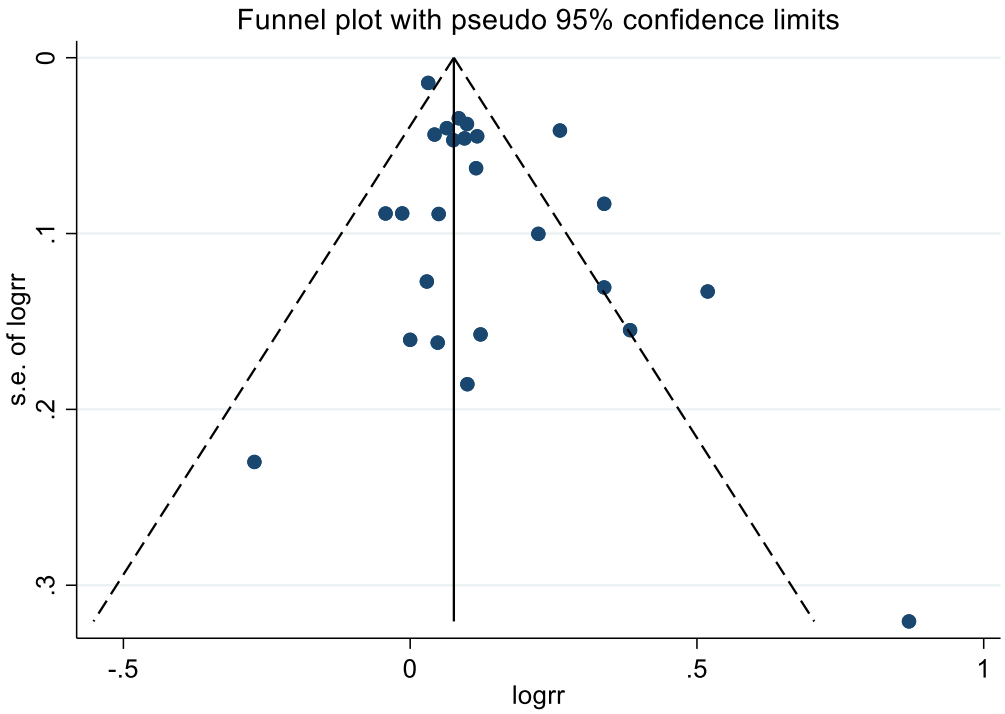
Study omitted	Estimate	[95% Conf.	Interval]
Almazeedi, 2020	1.4554933	1.3114676	1.6153359
McCarty, 2020	1.4626863	1.3171904	1.6242536
Mccullough, 2020	1.4678108	1.3216922	1.6300832
Salazar, 2020	1.4587231	1.3135937	1.6198866
Arshad, 2020	1.4721547	1.3261496	1.6342347
Stony Brook COVID-19, 2020	1.4625834	1.3163379	1.6250767
Rosenberg, 2020	1.4645751	1.317825	1.6276668
Price-Haywood, 2020	1.4633511	1.3166041	1.6264545
Salacup, 2020	1.4560728	1.3184568	1.6080528
Hashemi, 2020	1.454875	1.3098872	1.6159111
Biscarini, 2020	1.4580365	1.3122839	1.6199776
Fried, 2020	1.4623791	1.3147643	1.6265674
Tehrani, 2020	1.4557939	1.3104337	1.6172782
Kim, 2020	1.4611641	1.3138261	1.6250254
Mendy, 2020	1.4512935	1.3070762	1.6114231
Rodrıguez-Molinero, 2020	1.4539628	1.3089103	1.61509
Crouse, 2020	1.4545934	1.3092545	1.6160662
Hajifathalian, 2020	1.4535341	1.3084871	1.6146597
Berenguer, 2020	1.4570931	1.3097397	1.6210245
Castelnuovo, 2020	1.4555653	1.3092505	1.6182317
Portoles, 2020	1.4520395	1.3071913	1.6129383
Ramos-Rincon, 2020	1.4555131	1.3086935	1.618804
Lombardi, 2020	1.4531474	1.3074876	1.6150345
Lunski, 2020	1.4534943	1.3069357	1.616488
Mallow, 2020	1.4557445	1.3068982	1.6215434
Rossi, 2020	1.4502716	1.3058093	1.6107157
Nakeshbandi, 2020	1.4531265	1.3068702	1.615751
Parker, 2020	1.4498988	1.3056446	1.610091
Docherty, 2020	1.4553313	1.3059231	1.6218332
Mirani, 2020	1.450893	1.3057793	1.6121336
Zhang, 2020	1.4518838	1.3054672	1.6147219
Sands, 2020	1.4500164	1.3040735	1.6122923
Núñez-Gil, 2020	1.4474576	1.3029506	1.6079915
Wollenstein-Betech, 2020	1.4460288	1.2970564	1.6121114
Xie, 2020	1.4454393	1.3012034	1.6056634
Gutierrez, 2020	1.4410919	1.2967353	1.6015188
Sousa, 2020	1.447451	1.3037943	1.6069363
Pettit, 2020	1.4440517	1.2996014	1.6045576
DF de Souza, 2020	1.4449612	1.3009996	1.6048528
Shah, 2020	1.4426155	1.2983586	1.6029004

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Imam, 2020	1.4404271	1.295764	1.6012409
Calmes, 2020	1.4430385	1.2988892	1.6031855
Rottoli, 2020	1.4371952	1.2937157	1.5965873
Klang, 2020	1.4298403	1.2871754	1.5883176
Nachega, 2020	1.4382833	1.2949491	1.5974829
Alamdari, 2020	1.4401046	1.2969043	1.5991166
Coppelli, 2020	1.4412719	1.2981442	1.6001803
Giacomelli, 2020	1.4355083	1.2928371	1.5939237
Yazdanpanah, 2020	1.4316087	1.2893935	1.5895095
Gazzaruso, 2020	1.4244035	1.2831527	1.5812036
Mather, 2020	1.4267011	1.2856641	1.5832098
Steinberg, 2020	1.4363017	1.2941382	1.5940822
Escalera-Antezana, 2020	1.4401749	1.2977738	1.5982014
Abohamr, 2020	1.3768033	1.2478951	1.5190276
Combined	1.4483488	1.3049973	1.6074472

Supplementary Figure 6: Influence analysis of obesity and mortality risk in patients with COVID-19

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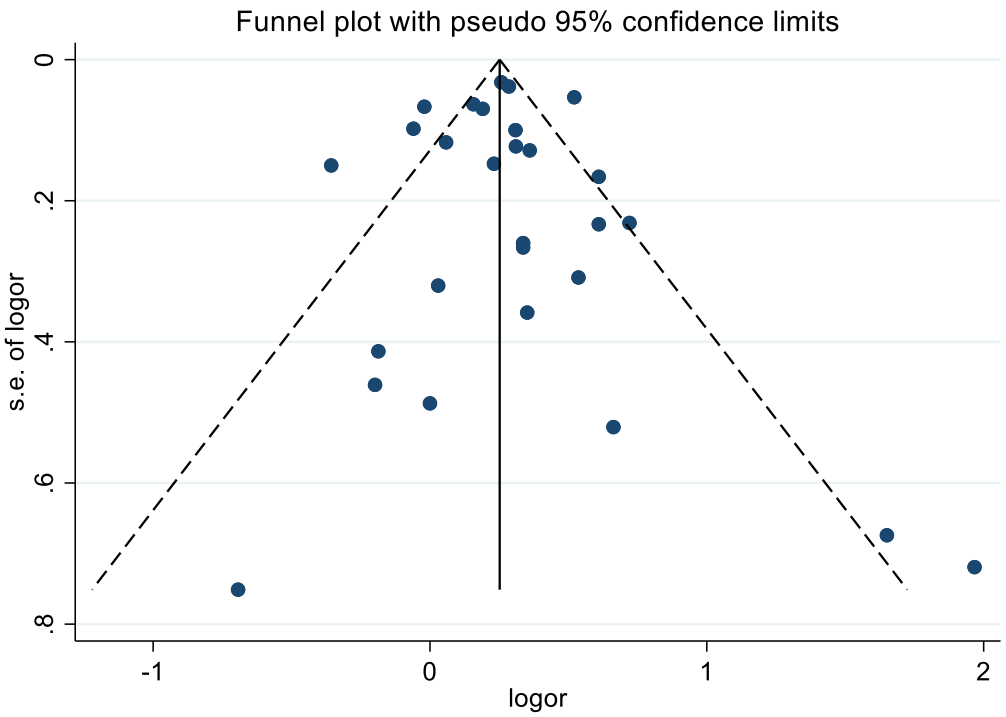


Supplementary Figure 7: Funnel plot of BMI per 5kg/km^2 and mortality risk in patients with COVID-19

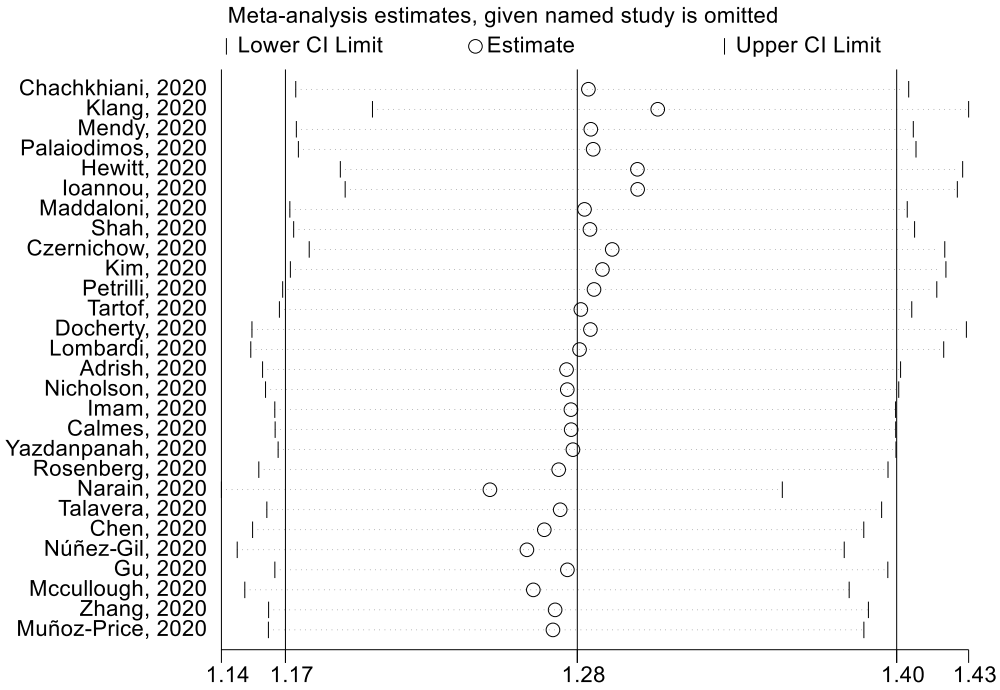
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Study omitted	Estimate	[95% Conf.	Interval]
Salazar, 2020	1.1262932	1.0773913	1.1774148
Haase, 2020	1.1295877	1.0794542	1.1820494
Li, 2020	1.1284989	1.0781709	1.1811761
Calmes, 2020	1.1250597	1.0753614	1.177055
Hajifathalian, 2020	1.1252736	1.0753136	1.1775547
Ferrando-Vivas, 2020	1.1313956	1.0795149	1.1857696
Anderson, 2020	1.1291318	1.0770124	1.1837733
Seiglie, 2020	1.1241773	1.0744675	1.176187
Lombardi, 2020	1.1259074	1.0754082	1.1787779
Petrilli, 2020	1.1278753	1.0753478	1.1829686
Kabarriti, 2020	1.126538	1.0745648	1.1810251
Santos, 2020	1.1266743	1.0735707	1.1824048
Gupta, 2020	1.1251055	1.0731854	1.1795374
Wang, 2020	1.1251912	1.0726496	1.1803064
Palaiodimos, 2020	1.1231542	1.073586	1.175011
Nakeshbandi, 2020	1.1232269	1.0722753	1.1765996
Klang, 2020 (b)	1.1233695	1.0716629	1.1775709
Gu, 2020	1.1227893	1.0731335	1.1747427
Shah, 2020	1.1186447	1.0692952	1.170272
Czernichow, 2020	1.1065391	1.0623916	1.1525211
Rottoli, 2020	1.116652	1.0682974	1.1671954
Gayam, 2020	1.1111181	1.0643468	1.1599447
Baronio, 2020	1.1171073	1.0688614	1.1675309
Klang, 2020 (a)	1.1115534	1.0657233	1.1593543
Muñoz-Price, 2020	1.1179643	1.0707144	1.1672993
Chachkhiani, 2020	1.1227052	1.0738231	1.1738126
Salazar, 2020	1.1227052	1.0738231	1.1738126
Ghany, 2020	1.1227052	1.0738231	1.1738126
Zimering, 2020	1.1227052	1.0738231	1.1738126
Tartof, 2020	1.1227052	1.0738231	1.1738126
Ioannou, 2020	1.1227052	1.0738231	1.1738126
Recinella	1.1227052	1.0738231	1.1738126
Harmouch, 2020	1.1227052	1.0738231	1.1738126
Combined	1.1227053	1.0738231	1.1738127

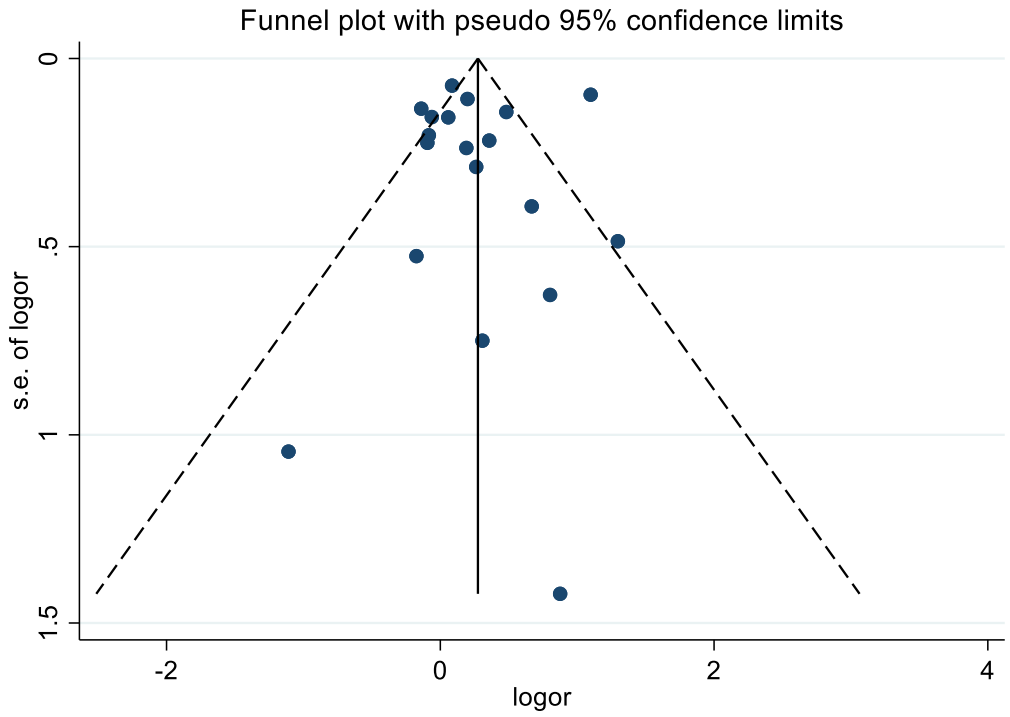
Supplementary Figure 8: Influence analysis of BMI per 5kg/km^2 and mortality risk in patients with COVID-19



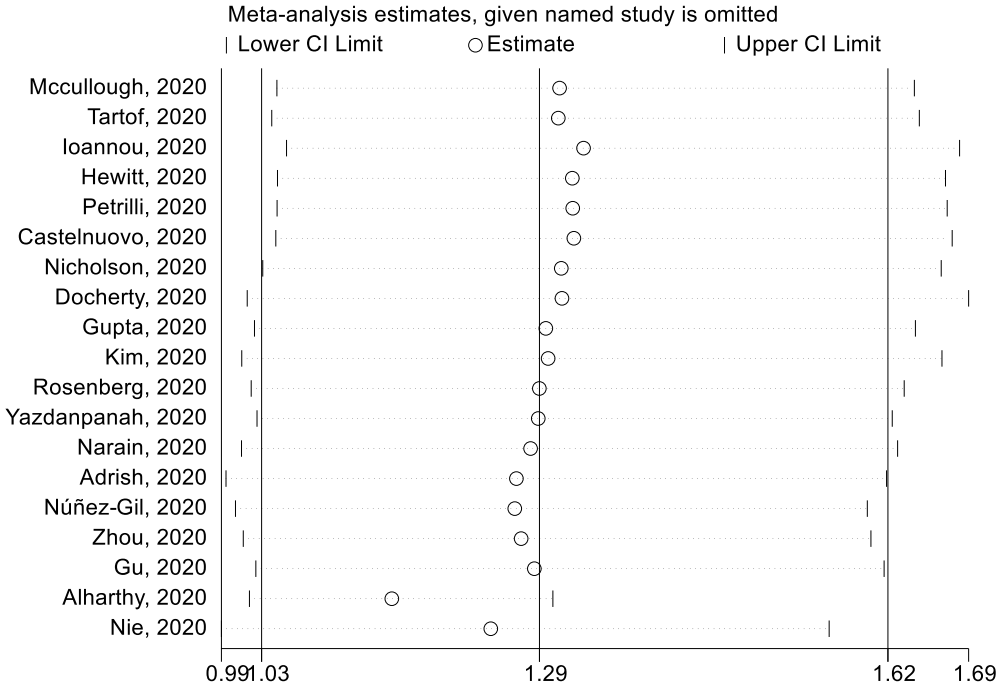
Supplementary Figure 9: Funnel plot of ever smoking vs never smoking and mortality risk in patients with COVID-19



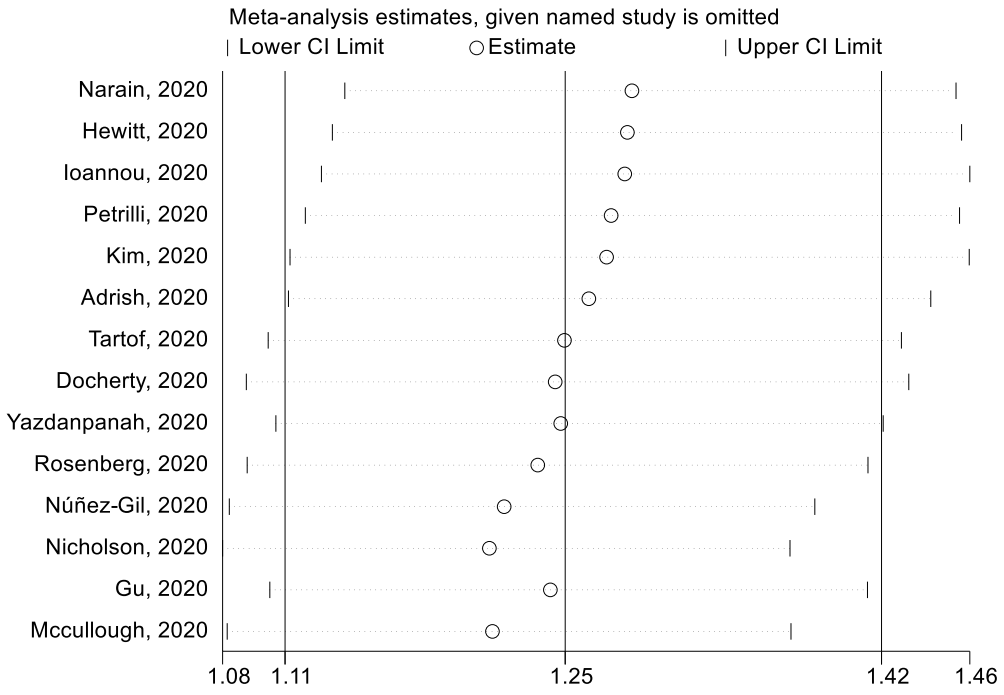
Supplementary Figure 10: Influence analysis of ever smoking vs never smoking and mortality risk in patients with COVID-19



Supplementary Figure 11: Funnel plot of current smoking vs never smoking and mortality risk in patients with COVID-19

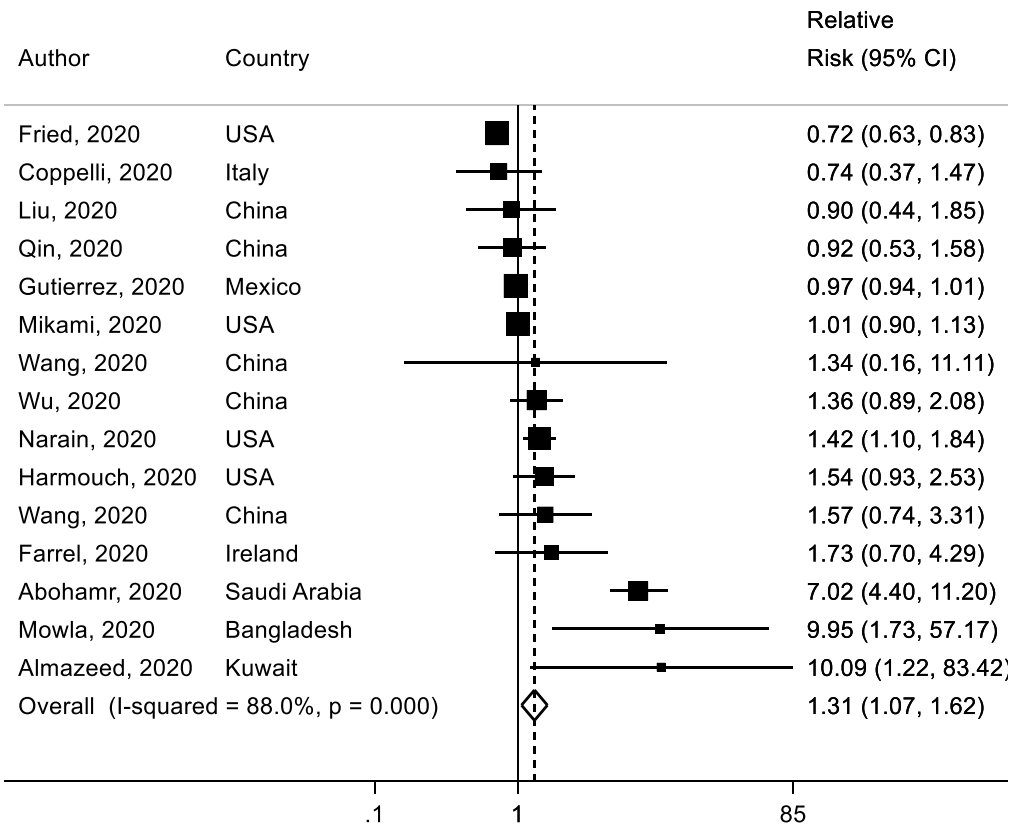


Supplementary Figure 12: Influence analysis of current smoking vs never smoking and mortality risk in patients with COVID-19



Supplementary Figure 13: Influence analysis of former smoking vs never smoking and mortality risk in patients with COVID-19

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Supplementary Figure 14: Association between undefined smoking status (versus never smoking) and mortality risk in patients with COVID-19

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Page 1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Page 3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Pages 5-6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Page 6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Page 6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Pages 6-7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Page 6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Page 6, supplement text
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Pages 6-7, Figure 1
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Page 7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Page 6-7, Table S1
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Page è
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Pages 7-9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Pages 7-9

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Page 8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Pages 7-9
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Page 9, Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Page 9, Table S1
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Page 9, Table S2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Table S1, Figures 2-3, 5-7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Pages 9-12
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Table S2, Pages 9-12
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Pages 9-13, Table 1-2, Figures S1-14
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Pages 13, 16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Page 16-17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Page 17
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Page 18

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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